HANDBOOK FOR MARINE RADIO COMMUNICATION

FOURTH EDITION



G. D. LEES
and W. G. WILLIAMSON

LLP

Handbook for Marine Radio Communication

4th edition

By
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and
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Foreword

The twentieth century's astonishing developments in maritime communications meant that from the outset regulation of procedures was essential. However even for the specialist the international standards made difficult reading. It was this that caused the British Post Office in 1908 to produce the first readable compendium, known as the Handbook for Wireless Operators. In its various editions it was the standard reference text for the next 90 years.

That the adoption of the automated Global Maritime Distress and Safety System in 1999 led to the demise of the specialist radio officer was no surprise. However, the decision to no longer publish a communicators' handbook was seen by many of us in the colleges as unwise, even foolhardy. We were especially mindful that interpretation of the regulations could be daunting to the non-specialist, even though licensed.

Graham Lees and William Williamson have done great service to maritime radio communications by publishing their unique successor to the traditional handbook. Skilled practitioners of examining and teaching with long experience in marine radio, they have produced a compilation of what the present day communicators need to know.

It is a measure of the rapid evolution in this field that this is the authors' fourth edition of the Handbook for Marine Radio Communication since 1993. In this volume, whilst retaining the trusted formula, they take the communications theme to new levels as they embrace the latest developments in satellites and automated information systems. They also look to what the future will hold, with the Galileo navigation system for example.

Each licensed operator of marine radio equipment should use this book as a standard work of reference. At the very least, it should be beside each operating console and in the libraries of training centres. In commending it, I am confident it will make a serious contribution to safety at sea.

PROFESSOR BRIAN N. COTTON,
C.ENG.
Editor of AMERCNews
Chairman of the Radio Officers' Association

Preface to 4th edition

The Global Maritime Distress and Safety System is now a well established maritime communications system. Operational experience gained over a number of years has led to changes in the regulations relating to GMDSS and innovative ways of using the system have also occurred. Major changes in technology have been introduced to GMDSS since the last edition of this work was published, for example Inmarsat has launched their Fleet services with priority preemption. The IMO has advocated the use of GMDSS for communication during piracy attacks and there has been significant changes in the maintenance and testing of EPIRBs. In this edition the distress chapter and satellite communications chapters have been amended to reflect these changes, and in view of their importance, a separate chapter has been devoted to EPIRBs and other sub-systems.

Regulations have not been included in full in this book but a number of sections are based on the provisions of the following:

(a) ITU Radio Regulations;
(b) International Convention for Safety of Life at Sea SOLAS (1974) and amendments (1988);
(c) ITU Manual for use by the Maritime Mobile and Maritime Mobile-Satellite Services (2002);
(d) Merchant Shipping (Radio Installations) Regulations 1998;
(e) The Merchant Shipping (Radio) (Fishing Vessels) Rules 1974;
(f) The Merchant Shipping (Radio) (Fishing Vessels) (Amendment) Rules 1982;
(g) Various Merchant Shipping Notices and publications issued by the Maritime and Coastguard Agency;
(h) Various publications issued by Ofcom;
(i) Inmarsat Maritime Communications Handbook. (Issue 4 of May 2002);

and other relevant statutory regulations and provisions. Readers should consult with appropriate international and national regulations for definitive interpretation of the rules and regulations.

We have taken the opportunity to outline the role played by AMERC in conducting the operation of GMDSS examinations in the UK and elsewhere. The syllabus for ENEM and GMDSS Radio Maintenance courses and the examination structure leading to the ETO qualifications has recently been finalised and we are grateful to AMERC for allowing us to include it in this edition. As automated communication system are now firmly in place within the maritime industry, we anticipate that GMDSS will continue to evolve and play an important part in Safety of Life at Sea for many years to come.

We are indebted to many people and organisations for their contribution and assistance in the preparation of this edition, including the following:

Prof. Brian Cotton, Chairman of the ROA and editor of *AMERC News*.
Phil Davies, Short Course Manager, the Lairdside Maritime Centre, Birkenhead.
Andy Fuller, IMSO, London.
Lynda Goulding, The EPIRB Register, HM Coastguard, Falmouth.
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Staff at Lairdside Maritime Centre, Birkenhead.

We would also like to extend our grateful thanks to the Maritime and Coastguard Agency for their kind permission to reproduce important sections from various M Notices.

Finally, we have improved the layout and increased the size of the book in the hope that it will be easier for readers to access all the reference material in this work.

W. G. WILLIAMSON
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July 2004

About the authors

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Born in Liverpool and raised in Bebington Wirral. Educated at Calday Grange Grammar School, West Kirby, and trained as Radio Officer at Riversdale College of Technology, Liverpool. Spent five years serving as Radio Officer followed by a further six years as Radio and Electronics Officer on a wide range of UK registered vessels including passenger ships, general cargo, oil and chemical tankers, gas carriers and car/bulk cargo carriers.

Started teaching marine radio and electronics at Riversdale College in 1976 and became Senior Lecturer in charge of HND and GMDSS courses. After completing 21 years teaching has now taken early retirement to concentrate on technical writing and advisory work specialising in the training needs of those involved in the marine radio communications and electronic navigational aids industries. In addition, continues role as GMDSS examiner on behalf of MCA/AMERC.

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Born Glasgow 1943 and joined the Merchant Navy as a Radio Officer following initial training at the Glasgow Wireless College. He spent seven years at sea during which time he served on a variety of vessels including general cargo ships, tramps and tankers. He continued his involvement with the marine industry as a Marine Electronic Engineer based at Liverpool, gaining wide servicing experience on UK and foreign flag vessels. During this period he served for some years as an executive member of the Radio Officers' Union.

After 20 years with the Marconi Marine Company, he joined the staff at Riversdale College of Technology in 1980 as a lecturer primarily concerned with Radio Officer training. Following reorganisation became a Senior Lecturer at Liverpool Community College with particular responsibilities for all GMDSS courses, GOC, ROC and LRC until he retired in 2001. He currently teaches part time on GMDSS courses at the Lairdside Maritime Centre. He remains an

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List of acronyms and abbreviations

AAIC	Accounting Authority Identification Code
A/C rain	Anti-clutter rain (control)
ADE	Above Deck Equipment
Admin	Administration
AGC	Automatic Gain Control
AIRS	Automatic Identification and Reporting System
AIS	Automatic Identification System
ALRS	Admiralty List of Radio Signals
AM	Amplitude Modulation
AMERC	Association of Marine Electronic and Radio Colleges
AMVER	Automated Mutual-assistance Vessel Rescue
AOR-E	Atlantic Ocean Region - East
AOR-W	Atlantic Ocean Region - West
ARPA	Automatic Radar Plotting Aid
ARQ	Automatic repetition request. Mode of telex operation.
ASF	Additional Secondary Factor
ASGD	Alarm Signal Generating Device
ASM	At sea maintenance
ATO	Authority to Operate
AUSREP	Australian Reporting System for shipping
BBC	British Broadcasting Corporation
BBS	Bulletin Board Service
BDE	Below Deck Equipment
BFO	Beat Frequency Oscillator
BIH	Bureau International de l'Heure
BITE	Built in test equipment
BS	British Standard
BT	British Telecom
BTEC	Business and Technology Education Council
BQ	Reply to a "RQ"
CAA	Civil Aviation Authority
C/A	Coarse/Acquire (code)
CCIR	International Radio Consultative Committee.
CCITT	International Consultative Committee for Telegraph and Telephone
CEPT	Conference of European Postal and Telecommunications Administrations
CES	Coast Earth Station.
Ch.	Channel
CIS	Commonwealth of Independent States

cm	centimetre
COMSAR	Committee on Radiocommunication and Search and Rescue
COSPAS-SARSAT	Search and Rescue Satellite System
CQ	General call to all stations
CR	Carriage return
CSR	Coastal Station Radio
CSS	Coordinator Surface Search
CW	Continuous wave
DE	Signal used in telex as an abbreviation meaning "This is"
DETR	Department of the Environment, Transport and the Regions
DF	Direction Finding (or Finder)
DGPS	Differential Global Positioning System
DHSD	Duplex High Speed Data
DMG	Distress message generator
DNIC	Data Network Identification Code
DNS	Decca Navigator System
DOE	Duplication of equipment
DRCS	Distress Radio Call System
DOS	Disc operating system
DRP	Digital Receiver Processor
DR	Dead reckoning
DSB	Double sideband
DSC	Digital Selective Calling
D.Tp.	Department of Transport. Now the Department of the Environment, Transport and the Regions
DTI	Department of Trade and Industry
DUTI	Difference Universal Astronomical Time
DVM	Digital voltmeter
ECDIS	Electronic chart display
EGC	Enhanced Group Call.
EGNOS	European Geostationary Navigation Overlay System
ELT	Emergency Locator Transmitters. Used with the COSPAS-SARSAT system.
E-Mail	Electronic mail
ENEM	Electronic Navigational Equipment Maintenance
EOS	End of sequence
EPIRB	Emergency Position Indicating Radio Beacon
ETA	Estimated time of arrival
ETO	Electro Technical Officer
ETSI	European Technical Standards Institution
EU	European Union
EUMETSAT	European Meteorological Satellite Organisation
FAX	Facsimile
FEC	Forward Error Correction. Mode of telex operation for broadcast purposes.
FDR	Federal Democratic Republic (of Germany)
FM	Frequency Modulation
Freq	Frequency
FSK	Frequency shift keying
FTC	Fast Time Constant

FU	Follow up
GA+	Go ahead (used with telex)
GEOREP	Georgian ship reporting system
GEOSAR	Geostationary Search and Rescue Satellite
GHz	gigahertz (1 000 000 000 Hertz)
GLONASS	Global Navigation Satellite System
GMDSS	Global Maritime Distress and Safety System
GMT	Greenwich Mean Time
GOC	General Operator's Certificate
GPS	Global Positioning System
GRI	Group Repetition Interval
g.t.	Gross tons
h	Hour
HBW	Horizontal beamwidth
Hex	Hexadecimal
HF	High Frequency (3-30 MHz)
HF DSC	High Frequency Digital Selective Calling
HM	Her Majesty
HMCG	Her Majesty's Coastguard
HNC	Higher National Certificate
HND	Higher National Diploma
HSD	High Speed Data
IAMSAR	International Aeronautical and Maritime Search and Rescue Manual
ICAO	International Civil Aviation Organisation
IC	Integrated circuit
ICC	International Chamber of Commerce
IDD	International Direct Dialling
ID	Identity
IF	Intermediate Frequency
IMB	International Maritime Bureau
IMF	International Monetary Fund
IMN	Inmarsat Mobile Number
IMO	International Maritime Organisation
IMOT	Israeli ship reporting system
IOC	International Oceanographic Commission
IODC	International operator direct dialling
Inmarsat	International Maritime Satellite (organisation)
INSPIRES	Indian ship reporting system
IMSO	International Mobile Satellite Organisation
IOR	Indian Ocean Region
IRS	Information Receiving Station
ISDN	Integrated Switched Digital Network
ISS	Information Sending Station
ITU	International Telecommunication Union
JASREP	Japanese ship reporting system
kHz	kilohertz (1 000 Hertz)
km	kilometre
kW	kilowatt (1 000 Watts)
LAN	Local area network

LES	Land Earth Station
LEOSAE	Low Earth Orbiting Satellite
LF	Line Feed
LOP	Line of Position
LRC	Long Range Certificate
LS	Letter shift
LSB	Lower sideband
LUT	Local User Terminal. Type of station used in the COSPAS-SARSAT system.
m	metre
MAP	Manual Assistance Point. Used for operator assistance on telex.
MAREP	British ship reporting system
MCA	Maritime and Coastguard Agency
MCC	Mission Control Centre. Type of station used in the COSPAS-SARSAT system.
MEDEVAC	Medical evacuation
METAREA	Meteorological Area (Used in WWNWS)
MES	Mobile Earth Station
MF	Medium Frequency (300-3000 kHz)
MF/HF	Medium Frequency/High Frequency
MGN	Marine Guidance Note
MHz	Megahertz (1 000 000 Hertz)
MID	Maritime Identification Digits
Mins	Minutes
MMSI	Maritime Mobile Service Identities
MPDS	Mobile Packet Data Service
MRCC	Maritime Rescue Coordination Centre (Often referred to as an RCC)
MRGC	Marine Radiocommunication General Certificate
MRSC	Maritime Rescue Sub-Centre
MSA	Maritime Safety Agency
MSC	Maritime Safety Committee
MSI	Maritime Safety Information.
MSN	Merchant Shipping Notice
NAC	National Administration Centre
NAVAREA	Navigational area in the WWWS
NAVSTAR	Alternative name for GPS system
NAVTEX	Navigational telex
NCS	Network Coordination Station.
NBDP	Narrow Band Direct Printing, also known as radiotelex
NELS	Northwest European Loran-C System
NFU	Non follow up
NiCad	Nickel cadmium
NNSS	Navy Navigation Satellite System
OBS	Prefix for meteorological messages
OCC	Operations Control Centre. Coordinating centre for Inmarsat.
Ofcom	Office of Communication
OOW	Officer on Watch
OPT	Operation and performance testing

OSC	On-scene commander
OTF	Optimum Transmitting Frequency
P	Precision (code)
PAD	Packet Assembly/Disassembly
PC	Personal computer
pdf	Picture definition format
PIN	Personal Identification Number
PLB	Personal Locator Beacon. Used with the COSPAS-SAESAT system.
POR	Pacific Ocean Region
PRF	Pulse repetition frequency
PSDN	Public Switched Data Network
PSTN	Public Switched Telephone Network
PTS	Proceed to select
PVT	Performance verification test
RA	Radiocommunications Agency (this is now Ofcom)
RACON	Type of radar beacon
RAMARK	Type of radar beacon
RCC	Rescue Coordinating Centre. (Shortened version of MRCC)
REC	Regional Examination Centre
RF	Radio frequency
ROC	Restricted Operator's Certificate
RQ	Request, or indication of a request
RT	Radiotelephony
RTL	Radiotelex Letter
RTT	Radio Teletype
sec.	second
SA	Selective availability
SAR	Search and Rescue
SARSAT	Search and Rescue Satellite Aided Tracking
SART	Search and Rescue Transponder. A homing device used in GMDSS.
SBAS	Satellite Based Augmentation System
SBM	Shore Based Maintenance
SCOTVEC	Scottish Technician and Vocational Education Council
SDR	Special drawing right
Selcall	Selective calling
SELFEC	Selective Forward Error Correction. One-way communication system used on telex.
SES	Ship Earth Station (Used before MES was adopted)
SFU	Store and forward unit
SHIPPOS	Danish ship reporting system
SINGREP	Singapore ship reporting system
SITREP	Situation report
SLT	Ship Letter Telegram
SMC	SAR Mission Coordinator
SOLAS	Safety Of Life At Sea.
SPOC	Search and Rescue Point of Contact
SRC	Short Range Certificate
SRR	Search and Rescue Region

SSAS	Ship Security Alert System
SSB	Single sideband
SSFC	Sequential Single Frequency Code
SSRS	Saudi Arabian ship reporting system
STCW	Standards of Certification and Training
SURNAV	French ship reporting system
SURPIC	Surface Picture
TAI	International Atomic Time
TDM	Time Division Multiplex. Relates to several communications all allocated time slots in which to transmit or receive.
TLX	Telex
TOR	Telex Over Radio
TR	Trade Route. Relates to details of ships name, position etc. to assist with traffic routing at coast stations.
TV	Television
UHF	Ultra High Frequency
UK	United Kingdom
UPS	Universal power source
USA	United States of America
USSR	Union of Soviet Socialist Republics
USB	Upper sideband
USCG	United States Coast Guard
UTC	Coordinated Universal Time
UT or UTI	Universal Astronomical Time
VBW	Vertical beamwidth
VDU	Visual Display Unit
VHF	Very High Frequency (30-300 MHz).
VMS	Vessel Monitoring System
VTS	Vessel Traffic Service
W	Watt
WAAS	Widea Area Augmentation System
WARC	World Administrative Radio Conference
WARC	World Radio Conference (Shortened version of WARC)
WMO	World Meteorological Organisation
WRU	Who are you? (used with telex)
WT	Wireless Telegraphy. Radiocommunication using Morse.
WWNWS	World Wide Navigational Warning Service
YTD	Yacht Telephone Debit

Glossary of terms and definitions

Aircraft station. A mobile station in the aeronautical mobile service, other than a survival craft station, located on board an aircraft

Call attempt. One or several call sequences directed to the same receiving station on one or more frequencies and within a few minutes. The call is considered to be unsuccessful if no reply is received within this time period.

Class of emission. The set of characteristics of an emission, designated by standard symbols, e.g. type of modulation of the main carrier, modulating signal, type of information to be transmitted, and also, if appropriate, any additional signal characteristics.

Coast Station. A land station in the maritime mobile service.

Controlling operator. The first land-based operator handling the radiotelegram, radiotelex call or RT call in the direction from the mobile station.

COSPAS-SARSAT. A search and rescue beacon locating system based on low orbiting satellites in near polar orbits which are designed to respond to transmissions from beacons on 121.5 MHz and 406 MHz.

Distress Alerting. Rapid reporting of a distress situation to authorities who can provide coordinated assistance to the station in distress.

Digital Selective Calling (DSC) System. Used in GMDSS to transmit distress alerts from ships. It may also be used by coast stations to acknowledge distress alerts and for relaying distress alerts from ship or coast stations.

Duplex operation. Operating method in which transmission is possible simultaneously in both directions of a telecommunication channel.

Earth station. A station located either or the earth's surface or within the major portion of the earth's atmosphere and intended for communication with one or more space stations or with one or more stations of the same kind by means of one or more reflecting satellites or other objects in space.

Emergency position-indicating radiobeacons (EPIRB). A station in the mobile service, the emissions of which are intended to facilitate search and rescue operations.

Harmful interference. Interference which endangers the functioning of a radionavigation service or of other safety services or seriously degrades, obstructs, or repeatedly interrupts a radiocommunication service operating in accordance with the ITU Radio Regulations.

Interference. The effect of unwanted energy due to one or a combination of emissions, radiations, or inductions upon reception in a radiocommunication system, manifested by any performance degradation, misinterpretation, or loss of information which could be extracted in the absence of such unwanted energy.

Land Earth Station. An earth station providing a feeder link to the maritime mobile satellite service operating in the fixed satellite service (or in the maritime mobile satellite service) located at a specified fixed point on land.

Locating signals. Transmissions intended to facilitate the finding of a ships in distress or the location of survivors.

Maritime mobile satellite-service. A mobile-satellite service in which mobile earth stations are located on board ships; survival craft stations and EPIRB stations may also participate in this service.

Maritime mobile service. A mobile service between coast stations and ship stations, or between ships, or between associated on-board communication stations; survival craft station and EPIRB stations may also participate in this service.

MRCC. Maritime Rescue Coordinating Centre. A unit responsible for promoting efficient organisation of search and rescue service and for coordinating the conduct of SAR operations within a SRR.

Maritime Safety Information (MSI). Navigational and meteorological warnings, meteorological forecasts, distress alerts and other urgent information broadcasting to ships.

Mobile earth station. A mobile earth station in the maritime mobile satellite service located on board ship.

Multiplex. Simultaneous transmission of several signals along a single path (or channel) without losing the information contained in each individual signal.

NAVAREA. Short title of an area in the world wide navigational warning service.

NAVTEX. Narrow band direct printing telegraphy system for transmission of navigational and meteorological warnings and urgent information to shipping.

On-board communication station. A low powered mobile station in the maritime mobile service intended for use for internal communications on board a ship, or between a ship and its lifeboats and liferafts during drills or operations, or for communication within a group of vessels being towed or pushed, as well as for line handling and mooring instructions.

Paired frequencies. Frequencies which are associated in pairs where each pair consists of one transmitting frequency and one receiving frequency.

Port operations service. A maritime mobile service in or near a port, between coast stations and ship station or between ship stations, in which messages are restricted to those relating to the operational handling, the movement and safety of ships and, in emergency, to the safety of persons. It does NOT include public correspondence.

Port station. A coast station in the port operations service.

Public correspondence. Any telecommunication which the offices and stations must, by reason of their being at the disposal of the public, accept for transmission.

Radiobeacon station. A station in the radionavigation service the emissions of which are intended to enable a mobile station to determine its bearing or direction in relation to the radiobeacons station.

Radiotelegram. A telegram, originating in or intended for a mobile station or a MES, transmitted on all or part of its route over the radiocommunication channels of the mobile service or the mobile-satellite service.

Radiotelephone Call. A telephone call, originating in or intended for a mobile station or a MES, transmitted on all or part of its rout over the radiocommunication channels of the mobile service or the mobile-satellite service.

Radiotelex Call. A telex call originating in or intended for a mobile station or a MES, transmitted on all or part of its route over the radiocommunication channels of the mobile service or the mobile-satellite service.

Rescue Coordinating Centre (RCC). Shortened version of MRCC. A unit responsible for promoting efficient organisation of search and rescue service and for coordinating the conduct of SAR operations within a SRR.

SAR *coordinating communications.* Communications necessary the for the coordination of ships and aircraft participating in a search resulting from a distress incident.

Satellite emergency position-indicating radiobeacons. An earth station in the mobile satellite service, the emissions of which are intended to facilitate search and rescue operations.

Search and rescue region (SRR). An area of defined dimensions within which SAR services are provided.

Ship movement service. A safety service in the maritime mobile service other than a port operations service between coast stations and ship station or between ship stations, in

which messages are restricted to those relating to the movements of ships. It does NOT include public correspondence.

Ship station. A mobile station in the maritime mobile service located on board a vessel which is not permanently moored, other than a survival craft station.

Simplex operation. Operating method in which transmission is made possible alternately in each direction of a telecommunication channel, for example by means of manual control.

Space station. A station located on an object which is beyond, is intended to go beyond, or has been beyond, the major portion of the earth's atmosphere.

Special Drawing Right. The monetary unit of the International Monetary Fund (IMF).

Standard frequency and time signal service. A radiocommunication service for scientific, technical and other purposes, providing the transmission of specified frequencies, time signals, or both of stated high precision, intended for general reception.

Survival craft. A craft capable of sustaining the lives of persons in distress from the time of abandoning the ship.

Survival craft station. A mobile station in the maritime mobile service or in the aeronautical mobile service intended solely for survival purposes and located in any lifeboat, liferaft or other survival equipment.

Telecommand. The use of telecommunication for the transmission of signals to initiate, modify or terminate functions of equipment at a distance.

Telecommunication. Any transmission, emission or reception of signs, signal, writing, images and sounds or intelligence of any nature by wire, radio, optical or other electromagnetic systems.

Telegraphy. A form of telecommunication which is concerned in any process providing transmission and reproduction at a distance of documentary matter, such a written or printed matter or fixed images, or the reproduction at a distance of any kind of information in such a form.

Telephony. A form of telecommunication set up for the transmission of speech or, in some cases, other sounds.

UTC. Coordinated universal time. A time scale based on the second as defined and recommended by the CCIR and maintained by the International Time Bureau (BIH). This was formerly expressed in Greenwich Mean Time (GMT).

World wide navigational warning service (WWNWS).A service established by the IMO and the International Hydrographic Organisation for the purpose of coordinating transmissions of radionavigation warnings in geographical areas.

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CHAPTER 1

Global Maritime Distress and Safety System (GMDSS)

1.1 GMDSS: OVERVIEW

Changes to the 1974 International Safety of Life at Sea Convention (SOLAS) Convention were made in 1988 that led to the introduction of the Global Maritime Distress and Safety System. The new Chapter 4 of this convention introduced the most radical change in marine radiocommunication in its hundred year history.

Ships in distress traditionally relied on other ships to come to their assistance after a distress message has been transmitted, i.e. ship to ship alerting. All ships were, and still are, obliged to assist any vessel in distress under existing international agreements. Although this system proved effective in congested waters it had certain limitation in less populated sea areas where restricted transmission range could be a disadvantage.

The GMDSS concept overcomes some of these limitations by introducing automatic distress alerting techniques to existing terrestrial and satellite communications. Such methods offer advantages for alerting shore authorities and locating vessels during emergencies while at the same time dispensing with the need for manual aural watchkeeping. The new system does not ignore or remove the traditional ship to ship distress alerting method described above but it does however place much greater emphasis on ship to shore distress alerting. Under the new system, rescue authorities ashore now assume the primary role of coordinating rescue operations at sea.

The GMDSS provides a comprehensive communication system available for use during distress and search and rescue (SAR) operations worldwide and maritime states had to agree to set up the necessary shore based infrastructure to provide this objective. Thus the primary object of GMDSS is to alert and inform shore based authorities, usually a Maritime Rescue Coordination Centre (MRCC), that a vessel is in imminent danger and requires immediate assistance. The MRCC can then coordinate search and rescue operations.

GMDSS lays down nine specific functions which all ships need to be able to perform irrespective of which sea area they will be operating in. These functions are:-

(a) transmitting ship to shore distress alerts (by at least two separate and independent means, each using a different radiocommunication service);
(b) transmitting and receiving ship to ship distress alerts;
(c) receiving shore to ship distress alerts;
(d) transmitting and receiving search and rescue coordinating communications;
(e) transmitting and receiving on-scene communications;
(f) transmission and receipt of locating signals;

(g) receipt of maritime safety information (MSI);
(h) transmitting and receiving general radio communications to and from shore based radio systems or networks;
(i) transmitting and receiving bridge to bridge communications.

It is mandatory for ships to be fitted with appropriate automated radiocommunication equipment in order that they can carry out these functions.

1.1.1 Search and Rescue Regions (SRRS)

The International Convention on Maritime Search and Rescue established an agreed international planned approach to SAR. To this end the world has been divided into areas for SAR purposes, also the organisation, cooperation, operating procedures and ship reporting systems have been standardised. SAR needed to be standardised internationally and the Safety of Life at Sea (SOLAS) Convention requires signatories to provide rescue services and coast watching services along their shorelines. The IMO Search and Rescue convention came into force in 1985. This provided coordination of SAR and assistance to anyone in distress at sea, irrespective of their location, nationality or circumstances. Ocean areas have been divided into search and rescue regions (SRRs) where a Maritime Rescue Coordination Centre (MRCC) is clearly identified as having primary responsibility in coordinating and controlling all maritime SAR activities. Extensive interconnecting communication links between RCCs allow for the rapid and efficient dissemination of information between these stations, thus using to full advantage the globally integrated satellite and terrestrial GMDSS communications.

1.1.2 **IAMSAR** Manual

The International Maritime Organisation (IMO) and the International Civil Aviation Organisation (ICAO) have published a joint ICAO/IMO search and rescue manual, called the International Aeronautical and Maritime Search and Rescue Manual (IAMSAR). This is to assist Governments in implementing the objectives of the International Convention on Maritime Search and Rescue 1979 and of Article 12(2) of the Convention on the High Seas 1958 which requires "every coastal State shall promote the establishment and maintenance of an adequate and effective search and rescue service regarding safety on and over the sea and — where circumstances so require - by way of mutual regional arrangements co-operate with neighbouring States for this purpose".

The IAMSAR Manual provides guidelines rather than provisions for a common maritime search and rescue policy, encouraging all coastal States to develop their organisations on similar lines and enabling adjacent States to co-operate and provide mutual assistance. This Manual also takes into account that maritime and aeronautical search and rescue organisations are complementary, the Manual has been aligned as closely as possible to ensure common policy in administrative and operational matters. It is published by the IMO and is available in four languages. It consists of three volumes.

Volume I: Organisation and Management. This covers the global SAR system, setting up and updating of SAR national and regional systems etc.

Volume II: Mission Coordination. Offers assistance to planning and coordination of SAR exercises and operations.

Volume III: Mobile Facilities. A reference source for use on board SAR rescue units, ships and aircraft. Details operations relating to SAR on scene coordination duties, and to emergencies involving their own unit.

1.2 GMDSS AREAS

With GMDSS, it is the vessel's area of operation or trading pattern which determines the type of radiocommunication equipment fitted on board. To this end four distinct sea areas have been designated. The GMDSS also switches the emphasis away from ship to ship distress alerting to ship to shore alerting using automatic means. It should be noted that although satellite communication is a major communication medium of GMDSS, it is not the exclusive medium. Other automatic means are available, such as digital selective calling (DSC). Radiotelephone and radiotelex services are also available for follow up distress communication and all three services use the conventional terrestrial maritime frequency bands.

Ships radio equipment has been considered in terms of the range of the communications system in use and the areas in which the ship will trade, thus four areas have been specified.

Area A1

An area within radiotelephone coverage range of at least one VHF coast station in which continuous DSC alerting is available, as may be defined by a Contracting Government.

Area A2

Excluding area Al, within radiotelephone coverage range of at least one shore based MF coast station in which continuous DSC alerting is available, as may be defined by a Contracting Government.

Area A3

This excludes areas Al and A2. It is the area within the coverage of an Inmarsat geostationary satellite in which continuous alerting is available (approximately 70° North to 70° South).

Area A4

The remaining sea area outside areas Al, A2 and A3.

1.3 GMDSS TERMINOLOGY

Some examples of the terminology used in the GMDSS are given below.

Distress alert transmission. This infers that a mobile unit, or a person is in distress or imminent danger and requires immediate assistance. In the terrestrial radiocommunication bands, a distress alert may be defined as a digital selective call using the distress call format. In the satellite service a distress alert may be

defined as a message using the distress format for relay through space stations. A satellite distress alert is automatically detected by a Land Earth Station and a satellite communication channel is immediately assigned. Therefore an end to end communication line from ship to rescue centre is established. The minimum information contained in a distress alert is the identification of the station in distress and its position. A mobile unit may be a ship, aircraft or other vehicle.

SAR coordinating communication. This relates to communication between the ship in distress and the Rescue Coordination Centres (RCCs) and ships and aircraft taking part in SAR activities. It also includes communications to and from RCCs by the on-scene coordinator in the distress area. Such communications may make use of satellite or terrestrial RT or radiotelex (Narrow Band Direct Printing or NBDP) systems depending on equipment fitted and the sea area in which the incident occurs. The worldwide interconnection of RCCs using SARNET is also included in the category of communication (see section 1.9.1).

On-scene communications. This type of communications generally relates to short range communication between (a) the distress ship and assisting units (b) communications associated with SAR activities. Normally RT is used on the designated safety frequencies in the MF and VHF bands. However, radiotelex (NBDP) using Forward Error Correction (FEC) may also be used on the designated safety frequency in the MF band.

Locating signals. These signals are sent from Emergency Position Indicating Radiobeacons (EPIRB) to greatly improve the chances of rescue. Also used are Search and Rescue Transponders (SARTs) that are used as homing devices to locate a distressed vessel and/or survivors by transmitting signals which can be received as distinctive signals on the searching ship or aircraft's radar (see Chapter 4).

Maritime Safety Information (MSI). GMDSS provides for dissemination of important navigational, meteorological information to shipping by automatic means on different frequency bands to ensure maximum coverage. Also includes shore to ship distress/safety alerts. For further details see sections 4.16 to 4.18.

General radiocommunication. This is designed to cope with what, by far, will be the most widely used of the nine functions listed in section 1.1, namely general commercial radio traffic, either ship to shore, shore to ship or intership. This may be in the form of telephone, radiotelex, email, data exchange etc on all the necessary frequencies to provide a good quality service.

Bridge to bridge communications. A VHF RT service available for intership communication on a worldwide basis that must include Ch.13 for safety of navigation. Ch.13 the ONLY channel that can be used within GMDSS without transmitting a preceding DSC call on Ch.70 and GMDSS ships should, where practicable, maintain a watch on this channel. This channel may also be used for ship movement and port operations communications subject to national regulations of the relevant administration.

1.4 GMDSS IMPLEMENTATION

The International Convention for Safety of Life at Sea (1974) was amended for GMDSS in 1988, and came into force on 1st February 1992. Full implementation of GMDSS regulations occurred on the 1st February 1999, applying to all ships

over 300 gross tons and all passenger ships on international voyages subject to the 1974 SOLAS convention. The objective of the system is for all ships to have compulsory communication equipment appropriate to their area of operations. The new GMDSS equipment permits rapid automatic alerting of shore authorities of distress situations.

Subject to the following provisions of this regulation, these Regulations apply to:

(l)(a) sea-going UK ships wherever they may be except while they are within the Great Lakes of North America and their connecting and tributary waters as far east as the lower exit of the St Lambert Lock at Montreal in the Province of Quebec, Canada; and

(b) other sea-going ships while they are within UK waters.

(2) These Regulations shall not apply to:

(a) troopships not registered in the UK:

(b) ships not propelled by mechanical means;

(c) pleasure craft;

(d) fishing vessels;

(e) cargo ships of less than 300 tons; and

(f) craft to which the Merchant Shipping (High-Speed Craft) Regulations 1996 (a) apply.

(3) Every ship the keel of which was laid or which was at a similar stage of construction before 1st February 1995 shall:

(a) subject to paragraph 6 of the Regulations (relating to performance standards), until the 1st February 1999 comply with either -

(i) the requirements of Part II of the Merchant Shipping (Radio Installations) Regulations 1998; or

(ii) the requirements of Part III of those regulations, regulations ll(l)(d) (NAVTEX) and 11(1)(f) (satellite EPIRB), and the requirements of the Merchant Shipping (Life-Saving Appliances) Regulations 1980 (b) or the Merchant Shipping (Life-Saving Appliances) Regulations 1986 (c) (as appropriate) relating to the carriage of radar transponders; and

(b) on or after 1st February 1999 comply with the requirements of Part II of Merchant Shipping (Radio Installations) Regulations 1998.

(4) Subject to paragraph (6) below, every ship the keel of which was laid or which was at a similar stage of construction on or after 1st February 1995 shall comply with the requirements of Part II of these Regulations.

(5) Subject to paragraph (6) below, regulations 10, 11(4) and 19(2) (a) shall apply to only UK passenger ships of Classes I, II and II(A) within the meaning of the 1984 Regulations wherever they may be and other passenger ships of those Classes while they are within UK waters.

(6) Any passenger ship the keel of which was laid or which was at a similar stage of construction before 1st July 1997 to which regulations 10 or 11(4) apply need not comply with the requirements of those regulations (as applicable) until the date of the first periodical survey after the date these Regulations come into force in respect of the ship.

1.5 SHIPBOARD EQUIPMENT FOR GMDSS

Part C of the SOLAS Amendments, 1988 and Part II of the Merchant Shipping (Radio Installations) Regulations 1998 provide information on the equipment required by GMDSS ships and the alternative arrangements permitted in the various sea areas. The following list is a general guide only, to allow operators to understand the range of equipment they may encounter on board ship.

Area A1

(a) VHF radiotelephone
(b) VHF DSC on Ch.70
(c) VHF DSC watch receiver
(d) SART (One)
(e) NAVTEX receiver
(f) Enhanced Group Call (EGC) equipment required if outside NAVTEX coverage
(g) Satellite EPIRB float free, or VHF EPIRB capable of transmitting a DSC distress alert on Ch.70 VHF
(h) two way VHF RT equipment for survival craft

Area A2

(a) VHF radiotelephone
(b) VHF DSC on Ch.70
(c) VHF DSC watch receiver
(d) SART (One)
(e) NAVTEX receiver
(f) Enhanced Group Call (EGC) and printer equipment required if outside NAVTEX coverage
(g) Satellite EPIRB float free
(h) two way VHF RT equipment for survival craft
(i) MF radiotelephone with DSC
(j) MF watch receiver dedicated to 2 187.5 kHz
(k) MF DSC encoder/decoder (this may be combined with items i) and j) above)

Area A3

(a) VHF radiotelephone
(b) VHF DSC on Ch.70
(c) VHF DSC watch receiver
(d) SART (One)
(e) NAVTEX receiver
(f) Enhanced Group Call (EGC) and printer required if outside NAVTEX coverage
(g) Satellite EPIRB float free
(h) two way VHF RT equipment for survival craft
(i) MF radiotelephone with DSC
(j) MF watch receiver dedicated to 2 187.5 kHz on item (m) below
(k) MF DSC encoder/decoder (this may be combined with items (i) and (j) above)

(l) Inmarsat MES
OR instead of (l) above:
(m) MF/HF plus DSC (scanning watch) receiver plus narrow band direct printing (NBDP)

Area A4

(a) VHF radiotelephone
(b) VHF DSC on Ch.70
(c) VHF DSC watch receiver
(d) SART (One)
(e) NAVTEX receiver
(f) Satellite EPIRB float free
(g) two way VHF RT equipment for survival craft
(h) MF/HF radiotelephone with DSC
(i) MF/HF Radiotelex (NBDP)
(j) MF/HF scanning DSC watch receiver

Notes on equipment

1. VHF radiotelephones must be capable of transmitting and receiving DSC on Ch.70 and RT on Ch.6, Ch.13 and Ch.16.
2. The DSC Ch.70 VHF watchkeeping equipment may be separate or combined with item 1 above.
3. Ships sailing exclusively in area A1 may carry in lieu of a 406 MHz satellite EPIRB an EPIRB which is capable of transmitting a DSC distress alert on Ch.70 VHF and providing location by means of a 9 GHz transponder.
4. MF radio installation must be capable of transmitting and receiving DSC on 2 187.5 kHz and RT on 2 182 kHz for distress purposes.
5. The dedicated 2 187.5 kHz DSC watchkeeping receiver may be separate or combined with DSC equipment in item 4 above.
6. MF/HF installations must be capable of transmitting and receiving on all distress and safety frequencies in the 1 605 - 27 500 kHz using DSC, RT.
7. MF/HF DSC watchkeeping receivers must be capable of receiving on the frequencies of 2 187.5 kHz and 8 414.5 kHz and at least one of the other HF DSC frequencies (4 207.5, 6 312, 12 577 or 16 804.5 kHz). The receiver may be separate or combined with DSC equipment in item 6 above.
8. Ships constructed before 1st February 1997 and exclusively engaged on voyages within areas A2, A3 and A4 may be granted exemptions from carrying DSC Ch.70 VHF, provided such ships maintain where practicable a continuous listening watch on Ch.16 VHF. This watch shall be kept from the position at which the ship is normally navigated.
9. Amendment to SOLAS 74 came into force on 1st February 1995 for new and existing ships.

Provision of 9 GHz SARTs. All passenger ships and all cargo ships of 500 gross tonnage and above must be provided with at least two 9 GHz radar transponders (SARTs). At least one transponder must be provided for cargo ships of 300 gross tonnage and above but less than 500 gross tonnage. Such transponders must conform to performance standards not inferior to those adopted by the IMO.

Attention is drawn to the need to provide a means of mounting the transponder at least one metre above sea-level in all survival craft.

10. The two way VHF RT equipment for survival craft referred to above relate to VHF (waterproofed) "walkie-talkies." Requirement for ships of 300 to 500 gross tons: minimum 2.

For ships over 500 tons minimum requirement: 3.

Additional Notes

From 1st February 1995, all two-way VHF RT apparatus provided on all passenger ships and all cargo ships of 300 gross tonnage and above for use in survival craft, and which do not already conform to the performance standards adopted by the IMO8, must at least be compatible with equipment conforming with these standards. As a minimum, therefore, all such apparatus must operate on the VHF distress and safety frequency, 156.8 MHz (Ch. 16) and one other, and be provided with a battery capable of operating for 8 hours at its highest rated output with a duty cycle defined as 6 sec. transmission, 6 sec. reception above squelch opening level and 48 sec. reception below squelch opening level, or be provided with one or more batteries of the same capacity. Any new equipment supplied to fulfil this requirement must conform to the IMO performance standards.

All radio equipment and radar transponders installed on UK ships must be of a type which is approved by the Ofcom to a specification which incorporates the relevant IMO requirements. Similarly , all radar equipment must conform to a standard incorporating the IMO requirements and be approved by the MSA. Merchant shipping Notice. M 1714 (M + F) gives further details.

1.5.1 Position updating

From the 1st July 2002 an amendment to the International Convention for the Safety of Life at Sea was adopted. A new regulation (Regulation 17) was added.

All two-way communication equipment carried on board a ship to which this chapter applies which is capable of automatically including the ship's position in the distress alert shall be automatically provided with this information from an internal or external navigation receiver, if either is installed. If such a receiver is not installed, the ship's position and the time at which the position was determined shall be manually updated at intervals not exceeding 4 hours, while the ship is under way, so that it is always ready for transmission by the equipment.

1.5.2 Implementation of new SOLAS Regulations for passenger ships

With effect from 1st July 1997 a number of amendments to the International Convention for the Safety of Life at Sea 1974 (SOLAS 74) came into force relating to the enhancement of safety on passenger ships. This includes certain amendments to those provisions of SOLAS relating to radio installations. These amendments will be applied to all United Kingdom passenger ships of Class I, II and II (A) through, generally speaking, incorporation into or amendments to the Merchant Shipping (Radio Installations) Regulations, 1998.

SOLAS regulations. SOLAS IV/6.4

In passenger ships, a distress panel shall be installed at the conning position. This panel shall contain either one single button which, when pressed, initiates a distress alert using all radiocommunications installations required on board for that purpose or one button for each individual installation. The panel shall clearly and visually indicate whenever any button or buttons have been pressed. Means shall be provided to prevent inadvertent activation of the button or buttons. If the satellite EPIRB is used as the secondary means of distress alerting and is not remotely activated, it shall be acceptable to have an additional EPIRB installed in the wheelhouse near the conning position.

UK interpretation

Existing equipment provided with distinct alerting facilities; i.e. buttons on the front panel, which will transmit a distress alert on pressing without further action from the operator; i.e. without requiring the manual tuning of the relevant VHF transmitter, will be acceptable.

Electronic switching to enable/disable particular radio installations to be used for distress alerting time from a single button is acceptable providing there is means to indicate which installations are enabled/disabled at any point of time. Where multiple buttons are provided each must be clearly marked with details of the installations to which they refer. Multiple panels are acceptable providing they are co-located. Existing arrangements to prevent inadvertent activation of the distress alert facilities remain acceptable, as do the indications of activation, unless equipment is subsequently replaced.

For existing ships not provided with a distinct conning position; i.e. a position where the ships' navigating staff would normally sit during any passage, this panel, panels or equipment must be located;

(i) within the area used for the navigation of the ship, i.e. not in a separate cabin (unless access doors are removed) or up/down stairs;
(ii) illuminated by the electrical lighting required by regulation 9(l)(d) of the Merchant Shipping (Radio Installations) Regulations, 1998. This states: be provided with reliable, permanently arranged electrical lighting, independent of the main and emergency sources of electrical power, for the adequate illumination of the radio controls for operating the radio installation.

All new ships and existing ships provided with a distinct conning position are to be provided with dedicated distress alerting panels located within the main navigation console(s); the conning position is understood to be the location where the officer of the watch is likely to perform his or her duties.

For all ships an additional satellite EPIRB is to be provided in the wheelhouse if used to fulfil the requirements for a secondary means of distress alerting, unless the existing EPIRB is remotely activated. Alternatively, if fitted, an alternative radio communications system may be selected, e.g. Inmarsat or, if certified for operations only in sea area A1, VHF DSC.

1.5 *Global Maritime Distress and Safety System (GMDSS)*

SOLAS regulations. Provision of ship's position information. SOLAS IV/6.5

In passenger ships, information on the ship's position shall be continuously and automatically provided to all relevant radiocommunication equipment to be included in the initial distress alert when the button or buttons on the distress panel is pressed.

UK interpretation

With the sole exception of satellite EPIRBs, all radio installations to be used for generating distress alerts are to be provided with facilities to continuously and automatically update ships position for inclusion in any distress alert. This position information may be derived from any source suitable for use throughout all intended voyages. Existing equipment unable to support this facility is to be replaced.

SOLAS regulations. Distress alarm panel. SOLAS IV/6.6

In passenger ships, a distress alarm panel shall be installed at the conning position. The distress alarm panel shall provide visual and aural indication of any distress alerts transmitted or received on board and shall also indicate through which radiocommunication service the distress alerts have been made.

UK interpretation

For existing ships, provided the distress alarms are present within the area normally used for conning the ship, or immediately adjacent thereto, present arrangements remain acceptable. For new ships, a distinct panel is to be provided. This panel may be that used to comply with new SOLAS IV/6.4

SOLAS regulations. Aeronautical frequency VHFradio. SOLAS IV/7.5

Every passenger ship shall be provided with means for two-way on-scene radiocommunications for search and rescue purposes using the aeronautical frequencies 121.5 MHz and 123.1 MHz from the position from which the ship is normally navigated.

UK interpretation

Fixed or, recommended, portable equipment plus headset and boom mike may be used to fulfil this requirement. In view of the high noise levels encountered during helicopter operations, a suitable headset and microphone should be provided. Equipment is to conform to the relevant specification(s) adopted by the CAA and should be connected to the main, emergency (if provided) and reserve power supplies - fixed equipment only.

SOLAS regulations. Radio personnel. SOLAS IV/16.2

In passenger ships, at least one person qualified in accordance with paragraph 1 of the SOLAS regulations shall be assigned to perform only radiocommunication duties during distress incidents.

UK interpretation

This person's function is to maintain continuous, as far as is practicable, communications with the Rescue Coordination Centre (RCC), or on-scene coordinator as appropriate, during any emergency. This person's duties are to be clearly specified on the Muster List. No other duties may be performed by the person throughout the distress incident. This person, who may not be the ship's Master, must hold a minimum of a GMDSS General Operator's Certificate of Competency, except and unless the ships is certified to operate only in sea area A1.

1.5.3 EPIRBS on fishing vessels

The Fishing Vessels (Life-Saving Appliances) Regulations of January 1988 gave effect to the provisions of the Safety at Sea Act 1986 concerning EPIRBs for fishing vessel of 12 metres or more in length. The length referred to is registered length.

The Regulations require that from at least 15th January 1989 fishing vessels of 12 metres or more in length must carry a type 406 MHz EPIRB. EPIRBs fitted to fishing vessels in compliance with the Regulations shall comply with MPT 1259 (Performance specification for a satellite EPIRB operating at 406 MHz) as amended.

Following tank tests into the conditions for EPIRB automatic deployment, the following siting advice was derived. To provide the best conditions for automatic deployment, the EPIRB should be sited so that it can float free and clear regardless of the attitude of the vessel during or following capsize, e.g. on the wheelhouse roof. Some compromise on this form of siting may need to be made to enable access for alternative manual activation. If the EPIRB is placed on one side of the vessel or immediately behind the wheelhouse then the likelihood of correct deployment is much reduced.

Although there is no mandatory requirement or EPIRBs to be provided in fishing vessels of less than 12 metres in length, it is strongly recommended that they should be provided.

1.5.4 Recommended GMDSS equipment for small craft

In the table below, recommended GMDSS equipment is indicated by an R, while 0 indicates that the equipment fit is optional.

	Area of operation from coast in nautical miles				
Equipment	*Up to 5m*	*Up to 30m*	*Up to 60m*	*Up to 150m*	*Unrestricted*
A	R	R	R	R	R
B	O	R	R	R	R
C	O	O	0	R	R
D	None	None	O	R	R
E	None	None	O	0	R
F	None	O	R	R	R
G	None	O	O	R	R

Where:

Equipment A represents a hand held, waterproofed VHF radio which can also be used in the liferaft.

Equipment B represents a fixed VHF radio installation and fitted with DSC,

Equipment C represents a float free 406 MHz EPIRB (with 121.5 MHz homer),
Equipment D represents an MF SSB radio installation and fitted with DSC,
Equipment E represents an Inmarsat installation,
Equipment F represents a NAVTEX receiver,
Equipment F represents a Search and Rescue Transponder (SART).

Equipment Specification. In Europe for small craft, the minimum standard for DSC equipped radios for fixed use is EN 301 205. This standard approximates to the international specification for "Class D" DSC equipment.

Vessel Monitoring System (VMS)

A European Council directive exists which concerns the establishment of a satellite based Vessel Monitoring System (VMS) to monitor the position of community fishing vessels. See section 9.32.

Radio qualifications for small craft operators.

Two qualifications exist, the Short Range Certificate (SRC) which is for operators operating in GMDSS sea area A1, on craft which do not legally require GMDSS (i.e. voluntary fitted vessels). For voluntary fitted vessels that sail outside area Al the Long Range Certificate (LRC) is available. See sections 12.19 and 12.14.

1.6 MINIMUM GMDSS PERSONNEL REQUIREMENTS FOR SHIPS STATIONS AND MESs

In compliance with international agreements, Administrations must ensure that the personnel of ships stations and MESs are adequately qualified to ensure efficient operation of the station. They also must take steps to ensure the operational availability and maintenance of equipment for distress and safety communications in accordance with the relevant international agreements.

An adequately qualified person shall be available to act as a dedicated communications operator in cases of distress.

The personnel of ships stations and MESs for which a radio station is compulsory under international agreements and using techniques and frequencies outlined in Chapter N IX of the Radio Regulations shall, with respect to the provision of Article 55 include at least:

(a) for stations on board ships which sail beyond the range of VHF coast stations, taking into account the provisions of the Convention for the Safety of Life at sea; a holder of a first or second class radio electronic certificate or a general operator's certificate;
(b) for stations on board ships which sail within the range of VHF coast stations, taking into account the provisions of the Convention for the Safety of Life at sea; a holder of a first or second class radio electronic certificate or a general operator's certificate or a restricted operator's certificate.

The personnel of ship stations and MESs for which a radio installation is not compulsory under international agreements and using techniques and frequencies outlined in Chapter N IX shall be adequately qualified and certificated in accordance with the administration's requirements.

1.7 GMDSS SHIP EQUIPMENT : BRIEF DESCRIPTION

SART. Search and rescue transponder

These provide the main means of locating survival craft or ships in distress by rescue units after an alert has been received. Operating in the 9 GHz frequency band SARTs generate a series of response signals after being interrogated by normal marine or aircraft radar transmissions. These responses show up as distinctive marks on the rescue units radar display. SARTs are portable devices for use either on board a vessel or taken to survival craft when abandoning ship but they may also be permanently located in survival craft. Further details in sections 4.14 and 4.15.

NAVTEX

A narrow band direct printing telex system. NAVTEX forms an important component of the GMDSS, for in addition to promulgating maritime safety information such as routine weather and navigation warnings, it may also used as a means of alerting all vessels of a distress incident. Incoming messsages, broadcast by coast stations, are automatically printed out by the ships receiving equipment. Further details in section 4.17.

EPIRB/Satellite EPIRB

Emergency position-indicating radiobeacon/satellite emergency position indicating radio beacon. The signals transmitted from these beacons are primarily used to facilitate search and rescue teams locate survivors of distress incidents. Further details are given in Chapter 4.

DSC *controller*

An interfacing device used to modulate a transmitter by providing it with the correct digital coding information for DSC. It is also used in conjunction with receivers to convert the received digital signals by displaying them in written form (see Chapter 3).

EGC equipment

Enhanced group call, automated satellite equipment which receives signals sent from land earth stations addressed to all ships or to predetermined groups of ships in a designated geographical area. It is a service available via the Inmarsat satellite system (see section 4.18).

Inmarsat MES

Mobile earth station; communication equipment operating in the maritime mobile-satellite service which is located on board ship to transmit and receive signals via Inmarsat satellites (see Chapter 5).

1.8 GMDSS DISTRESS ALERT PROCEDURE

It is apparent from the previous sections that mobile stations in distress have a number of options available to them to alert shore based authorities that they

require assistance. The communication medium used will, to a large extent, depend on the area the vessel is in and on the propagation conditions at the time. For example:

Ships in A1 areas would transmit a ship to shore or ship to ship alert on Ch.70 VHF using DSC, followed by RT communication on Ch.16 VHF. Ships in A2 areas would transmit their alert on 2 187.5 kHz using DSC and indicating whether RT or radiotelex (NBDP) is to be used to transmit subsequent distress communications. However, it should be noted that RT is the preferred option.

For ships in areas A3 and A4, to make *ship/shore alerts,* a choice is available, once again depending on:

(i) the equipment fitted;
(ii) this equipments operational availability;
(iii) the existing propagation conditions.

Thus for example:

(a) A shore station could be alerted via an Inmarsat-A, B, C or F land earth station;
(b) A coast station could be alerted by using an HF DSC communication;
(c) A satellite EPIRB could be activated.

Such arrangements are designed to offer a high probability of a successful alert irrespective of the sea area involved, thus enabling a rapid response to the alert by rescue teams. It should be noted that ships in all areas are required to be fitted with EPIRBs, search and rescue radar transponders (SARTs) and two-way VHF RT equipment for survival craft under GMDSS carriage requirements. Thus in the event of abandoning ship without sending a distress alert, the EPIRB, SART and two way VHF RT equipment should be taken into the survival craft. The EPIRB provides a secondary method of distress alerting as well as being a location device. The SART is a radar operated location device while the two-way walkie-talkie VHF RT equipment allows survivors to communicate to rescue craft when within VHF communication range.

In addition to examples (a), (b) and (c) above, ships also have the option of making *ship/ship DSC distress alerts.* Such transmissions would be sent on either 2 187.5 kHz, or Ch.70 VHF, with subsequent RT communications on 2 182 kHz or Ch.16 VHF. However, ships should attempt to make a ship to shore alert in the first instance.

Any alert should contain information which gives the ships identification, its position and, if possible, the nature of the distress.

When alerts are received by a coast radio station or an Inmarsat Land Earth Station, they are immediately passed to a Maritime Rescue Coordination Centre (MRCC). The MRCC with its extensive communication capability, will relay the alert to search and rescue units and to other ships in the area and will control and coordinate rescue procedures. Relaying of alerts to other vessels in the area is carried out using satellite communications or by terrestrial communications on appropriate frequencies. Use is made of DSC "area call", NAVTEX, or Enhanced Group Call (EGC) facilities to notify only those ships in the immediate vicinity of the distress. Ships which have been alerted in this way then establish contact with the RCC via an appropriate medium (satellite or terrestrial communications).

1.9 ROLE OF THE MARITIME RESCUE COORDINATION CENTRE (MRCC)

As the name implies, MRCCs (often abbreviated to RCCs) are used to assist with maritime distress situations. Extensive communication links provide "end to end" connection between the distress vessel and competent rescue authorities ashore. Because of the very high priority status accorded to distress alerts and the use of automatic signalling systems, this direct connection link is rapidly established, usually within only a few seconds.

Comprehensive communication systems link an individual MRCC with either a coast station, a land earth station (Inmarsat LES), a COSPAS-SARSAT Mission Control Centre (MCC) or Local User Terminal (LUT). When an MRCC receives an original distress alert via one of these stations, it will relay details of the alert to SAR units and to other ships within the general area of the reported distress. This relayed message should indicate the distress vessels identification, its position and any other relevant information of practical use in rescue operations.

The MRCC which initially receives a distress alert, appropriately called First RCC, assumes responsibility for all further coordination of subsequent SAR operations. However this responsibility may be transferred to another MRCC which may be in a better position to coordinate rescue efforts. See Fig. 1.1

MRCCs are also generally involved in subsequent SAR coordinating communications. This may include communications between the designated "on-scene coordinator" on board SAR units within the general area of the distress incident.

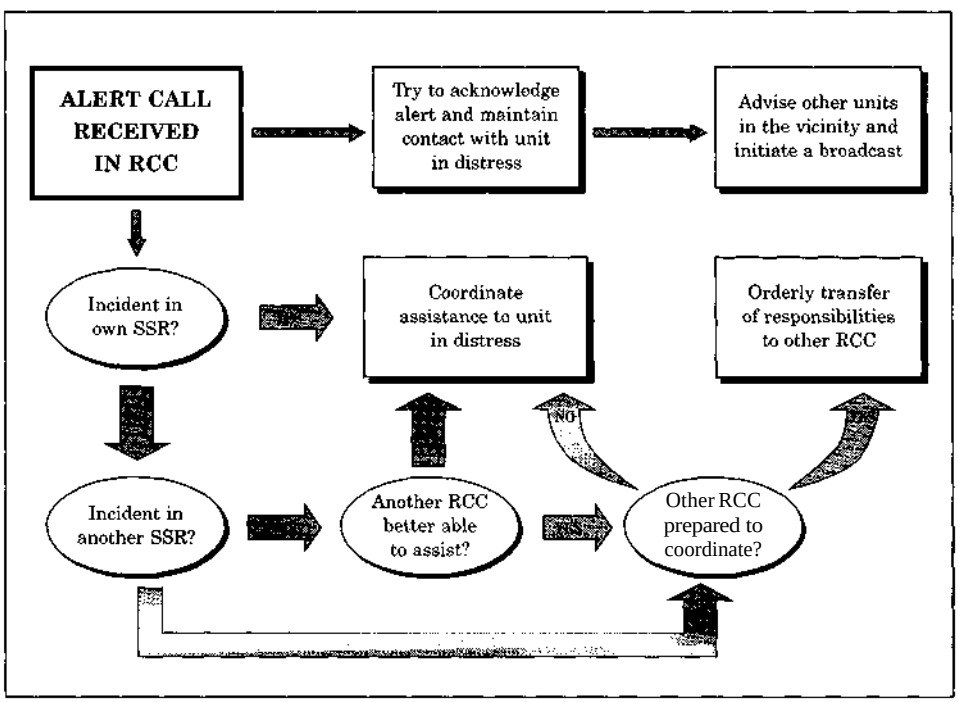


Fig. 1.1 The first Rescue Coordination Centre (RCC) actions on receipt of distress alert.

1.9.1 SARNET

SARNET is the use of Inmarsat-C equipment by MRCCs to communicate quickly with other MRCCs. The use of SARNET between MRCCs in northern Europe, including HMCG at Falmouth have proved very successful. SARNET is particularly useful when an MRCC has received a distress alert by HF DSC and which may be thousands of miles away. Rapid relaying of the distress information to a rescue centre nearer the scene of the incident is facilitated by use of this system. MRCCs worldwide are encouraged to consider installing Inmarsat-C terminals as standard equipment allowing MRCC-to-MRCC links to expand and further exploit this equipment for SAR purposes.

1.9.2 Registration database for GMDSS

All identifications used within the GMDSS for identifying vessels in distress are required to be held on database that can be accessed by MRCCs, etc, as necessary. An annex to Resolution 340 (WRC-97) makes administrations or organisations responsible for assigning these identities to make suitable arrangements for ensuring that these registrations of identities are made and maintained. This annex further states:

> Means shall be provided by the Bureau and administrations maintaining national databases to allow rescue coordination centres immediate access to this database on a 24 hour per day, 7-day per week basis.

Each registration database should include the following information:

- vessel name;
- maritime mobile service identity (MMSI);
- call sign;
- Emergency Position Indicating Radiobeacons (EPIRBs) identification code (if applicable);
- country (vessel flag State; may be derived from MMSI and call sign);
- vessel identification number (IMO number or national registration number);
- name, address, telephone and (if applicable) telefax number of emergency contact person ashore;
- alternative 24 - hour emergency telephone number;
- capacity of persons on board (passengers and crew);
- ship's radio installation (Inmarsat A, B, C, F, M, VHF digital selective calling etc.): and
- Inmarsat mobile earth station identities (INMs) if applicable.

1.10 GMDSS MASTER PLAN

When GMDSS amendments to the 1974 SOLAS convention were being drawn up, each Contracting Government had to agree to cooperate in setting up the necessary shore-based infrastructure for terrestrial and satellite communications. The amended regulations included a provision that:

> Each Contracting government undertakes to provide the Organisation (the IMO) with pertinent information concerning the shore-based facilities in the

Maritime Mobile Service, Mobile-Satellite Service and Maritime Mobile-Satellite Service, established for sea areas which it has designated off its coasts.

Thus the International Maritime Organisation (IMO) publishes a master plan which lists all the shore based communications facilities and services worldwide which are made available by contracting governments within the GMDSS. The plan is updated regularly and covers both space and terrestrial communication methods and includes details on the following twelve points.

(a) A list of VHF coast stations for sea area A1.
(b) A list of MF coast stations for sea area A2.
(c) A list of HF coast stations for sea area A3 and A4.
(d) A list of Inmarsat LESs.
(e) A list of COSPAS-SARSAT MCCs.
(f) Information concerning operational and planned NAVTEX services.
(g) Distress message routeing and the Inmarsat/MRCC ship to shore distress alerting networks.
(h) MESs commissioned for MRCC operations.
(i) An HF NBDP MSI broadcast schedule.
(j) A list of proposed and operational SafetyNET services.
(k) Maritime SAR regions, MRCC and associated shore-based facilities.
(1) Information concerning shore based facilities in the GMDSS.

The plan information is available to maritime, aeronautical, telecommunication, meteorological and SAR authorities etc, or any other interested parties and to training establishments.

1.11 AVAILABILITY AND MAINTENANCE OF GMDSS EQUIPMENT: TERMINOLOGY

It has been agreed internationally that certain requirements must be attained in order to ensure the availability of radio equipment on convention ships while at sea. The methods used to ensure such requirements are as follows:

(a) Duplication of equipment (DOE). Equipment duplication may be used as a means of contributing to the availability of equipment in the GMDSS.
(b) Shore based maintenance (SBM). Within the GMDSS, if availability is ensured by using a combination of methods which includes shore-based maintenance, an arrangement acceptable to the administration should be established to ensure adequate support of the ship for the maintenance and repair of its radio installations.
(c) At sea maintenance (ASM). Where ASM is used as an option to ensure availability of electronic equipment through a maintenance capability on UK ships, then the arrangements made must be acceptable to the Department of Transport.

1.12 AVAILABILITY AND SERVICEABILITY OF GMDSS EQUIPMENT

The methods to be used to ensure that availability and serviceability of equipment under the Global Maritime Distress and Safety System (GMDSS) are given in

Merchant Shipping Notice MSN 1690 (M). This Notice is an integral part of the Merchant Shipping (Radio Installations) Regulations 1998.

1.12.1 Requirements for ensuring availability

Every ship, subject to this Regulation, shall ensure availability by using at least one; and for area A3 or area A4 ships at least two, of the following methods:

(i) duplication of equipment;
(ii) shore based maintenance;
(iii) at-sea electronic maintenance capability;
(iv) any other method or combination approved by the Secretary of State.

Irrespective of the methods used to ensure that availability of the equipment, but subject to the exception of Regulation 49 (this relates to powers for detaining vessels not complying with regulations), a ship should not depart from any port unless and until the ship is able to perform all distress and safety communications.

Irrespective of the methods used by the ship, all manufacturers' instruction manuals and maintenance manuals for each piece of equipment required should be available on board. In the case of UK ships these manuals must be in the English language. Adequate tools, spare parts and test equipment appropriate to the methods used by the ship should be provided. The manuals, tool, spare parts and test equipment, as applicable, should be readily accessible.

The regulations for ensuring availability vary according to the GMDSS sea areas as is shown in the following sub sections.

1.12.2 Duplication of equipment ensuring availability

If availability is ensured by using duplication, either singly or in combination with other approved methods, the following radio installations, in addition to those required by Regulations 8, 9, 10, 11, 12 as appropriate, and complying with Regulation 5, should be available on board ships engaged on voyages in:

(i) *Sea Area A1.* A VHF radio installation complying with requirements of Regulation 8(l)(a);
(ii) *Sea Areas Al and A2.* A VHF radio installation complying with requirements of Regulation 8(l)(a) and an MF radio installation complying with the requirements of Regulation 10(l)(a);
(iii) *Sea Areas Al, A2 and A3.* A VHF radio installation complying with requirements of Regulation 8(l)(a) and either an MF/HF radio installation complying with the requirements of Regulation 11(1) Alternative B(a) and being able to fully comply with the watch requirements of Regulation 13(l)(c); or an Inmarsat MES complying with the requirements of Regulation 11(1) alternative A(a). The MF/HF installation or Inmarsat MES installed for duplication should also comply with regulation 11(2);
(iv) Sea *Areas Al, A2, A3 and A4.* A VHF radio installation complying with requirements of Regulation 8(l)(a) an MF/HF radio installation complying with the requirements of Regulation 11(1) Alternative B(a) and being

able to fully comply with the watch requirements of Regulation 13(l)(c). For ships operating in sea area A4 only occasionally and having originally installed an MF/HF radio installation, the additional MF/HF radio installation may be substituted by an Inmarsat MES complying with the requirements of Regulation 11(1) alternative A(a). The MF/HF installation or an Inmarsat MES installed for duplication should also comply with regulation 11(2).

The additional radio installations specified in paragraphs (i) to (iv) above should each be connected to a separate antenna and be installed and ready for immediate operation.

It should be possible to connect the additional radio installations specified in paragraphs (i) to (iv) above, i.e. the duplicated equipment, to the reserve source or sources of energy required by regulation 14(2) in addition to the appropriate radio equipment specified in that regulation, hereafter referred to as "basic equipment".

The capacity of the reserve source or sources of energy should be sufficient o operate the particular installation (i.e. the basic equipment or the duplicated equipment) with the highest power consumption, for the appropriate period specified in Regulation 14(2)(a) to 14(2)(c). However, the arrangement for the reserve source or sources of energy should be such that a single fault in this arrangement should not be able to affect the basic and the duplicated equipment.

If the reserve source of energy is a battery, the charging arrangements should be duplicated. Alternatively, a suitable alarm system should be installed to indicate a failure in the changing arrangement and provision should be made to connect the equipment direct to a suitable alternative supply.

1.12.3 Shore based maintenance for ensuring availability

If availability is ensured by using a combination of methods which include shore based maintenance, an arrangement acceptable to the Maritime and Coastguard Agency (MCA) should be established to ensure adequate support of the ship for maintenance and repair of its radio installations. Acceptable arrangements are:

(i) a signed declaration for the Owner or his Representative setting out the arrangement with a company or companies covering the trading area of the ship to provide maintenance and repair facilities on a call-out basis;
(ii) provision of facilities at the main base of ships engaged on a regular trading pattern.

Note: Records of Equipment (Forms DTP 30079 or DTP 30080) should include an indication of the types of arrangements for shore-based maintenance.

1.12.4 At sea maintenance for ensuring availability

If availability is ensured by using a combination of methods which includes at-sea electronic maintenance capability, adequate additional technical documentation, tools, test equipment and spare parts must be carried on board in order to enable the maintainer to perform tests and localise and repair faults in the radio equipment. The extent of this additional technical documentation, tools, measuring

equipment and spare parts to be carried on board should be consistent with the equipment installed. An indication of such approval should be entered in the Records of Equipment (Forms DTP 30079 or DTP 30080).

The person designated to perform function for at-sea electronic maintenance should hold an appropriate certificate as specified by the Radio Regulations or have equivalent at-sea electronic maintenance qualifications.

In the case of United Kingdom ships, for the purpose of the Regulations an officer or crew member shall be deemed qualified to carry out radio maintenance if he or she holds

(i) a Radio Maintenance Certificate granted by the Secretary of State , or
(ii) a certificate recognised by the Maritime and Coastguard Agency as being equivalent to the certificate in (i) above.

1.12.5 Acceptable combinations (or equivalent)

Any Inmarsat MES equipment providing the enhanced group calling (EGC) capability may be accepted as fulfilling the basic requirement of Regulation 8 (l)(e) for EGC reception.

Any Inmarsat MES or any MF/HF radiotelephone installed for duplication may also satisfy the basic requirements of Regulations 10(l)(c), 11(1) alternative A(d) and 11(1) alternative B(c) for another radio service to initiate the transmission of ship-to-shore distress alerts.

Any Inmarsat MES or any MF/HF radiotelephone may be installed for duplication irrespective of whether the basic radio installation complies with Regulation 11(1) alternative A(a) or Regulation 11(1) alternative B(a).

1.13 AVAILABILITY OF GMDSS EQUIPMENT: FURTHER REQUIREMENTS

As outlined in the previous section, every GMDSS ship subject to the Merchant Shipping (Radio Installations) Regulations 1998 shall ensure availability as required by the Merchant Shipping Notice MSN 1690 (M). However, certain conditions must be met irrespective of the method used to ensure availability of equipment to function as per the appropriate regulations.

(a) Equipment shall be so designed that the main unit can be replaced readily, without elaborate recalibration or readjustment.
(b) Where appropriate, equipment shall be so constructed and installed that it is readily accessible for inspection or on-board maintenance purposes.
(c) Adequate information shall be provided on board to enable the equipment to be properly operated and maintained.
(d) Adequate tools, spare parts and test equipment suitable to the methods used by the ship should be provided.
(e) Required radio equipment shall be maintained to meet the recommended performance standards of such equipment.

On all UK ships, a person nominated by the Master, but usually a person holding an appropriate Radio Operators Certificate, shall, while the ship is at sea, carry out the daily, weekly and monthly tests required by Schedule 2 of the

Merchant Shipping (Radio Installations) Regulations 1998. If any of the radio equipment required by these Regulations is not in working order, the nominated person shall inform the Master and record details of the deficiencies in the Radio Log.

1.14 FALSE DISTRESS ALERTS

Regrettably, the high incidence of false distress alerts received at MRCCs worldwide continues despite wide publicity having been given to this subject. Manufacturers have introduced measures which require a deliberate action by an operator before a distress alert is transmitted by providing a protective cover over the distress button. Training establishments continue to stress the importance of understanding the consequences of transmitting false alerts but the problem persists. It is a matter of concern that the main cause of false alerts continues to be as the result of human error by the accidental or incorrect use of GMDSS equipment on board ship. Ship's masters can assist by verifying that any crew member likely to send a distress alert is fully conversant with the operation of all GMDSS equipment on board. It is also good practice to periodically instruct other crew members in the correct distress procedures in the event that the main GMDSS communicator is incapacitated by injury etc. To further reduce the chance of false alerts the routine test of GMDSS equipment should be conducted by the direct supervision of the person designated to operate the equipment during distress situations.

False alerts cause a severe strain on MRCC resources, for each alert has to be investigated and this can divert attention away from a real distress situation. Ships personnel must ensure that they take great care when operating GMDSS equipment to avoid making an inadvertent distress alert transmission.

The cause of problems associated with false alerts are enumerated below.

Accidental transmission of a VHF DSC alert on Ch. 70.
Accidental transmission of a MF DSC alert on 2 187.5 kHz.
Accidental transmission of an HF DSC alert.
Accidental activation of a distress alert using Inmarsat equipment.
Accidental activation of a satellite EPIRBs.

Other difficulties may arise which can lead to excessive and unnecessary DSC signalling and cause great confusion during distress situations. Such difficulties occur when ships incorrectly transmit an acknowledgement by DSC of a distress alert received on 2 187.5 kHz or on one of the HF DSC distress alerting frequencies. Ships must NOT send DSC distress relays when they have received a DSC distress alert on MF or VHF.

The Maritime Safety Committee (MSC) agreed the following measures to reduce the number of false distress alerts. That Governments:

- Ensure all GMDSS equipment being manufactured meets the latest IMO performance standards (dedicated and protected distress buttons)
- Require shipowners to ensure, when ordering equipment, that it complies fully with the latest IMO performance standards.

- Encourage manufacturers of GMDSS equipment to investigate, as a matter of urgency, modifying equipment not fitted with a dedicated and protected distress button, to be so fitted.
- Consider establishing requirements for GMDSS radio equipment not fitted with dedicated and protected distress button, to be modified to incorporate such facilities.
- Encourage manufacturers to co-operate so as to agree common standards and simplification of operating equipment design, especially relating to facilities for initiating and responding to distress alerts.
- Encourage shipowners and manufacturers to improve operating standards so as to minimise false distress alerts

1.14.1 Action to be taken by owners, masters and skippers

Masters should make sure that all who may be required to send a distress alert are instructed and knowledgeable in the operation of all radio equipment on the ship. Such instructions should also be given periodically on board the ship to all relevant crew members by the person responsible for communications.

To reduce the chance of false alerts, routine testing of GMDSS equipment should only be undertaken under the direct supervision of he person designated to have primary responsibility for radiocommunications during distress incidents.

See the next section for further guidance. The IMO guidelines for avoiding false distress alerts are given in Appendix 1.

1.15 FALSE DISTRESS ALERTS: REPORTING PROCEDURES

Accidental transmission of a VHF DSC alert on Ch. 70 observed

Stop the transmission immediately, then switch to Ch. 16 VHF and make an "all stations" broadcast. Indicate in this RT message your name, callsign, MMSI number, position and that you are cancelling the false alert sent at (quote) date/time (UTC).

An example of the format of a cancellation message sent by RT to "all stations" is given below. If considered necessary, this type of message may also be addressed to an individual MRCC.

ALL STATIONS, ALL STATIONS, ALL STATIONS
THIS IS
ALISTAMCA, ALISTAMCA, ALISTAMCA
ALISTAMCA, MMSI NUMBER 232445367 MY POSITION 50 DEGREES 21 MINUTES NORTH, 09 DEGREES 03 MINUTES WEST CANCEL MY DISTRESS ALERT OF 1545 HOURS TODAY TRANSMITTED ON CHANNEL 70.
MASTER

Accidental transmission of a MF DSC alert on 2 187.5 kHz observed

Stop the transmission immediately, then switch to 2 182 kHz and make an "all stations" broadcast. Indicate in this RT message your name, callsign, MMSI number, position and that you are cancelling the false alert sent at (quote) date/time (UTC).

Following specification changes recommended by the IMO, modern DSC equipment may include a cancel false alert facility using a self acknowledgement feature. The ship sends this acknowledgement using its own MMSI number which because of the close proximity to the receiver will be picked up and accepted by the DSC. This system has the advantage that it will be received over the same reception range as the original false alert, however a cancellation of the false alert transmitted by RT on 2 182 kHz should still be sent.

Note: Under normal circumstances, no action will be instigated against any vessel cancelling or reporting the inadvertent transmission of a distress alert. However, as there is a strict ban on such transmissions and because false alerts cause a severe strain on SAR resources, Governments may prosecute vessels or individuals which repeatedly violate the regulations regarding false alerts.

Accidental transmission of an HF DSC alert observed, e.g. on 8 414.5 kHz

Stop the transmission immediately, then switch to the associated HF band (in the example given 8 291 kHz would be used) and make an "all stations" broadcast. Indicate in this RT message your name, callsign, MMSI number, position and that you are cancelling the false alert sent at (quote) date/time (UTC).

Accidental activation of a distress alert using Inmarsat-C observed

Immediately notify the appropriate RCC by sending a distress priority message via the same LES that the original false alert was sent to. Indicate in this message your ship's name, callsign, Inmarsat-C identity number (ship's IMN), position and that you are cancelling the false alert sent at (quote) date/time (UTC).

Inmarsat-C: Performance Verification Test (PVT)

The distress alerting function on Inmarsat-C is tested during the Performance Verification Test. During the test, a special test distress message is substituted for the live distress message, but this substitution is only valid for a short period of time (often about one minute). Although much of the PVT is automated, it is desirable for the operator to remain in attendance at the terminal to test the distress alert at the correct time (usually prompted by the terminal). This reduces the possibility of initiating the distress test after the terminal has returned automatically to live test mode. If an Inmarsat alert is sent inadvertently then the ship should call back to the relevant Land Earth Station as follows "Cancel my Inmarsat-C distress alert (insert ships name and identity number) of (insert time).

Accidental activation of a satellite EPIRB

If you become aware that for any reason an EPIRB has accidentally activated, you must immediately notify the nearest coast station, or an appropriate LES or MRCC and cancel the alert. It is recommended that until the SAR authorities have been notified the EPIRB should be left on. This permits the beacon to be located and thus prevents unnecessary searching.

Recently, and for the first time since the introduction of GMDSS, a 406 MHz EPIRB Guard Receiver has become commercially available. This device will sound an audible alarm whenever your own ships EPIRB has activated, allowing the ship's staff to take remedial action by informing an MRCC. See section 4.11.1

1.15.1 Other problems: misuse of DSC acknowledgement

Ships acknowledging a DSC distress alert on Ch. 70 VHF

(a) Ships operating in GMDSS sea area A1 SHOULD NOT transmit a DSC acknowledgement even if prompted to do so by the DSC controller. Ships must assume that the alert will have been heard and will be acknowledged by a coast station. The correct procedure for ships receiving a DSC alert in this area is to switch to Ch.16 VHF and listen for further distress traffic and acknowledge this by RT. Vessels will then assist as appropriate under the normal rules of the sea.

Ships acknowledging a DSC distress alert on 2 187.5 kHz

(a) Ships operating in GMDSS sea area A2 SHOULD NOT transmit a DSC acknowledgement even though prompted to do so by the DSC controller. Ships must assume that the alert will have been heard and will be acknowledged by a coast station. The correct procedure for ships receiving a DSC alert in this area is to switch to 2 182 kHz and listen for further distress traffic and acknowledge this by RT. Vessels will then assist as appropriate under the normal rules of the sea.

(b) Ships operating outside GMDSS sea area A2 which receive a distress alert which is, beyond all doubt, in their vicinity should send an acknowledgement by RT on 2 182 kHz. If however additional DSC distress alerts giving the same information are received a DSC acknowledgement may be sent. RCCs should be informed about the situation via a coast station or LES. Vessels will assist as appropriate under the normal rules of the sea.

Ships acknowledging a DSC distress alert on HF

(a) Ships receiving an HF DSC distress alert SHOULD NOT transmit a DSC or RT acknowledgement, even though prompted to do so by the DSC controller. The correct procedure is for ships to listen for further distress traffic by RT or radiotelex (NBDP) on the appropriate HF band on which the DSC alert was received. If however, additional DSC distress alerts giving the same information are received, or it becomes obvious that the DSC distress alert has not been acknowledged by a coast station, the receiving ship must RELAY the distress alert to any shore station. This RT distress relay message must be sent to an individual MRCC or coast station and must NOT be sent to "all stations."

The IMO guidelines for avoiding false distress alerts are given in full in Appendix 1.

1.15.2 GMDSS equipment and operator competency

The extract below is taken from IMO COMSAR/Circ.17.

> Use of GMDSS equipment for transmission of general radiocommunications is one of the functional requirements in SOLAS Chapter IV, regulation 4. Regular use of GMDSS equipment helps to develop operator competency and ensure equipment availability. If ships use other radiocommunication systems for the bulk of their business communications, they should adopt a regular programme of sending selected traffic or test messages via GMDSS equipment to ensure operator competency and equipment availability and to help reduce the inci-

dence of false alerts. This policy extends to all GMDSS equipment suites including Digital Selective Calling on VHF, MF and HF, to the Inmarsat A, B, C and F systems, and to any duplicated VHF and long-range communication facilities.

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CHAPTER 2

Distress, Urgency and Safety Communications

2.1 GMDSS: GENERAL PROVISIONS

The Global Maritime Distress and Safety System (GMDSS) has been set up to provide an international communications network to assist vessels in distress. All distress communications in the maritime mobile service whether by radiotelephony, digital selective calling and satellite techniques and/or direct printing telegraphy (telex), must be conducted strictly in accordance with the correct procedures. This ensures that vessels in distress obtain help without delay.

Ships equipped for and operating under GMDSS conditions are obliged to comply with Chapter N IX of the ITU Radio Regulations. Some provisions of this chapter also apply to the aeronautical mobile service except in the case of special arrangements between the governments concerned.

The procedures specified in Chapter N IX of the ITU Radio Regulations are obligatory in the maritime mobile satellite service and for communications between stations on board aircraft and stations of the maritime mobile satellite service, wherever this service or stations of this service are specifically mentioned.

The International Convention for the Safety of Life at Sea, (SOLAS) 1974, prescribes which ships and which of their survival craft shall be provided with radio equipment, and which ships shall carry portable radio equipment for use in survival craft.

Although the procedures are obligatory for all stations using communications in the maritime mobile and maritime mobile satellite service they do allow:

(a) A mobile station or Mobile Earth Stations (MES) in distress to use any means at their disposal to attract attention, make known their position and obtain help.

(b) Any stations whether mobile (ship / aircraft) or land stations or coast stations involved in search and rescue duties to use, in exceptional circumstances, any means at their disposal to assist a mobile station or a mobile earth station in distress.

(c) Any coast station or LES may use, in exceptional circumstances, any means at their disposal to assist a mobile station or MES in distress.

When transmitting distress, urgency or safety messages by radiotelephony, transmissions should be made slowly and distinctly with clear pronunciation. The use of the International Code of Signals is recommended, including the use of abbreviations and the phonetic alphabet, particularly in situations where language difficulties arise.

Mobile stations of the maritime mobile service may communicate, for safety purposes, with stations of the aeronautical mobile service, but they must use the

approved frequencies and observe the correct procedures. Mobile stations of the aeronautical mobile service may communicate, for safety purposes, with stations of the maritime mobile service, but they must again use the approved frequencies and also observe the correct procedures.

Any station on board an aircraft required by national or international regulations to communicate for distress , urgency or safety purposes with stations of the maritime mobile for safety purposes must comply with GMDSS provisions. They must be able to transmit and receive:

(a) class J3E emissions on carrier frequencies 2 182 kHz and 4 125 kHz;
(b) class G3E emissions on Ch.16 VHF and optionally Ch.6 VHF.

2.2 DISTRESS FREQUENCIES

Within GMDSS a number of frequencies are specifically allocated for distress communication purposes depending on the communication medium being used. These frequencies should be used by ships, aircraft or survival craft when seeking assistance in the maritime mobile service. They should be used for the distress call and distress traffic and for calls preceded by either the urgency or safety signal.

Designated frequencies used for distress and safety calls using DSC are:

2 187.5 kHz
4 207.5 kHz
6 312 kHz
8 414.5 kHz
12 577 kHz
16 804.5 kHz
Ch.70 VHF

Designated frequencies used by ships for RT distress and safety traffic are:

2 182 kHz
4 125 kHz*
6 215 kHz
8 291 kHz
12 290 kHz
16 420 kHz
Ch. 16 VHF

* May be used by aircraft engaged in search and rescue (SAR) operations for distress/ safety communications to stations in the maritime mobile service.

Frequencies used exclusively for distress, urgency and safety traffic using radiotelex or narrow band direct printing (NBDP) are:

2 174.5 kHz
4 177.5 kHz
6 268 kHz
8 376.5 kHz
12 520 kHz
16 695 kHz

Frequencies used by coast stations to transmit high seas marine safety information using radiotelex (NBDP) with forward error correction are:

4 210 kHz
6 314 kHz
8 416.5 kHz
12 579 kHz
16 806.5 kHz
19 680.5 kHz
22 376 kHz
26 100.5 kHz

The aeronautical frequencies below are used for intercommunication purposes between mobile stations and to participating land stations during coordinated SAR operations are:

3 023 kHz
5 680 kHz

Frequencies used exclusively by coast stations to transmit meteorological, navigational warnings and other urgent information to ships using radiotelex (NBDP) for NAVTEX are:

518 kHz
4 209.5 kHz
490 kHz

490 kHz will be used exclusively for the transmission of NAVTEX type information by coast stations for "second or local language" broadcasts.

2.3 AERONAUTICAL FREQUENCIES

In the aeronautical VHF radiotelephony service (117.975–137 MHz) the emergency frequency is 121.5 MHz. Survival craft can also use this frequency for distress and urgency communications. Additionally this frequency may be used in the COSPAS-SARSAT emergency location system by emergency position-indicating radiobeacons (EPIRBs) for distress alert transmissions.

The auxiliary aeronautical frequency of 123.1 MHz is used by the service and by other mobile and land stations engaged in coordinated SAR operations. Maritime mobile stations may communicate with aeronautical stations on 121.5 MHz for distress and urgency purposes only, and on 123.1 MHz for coordinated SAR operations. Class A3E emissions are used on these frequencies.

2.3.1 Aeronautical use of 121.5 MHz

The aeronautical emergency frequency in the band 117.975 to 136 MHz is 121.5 MHz. It is available for use for emergency purposes, broadly outlined as follows:

(1) To provide a clear channel between aircraft in distress or emergency and a ground station when the normal channels are being utilised for other aircraft.
(2) To provide a VHF communication channel between aircraft and aerodromes, not normally used by international air services, in case of an emergency condition arising.

(3) To provide a common VHF communication channel between aircraft, either civil or military, and between aircraft and surface facilities, involved in common SAR operations, prior to changing to an appropriate frequency.
(4) To provide VHF air-ground communications between aircraft and suitably equipped vessels and survival craft.
(5) To provide frequencies for the operation of survival radio equipment and emergency position indicating radiobeacons (EPIRBs), emergency locator transmitter (ELT) and for communication between survival craft and aircraft engaged in search and rescue operations.

Note: Civil aircraft, when operating over ocean areas, are required to carry an emergency location beacon and/or survival craft radio equipment operating on the frequencies 121.5 and/or 243 MHz. Designated SAR aircraft are required to be capable of homing on these frequencies for locating a distress scene and survivors.

2.4 GMDSS: VHF CHANNEL USAGE

Ch. 6 VHF: Used for communication between ships and aircraft for coordinated SAR operations. May also be used by aircraft to communicate to ships for safety purposes.

Ch.13 VHF: Used for ship to ship RT communication relating to safety of navigation. This is the ONLY channel that can be used within GMDSS without transmitting a preceding DSC call on Ch.70. Commercial vessels should, where practicable, monitor this channel for communications relating to safety of navigation.

Ch.16 VHF: Used for distress and safety traffic on RT, may also be used by aircraft for safety purposes only. The World Administration Radio Conference of 1997 allows the use of Ch.16 for general calling.

Ch. 70 VHF: Used exclusively as a DSC calling channel in the maritime mobile service for all types of DSC calls (distress or routine).

2.5 SATELLITE SERVICE

A number of bands are allocated exclusively to particular services in the satellite service, such bands must also be protected to avoid harmful interference.

Band	*Use*
406–406.1 MHz.	Earth to space EPIRB transmissions
1 530-1 544 MHz	Maritime mobile service. Routine communications and also for distress and safety communications (space to earth).
1544–1 545 MHz	Space to earth distress/safety communications which includes: (a) Satellite feeder links to relay EPIRB transmissions to earth stations. (b) Narrow band (space to earth) links from space stations to mobile stations.
1 626.5-1 645.5 MHz	Maritime mobile service. Routine communications and also for distress and safety communications (earth to space).

Band	*Use*
1645.5-1646.5MHz	Earth to space distress/safety communications which includes: (a) Satellite EPIRB transmissions. (b) Distress alert relay of signals received by low polar orbiting satellites to geostationary satellites.

2.6 SURVIVAL CRAFT AND GMDSS

Survival craft using equipment for emergency radiocommunication, location or alerting signals have to comply with certain conditions:

(a) Where portable VHF RT equipment operating in the allocated band (156–174 MHz) is used, it must be able to transmit and receive on Ch.16 and at least one other channel in this band.

(b) Radar Search and Rescue Transponders (SARTs) transmitting locating signals must be able to operate in the frequency band 9 200-9 500 MHz (3 cm or X-band).

(c) Digital selective calling equipment if capable of operating on the indicated bands must be able to transmit on:
  (i) 2 187.5 kHz in the band 1 605-2850 kHz
  (ii) 8 414.5 kHz in the bands 4 000-27 500 kHz
  (iii) Ch.70 VHF in the band 156–174 MHz

2.7 DISTRESS AND SAFETY COMMUNICATIONS: PROTECTION OF FREQUENCIES

It is imperative that distress and safety communications maintain the highest possible integrity and the distress and safety frequencies are protected from harmful interference. Harmful interference being that interference which puts at risk the operation of safety services or radionavigation services or seriously degrades, obstructs, or repeatedly interrupts any radiocommunication services. Therefore to protect distress integrity and to prevent harmful interference it is absolutely forbidden to broadcast or cause harmful interference by any emissions on any frequency especially those allocated to distress working.

To further protect distress and safety communications, all other transmissions within particular bands are forbidden apart from transmissions on specific frequencies. For example, transmissions in the band 2 173.5-2 190.5 kHz are prohibited, apart from those on 2 182 kHz, 2 174.5 kHz, 2 177 kHz, 2 187.5 kHz and 2 189.5 kHz.

All emissions in the frequency band 156.7625-156.8375 MHz are forbidden which might cause interference to VHF Ch. 16 (156.8 MHz).

2.8 TRANSMITTER TESTS

Test transmissions should only be made when absolutely necessary and be kept to a minimum to avoid harmful interference especially on the distress and safety frequencies. Whenever practicable, artificial aerials (dummy load) should be used but where aerials do have to be used, tests should be conducted on low power and

coordinated with competent authorities. Testing on distress and safety calling frequencies should be avoided. Testing stations must always identify themselves and make it clear they are sending a test transmission. Before making a test operators should listen on the frequency to be used to ensure that no distress or safety communications are in progress.

2.9 GMDSS GENERAL WATCHKEEPING ARRANGEMENTS

Stations which assume watch-keeping responsibilities in GMDSS shall comply with the following conditions:

Land Earth Stations (LES). Maintain continuous automatic watch for distress alerts relayed from satellites.

Mobile Earth Stations (MES). Should maintain a watch for shore to ship distress alert relays except when they are occupied communicating on a working channel.

Coast Stations. Maintain automatic DSC watch on the frequencies and for times indicated in the ITU List of Coast Stations (published by the International Telecommunication Union, Geneva).

Ship Stations. Those stations complying with Chapter N IX of the Radio Regulations (GMDSS ships), while at sea shall:

(a) maintain automatic DSC calling watch on the correct distress and calling frequencies of the bands in which they are operating.
(b) Where equipped to do so ships maintain automatic reception on appropriate frequencies for transmissions of meteorological / navigational or other urgent warnings to ships.
(c) On Ch.16 until 1st February 2005.
(d) Where practicable ships should maintain a watch on Ch.13 VHF for communications relating to navigational safety.

2.10 DISTRESS AND SAFETY COMMUNICATION IN GMDSS: GENERAL OPERATIONAL PROCEDURES

Distress and safety communications use the MF, HF and VHF bands, i.e. on the terrestrial radiocommunication bands. Satellite communication bands are also used for distress and safety communications.

Distress alerts are transmitted either:

(a) via satellite with absolute priority using the general communication channels, or
(b) via satellite using exclusive distress and safety frequencies, or
(c) using DSC on the appropriate distress and safety frequencies in the MF, HF or VHF bands.

It should be noted that all stations receiving a DSC distress alert must immediately cease any transmission which may interfere with distress traffic. Stations must continue the watch until the distress alert has been acknowledged.

2.10.1 The authority of the master

The transmission of a distress alert by whatever method can only done on the authority of the master or person responsible for the ship, aircraft or other vehicle.

Therefore GMDSS operators must be receive a formal command the master or person responsible for the ship, aircraft or other vehicle before sending out the distress alert.

2.11 DISTRESS ALERTING

A distress alert transmission infers that a mobile unit (ship, aircraft or other vehicle) or a person* is in distress or imminent danger and requires immediate assistance. In general, distress incidents fall into two main categories:

(a) coastal, in which some or all of the following may be available to assist: ships, aircraft, helicopters and shore-based life-saving facilities;
(b) ocean, in which ships and long range aircraft may be available although, in the more remote ocean areas, only ships may be available.

In either case, a distress alert may be defined as:

(a) *In the terrestrial radiocommunication bands:*
a digital selective call using the distress call format, or
(b) *In the satellite service:*
a message using the distress format for relay through space stations.

The distress alert must include the identification of the station in distress and its position. The alert may contain the course and speed of the mobile unit and the time that this information was recorded. Other information such as the nature of the distress and type of assistance required or anything else which may facilitate rescue may be included in the alert.

Ship-to-shore distress alerts are used to warn Maritime Rescue Coordination Centres (MRCC) that a vessel is in distress. This alerting is routed via LES using satellite transmissions from MES or EPIRB. Alternatively, using the terrestrial service, alerts from ship stations or EPIRBs are made to coast stations. Ship to ship alerts warn other vessels in the vicinity of the distressed vessel about a distress situation using DSC techniques in the MF and VHF terrestrial bands. The HF band may also be used.

* In the case of a person in distress, the applications of the following procedures may need to be adapted to meet the requirements of the particular set of circumstances encountered.

2.11.1 Sending a DSC distress alert

To transmit a DSC alert on MF or VHF the following procedure is used:

(a) Tune the transmitter to DSC distress channel (2 187.5 kHz MF or Ch.70 VHF).
(b) If time permits select the nature of distress on the DSC controller.
(c) Insert by keying in according to manufacturers instructions:
(i) the type of subsequent communication (usually RT).*
(d) Transmit the DSC alert.
(e) When a DSC alert acknowledgement is received from a coast station or MRCC, tune RT transmitter/receiver to the distress frequency in the same band (2 182 kHz MF or Ch.16 VHF). Most modern equipment will automat-

ically switch the transmitter/receiver to the correct RT frequency or channel. Transmit the RT distress call and message in the format given in the following sub section. It should be noted that this alert will be repeated automatically approximately every four minutes until the acknowledgement is received by the equipment or it is manually switched off.

* The following new SOLAS regulation 18 has been added after existing regulation 17. All two-way communication equipment carried on board a ship to which this chapter applies (Chapter IV) which is capable of automatically including the ship's position in the distress alert shall be automatically provided with this information from an internal or external navigational receiver, if either is installed. If such a receiver is not installed, the ship's position and the time at which the position was determined shall be manually updated at intervals not exceeding 4 hours while the ship is under way, so that it is always ready for transmission by the equipment.

Although there is provision for the circumstances where a navigation receiver is not installed, regulation 19, section 2.1.6 of Chapter V of SOLAS requires from 1st July 2002 the carriage of a Global Navigation Satellite System receiver.

2.11.2 Radiotelephony distress message

The RT distress message must be made in the form:

MAYDAY;
the nine digit MMSI **and** the call sign or other identification of the ship;
the ship's position if not included in the DSC distress alert;
the nature of the distress;
the type of assistance required;
any other information which might facilitate the rescue.

Example:

MAYDAY
268026000 HELENA/KBVA
56 DEGREES 21 MINUTES NORTH 08 DEGREES 34 MINUTES WEST
ON FIRE AND DRIFTING
REQUIRE IMMEDIATE ASSISTANCE
WIND NORTHWESTERLY FORCE SIX.
OVER

2.11.3 Radiotelex distress message

If subsequent communication by radiotelex (NBDP) is indicated in the DSC distress alert, then radiotelex distress communications should be made in the Forward Error Correction (FEC) or broadcasting mode. The distress message contains the same information as for RT, i.e.

the distress signal MAYDAY;
the name, or other identification (e.g. MMSI) of the mobile station in distress;
the position of the vessel in distress;
the nature of the distress and the type of assistance required;
any other information which might facilitate rescue.

Example:
MAYDAY
206917112 MOULAY/EWBF
35.18 NORTH143.26 EAST
ON FIRE
REQUIRE IMMEDIATE ASSISTANCE
INTEND ABANDONING SHIP
OVER

2.12 SHORE TO SHIP DISTRESS ALERT RELAY

A coast station or an RCC which receives a distress alert shall initiate the transmission of a shore to ship distress alert relay. This distress alert relay will be addressed to all ships, to selected ships or to an individual ship using the satellite or terrestrial service. Such alert relay messages must contain the identification of the mobile unit in distress, its position and any additional information which facilitates rescue operations.

2.13 DISTRESS ALERT BY A STATION NOT ITSELF IN DISTRESS

Where a station in the mobile or mobile satellite service learns another mobile unit is in distress it must transmit a distress alert when:

(a) the mobile unit in distress is not in a position to transmit the distress alert itself;
(b) the master or person responsible for the mobile not in distress or the person responsible for the land station considers further help is required.

A station transmitting a distress alert relay must make it very clear that it is not itself in distress.

2.13.1 DSC distress relay

These calls may be sent by coast stations to alert ships that a distress has occurred in a particular area. It is sent when ships may not have received the original distress alert.

ITU recommendation ITU-R M.541-8 on operational procedures for the use of DSC equipment in the Maritime Mobile Service identifies only TWO situations in which a ship would transmit a distress relay call (distress relay alert):

(1) on receiving a distress alert on a HF channel, which is not acknowledged by a coast station within 5 minutes. The distress relay call should be address to an appropriate coast station; and
(2) on knowing that another ship in distress is not itself able to transmit the distress alert and the Master of the ship considers that further help is necessary. The distress relay call should be addressed to "all ships" or to an appropriate coast station.

In NO case is a ship permitted to transmit a DSC distress relay call on receipt of a DSC distress alert on either VHF or MF channels. Distress relay calls on HF channels should be initiated manually.

2.13.2 **Radiotelephony** distress **relay**

When using RT this call consists of:

the signal MAYDAY RELAY pronounced as the French expression "m'aider relais" spoken three times;
the words THIS IS (or DE spoken as DELTA ECHO in case of language difficulties);
the call sign or other identification of the transmitting station spoken three times.

Example:

MAYDAY RELAY MAYDAY RELAY MAYDAY RELAY
THIS IS
LASHIO, LASHIO, LASHIO

This call should be followed by the repeated transmission of the distress message received from the station in distress.

Thus an example of the full RT procedure for MAYDAY RELAY, call and message, is as follows:

MAYDAY RELAY MAYDAY RELAY MAYDAY RELAY
THIS IS
LASHIO, LASHIO, LASHIO/XYNP
MAYDAY
BLEKKAARD/OSBM
47 DEGREES 35 MINUTES NORTH 14 DEGREES 54 MINUTES WEST
STRUCK UNIDENTIFIED OBJECT AND TAKING IN WATER
REQUIRE IMMEDIATE ASSISTANCE
WIND SOUTHWESTERLY FORCE FIVE
OVER

When a station of the mobile service transmits a distress message under the conditions outlined above, it shall take all necessary steps to notify the authorities who may be able to render assistance.

2.14 ACKNOWLEDGEMENT OF RECEIPT OF DISTRESS ALERTS

Satellite service

Acknowledgement of receipt of an alert from a MES must be sent immediately.

DSC

Acknowledgement of receipt of a distress alert by DSC on terrestrial bands must be in accordance with relevant recommendations. Once the distress ship receives a DSC distress acknowledgement call, usually from a coast station, it should then commence the distress traffic by radiotelephony (RT) on the distress traffic frequency (2 182 kHz on MF or Ch. 16 VHF).

Ships receiving a DSC distress alert, are obliged to acknowledge by RT on the appropriate frequency or channel and must NOT acknowledge using DSC. It is also worth repeating here the ITU recommendation ITU-R M.541-8 on opera-

tional procedures for the use of DSC equipment in the Maritime Mobile Service. In NO case is a ship permitted to transmit a DSC distress relay call on receipt of a DSC distress alert on either VHF or MF channels.

Radiotelephony

Acknowledgements from a ship or MES are given after the distress message has been received and must take the standard RT form:

the distress signal MAYDAY;
the call sign or other identification of the station sending the distress message, spoken three times;
the words THIS IS (or DE spoken as DELTA ECHO in the case of language difficulties);
the call sign or other identification of the station acknowledging receipt, spoken three times;
the word RECEIVED (or RRR spoken as ROMEO ROMEO ROMEO) in the case of language difficulties);
the distress signal MAYDAY.

Example:

MAYDAY
HELENA, HELENA, HELENA
THIS IS
JANPUR, JANPUR, JANPUR
RECEIVED MAYDAY OVER

Radiotelex

Acknowledgements must be given in the form:

the distress signal MAYDAY;
the call sign or other identification of the station sending the distress alert;
the word DE (meaning "this is" or "from")
the call sign or other identification of the station acknowledging receipt of the distress alert;
the signal RRR;
the distress signal MAYDAY.

Example:

MAYDAY
KBVA
DE
VWEZ
RRR MAYDAY

An LES receiving a distress alert from an MES by telex, simply acknowledges receipt by calling back the MES in distress. This communication would also be by telex, using the procedure shown in the example above.

2.15 ACKNOWLEDGING RECEIPT OF DISTRESS ALERTS BY COAST STATIONS, LESs OR RCCs

Coast stations or land earth stations, on reception of the initial distress alert must forward the alert to the appropriate RCC as soon as possible. The alert must be acknowledged as soon as possible by a coast station or RCC via a coast station or appropriate LES.

Where a coast station uses DSC to acknowledge a distress alert the acknowledgement must be made on the same distress calling frequency used for the original alert and be addressed to all ships. Details of the identity of the ship in distress must be included in the acknowledgement message.

2.16 ACKNOWLEDGING RECEIPT OF DISTRESS ALERTS BY SHIP STATIONS OR MESs

A ship or MES in receipt of a distress alert shall, as soon as possible, inform the master or person responsible for the ship of the contents of the distress alert.

(i) In areas where reliable communications with one or more coast stations are practicable, ship stations in receipt of a distress alert should defer acknowledgement for a short interval so that receipt may be acknowledged by a coast station.

(ii) Ships operating in areas where reliable communications with a coast station are not practicable which receive a distress alert from a ship station which is, beyond doubt, in their vicinity, shall, as soon as possible and if appropriately equipped, acknowledge receipt and inform a RCC through a coast station or LES.

Ships acknowledging receipt of a distress alert under the circumstances outlined above should acknowledge using RT on the distress and safety frequency on the band used for the alert.

(iii) Ships receiving an HF distress alert however, shall not acknowledge it but should set watch on the RT frequency indicated in the distress alert. They may also set watch on the radiotelex (NBDP) frequency associated with the distress alert frequency, if appropriate. If the distress alert is not acknowledged by a coast station within three minutes then the receiving ship station must relay the distress alert.

A ship station in receipt of a shore to ship distress alert should establish communication as directed and render such assistance as required and appropriate.

Generally, every mobile station which acknowledges receipt of a distress message shall, on the order of the master or person responsible for the ship, aircraft or other vehicle, transmit as soon as possible, the following information, in the order shown:

(a) its identity;
(b) its position;
(c) the speed at which it is proceeding towards, and the approximate time it will take to reach the mobile station in distress.

Notes: Paragraph (i) above refers to ship to shore alerting (using VHF or MF) in GMDSS sea areas A1 and A2.

Paragraph (ii) above refers to ship to ship alerting (using VHF or MF) in mid ocean well outside GMDSS areas Al and A2. This is an option ships have under GMDSS and does not refer to ship to shore alerting using HF in GMDSS areas A3 or A4.

When a receiving ship acknowledges a ship to ship distress alert, it is accepting responsibility for the distress. It is therefore incumbent upon that ship to inform a Maritime Rescue Coordination Centre of the situation by any communication means available.

Paragraph (iii) above refers to ship to shore alerting using HF in GMDSS areas A3 or A4. Ships should not acknowledge received DSC distress alerts. If the distress alert is repeated, receiving ships should inform an MRCC using the mayday relay format.

2.17 PREPARATIONS FOR HANDLING DISTRESS TRAFFIC

On receipt of a transmitted DSC distress alert ship and coast stations must set watch on the radiotelephony distress and safety frequency associated with the frequency on which the distress alert was received. For example if the DSC distress alert was transmitted on 2 187.5 kHz then a listening watch must be set on the RT distress frequency of 2 182 kHz, if the DSC alert was sent on 6 312 kHz receiving stations would listen out on 6 215 kHz. Similarly ship and coast stations with radiotelex (NBDP) equipment must set watch on the radiotelex distress frequency if the distress alert message indicates that radiotelex is to be used for subsequent communications. If practicable, in the latter case, stations should also set watch on the appropriate RT frequency associated with the distress alert frequency. See Appendix 8 for list of GMDSS distress and safety frequencies.

2.18 DISTRESS TRAFFIC

Distress traffic is defined as all messages relating to the immediate assistance required by the ship in distress. It also includes search and rescue (SAR) communications and on-scene communications. All distress traffic should take place as far as possible on the frequencies set aside in each band for this purpose.

The radiotelephony distress signal is the word MAYDAY, pronounced as in the French expression "m'aider".

The MAYDAY signal must ***always*** precede all RT distress traffic transmission.

When using radiotelex for distress traffic, forward error correction (FEC) techniques must be used. All messages must be preceded by at least one carriage return, a line feed signal, a letter shift signal and the distress signal (MAYDAY). Distress communication should be established by the ship in distress and should be made in the broadcast forward error correction (FEC) mode. ARQ mode may be subsequently used where it is advantageous to do so.

2.19 IMPOSING SILENCE

The RCC responsible for controlling SAR operations also coordinates the distress traffic relating to the incident, it may, however, appoint another station to do this.

Controlling stations may impose silence on stations which interfere with distress or SAR traffic, this may be addressed to all stations or to an individual station. The procedure used is:

(a) in radiotelephony, the signal SEELONCE MAYDAY, pronounced as in the French expression "silence, m'aider";
(b) in radiotelex (NBDP) normally using FEC mode the signal SILENCE MAYDAY. However, the ARQ mode may be used when it is advantageous to do so.

All stations which are aware of distress traffic but are not taking part in it are forbidden to transmit on the frequencies being used for distress traffic until they receive a signal indicating normal working may be resumed.

A station in the mobile service which is able to continue its normal service while following distress traffic may do so. This is when the distress traffic is well established and provided the mobile station does not interfere with the distress traffic as previously indicated.

2.20 RESUMPTION OF NORMAL WORKING

When distress traffic has ceased on frequencies which have been used for distress traffic the RCC or station controlling SAR operations must initiate a message for transmission on those frequencies indicating that distress traffic has finished.

On radiotelephony this message consists of:

the distress signal MAYDAY;
the call "Hello all stations" or CQ (spoken as CHARLIE QUEBEC) spoken three times;
the words THIS IS (or DE spoken as DELTA ECHO in the case of language difficulties);
the call sign or other identification of the station sending the message;
the time of handing in the message;
the name and call sign of the mobile station which was in distress;
the words SEELONCE FEENEE pronounced as the French words "silence fini".

Example:

MAYDAY
ALL STATIONS, ALL STATIONS, ALL STATIONS
THIS IS
CLYDE COASTGUARD
1536 UTC YUCATAN/XCGC
SEELONCE FEENEE

When using radiotelex (NBDP) the message consists of:

the distress signal MAYDAY;
the call CQ;
the word DE;
the call sign or other identification of the station sending the message;
the time of handing in the message;

the name and call sign of the mobile station which was in distress;
the words SILENCE FINI.

Example:

MAYDAY
CQ DE OXZ
1536 UTC YUCATAN/XCGC
SILENCE FINI

2.21 ON-SCENE COMMUNICATIONS: ON-SCENE COORDINATOR

On-scene communications are those between the mobile unit in distress and other mobile units assisting in the distress. It also includes communications between mobile units and the unit coordinating SAR operations. This could be the on-scene coordinator (OSC), the commander of a rescue unit designated to coordinate search and rescue operations within a specified search area by the appropriate MRCC or MRSC to locate and rescue survivors. Until such time as an OSC has been designated, the first rescue unit arriving at the scene of the action should automatically assume the duties and responsibilities of an OSC.

The OSC shall be responsible for the following duties.

1. Coordinate operations of all SAR facilities on-scene.
2. Receive the search action plan or rescue plan from the SAR Mission Coordinator (SMC) or plan the search or rescue operation, if no plan is otherwise available.
3. Modify the search action or rescue action plan as the situation on-scene dictates, keeping the SMC advised (do in consultation with the SMC where practicable).
4. Coordinate on-scene communications.
5. Monitor the performance of other participating facilities.
6. Ensure operations are conducted safely, paying particular attention to maintaining safe separations among all facilities, both surface and air.
7. Make periodic situation reports (SITREPs) to the SMC. SITREPs should include but not be limited to:
   weather and sea conditions
   the results of search to date
   any actions taken
   any future plans or recommendations.
8. Maintain a detailed record of the operations:
   on-scene arrival and departure times of SAR facilities, other vessels and aircraft engaged in the operation
   areas searched
   track spacing used
   sighting and leads reported
   actions taken
   results obtained
9. Advise the SMC to release facilities no longer required.
10. Report the number and names of survivors to the SMC.

11. Provide the SMC with the names and designations of facilities with survivors aboard.
12. Report which survivors are in each facility.
13. Request additional SMC assistance when necessary (for example, medical evacuation of seriously injured survivors).

For further information on specific SAR duties and operations refer to the International Aeronautical and Maritime Search and Rescue Manual 1998 Edition and published by IMO and ICAO (ISBN sales number IMO-962E).

2.21.1 Standard format for Search and Rescue Situation Reports (SITREPs)

Situation reports (SITREPs) should be compiled as follows:

Short Form

To pass urgent essential details when requesting assistance, or to provide the earliest notice of a casualty.

TRANSMISSION PRIORITY (distress/urgency etc.)
DATE AND TIME (UTC or local date time group)
FROM (originating RCC)
TO
SAR SITREP (NUMBER) (to indicate nature of message and completeness of sequence of SITREPs concerning the casualty)
IDENTITY OF CASUALTY (name, call sign, flag State)
POSITION (latitude/longitude)
SITUATION (type of message, distress or urgency; date/time; nature of distress/urgency, for example, fire, collision, medical)
NUMBER OF PERSONS AT RISK
ASSISTANCE REQUIRED
COORDINATING RCC

Full Form

To pass amplifying or updating information during SAR operations, the following additional sections should be used as required.

DESCRIPTION OF CASUALTY (physical description, owner/charterer, cargo carried, passage from/to, life-saving appliances carried etc.)
WEATHER ON-SCENE (wind, sea/swell state, air/sea temperature, visibility, cloud cover/ceiling, barometric pressure)
INITIAL ACTIONS TAKEN (by distress craft and RCC)
SEARCH AREA (as planned by RCC)
COORDINATING INSTRUCTIONS (OSC designated, units participating, communications, etc.)
FUTURE PLANS
ADDITIONAL INFORMATION/CONCLUSIONS (include time SAR operation terminated)

Notes: Each SITREP concerning the same casualty should be numbered sequentially.

If help is required from the addressee, the first SITREP should be issued in short form if remaining information is not readily available.

When time permits, the full form may be used for the first SITREP or to amplify it.

Further SITREPs should be issued as soon as other relevant information has been obtained, particularly changes to on-scene weather. Information already passed should not need repetition.

During prolonged operations "no change" SITREPs, when appropriate, should be issued at intervals of about three hours to reassure the recipients that nothing has been missed.

When the incident has been concluded, a final SITREP should be issued as confirmation.

2.21.2 On-scene communications: Use of frequencies

The preferred frequencies for RT on-scene communication are 2 182 kHz and Ch.16 VHF, using simplex transmissions so that all on-scene mobiles share the relevant distress information.

The radiotelex (NBDP) frequency of 2 174.5 kHz may also be used for ship-to-ship on-scene communications but if telex is selected then forward error correction (FEC) mode must be used.

In addition to 2 182 kHz and Ch.16 VHF, the frequencies listed below may be used for on-scene communications between ships and aircraft.

3 023 kHz*
4 125 kHz
5 680 kHz*
Ch.6 VHF

* Ships can only listen to aircraft using these frequencies.

Once an on-scene frequency has been established, a continuous aural or teleprinter watch is maintained by all participating on-scene mobile units on that frequency.

Mobile stations of the maritime mobile service may communicate with stations of the aeronautical service on the aeronautical emergency frequency 121.5 MHz for distress and urgency only, and on the the aeronautical auxiliary frequency of 123.1 MHz for coordinated SAR operations, using class A3E emissions for both frequencies. They shall then comply with any special arrangements between the governments concerned by which the aeronautical mobile service is regulated.

2.21.3 Shore-to-ship distress communication with MRCC

Severe difficulties can be faced by MRCCs in contacting ships in a distress situation due to satellite communication links being jammed with other traffic.

The International Maritime Organisation (IMO) has issued guidance to MRCCs to assist them in contacting ships fitted with Inmarsat Mobile Earth Stations.

Ships Masters should ensure, that if in distress or assisting in SAR operations, that measures are taken to end all non-emergency communications from the ship to allow them to offer full assistance to the MRCC in maintaining communication. Inmarsat Fleet services have pre-emption in both directions. See sections 5.8 and 5.9.

2.22 LOCATING AND HOMING SIGNALS

Locating signals are used to assist with finding the distress mobile or for the location of survivors. Signals can be transmitted by:

(a) the mobile station in distress,
(b) by survival craft,
(c) by satellite EPIRBs,
(d) by search and rescue transponder (SART),
(e) by units searching for the mobile in distress.

Locating signals may be transmitted in the frequency bands:

117.975-136 MHz
156–174 MHz (VHF)
406–406.1 MHz
9 200-9 500 MHz (X-band radar)

Homing signals are transmitted by mobile units in distress or by survival craft to provide searching units with a locating signal used to determine the bearing of the transmitting station.

2.23 CHARGING FOR DISTRESS COMMUNICATION

The Inmarsat council defined what constituted distress and safety calls and confirmed that calls conforming to these definitions would not be charged for the use of satellites. This took effect from 1st January 1992, a month before GMDSS was introduced. The definition of ship to shore distress and safety traffic is:

(a) distress alerts;
(b) SAR coordination communications including communication subsequent to the initial distress alerts, by voice or telex, relating to the immediate assistance required by a ship which is in distress and in grave and imminent danger, provided that they are made to associated rescue coordination centres via code 39 (Maritime assistance);
(c) urgent navigational/meteorological danger reports by telex communications via code 42 (Navigational hazards and warnings);
(d) medical assistance for grave and imminent danger, by voice or telex via code 38 (Medical assistance) to an agency recognised by national maritime administrations.

Equivalent waiver arrangements cover ship-to-shore traffic. It was also agreed that reduced charges will apply for ship to shore medical advice communications via code 32 (Medical advice).

No charge is raised, against the maritime mobile station, for transmission in the maritime mobile service provided that the communications are:

(a) distress calls/messages or replies to such calls/messages;
(b) messages originating in maritime mobile stations requiring shore rescue authority support, e.g. man overboard, oil pollution or piracy attack;
(c) messages originating in maritime mobile stations notifying the presence of dangers to navigation (e.g. icebergs, derelicts, mines) or announcing cyclones, storms or fog;
(d) messages originating in maritime mobile stations notifying sudden changes in the position of buoys, the working of lighthouses, devices connected with buoyage, etc; or
(e) messages originating in maritime mobile stations notifying unexpected phenomena threatening air navigation or the sudden occurrence of obstacles at airports or special landing/berthing places at sea.

Any land station and landline charges for distress and safety communications will be billed to the appropriate authority.

2.24 OPERATIONAL PROCEDURES FOR URGENCY AND SAFETY MESSAGES IN THE GMDSS

Urgency and safety communications are defined as those relating to:

(a) navigational and meteorological warnings and urgent information;
(b) ship to ship safety of navigation communications;
(c) ship reporting communications;
(d) support communications for SAR operations;
(e) other urgency or safety messages;
(f) communications relating to the navigation, movements and needs of ships and weather observation messages destined for an official meteorological service.

2.24.1 **Urgency communications**

The use of the urgency call format implies that the station making the call has a very urgent message to transmit concerning the safety of the mobile unit or a person. The urgency signal or urgency call format must only be sent on the authority of the master or person responsible for the mobile unit or by a land earth station or coast station with approval of the responsible authority. The urgency signal must be transmitted on one or more of the authorised distress and safety frequencies or via the maritime mobile satellite service.

In the terrestrial service the announcement of the urgency message must be made on one or more of the authorised distress and calling frequencies using DSC and the urgency call format.

If the maritime mobile satellite service is to be used a separate urgency announcement is not required, as selection of URGENT PRIORITY gains priority access to the system.

The urgency signal in radiotelephony consists of the words PAN PAN each word being pronounced as in the French word "panne".

To transmit a DSC urgency message announcement, the following procedure is used:

(a) Tune the transmitter to DSC distress channel (2 187.5 kHz MF, or Ch.70 VHF).
(b) Insert by keying into the DSC controller according to manufacturers instructions:
  (i) The all ships call or the MMSI of an individual station.
  (ii) The call category (urgency).
  (iii) The frequency (or channel) on which the subsequent urgency communication will be transmitted. (2 182 kHz MF, or Ch.16 VHF).
  (iv) The type of communication to be used for the subsequent urgency transmission (usually RT but could be telex).
(c) Transmit the DSC call.
(d) Tune RT transmitter/receiver to the frequency or channel given in the DSC call (2 182 kHz MF, or Ch.16 VHF).

Transmit the urgency message using standard RT procedures for this type of message, as follows:

PAN PAN, PAN PAN, PAN PAN
ALL STATIONS, or called station repeated three times
THIS IS
the 9 digit MMSI AND the call sign or other identification of own ship repeated three times
the text of the urgency message

Example:

PAN PAN, PAN PAN, PAN PAN
ALL STATIONS, ALL STATIONS, ALL STATIONS
THIS IS
352251000 ORAMKI, 352251000 ORAMKI, 352251000 ORAMKI

In radiotelex (NBDP) the urgency message must be preceded by the urgency signal and the identification of the transmitting station.

Example:

PAN PAN DE ZPVG

When an urgency message requesting action has been transmitted the transmitting station must cancel this message as soon as it knows that the action is no longer required.

Urgency communication using radiotelex (NBDP) should be established in the broadcast FEC mode. ARQ mode may be subsequently used where it is advantageous to do so.

All messages must be preceded by at least one carriage return (CR), a line feed signal (LF), and the urgency signal (PAN PAN).

In the maritime mobile service the message shall be transmitted on a working frequency:

(a) in the case of a long message or medical call, or
(b) in areas of heavy traffic in the case of the repetition of messages transmitted on the frequencies above. An indication to this effect should be given at the end of the call.

The urgency signal shall have priority over all other communications, except distress. All stations which hear it shall take care not to interfere with the transmission of the message which follows the urgency signal.

In the maritime mobile service, urgency messages may be addressed either to all stations or to a particular station. Messages preceded by the urgency signal should as a general rule be drawn up in plain language.

Mobile stations which hear the urgency signal shall continue to listen for at least three minutes. At the end of this period, if no urgency message has been heard, a land station should, if possible, be notified of the receipt of the urgency signal after which normal service may be resumed.

However, land and mobile stations which are in communication on frequencies other than those used for the transmission of the urgency signal and message may continue their normal work without interruption provided the urgency message is not addressed to "all stations".

When the urgency signal has been sent before transmitting a message to "all stations" which calls for action by the stations receiving the message, the station responsible for its transmission shall cancel it as soon as it knows that action is no longer necessary. This cancellation message shall likewise be addressed to "all stations".

Example of an RT urgency call and message made after an initial DSC urgency call:

PAN PAN PAN PAN PAN PAN
ALL SHIPS ALL SHIPS ALL SHIPS
THIS IS
002320018 HOLYHEAD COASTGUARD, 002320018 HOLYHEAD COASTGUARD, 002320018 HOLYHEAD COASTGUARD
YACHT HAMAJO LAST SIGHTED 15 MILES SOUTHWEST OF CALF OF MAN NOW 72 HOURS OVERDUE DUBLIN ALL VESSEL KEEP SHARP LOOK OUT AND REPORT ANY SIGHTINGS OF THIS VESSEL OVER

2.24.2 Safety communications

The safety call format implies that the station making the call has an important navigational or meteorological warning to transmit. In the terrestrial service the announcement of the safety message must be made on one or more of the authorised distress and calling frequencies using DSC. A separate announcement need not be made if the message is to be transmitted through the maritime mobile satellite service.

To transmit a DSC safety message announcement, the following procedure is used:

(a) Tune the transmitter to DSC distress channel (2 187.5 kHz MF, or Ch.70 VHF).

(b) Insert by keying into the DSC controller according to manufacturers instructions:

  (i) The appropriate call format either, all ships, area call or individual call (in the latter case insert the MMSI of the called station).

  (ii) The call category (safety).

(iii) The frequency (or channel) on which the subsequent safety communication will be transmitted (2 182 kHz MF, or Ch.16 VHF).

(iv) The type of communication to be used for the subsequent safety transmission (usually RT but could be telex).

(c) Transmit the DSC call.

(d) Tune RT transmitter/receiver to the working frequency or channel given in the DSC call.

Transmit the safety message using standard RT procedures for this type of message, as follows:

SECURITE, SECURITE, SECURITE
ALL STATIONS, or called station repeated three times
THIS IS
the 9 digit MMSI AND the call sign or other identification of own ship repeated three times
the text of the safety message.

Example:

SECURITE, SECURITE, SECURITE
ALL STATIONS, ALL STATIONS, ALL STATIONS
THIS IS
257966140 GARGANA, 257966140 GARGANA, 257966140 GARGANA
CONTAINER FLOATING IN THE SEA IN POSITION 58 DEGREES 21 MINUTES NORTH 20 DEGREES 04 21 MINUTES EAST AT 1430 HOURS
VESSELS KEEP SHARP LOOK OUT OVER

The safety signal must be transmitted on one or more of the authorised distress and safety frequencies or via the maritime mobile satellite service. The safety signal in radiotelephony consists of the word SECURITE pronounced (SAY-CURE-E-TAY) as in French.

In radiotelex (NBDP) the safety message must be preceded by the safety signal and the identification of the transmitting station

Example:

SECURITE DE EKYL

Safety communication using radiotelex (NBDP) should be established in the broadcast (FEC) mode. ARQ mode may be subsequently used where it is advantageous to do so. All messages must be preceded by at least one carriage return, a line feed signal, a letter shift signal and the safety signal (SECURITE).

Messages containing information concerning the presence of dangerous ice, dangerous wrecks, or any other imminent danger to marine navigation, must be transmitted as soon as possible to other ship stations in the vicinity, and to the appropriate authorities at the first point on the coast with which contact can be established. These transmissions must be preceded by the safety signal.

All stations hearing the safety signal shall listen to the safety message until they are satisfied that the message is of no concern to them. They shall not make any transmission likely to interfere with the message.

2.25 MEDICAL TRANSPORTS

The term medical transports is defined in the 1949 Geneva Conventions and Additional Protocols. It refers to any means of transportation by land, water or air, whether military or civilian, permanent or temporary, assigned exclusively to medical transportation and under the control of a competent authority of a party to a conflict or of neutral States and of other States not parties to an armed conflict, when these ships, craft and aircraft assist the wounded, the sick and the shipwrecked.

For the purpose of announcing and identifying medical transports protected under the Conventions previously mentioned, urgency procedures are used. When RT is used, the urgency signal is followed by the single word MAY-DEE-CAL pronounced as in the French word "medical".

Example:

PAN PAN MAY-DEE-CAL

Where radiotelex (NBDP) is used, the urgency signal is followed by the word MEDICAL.

Example:

PANPANMEDICAL

The use of such a signal indicates that the message which follows concerns a protected medical transport. The message contains the following information :

(a) call sign or other recognised means of identification of the medical transport;
(b) the position of the medical transport;
(c) number and type of vehicles in the medical transport;
(d) intended route;
(e) estimated time en route and of departure and arrival, as appropriate;
(f) any other information, such a flight altitude, radio frequencies guarded, languages used and secondary surveillance radar modes and codes.

Standard maritime radar transponders may be used for the identification and location of medical transports at sea, while aircraft medical transports may use secondary surveillance radar for a similar purpose.

The use of radiocommunications for announcing and identifying medical transports is optional; however, if they are used, the provisions of the Radio Regulations shall apply. Thus a DSC call can be made announcing that a medical transport message will be broadcast on an appropriate distress and safety frequency or channel.

2.26 INTERSHIP SAFETY OF NAVIGATION COMMUNICATION

Intership navigation safety communications are defined as VHF RT communications between ships whose purpose contributes to the safe movement of ships. Ch.13 VHF is used for bridge to bridge RT (voice) communication within the GMDSS for safety of navigation. It is the ONLY channel that can be used within GMDSS *without transmitting* a preceding DSC call on Ch.70. Commercial vessels

should, where practicable, monitor this channel for communications relating to safety of navigation.

2.27 OTHER FREQUENCIES USED FOR DISTRESS AND SAFETY

Radiocommunications for distress and safety purposes can be carried out on any appropriate communications frequency including those used for public correspondence. The maritime mobile satellite bands 1 530-1 544 MHz and 1 626.5-1 645.5 MHz may be used for distress and safety radiocommunications as well as distress alerting purposes.

2.28 PIRACY AND ARMED ROBBERY AT SEA: USE OF RADIO

The risk of piracy in international waters and of armed robbery against ships at anchor or when underway through coastal State's territorial waters has increased in certain areas of the world. The number and severity of such incidents has increased significantly in recent years particularly to shipping in South East Asia waters. Such attacks pose a real threat not only to those on board ship but also to the interests of coastal States. Attacks continue to occur in the Indian Sub-Continent, while the number of attacks in Africa has seen a major increase. Vessels under way in seas off the North East coast of Somalia and in the Red Sea/Gulf of Aden are at risk from hijacking. Vessels and their crews risk being held for ransom by armed militia. The use of the ship's radio equipment is encouraged to report such incidents and summon assistance if necessary. Marine Guidance Notice MGN 241 (M) issued in November 2002 offers recommendations to counter such attacks. Among these recommendations are those relating to radio procedures which are given below.

2.28.1 **Radio procedures**

Radio personnel

A suitably qualified radio operator should be on duty at all times when ships are in, or approaching, areas where attacks occur. This duty should not be performed by the Master though, on occasions, this may be unavoidable. Since the mandatory introduction of GMDSS in February 1999, the Navigational Officer on Watch (OOW) often carries out the duties of a Radio Operator. To ensure that the vessel's bridge is adequately manned when transiting potentially hazardous waters, it is advisable that a duly qualified, dedicated Radio Operator perform Radio Watch duty. This contingency allows the OOW and the master to concentrate on navigational duties and maintaining the extra vigilance that is required when operating in high-risk areas.

Prior to entering areas where attacks have occurred Radio Operators should practice and perfect all appropriate radio operational procedures and ensure all transmitters, including satellite mobile earth stations, are fully operational and available for immediate use on distress and safety frequencies. Where an Inmarsat mobile earth station is provided it may prove useful to draft and store "standard messages" for ready use in an emergency in either the equipment's memory or on computer disk. A special code for piracy/armed robbery attack is

available for use on Digital Selective Calling (DSC) equipment. Where practicable and appropriate DSC equipment should be modified to incorporate this facility. Masters should ensure that all procedures to generate a distress alert on any communications equipment are clearly marked on, or near, the equipment and all appropriate crew members briefed on their operation.

Masters should bear in mind the possibility that attackers may be monitoring ship-to-shore communications and using intercepted information to select their targets. Caution should, therefore, be exercised when transmitting information on cargo or valuables on board in areas where attacks occur.

2.28.2 Radio watchkeeping and responses

A continuous radio watch should be maintained with the appropriate shore or naval authorities when in areas where attacks occur. Continuous watch should also be maintained on all distress and safety frequencies, particularly VHF Ch. 16 and 2 182 kHz. Ships should also ensure all maritime safety information broadcasts for the area are monitored. As it is anticipated that Inmarsat's enhanced group calling system (EGC) will normally be used for such broadcasts using the SafetyNET service, owners should ensure a suitably configured EGC receiver is continuously available when in, or approaching, areas where there is a risk of attack. Owners should also consider fitting a dedicated receiver for this purpose, i.e. one that is not incorporated into a mobile earth station used for commercial purposes, to ensure no urgent broadcasts are missed.

(Masters should note that the IMB Piracy Reporting Centre broadcasts daily status reports to ships in Indian, Atlantic and Pacific Ocean Regions on the SafetyNET of Inmarsat at 0001 UTC each day.)

The International Maritime Organisation (IMO) recommends in MSC Circular 597, issued August 1992 and supplemented by an Addendum issued in May 1993, that reports on pirates or armed robbers should be made to the relevant Maritime Rescue Coordination Centre (MRCC) for the area. Information on RCCs may be found in the Search and Rescue Section of Volume 5 of Admiralty List of Radio Signals. MSC Circular 597 also recommends that governments should arrange for MRCCs to be able to pass reports of attacks to the appropriate law enforcement agencies or naval authorities. The IMO subsequently published MSC Circular 622/Rev 1 in July 1999. This circular gives detailed recommendations to governments to assist in the prevention and suppression of piracy and armed robbery attacks against ships. In may 2002 the IMO published MSC Circular 623/Rev 3 as an equivalent guide to ship owners and ship operators.

In the event Masters are unable to contact the relevant MRCC, it is recommended that they report the incident to the IMB Piracy Reporting Centre, which in turn, will pass the message to the appropriate authorities. Contact details are listed in section 2.28.5

If suspicious movements are identified which may result in an imminent attack, the ship is advised to contact the relevant MRCC. Where the Master believes these movements could constitute a direct danger to navigation consideration should be given to broadcasting an "All Stations" (CQ) "danger message" as a warning to other ships in the vicinity as well as advising the appropriate MRCC. A danger message should be transmitted in plain language on a VHF working

channel following an announcement on VHF Channel 16, and/or transmission of a DSC call on VHF channel 70 using "safety" priority. All subsequent messages should be preceded by the safety signal (Securite).

When, in his opinion, there is conclusive evidence that the safety of his ship is threatened, the Master should immediately contact the relevant MRCC and, if considered appropriate, authorise broadcast of an "All stations" "Urgency message" on VHF Channel 16, 2 182 kHz, or any other communications service considered appropriate, e.g. Inmarsat. All such messages should be preceded by the Urgency signal (PAN PAN) and/or a DSC Urgency call on VHF Channel 70 or 2 187.5 kHz using the "all ships urgency" category. If the Urgency signal has been used and an attack does not, in fact, develop the ship should cancel the message as soon as it knows that action is no longer necessary. This cancellation message should likewise be addressed to "all stations".

Should an attack occur and, in the opinion of the Master, the ship or crew are in grave or imminent danger requiring immediate assistance, he should immediately authorise the broadcast of a distress message, using the radio communication systems most appropriate for the area taking into account its GMDSS designation: A1, A2, A3 or A4. The appropriate RCC should acknowledge receipt and attempt to establish communications. To minimise delay, if using a mobile earth station, ships should ensure the coast earth station associated with the RCC is used.

Masters should bear in mind that the distress signal is provided for use only in case of imminent danger and its use for less urgent purposes might result in insufficient attention being paid to calls from ships really in need of immediate assistance. Care and discretion must be employed in its use, to prevent its devaluation in the future.

Where the transmission of the distress signal is not fully justified use should be made of the urgency signal. The urgency signal has priority over all communications other than distress.

2.28.3 **Standard Message** Formats

The standard message formats for:

(a) initial messages — piracy attack alert, and

(b) piracy attack/sighting/suspicious act reports which were agreed by the IMO Sub-Committee on Radio Communication in January 1993 and updated by MSC Circular 622/Rev 1 published in June 1999, are set out in section 2.28.6

In addition, guidance for the use of radio signals by ships under attack or threat of attack from pirates or armed robbers is available in Maritime Safety Committee (MSC) Circular 805 published in June 1997. This circular recommends that a "Piracy/Armed Robbery Attack Message" should be sent through Inmarsat-C or on an available DSC or other distress and safety frequency. Given that some pirates may carry equipment capable of detecting all radio signals, including satellite communications, this circular also recommends that communication should not be attempted if a ship has been boarded and its crew specifically ordered to maintain radio silence.

2.28.4 **Secreted VHF transceiver**

As a result of communications equipment being damaged in the past by attackers to prevent an early alarm being raised, particularly when attacks have taken place off port, owners and Masters are recommended to secrete a VHF transceiver on the ship to allow contact to be established with the shore authorities if the main communications equipment is put out of action. Consideration could also be given to the installation of handheld Iridium telephones. These sets have global coverage, unlike the traditional VHF transceiver, and would allow the ship's Master to inform, and converse with, more distant authorities as well as the authorities in the region of the attack.

2.28.5 Piracy Reporting Centre

The latest information on piracy attacks and the regions of greatest risk may be obtained free of charge from the ICC International Maritime Bureau's Piracy Reporting Centre at Kuala Lumpur.

The centre operates 24 hours a day and can be contacted on the following Anti-Piracy HELPLINE numbers:

Telephone: ++ 60 3 2078 5763
FAX: ++ 60 3 2078 5769
TELEX: MA 31880 IMBPCI
E-mail: *imbkkl@icc-ccs.org.uk*
INTERNET: *www.icc.ccs.org*

The centre issues status reports and warning messages on the SafetyNET service of Inmarsat-C at 0001 UTC each day. The centre also posts a weekly update of attacks on the internet at *www.icc.ccs.org.* This update, posted every Tuesday, is compiled from the centre's daily status bulletins to ships at sea.

2.28.6 **Initial message: Piracy attack/armed robbery report**

1. Vessel's name and call sign/Inmarsat identity (IMN), plus ocean region code, IMO number and MMSI.
   MAYDAY/DISTRESS ALERT (see Note).
   URGENCY SIGNAL
   PIRACY/ARMED ROBBERY ATTACK.
2. Vessel's position (and time of position UTC) - including course and speed.
3. Nature of event.

Note: It is expected that this message will be a distress message because the vessel or persons will be in grave or imminent danger when under attack. Where this is not the case, the MAYDAY/DISTRESS ALERT is to be omitted.

Use of distress priority (3) in the Inmarsat system will not require MAYDAY/DISTRESS ALERT to be included.

2.28.7 **Piracy/armed robbery attack/suspicious act report**

1. Vessel's name, call sign and IMO number.
2. Reference initial PIRACY/ARMED ROBBERY ALERT.

3. Position of incident.
4. Date/time of incident (UTC).
5. Details of incident, e.g.
   Method of attack.
   Description of suspect craft.
   Number and brief description of pirates, including weapons carried and/or language spoken.
   Injuries to crew.
   Damage to ship.
   Brief details of stolen property/cargo.
6. Last observed movements of pirate/suspect vessel, e.g. Date/time/course/position/speed.
7. Assistance required.
8. Preferred communications with reporting vessel, e.g. Appropriate Coast Radio Station.
   HF/MF/VHF. Inmarsat identification (IMN), plus the ocean region code, MMSI.
9. Date/time of report (UTC).

2.28.8 After attack reports

A post attack report should be sent immediately to the relevant RCC for onward transmission to the law enforcement agencies or naval authorities of the coast state. The format of report to the IMO, required under the relevant IMO Assembly Resolutions on acts of piracy and armed robbery at sea is given below.

1. Number
2. Name
   Type of ship
   Flag
   Gross tonnage
3. Date
   Time
4. Position of incident. The position should be as accurate as possible including latitude and longitude coordinates or as bearing and distance from conspicuous landmark.
5. Details of the incident
6. Consequences for crew, ship,cargo
7. Action taken by the Master and the crew
8. Was the incident reported to the coastal Authority? If so, to whom?
9. Reporting State or international organisation
10. Action taken by the coastal State.

As well as the information on the identity and location of the ship, any injuries to crew members or damage to the ship should be reported as should the direction in which the attackers departed together with brief details of their numbers and, if possible a description of their craft. If the crew have apprehended an attacker that should also be reported in this signal.

2.29 FUTURE DEVELOPMENTS: THE SHIP SECURITY ALERT SYSTEM (SSAS)

MSC /Circ.1072 gives guidance on the design of ship security alert systems provided to comply with SOLAS regulations. The intention of this security alert system is to be able to send a covert signal or message from a ship that is not obvious to anyone on the ship who is not aware of the alert mechanism. It is of use therefore in circumstances where a ship wishes to inform a person ashore of a problem with a minimum number of persons onboard being aware of the action. The procedures for the security alert are agreed with the ship's Administration and as part of the ship security plan and ideally should be individual to the ship. It *is* not intended that the ship security alert procedures should be to an internationally agreed standard or conform to any particular format for all ships.

Three methods of achieving the alert have been identified as follows:

- A system employing proprietary tracking equipment provided by traffic service providers (e.g. Pole Star, ShipLoc and Transas). The ship carries a concealed equipment box working over a satellite service on its upper deck which transmits a position report at, typically, 6-hourly intervals. Interruption of power to the equipment or arming of the equipment by means of sensors or manual buttons causes the equipment to transmit a different format of position report. The tracking service providers monitor the transmission reports and inform the Company when the transmission format changes.
- A system may utilise modifications of GMDSS equipment.* Some GMDSS equipment is not very suitable as it is optimised for "all station" calling and may involve manual setting of frequencies etc. and provides confirmation on the ship of messages sent. In these types of systems the ship security alert contains identifiers to ensure that is is not possible to confuse it with a GMDSS distress, urgency safety alert.

  * Inmarsat are understood to be developing modifications to existing equipment that will allow for this service to be implemented. *(Note:* COSPAS-SARSAT are also understood to be considering offering a service over the COSPAS-SARSAT system.)
- A system in current use involves the exchange of messages containing key works between a ship and, typically, the Company. These messsages may be by speech or data communications. Ship equipment which may be used includes cellular phones in coastal areas and satellite services away from coastal areas. It may be possible to use GMDSS VHF/MF/HF equipment in areas where there are coastal facilities for receiving addressed calls.

The above list is not intended to be exhaustive and is not intended to inhibit future developments.

The ship security alert system requires a minimum of two activations points, one of which should be on the bridge. These will typically be fixed or portable telephone handsets, fixed or portable keypads or fixed or portable buttons. Measures should be incorporated in the activation points to avoid their inadvertent operation and the generation of false alerts.

The SSAS has to conform to performance standards not inferior to IMO Resolution MSC.136 (76), and shall be capable of being activated from the navigation bridge and at least one other location.

The SSAS when activated should:

(a) initiate and transmit a ship to shore security alert to a competent authority;
(b) not send the ship security alert to any other ships;
(c) not raise any alarm on board the ship; and
(d) continue the ship security alert until deactivated and/or reset.

2.30 MEDICAL ADVICE: GENERAL INFORMATION

Numerous coast stations throughout the world offer a 24-hour medical advice service to shipping. Ships can seek a medical consultation about illness or injury by making a radiotelephone call, or sending a telex to the appropriate coast station. This radiomedical consultation is given free of charge but incurs no responsibility. The exchange of telex or telephone calls relating to the consultation are also free of charge. Administrations undertake to forward medical advice messages with the least possible delay. In situations where a radiomedical consultation is required very rapidly the use of the urgency signal "PAN PAN" (sent three times) is recommended to gain priority.

When requesting medical assistance for an ill or injured person, information concerning (a) the patient and (b) the ship will be required by shore authorities. Although use of the English language is normal, some administrations may specify the language to be used in the exchange of messsages. Where this causes difficulty the use of the International Code of Signals, Volume II (medical section) is highly recommended and messages should be given in the correct format.

Information about the patient

Description of patient
Previous health
Localisation of symptoms, infections or diseases
General symptoms
Particular symptoms
Diagnosis

Information about the vessel

Name and nationality
Call sign
Identification number. (MMSI or IMN if using the Inmarsat system).
Position
Next port and ETA
Nearest port.

Language to be used

English
State other language to be used
Medical section (Volume II), International Code of Signals.

Request for additional information from shore medical authorities

Once the relevant information about the casualty or patient has been sent to the shore medical authorities they may request additional information from the

ship's staff to assist them make a diagnosis. This will be in the form of "follow up" questions such as that given in the example below.

Comment or question	*Code*
I cannot make a diagnosis	MQP
Please answer the following question(s)	MQC
Temperature taken in the mouth is?	MBR RQ
Vomiting is present?	MEM RQ

Ships should reply to this request in the same order as it has been demanded and once this additional information has been considered by the medical authorities medical advice will be sent to the ship. Once again this will be done using appropriate code groups from the medical section of the International Code of Signals.

Operators should be aware of the procedural signals "C", "N" and "RQ" when using the International Code of Signals. These signal when used immediately after the main three digit code changes its meaning into affirmative (correct), negative (no) and interrogative (question) respectively, see examples below.

MDL	Pain is severe.
MDL N	Pain is not severe.
MDL RQ	Is pain is severe?
MDL C	Affirm pain is severe.

2.31 MEDEVAC: GENERAL INFORMATION

In evaluating the need for medical evacuation (MEDEVAC), the risks should be given deep consideration and only attempted when the patient is in a serious condition. The benefits of such evacuations must be weighted against the inherent dangers posed to the patient being evacuated and to the crew of the rescue craft. To assist in making a decision on the necessity for evacuation, medical advice can be obtained from an MRCC or coast radio station, such stations will normally transfer the call to a doctor. Medical advice messages should be addressed to the nearest coast station offering this service appropriate to the area in which the vessel is sailing and should be signed by the master or person responsible for the ship. Details of station offering medical advice is contained in Admiralty List of Radio Signals, Volume I. It should be noted that the final decision whether to conduct an evacuation or not resides in the person in command of the rescue craft.

2.31.1 MEDEVAC: Information to be furnished to MRCC

The information furnished to the MRCC or coast radio station should include the items outlined below. In certain cases it may also be necessary to amplify this message by including additional information.

Information about the vessel

(a) Name, nationality and radio call sign or identification number. (MMSI or IMN if using the Inmarsat system).
(b) Position of vessel and next port of destination.
(c) ETA, course and speed.

(d) Additional pertinent remarks. This could include details of the type of vessel, the colour of the hull or other distinguishing features.

Information about nearest port

(a) Name, address and phone number of vessel's agent.
(b) Last port of call, next port of call and ETA of next port of call.
(c) Communications and homing signal available.

Information about the patient

(a) Patient's name, age, gender, nationality and language.
(b) Patient's respiration, pulse rate, temperature and blood pressure.
(c) Location of pain.
(d) Nature of illness or injury, including apparent cause and related history.
(e) Symptoms (visible and those felt by the patient).
(f) Type, time, form and amounts of all medications given.
(g) Time of last food consumption.
(h) Ability of patient to eat, drink, walk or be moved.
(i) With accident cases, how the accident occurred.
(j) Whether the vessel has a medicine chest and refer to appropriate class of drugs and whether a physician or other medically trained person is aboard.
(k) Whether a suitable clear area is available for helicopter hoist operations or landing.
(1) Additional pertinent remarks.

Once again in cases of language difficulties, consideration should be given to using the medical section of the IMO's International Code of Signals.

Once the doctor ashore has all the available information a medical assessment of the seriousness of the situation can be made. The doctor will normally give advice to the ship's staff as to immediate care of the patient or casualty. On completion of the communications the doctor will advise the SAR authorities as to the most appropriate evacuation method and whether helicopter evacuation is desirable. The MRCC will make appropriate arrangements to maintain communication with the ship.

2.31.2 Radiocommunication with helicopters

Helicopters normally use the aeronautical VHF and UHF radiotelephony bands and will not be capable of working on the maritime MF RT band. However, some large helicopters are able to communicate on 2 182 kHz the GMDSS RT distress and safety frequency.

Successful helicopter medevac operations depend on good communications being established between ship and helicopter and between ship and other interested authorities. In situations where direct communication between the ship and the helicopter offering assistance cannot be established on either VHF or 2 182 kHz possibilities for relaying information exist. An MRCC or coast radio station can be used or alternatively if a lifeboat is in the vicinity it too could be used as a relaying station.

As previously stated in section 1.5.1, under the SOLAS regulations passenger ships have to be equipped with aeronautical frequency VHF radios. Thus every passenger ship will have the means for two-way on-scene radiocommunications for search and rescue purposes using the aeronautical frequencies 121.5 MHz and 123.1 MHz to communicate with helicopters.

For further general information about helicopter evacuation procedures reference should be made to the International Aeronautical and Maritime Search and Rescue Manual, Volume III. Section 2, titled "Rendering Assistance" is relevant as are the more detailed procedures for medical evacuation contained in Section 4, entitled "On-board Emergencies". The Guide to Helicopter/Ship Operations published by the International Chamber of Shipping (ICS) may also be referred to as an information source.

2.32 MEDICAL ADVICE VIA INMARSAT

Provision *is* made in GMDSS for ships to obtain medical advice or assistance at any time by utilising the Inmarsat satellite communication system. Vessels can call the appropriate national authority by telex or telephone.

For example, medical advice from Goonhilly is obtained by using the following procedures:

(a) select LES identification code (e.g. 02);
(b) select duplex channel;
(c) *select ROUTINE PRIORITY;*
(d) initiate call;
(e) when clear to do so either:
Select 32+ for telex or key 32# for telephone.

This connects the ship with the relevant medical authorities directly or to a special operator who will ensure that the message is passed to the medical authorities.

A standard message format as shown below should always be used for this type of message:

(a) the word *MEDICO;*
(b) name of ship;
(c) ship's identification number (IMN) and call sign;
(d) ship's position;
(e) condition of the ill or injured patient;
(f) symptoms;
(g) any other information considered relevant.

For calls to medical authorities other than those associated with a LES whose telex or telephone number is known the normal ROUTINE call procedure is used.

2.33 MEDICAL ASSISTANCE VIA INMARSAT

Medical assistance can be obtained from land earth stations if the condition of the patient warrants it. Ships should use the calling procedure outlined below once again using Goonhilly as the LES:

(a) select LES identification code (e.g. 02);
(b) select duplex channel;
(c) select ROUTINE PRIORITY;
(d) initiate call;
(e) when clear to do so either:
Select 38+ for telex or key 38# for telephone.

Details for evacuation of the patient, having a doctor sent to the ship, or having an ambulance on standby for the ships arrival etc can then be arranged between the ship and medical authorities ashore.

2.34 MEDICAL ADVICE VIA UK COASTGUARD STATIONS

In the UK, rescue centres operated by HMCG are responsible for the service to provide medical advice calls from ships at sea. In the UK, the two officially designated medical centres providing this service are located at the Queen Alexandra Hospital, Portsmouth, and at the Aberdeen Royal Infirmary. For situations where illness or injuries occur at sea, the radio medical advice is provided free of charge and such advice is considered as a supplement to the first aid training of ship's crew.

Medical messages are transmitted by radiotelephony on 2 182 kHz or Ch.16 VHF after an initial DSC urgency alert on 2 187.5 kHz (MF) or Ch.70 VHF. Alternatively the appropriate Inmarsat service can be accessed.

International regulations allows the urgency signal ("PAN PAN" repeated three times) to be used to precede a medical call. Calls may be addressed to "All stations" as shown below.

PAN PAN, PAN PAN, PAN PAN,
ALL STATIONS, ALL STATIONS, ALL STATIONS
THIS IS
HARPULA, HARPULA, HARPULA
CALLSIGN MIKE ECHO NOVEMBER DELTA
IN POSITION 58°33' NORTH 009°41' WEST
I REQUIRE MEDICAL ADVICE
OVER

Alternatively where the ship is close to a Coastguard or coast radio station can be called directly as in the example below.

PAN PAN, PAN PAN, PAN PAN,
HM COASTGUARD, HM COASTGUARD, HM COASTGUARD
THIS IS
FINTRY BAY, FINTRY BAY, FINTRY BAY
CALLSIGN GOLF NOVEMBER CHARLIE FOXTROT
IN POSITION 53°13' NORTH 003°44' WEST
I REQUIRE MEDICAL ADVICE
OVER

All UK Coastguard stations are fully equipped with equipment to connect ships to a designated medical contact in its area and will direct ships as to the working frequency or channel to be used. Thus ships will be able to be connected directly

to a doctor by telephone. This direct radiotelephone connection with a doctor is the preferred communication method when medical advice is required.

The general format used for medical messages is outlined below:

(a) ship's name, callsign and nationality;
(b) ship's position, next port of call, ETA and nearest port if required to divert;
(c) patient's details, e.g. name, sex, age medical history, etc;
(c) brief details of of the patient's illness or injury
(d) medication available on board the vessel.

2.35 MEDICAL ASSISTANCE VIA UK COASTGUARD STATIONS

Ships who have sought medical advice via UK Coastguard stations may find that the medical authorities ashore recommend that further medical assistance is necessary. Particularly in situations where the ship is in mid ocean the Coastguard should be advised when the ship is likely to be within range of a helicopter in case of medical evacuation. HM Coastguard will normally coordinate the rescue services irrespective of the form of evacuation used, e.g. by lifeboat or helicopter, sending a medical team or doctor to the ship by helicopter, or arranging an ambulance to standby for the ships arrival, etc.

CHAPTER 3

Digital Selective Calling (DSC)

3.1 DIGITAL SELECTIVE CALLING: GENERAL INTRODUCTION

DSC forms an vital subsystem within the GMDSS, being used as the primary alerting system of the MF, HF and VHF terrestrial services. Indeed, the terrestrial elements of GMDSS adopted by the 1998 amendments to the International Convention for SOLAS, (1974) are based on the use of DSC for distress and safety communications. As such, any ship or coast station receiving a DSC distress alert becomes aware that an important communication relating to distress will follow on a radiotelephone or radiotelex distress channel/frequency.

The use of DSC within the GMDSS is now well established. However, it is worth noting that a distinction has to be made between the use of DSC for distress, urgency and safety calls and its use as a calling mechanism in establishing general communications, i.e. routine or business traffic. Procedures differ slightly depending on the frequency bands in use, for example on the VHF band, *ALL* DSC calls are made on Ch.70. On the MF and HF bands specific frequencies are used for distress, urgency and safety while separate dedicated frequencies are used for routine DSC calling, e.g. 2 177 kHz is used for routine ship-to-ship calling. While ship-to-ship DSC calling is possible using the HF bands, in practice it is seldom used. To emphasise these different uses of DSC, this chapter has been divided into two parts, with Part I dealing with distress, urgency and safety issues while Part II deals with routine DSC calling procedures.

Watchkeeping on commercial frequencies can be set up by the ship's operators by selecting the correct frequencies to be scanned by the DSC receiver. This action must not of course interrupt the statutory monitoring of the various DSC distress frequencies by the dedicated DSC watchkeeping receiver. Thus, generally, a separate DSC receiver is required to monitor or scan ship-to-ship and commercial DSC frequencies in addition to the dedicated DSC distress watchkeeping receiver.

As the name suggests, Digital Selective Calling is simply a calling system which transmits packets of digitised data. The five categories of calls are: Distress, Urgency, Safety, Ship's Business and Routine. The calling system employs an error detecting code. Thus the system can be used to make calls from ship-to-shore, shore-to-ship or ship-to-ship. In addition to routine calls to specific stations the system can also be used to make other types of calls. These include calls to "all ships" (including urgency and safety), calls to ships in a particular geographical area, or calls to specific fleets of ships.

A feature of DSC is the inclusion of telecommands that permit the operator to include additional information into a DSC call. For example, the appropriate channel or transmission frequencies to be used for subsequent communication or

mode of emission to be used can be included as a telecommand. In the initial DSC call therefore, the transmission mode of the subsequent communication will be included, which allows the receiving station to identify whether radiotelephony or radiotelex is to be used. A second telecommand indicates to the receiving station the appropriate frequency to which to tune for subsequent traffic. Some modern radiocommunication equipment permits full automatic DSC operation on board ship. With this system, when used in conjunction with automatically tunable transmitters and receivers, unattended operation is possible. This provides automatic call acknowledgements on receipt of a DSC call and automatic transfer to the appropriate working frequencies or channels.

The advantages of DSC become apparent when, for example, a coast station requires to contact a ship. The coast station can make a DSC transmission to a specific ship on a national calling frequency. Although all ships within receiving range of the coast stations transmission will be able to pick up the signals, only the specified ship will respond to the call and display a "received call" indication. The reason being that each ship or coast station has a Maritime Mobile Service Identity (MMSI), this is a nine-digit code which uniquely identifies the ship or coast station or a group of stations. Maritime Identification Digits (MID) contained within the MMSI codes give an indication of nationality of the station. For example, codes commencing with the numbers 232, 233 and 234 indicate UK stations, 227 those of French stations (see Appendix 7). All coast station codes start with digits 00.

Examples

Mersey Mammoth/GHPK	MMSI = 232000580
Holyhead Coast Guard	MMSI = 002320018

The MMSI coding system also has to cater for the inclusion of a code which will allow "all ships" to respond to alerting and safety calls. Other codes are used:

(a) to call ships with a common interest, i.e. ships belonging to a particular company or nationality;
(b) to call groups of ships in a particular geographical area.

Various technical methods are employed in the DSC controller to eliminate the reception of false alert calls. The system is very rapid, an individual DSC call can vary between 0.45 to 7.2 seconds depending on the type of call and whether the transmission is made on MF or VHF.

3.2 DSC DIGITAL CODING

The technical characteristics of equipment used for digital selective calling shall be in conformity with the relevant CCIR Recommendations.

The call information is in the form of seven unit binary combinations producing 128 characters. The characters are coded in a ten unit error detecting code, each character is sent twice but separated in time (time spread or multiplex) and a message check character added at the end of the call. These methods allow error correction to be obtained. Error correction is essential to overcome the problems of fading and interference that can occur on radio circuits which can mutilate the transmitted digital code causing incorrect data to be received.

PART I - DSC DISTRESS ALERTING

3.3 DISTRESS ALERTING BY DSC: GENERAL INFORMATION

Operational experience has shown that, since the introduction of GMDSS in 1991, many hundreds of DSC distress alerts received by shore stations are false. Ships personnel are urged that they take utmost care when using DSC equipment to avoid "finger trouble" and to prevent the accidental transmission of a false alert. Shore station personnel have to spend considerable time and effort investigating such alerts, which could divert attention from genuine distress alerts (see Appendix 1). Also the procedures used to acknowledge a received DSC distress alert must be rigidly adhered to (see section 3.6)

DSC alerting equipment provides for automatic alerting without the need for aural watch and can operate under conditions of poor signal and high noise interference. Some DSC controllers automatically switch the transmitter to the correct DSC frequency when the distress condition is selected.

DSC controllers may be interfaced with electronic navigating equipment to continuously update the ship's position. However, where this facility is not connected, ships navigating staff are recommended to manually enter the ships position at regular intervals (see Chapter 11). This ensures that in the event of an emergency recent positional information will be included in the distress alert.

DSC distress alert calls or relayed distress alert calls are repeated to increase the probability of reception. Error correction is included by transmitting each character twice and by the inclusion of an overall message check character. On receipt of a DSC call, a display or printout of the message is produced which includes information about the addressed station plus identity of the transmitting station. When distress and safety calls are received, in addition to the printout of the message, audible or visual alarms are activated to draw the attention of the operator. All DSC distress calls are addressed to "all stations" and will be received by all suitably equipped units within the propagation range of the frequency band in use.

Distress alerts can be sent on any DSC distress and safety frequency in the MF, HF or VHF bands including:

MF band	*HF bands*	*VHF band*
2 187.5 kHz	4 207.5 kHz	156.525 MHz (Ch.70)
	6 312 kHz	
	8 414.5 kHz	
	12 577 kHz	
	16 804.5 kHz	

Note: Ch.70 VHF is used for all calls, i.e. distress/safety and routine calls.

Two types of distress call attempt exist:

(a) Single frequency call attempt: This is five consecutive DSC distress calls on ONE frequency in the MF and HF bands as indicated above; it takes approximately 35 seconds to complete. To avoid call collision and loss of acknowledgements, this call attempt may be transmitted on the same frequency again after a random delay of between 3.5 to 4.5 minutes from the beginning of the initial call. This allows not only acknowledgements but prevents mutual interference if two stations start their transmissions at the same time.

On MF and HF, single frequency call attempts may be repeated on different frequencies after a random delay of between 3.5 and 4.5 minutes from the beginning of the initial call. However, if a station is capable of receiving acknowledgements continuously on all distress frequencies except for the transmit frequency in use, then single frequency call attempts may be repeated on different frequencies without this delay.

(b) Multi-frequency call attempt: Up to six consecutive DSC distress calls spread over the six DSC distress frequencies, (one in the MF and five in the HF bands). Stations transmitting multi-frequency distress call attempts should be able to receive acknowledgements continuously on all distress frequencies except for the transmit frequency in use, or be able to complete the call attempt within one minute.

Multi-frequency call attempts may be repeated after a random delay of between 3.5 and 4.5 minutes from the beginning of the previous call attempt.

Alerts give immediate and absolute priority of communication to the ship in distress. Any ship receiving a distress alert should immediately cease transmission so that no interference occurs.

The receiving ship station should then listen on the appropriate frequency band until the distress alert has been acknowledged. For example, if the distress alert was received on Ch.70 VHF, then voice acknowledgement by RT would be made on Ch.16 VHF. Perhaps, instead, the original DSC alert was received on the 8 MHz band: in this instance the radiotelephone watch should be set on the appropriate frequency in the 8 MHz band (8 291 kHz). Ships operators are likely to hear the distress ship in communication with a coast station on this frequency. If this is not heard, then the receiving ship station should relay details of the distress incident ashore by any means. Ship-to-shore alerts are connected to the appropriate RCC via terrestrial coast radio stations.

Ship-to-ship distress alerts using DSC are generally transmitted in the MF or VHF bands.

3.4 DSC DISTRESS ALERTS

When a DSC call is made, the message is always in the same format as that shown below. The information is inserted either automatically or manually before transmission. If circumstances prevent the manual loading of any information, what is known as "default" information is included automatically.

Format specifier	DISTRESS (automatically included)
Self Identity	The nine digit MMSI which identifies the ship in distress transmitting the message (automatically included)
Nature of Distress	This can be one of the following*

1. Fire or explosion
2. Flooding
3. Collision
4. Grounding
5. Listing and in danger of capsizing
6. Sinking
7. Disabled and adrift

8. Undesignated distress. This used as "default" information.
9. Abandoning ship
10. EPIRB emission (used only for VHF DSC EPIRBs)
11. Piracy/Armed robbery attack

Distress Coordinates	Ship's position in latitude and longitude automatically included if the ship has a navigation interface to position fixing equipment which constantly updates the information prior to the emergency. Default information is sent if no positional information available.
Time	This is the time the distress position is valid. Time omitted if no time information available.
Subsequent communication	This identifies whether the subsequent distress message will be sent by RT or NBDP (RT is preferred first choice).

*The nature of distress may be limited to "Undesignated distress" in some DSC controllers.

3.5 FORMAT OF DSC DISTRESS ALERT ACKNOWLEDGEMENT BY COAST STATIONS

Acknowledgements, using DSC are usually SENT BY A COAST STATION. Distress acknowledgement calls are addressed to "all ships" and are transmitted on the same DSC frequency as the original distress alert. Such calls include the identification of the ship in distress. Except in some very special circumstances, receiving ships DO NOT acknowledge by DSC. Instead, the acknowledgement should be made by radiotelephony using the standard RT distress procedures. See section 3.6.

Coast stations should set watch on the appropriate radiotelephony frequency or channel indicated in the "mode of subsequent communication" telecommand given in the received distress call. It is recommended that RT should be used for all subsequent communication. If however, the "mode of subsequent communication" telecommand indicates telex, coast stations should set watch on an appropriate telex distress frequency. In both circumstances the RT and telex frequencies should be those associated with the frequency on which the distress call was received.

An example of the DSC call composition for a distress acknowledgement call is given below.

Format specifier	ALL SHIPS (automatically included)
Category	DISTRESS (automatically included)
Self identification	The nine digit maritime MMSI of the station acknowledging the alert (automatically included)
Telecommand	DISTRESS ACKNOWLEDGEMENT (automatically included)
Distress ship	The nine digit maritime MMSI of the ship in distress identification
Nature of Distress	Identical to the information in the received distress call

Distress coordinates	Identical to the information in the received distress call
Time	Identical to the information in the received distress call
Subsequent communication	Identical to the information in the received distress call

3.6 ACKNOWLEDGEMENT OF RECEIVED DSC DISTRESS ALERT BY SHIP STATIONS

3.6.1 For ships in ALL GMDSS sea areas

Ships or mobile earth stations in receipt of a distress alert shall, as soon as possible, inform the master, or person responsible for the ship, of the contents of the distress alert.

3.6.2 For ships operating in a GMDSS sea area **A1**

Ships operating in a designated GMDSS sea area Al will be within VHF RT coverage of at least one VHF coast station (possibly more), providing continuous DSC distress alert watchkeeping. Ships receiving a DSC distress alert on Ch.70 VHF should assume the alert has been heard by a coast station. Ships operators should then follow the procedure below.

(a) *Do not transmit a DSC acknowledgement,* even though the equipment may prompt operators to do so. The coast station will acknowledge the distress alert by DSC.

(b) Prepare the transceiver for RT distress communication by tuning to the distress channel in the same band in which the DSC alert was received, i.e. Ch.16 VHF.

(c) Listen to the RT distress call and message from the distress ship, then acknowledge the distress alert *using radiotelephony (RT)* as follows:

MAYDAY
the nine digit MMSI (or the call sign) of the ship in distress sent three times
THIS IS
the nine digit MMSI (or the call sign) of own ship sent three times
RECEIVED MAYDAY

(d) If however, it appears that no other station has received the DSC distress alert from the distress ship, *and the DSC distress alert continues to be received,* the receiving ship should relay the information about the distress alert to an MRCC station by any practicable means.

It should be noted that the relevant international radio rules state: "In areas where reliable communications with one or more coast stations are practicable, ship stations in receipt of a distress alert should defer acknowledgement for a short interval so that receipt may be acknowledged by a coast station." In effect, "areas where reliable communications with one or more coast stations are practicable" means in this case, within GMDSS sea area Al, where by definition, there will be at least one coast station within range to acknowledge the receipt of the distress alert . It is worth repeating that the acknowledgement is done using RT procedures.

The IMO Circular COMSAR/Circ.25 of 2001 amended procedures for distress relays and and acknowledgements. Ships receiving a DSC Distress Alert on Ch. 70 VHF are not permitted to relay the DSC call under any circumstances, however they may relay information by other means.

3.6.3 For ships operating in a GMDSS sea area A2

Ships operating in a designated GMDSS sea area A2 will be within RT coverage of at least one MF coast station providing continuous DSC distress alert watchkeeping. Ships receiving a DSC distress alert on 2 187.5 kHz should assume the alert has been heard by a coast station. Ships operators should then follow the procedure below.

(a) *Do not transmit a DSC acknowledgement,* even though the equipment may prompt operators to do so. The coast station will acknowledge the distress alert by DSC.

(b) Prepare the transmitter and receiver for RT distress communication by tuning to the distress frequency in the same band in which the DSC alert was received, i.e. 2 182 kHz.

(c) Listen to the RT distress call and message from the distress ship, then acknowledge the distress alert *using radiotelephony (RT)* as follows:

MAYDAY
the nine digit MMSI (or the call sign) of the ship in distress sent three times
THIS IS
the nine digit MMSI (or the call sign) of own ship sent three times
RECEIVED MAYDAY

(d) If however, it appears that no other station has received the DSC distress alert from the distress ship, *and the DSC distress alert continues to be received,* the receiving ship should relay the information about the distress alert to an MRCC station by any practicable means.

Thus ships which receive a DSC distress alert from another ship should delay RT acknowledgement to permit time for COAST STATIONS to acknowledge. As with the previous sub section, "areas where reliable communications with one or more coast stations are practicable" means in this particular case, within GMDSS sea area A2 where again by definition, there will be a coast station within range to acknowledge the receipt of the distress alert by DSC.

Changes introduced by COMSAR/Circ.25 of 2001 do not permit ships receiving a DSC Distress Alert on 2 187.5 kHz to relay this call by DSC under any circumstances, however they may relay information by other means.

Note: Where ships receive an alert on 2 187.5 kHz, modern equipment will not allow the operator to acknowledge it until 2 minutes 45 seconds have elapsed. Also if, while waiting for this interval, a DSC acknowledgement is received from an MRCC, the ships equipment automatically prevents a DSC acknowledgement being sent.

3.6.4 For ships operating outside a GMDSS sea area A2 receiving a DSC distress alert on HF

Ships receiving a distress alert on one of the HF DSC distress frequencies should *not acknowledge by DSC or RT.* Operators receiving such a call, must listen on the radiotelephone or radiotelex distress and safety frequencies associated with the calling frequency on which the DSC alert was received. (See Appendix 8). Ships should then wait for at least THREE minutes for a DSC acknowledgement by a coast station on the selected HF band.

If after three minutes *NO* coast station acknowledgement is received, and no HF RT distress communications is heard between the distress ship and coast station, receiving ships should then transmit a DSC distress relay alert. This informs any RCC that a distress alert has been received and provides the RCC with details about the ship in distress. This message can be sent by any means, on any suitable GMDSS distress and safety frequency.

Occasionally ships will receive a DSC distress alert relay from a coast station on HF that has been addressed to ships in a particular geographical area. Ships should not acknowledge this by DSC but rather should acknowledge by RT on the appropriate RT distress frequency in the same band that the DSC distress relay call was made.

3.6.5 For ships operating outside a GMDSS sea area A2

Ships operating in GMDSS areas A3 and A4 which receive a SHIP-TO-SHIP DSC distress alert on 2 187.5 kHz or Ch.70 VHF, which is, beyond doubt, in its vicinity, should send an acknowledgement as soon as possible by RT on 2 182 kHz or Ch.16 VHF. However if no other station appears to have received the DSC distress alert and subsequent DSC distress alerts are received from the same ship, ie the same alert continues to be repeated, ships should acknowledge *using DSC.* This terminates the call of the distress ship. The receiving ship should then inform a coast station or LES of the alert details as soon as possible using any communication means practicable.

3.6.6 For ships receiving a distress alert relay via a coast radio station

Coast radio stations are required to acknowledge receipt of a distress alert as soon as possible under the conditions of the Radio Regulations. They will then usually rebroadcast the distress information as a DSC distress relay call, addressed either to, "all ships", ships in a specified geographical area, a group of ships or to an individual ship as appropriate.

Ships which receive such a DSC distress relay call, should acknowledge receipt by RT on an appropriate distress channel or frequency in which the distress relay call was made. This would normally be 2 182 kHz or Ch.16 VHF, but could be one of the HF RT terrestrial distress frequencies.

The format of this acknowledgement message is as follows:

MAYDAY
the nine digit MMSI (or the call sign) of the calling coast station
THIS IS
the nine digit MMSI (or the call sign) of own ship
RECEIVED MAYDAY

3.7 DISTRESS TRAFFIC

On receipt of a DSC distress acknowledgement (usually from a coast station), the ship in distress should commence the distress traffic by RT on the distress traffic frequency (2 182 kHz on MF or Ch.16 VHF).

3.8 ALERTING BY DSC: DISTRESS ALERT RELAY

These are sent under the following conditions:

(a) By coast stations to alert ships that a distress has occurred in a particular area. This is sent when ships may not have received the original distress alert. (This could be because the original DSC distress was sent on a non-standard frequency or had been sent on the correct frequency but DSC had not been used).

(b) By a ship to an appropriate coast station. For example, when the ship had received a DSC call on an HF frequency but that alert had not been acknowledged by a coast station within THREE minutes.

Format Specifier	*Either ALL* SHIPS or SHIPS IN A PARTICULAR GEOGRAPHICAL AREA *or* INDIVIDUAL STATION
Address	If for "all ships" no address needed. For ships in a geographical area, the area will be defined. For an individual ship the MMSI of that ship is used.
Category	DISTRESS
Self Identification	Nine digit MMSI of the transmitting station
Telecommand	DISTRESS RELAY
Distress ship ID	Nine digit MMSI of station in distress
Nature of distress	Identical to the information in the received distress call
Distress coordinates	Identical to the information in the received distress call
Time	Identical to the information in the received distress call
Subsequent communication	Identical to the information in the received distress call

Either a single frequency or multi frequency call attempt can be made for a distress relay. Distress relay calls transmitted by coast stations, or by ship stations addressed to "all ships", should be acknowledged by ship stations *using radiotelephony (RT).* Distress relay calls transmitted by ships should be acknowledged by a coast station transmitting a distress relay acknowledgement call using DSC on an appropriate frequency.

For DSC distress relays to ships in a particular geographical area, the print out and alarm will not activate if the ship is outside the geographical coordinates of the address. It is thus important that operators on ships which do not have a GPS interfaced with the DSC to manually update their position on a regular basis.

3.9 ACKNOWLEDGEMENT OF A DSC DISTRESS RELAY RECEIVED FROM A COAST STATION

Coast stations will, after having received and acknowledged a DSC distress alert, normally retransmit the information received as a DSC distress relay call, addressed to all ships, all ships in a specific geographic area, a group of ships or a specific ship.

Ships receiving a distress relay call transmitted by a coast station should acknowledge receipt of the call by RT on the distress traffic channel in the same band in which the relay call was received, i.e. 2 182 kHz on MF or Ch.16 VHF. The acknowledgement is transmitted as follows:-

MAYDAY
the nine digit MMSI or the call sign or other identification of the calling coast station
THIS IS
the nine digit MMSI or the call sign or other identification of own ship
RECEIVED MAYDAY

3.9.1 Acknowledgement of a DSC distress relay received from another ship

Ships receiving a distress relay alert from another ship should follow the same procedure for acknowledgement of a distress alert, that is use RT. See section 3.6.

3.10 SENDING A DSC DISTRESS ALERT ON HF

The procedures for DSC communication on HF are broadly similar to that previously described for MF, however propagation characteristics of HF waves should be considered when choosing the transmission frequency. As a general rule the 8 MHz distress channel in the maritime band (8 414.5 kHz) is an appropriate first choice. In all cases the subsequent communication (usually RT), should be made on the same HF frequency band as used for the original alert. Thus, if 8 414.5 kHz had been used for the distress alert, then 8 291 kHz would be used for subsequent RT distress traffic. All the distress communications which follow should be carried out using the standard RT procedures described in Chapter 2.

As previously described the HF DSC alert may be either a single frequency attempt or a multi-frequency attempt, (see section 3.3). However to avoid doubt by receiving stations about which band to establish subsequent communication on, DSC distress alerts should be made on one HF band at a time. Station should wait a few moments to establish subsequent RT communication with a coast station before repeating the DSC distress alert on another HF band.

3.11 DSC URGENCY AND SAFETY CALLS TO "ALL SHIPS"

"All ship" calls by DSC on the distress and safety calling frequencies, should be used by coast or ship stations to advise shipping of the priority of impending transmissions of vital navigational and safety messages, except where the transmissions take place at routine times. The call should indicate the working frequency which will be used for the subsequent transmission of a very important navigational or safety message.

The announcement and identification of medical transports should be carried out by DSC techniques, using appropriate distress and safety calling frequencies. Such calls should use the category "urgency", telecommand "medical transport" and be addressed to "all ships".

3.12 UEGENCY MESSAGES AND DSC

The transmission procedure of urgency messages is in two parts:

(a) the announcement of the urgency message;
(b) the transmission of the urgency message.

The announcement is made by transmitting a DSC urgency call on the DSC distress and safety channel (2 187.5 kHz MF, or Ch.70 VHF).

The urgency message is transmitted on the distress channel (2 182 kHz or Ch.16 VHF) using standard RT procedure for urgency messages. The DSC urgency call can be made to all stations or to an individual station. The frequency on which the subsequent urgency message will be transmitted should be included in the DSC urgency call.

To avoid overloading RT distress frequencies, in circumstances where a long urgency message may have to be transmitted, e.g. a medical message, a channel/frequency other than Ch.16 VHF or 2 182 kHz may be used. Ships operators must select the appropriate telecommand on the DSC equipment to insert the alternate RT working frequency they propose to use.

3.12.1 **Reception of urgency messages by DSC**

Ships receiving a DSC urgency call announcing an urgency message addressed to all ships shall NOT acknowledge the receipt of the DSC call, but should tune the RT receiver to the frequency indicated in the call and listen to the urgency message.

3.13 SAFETY MESSAGES AND DSC

The transmission procedure for safety messages is similar to urgency messages, that is:

(a) the announcement of the safety message;
(b) the transmission of the safety message.

The announcement is made by transmitting a DSC safety call on the DSC distress and safety channel (2 187.5 kHz MF, or Ch.70 VHF). The safety message is transmitted on an RT frequency or channel (usually 2 182 kHz or Ch.16 VHF), using standard RT procedure for safety messages. The DSC safety call can be made to all stations, all stations in a specific geographical area, or to an individual station.

As with DSC urgency calls, to avoid overloading RT distress frequencies, in circumstances where a long safety message may have to be transmitted, a channel/frequency other than Ch.16 VHF or 2 182 kHz may be used. Ships operators must include the frequency on which the subsequent safety message will be transmitted in the DSC safety call.

3.13.1 **Reception of** safety **messages by DSC**

Ships receiving a DSC safety call announcing an safety message addressed to all ships shall NOT acknowledge the receipt of the DSC call, but should tune the RT receiver to the frequency indicated in the call and listen to the safety message.

3.14 DSC WATCHKEEPING ARRANGEMENTS

HF coast stations which offer a watch facility on the HF DSC safety frequencies for GMDSS sea areas A3 and A4 are given in the GMDSS Master Plan. At the time of publication of this edition, approximately 100 stations worldwide monitor these HF DSC distress frequencies. It is known that additional stations are planned to come into operation within the next few years.

As per Regulation 12 of the Chapter IV of the Amendments to the 1974 SOLAS Convention (concerning radiocommunications for GMDSS), every ship, while at sea, shall maintain a continuous watch:

(a) On DSC Ch.70 VHF, if the ship, in accordance with the requirements of regulation 7.1.2 (of the Amendments to the 1974 SOLAS Convention), is fitted with a VHF radio installation.

(b) On the distress and safety DSC frequency of 2 187.5 kHz, if the ship, in accordance with the requirements of regulation 9.1.2 or 10.1.3 (of the Amendments to the 1974 SOLAS Convention), is fitted with an MF radio installation.

(c) On the distress and safety DSC frequencies of 2 187.5 kHz and 8 414.5 kHz and at least one of the other HF DSC frequencies (4 207.5, 6 312, 12 577 or 16 804.5 kHz), the frequency chosen will be appropriate to the ships position and time of day, if the ship, in accordance with the requirements of regulation 10.2.2 or 11.1 (of the Amendments to the 1974 SOLAS Convention), is fitted with an MF/HF radio installation. This watch may be kept by means of a scanning receiver.

(d) For satellite shore to ship distress alerts, if the ship, in accordance with the requirements of regulation 10.1.1 (of the Amendments to the 1974 SOLAS Convention), is fitted with an Inmarsat MES.

3.15 DEDICATED DSC WATCHKEEPING RECEIVERS

Ships complying with GMDSS regulations will have to carry dedicated DSC watch receivers. These are used to keep continuous watch on the relevant DSC distress frequencies.

A distinction must be made between MF/HF receivers which can be programmed to scan commercial, "public correspondence" DSC channels and the dedicated DSC watch receivers used exclusively to respond to incoming DSC distress alerts. Equipment carriage requirements are outlined in Part C of the SOLAS amendments, 1988 and Part II of the Merchant Shipping (Radio Installations) Regulations 1998, but in terms of watchkeeping receivers, a brief outline is given below.

— *Area 1:* Ships must carry a VHF DSC watch receiver (operating on Ch.70 VHF).

— *Area 2:* Ships must carry a MF DSC watch receiver dedicated to 2 187.5 kHz.

— *Area 3:* Ships carrying the MF/HF telex option instead of Inmarsat MES, must carry an HF scanning watch receiver. A dedicated MF DSC watch receiver (2 187.5 kHz), must also be carried but this may be incorporated into the HF scanning receiver.

— *Area 4:* Ships must carry a MF/HF scanning DSC watch receiver.

An MF/HF scanning receiver is a type which sequentially switches between 2 187.5 kHz and 8 414.5 kHz and the other four HF DSC distress and safety frequencies. The receiver is able to detect a special "dot pattern" at a particular speed (100 baud) which is sent at the beginning of every DSC call. Should a distress alert be intercepted on a particular frequency band the call will be processed and an audible alarm will sound. When the receiver is connected to an associated DSC controller the incoming call will be displayed. Some types of equipment may have a printer fitted to automatically print a hard copy of any incoming calls. The equipment can be manually programmed to scan any of the HF maritime DSC distress and safety calling frequencies in addition to 2 187.5 kHz and 8 414.5 kHz. This arrangement meets a requirement which satisfies HF DSC watchkeeping needs. It performs this scanning operation within two seconds before repeating the operation.

3.16 UK DISTRESS AND SAFETY WATCHKEEPING ARRANGEMENTS

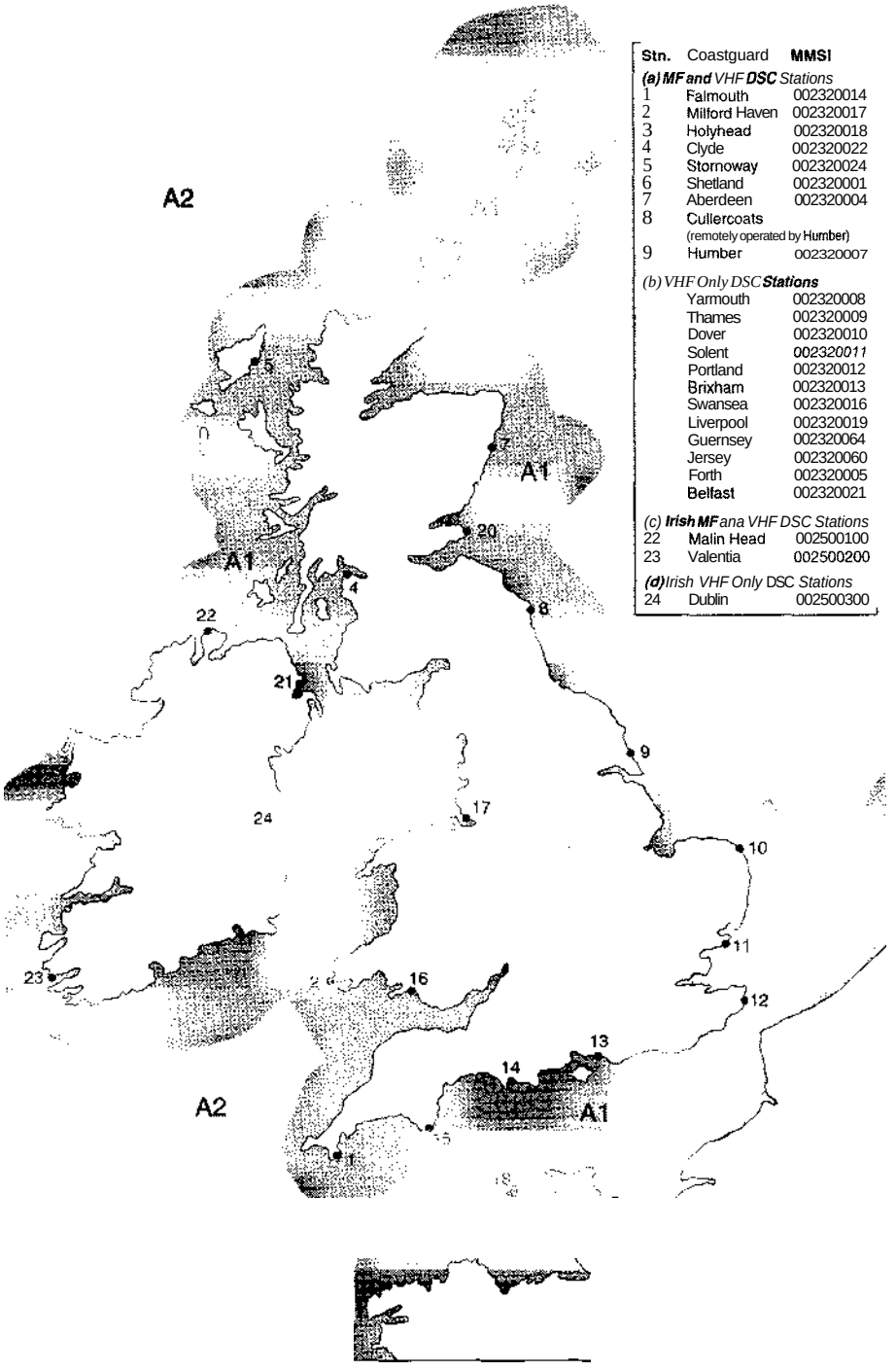
HM Coastguard has the principal responsibility for the coordination of search and rescue activities in UK waters. This includes coordinating SAR operations involving Royal National Lifeboat Institution craft, Royal Navy and Royal Air Force helicopters and fixed wing aircraft, merchant ships, commercial aircraft and ferries around UK waters, extending 1,000 miles into the North Atlantic.

There are six Maritime Rescue Coordination Centres (MRCCs) and fifteen Maritime Rescue Sub Centres (MRSCs) around UK. Nine of the stations listed below maintain continuous distress and safety watch using digital selective calling (DSC) on 2 187.5 kHz in the MF band. Follow-up communications would normally be conducted by RT on 2 182 kHz.

Monitoring Station	*MMSI*
MRCC Aberdeen	002320004
MRSC Shetland	002320001
MRSC Stornoway	002320024
MRCC Clyde	002320022
MRSC Holyhead	002320018
MRSC Milford Haven	002320017
MRCC Falmouth	002320014
MRSC Humber	002320007
MRSC Cullercoats (remotely operated by Humber)	

3.17 DSC DISTRESS SUMMARY

A ship transmits a distress alert using DSC and this alert call contains information relating to the ships MMSI, position, time, and nature of distress. Alert calls are sent primarily to shore stations, for example for sea areas A1 on Ch.70 VHF, for sea areas A2, on 2 187.5 kHz and for sea areas A3 and A4 on a suitable HF DSC distress and safety frequency. Additionally, ship to ship alerts may be transmitted on MF (2 187.5 kHz), or on VHF (Ch.70).



Stn.	Coastguard	MMSI
(a) MF and VHF DSC Stations		
1	Falmouth	002320014
2	Milford Haven	002320017
3	Holyhead	002320018
4	Clyde	002320022
5	Stornoway	002320024
6	Shetland	002320001
7	Aberdeen	002320004
8	Cullercoats (remotely operated by Humber)	
9	Humber	002320007
(b) VHF Only DSC Stations		
	Yarmouth	002320008
	Thames	002320009
	Dover	002320010
	Solent	002320011
	Portland	002320012
	Brixham	002320013
	Swansea	002320016
	Liverpool	002320019
	Guernsey	002320064
	Jersey	002320060
	Forth	002320005
	Belfast	002320021
(c) Irish MF ana VHF DSC Stations		
22	Malin Head	002500100
23	Valentia	002500200
(d) Irish VHF Only DSC Stations		
24	Dublin	002500300

Fig. 3.1 Approximate limits of GMDSS sea areas Al and A2 around UK and Irish waters.

Coast stations acknowledge DSC distress alerts from ships using a DSC message addressed to "all ships"; this will be sent as soon as practicable on VHF. On MF/HF this will be sent after a minimum delay of 1 minute, but a maximum delay of 2.75 minutes is possible. The delay allows sufficient time for single or multiple frequency attempts to be completed by the distress ship and for the coast station to react to the alert. Reception of the distress acknowledgement will automatically cancel repetitions of the distress alert from the distress ship. The coast station then sets watch on the appropriate radiotelephony or radiotelex frequency.

3.17.1 Full distress procedure using DSC and RT on VHF

(a) By ship in distress in GMDSS area A1

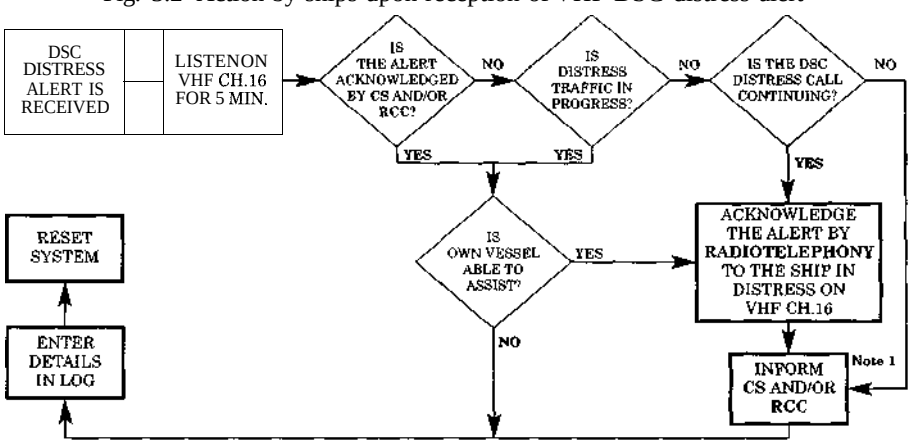
(i) Transmit a DSC distress alert on Ch.70 VHF.
(ii) Normally a DSC acknowledgement should be received from a coast station.
(iii) Transmit a MAYDAY call and message on Ch.16 VHF.

(b) By ship receiving a VHF DSC distress alert in GMDSS area Al

(i) View the details of the distress alert (MMSI of ship in distress, etc).
(ii) Do NOT acknowledge by DSC, but switch to Ch.16 VHF and wait for the MAYDAY call and message which should follow.
(iii) Acknowledge the MAYDAY message on Ch.16 VHF using standard RT procedures.
(iv) If, however, it appears that no other station has received the DSC distress alert from the distress ship, *and the DSC distress alert continues to be received,* the receiving ship should relay the information about the distress alert to an MRCC station by any practicable means. See Fig. 3.2.

Note: Ships which receive a DSC distress alert from another ship should delay their RT acknowledgement to permit time for sea area Al COAST STATIONS to acknowledge.

Fig. 3.2 Action by ships upon reception of VHF DSC distress alert



REMARKS:

Note 1: Appropriate or relevant RCC and/or Coast Station shall be informed accordingly. If further DSC alerts are received from the same source and the ship in distress is beyond doubt in the vicinity, a DSC acknowledgement may, after consultation with an RCC or Coast Station, be sent to terminate the call.

Note 2: In no case is a ship permitted to transmit a DSC distress call on receipt of a DSC distress alert on VHF channel 70.

CS - Coast Station. RCC = Rescue Coordination Centre

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3.17.2 Full GMDSS distress procedure using DSC and RT on MF

(a) By ship in distress in GMDSS area A2

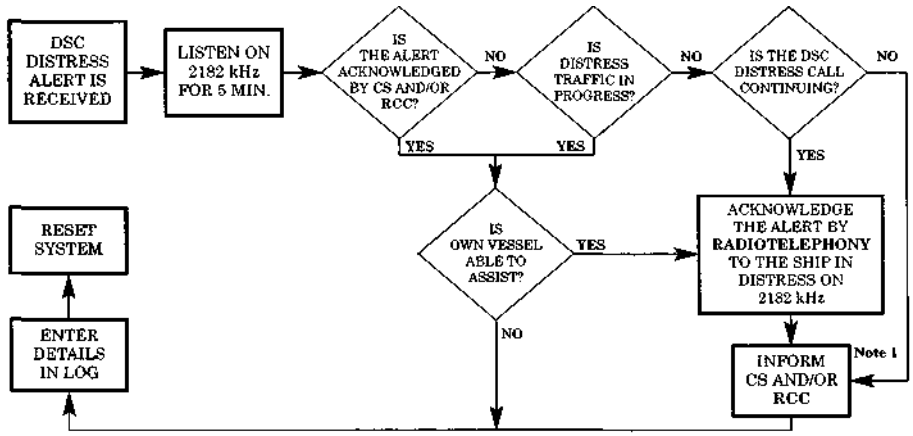
(i) Transmit a DSC distress alert on 2 187.5 kHz.
(ii) Normally a DSC acknowledgement should be received from a coast station.
(iii) Transmit a MAYDAY call and message on 2 182 kHz.

(b) By ship receiving a distress alert in GMDSS area A2

(i) View the details of the distress alert (MMSI of ship in distress etc).
(ii) Do NOT acknowledge by DSC but switch to 2 182 kHz and listen for the RT MAYDAY call and message which should follow.
(iii) Acknowledge the MAYDAY message on 2 182 kHz using standard RT procedures.
(iv) If, however, it appears that no other station has received the DSC distress alert from the distress ship, *and the DSC distress alert continues to be received,* the receiving ship should relay the information about the distress alert to an MRCC station by any practicable means. See Fig. 3.3

Note: Ships which receive a DSC distress alert from another ship should delay their RT acknowledgement to permit time for sea area A2 COAST STATIONS to acknowledge. As noted previously, ships operators should also be aware that when ships receive an alert on 2 187.5 kHz, modern equipment will not allow the operator to acknowledge it until 2 minutes 45 seconds have elapsed. Also if, while waiting for this interval, a DSC acknowledgement is received from an MRCC, the ships equipment automatically prevents a DSC acknowledgement being sent.

Fig. 3.3 Action by ships upon reception of MF DSC distress alert



REMARKS:
Note 1: Appropriate or relevant RCC and/or Coast Station shall be informed accordingly. If further DSC alerts are received from the same source and the ship in distress is beyond doubt in the vicinity, a DSC acknowledgement may, after consultation with an RCC or Coast Station, be sent to terminate the call.
Note 2: In no case is a ship permitted to transmit a DSC distress call on receipt of a DSC distress alert on either VHF or MF channels.

CS = Coast Station, RCC = Rescue Coordination Centre

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3.17.3 Full GMDSS distress procedure using DSC and RT on HF

(a) By ship distress in GMDSS areas A3 or A4

(i) Transmit a DSC distress alert on one of the five HF bands, 8 414.5 kHz would be an appropriate first choice.

(ii) Normally a DSC acknowledgement should be received from a coast station.

(iii) Transmit a MAYDAY call and message on the associated HF RT band. Thus if 8 414.5 kHz had been used for the DSC distress alert, 8 291 kHz would be used for the MAYDAY message.

(b) By ship receiving an HF DSC distress alert in GMDSS areas A3 or A4

(i) View the details of the distress alert (MMSI of ship in distress etc).

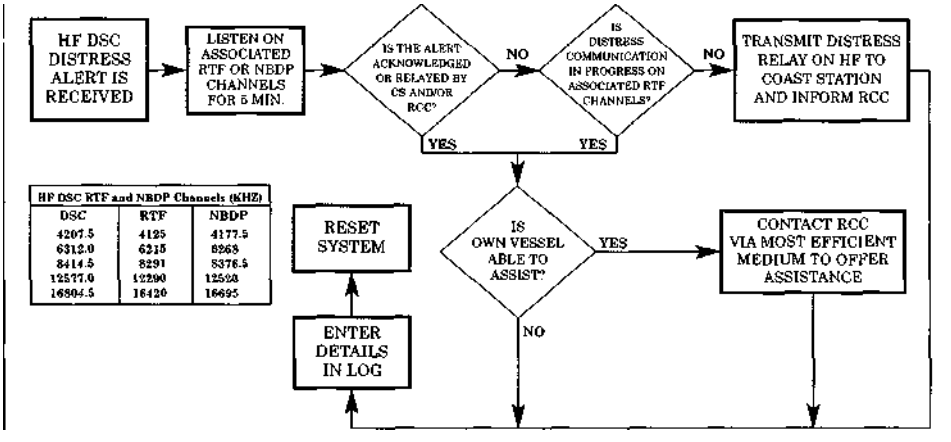
(ii) Do NOT acknowledge the DSC distress alert. Switch to the associated HF RT band, and listen to the MAYDAY call and message which should follow.

(iii) Wait for at least THREE minutes for an acknowledgement of the MAYDAY message by a COAST STATION on the selected HF RT band.

(iv) If after three minutes, no acknowledgement from a coast station is received, transmit a DSC MAYDAY RELAY alert and inform any RCC that a MAYDAY message has been received and give the details. This relay message can be sent by any means, on any suitable GMDSS distress and safety frequency. See Fig. 3.4

Note: Modern DSC equipment does not allow the operator to acknowledge a received HF DSC distress alert. The correct procedure is to establish communication with an MRCC using the Mayday Relay procedures outlined above. If the receiving vessel is able to assist it communicates this information directly to an MRCC and awaits instructions.

Fig. 3.4 Action by ships upon reception of HF DSC distress alert



Note 1: If it is clear the ship or persons in distress are not in the vicinity and/or other crafts are better placed to assist, superfluous communications which could interfere with search and rescue activities are to be avoided. Details should be recorded in the appropriate logbook.

Note 2: The ship should establish communications with the station controlling the distress as directed and render such assistance as required and appropriate.

Note 3: Distress relay calls should be initiated manually.

CS = Coast Station. RCC = Rescue Coordination Centre

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3.17.4 Ship to ship distress alerting in GMDSS areas A3 or A4

An alternative distress alerting procedure using DSC and RT on MF or VHF is available in these areas.

(a) By distress ship

(i) Transmit a ship to ship DSC distress alert on Ch.70 VHF or 2 187.5 kHz in the MF band.

(iii) Transmit a MAYDAY call and message either on Ch.16 VHF or 2 182 kHz in the MF band.

(ii) Expect an RT acknowledgement from a ship station. If none received repeat the DSC distress alert as often as necessary.

(b) By ship receiving a ship to ship distress alert

(i) View the details of the distress alert (MMSI of ship in distress etc).

(ii) Listen on Ch.16 VHF or 2 182 kHz MF and acknowledge the distress message by RT.

(iii) If no RT distress message heard and/or DSC distress alert is repeated, acknowledge distress alert by DSC.

(iv) Inform any RCC that a MAYDAY message has been received and give the details. This relay message can be sent by any means, on any suitable GMDSS distress and safety frequency.

PART II - ROUTINE DSC CALLING PROCEDURE: MARITIME MOBILE SERVICE

3.18 CONDITIONS RELATING TO DSC EQUIPMENT

Conditions to be observed in the Maritime Mobile Service in regard to ships using DSC equipment.

(a) Bands between 415–526.5 kHz.

Ships with DSC equipment must be able to send and receive class F1B or J2B emissions on at least two DSC channels necessary for their service.

(b) Bands between 1 605-4 000 kHz

Ships equipped for DSC operations in this band must be able to:

(i) send and receive class F1B or J2B emissions on 2 187.5 kHz

(ii) in addition, send and receive class F1B or J2B emissions on other DSC frequencies in this band necessary for their service.

(c) Bands between 4 000-27 500 kHz

Ships equipped for DSC operations in this band must be able to : -

(i) send and receive class F1B or J2B emissions on the frequencies designated for DSC calling in each of the maritime HF bands in which they are operating.

(ii) send and receive class F1B or J2B emissions on an international calling channel in each of the HF maritime mobile bands necessary for their service.

(iii) send and receive class F1B or J2B emissions on other DSC channels in each of the HF maritime mobile bands necessary for their service.

(d) VHF band (156–174 MHz)

All ship stations equipped with DSC apparatus to work in this band shall be able to send and receive class G2B emissions on Ch. 70 VHF.

3.19 DSC: USE OF FREQUENCIES

The provisions described in the section below are applicable to calling and acknowledgement, when DSC techniques are used, except in cases of distress, urgency and safety, to which the provisions of Chapter N IX of the ITU Radio Regulations apply.

The technical characteristics of the digital selective-calling equipment shall be in accordance with the relevant CCIR Recommendations.

The frequencies on which coast stations provide services using digital selective calling techniques shall be indicated in the ITU List of Coast Stations, which shall also supply any other useful information concerning such services.

3.20 MF DSC: BANDS BETWEEN 415 kHz AND 526.5 kHz

The class of emission to be used for digital selective calling and acknowledgement in the authorised bands between 415 kHz and 526.5 kHz shall be F1B. Transmissions of digital selective calls and acknowledgements by ship stations shall be limited to a mean power of 400 watts.

3.20.1 MF DSC: Call and acknowledgement

For call and acknowledgement by DSC techniques, an appropriate channel shall be used.

The international digital selective-calling frequency 455.5 kHz may be assigned to any coast station. In order to reduce interference on this frequency, it may be used as a general rule by coast stations to call ships of another nationality, or in cases where it is not known on which DSC frequencies within these bands the ship station is maintaining watch.

The international DSC frequency 458.5 kHz may be used by any ship station. In order to reduce interference on this frequency, it shall only be used when calling cannot be made on national frequencies assigned to the coast station.

The frequency to be used for transmission of acknowledgement shall normally be the frequency paired with the calling frequency used.

3.20.2 MF DSC: Watch

A coast station providing international public correspondence service using DSC techniques within the MF band should, during its hours of service, maintain automatic DSC watch on appropriate national or international calling frequencies. Details of the hours and frequencies will be indicated in the ITU List of Coast Stations or ALRS Volume 1.

Ship stations equipped with DSC apparatus for working in the authorised maritime MF bands should, when within the coverage area of coast stations providing DSC services in these bands, maintain an automatic DSC watch on one or

more appropriate DSC frequencies within these bands. Ships should take into account the DSC frequencies operated by the coast stations.

3.21 BANDS BETWEEN 1 605 kHz AND 4 000 kHz

The class of emission to be used for digital selective-calling and acknowledgement in the bands between 1 605 kHz and 4 000 kHz will be FIB or J2B. In Region 1, transmissions of digital selective calls and acknowledgments by ship stations shall be limited to a mean power of 400 Watts.

3.21.1 Bands between 1 605 kHz and 4 000 kHz: Call and acknowledgement

When making their call to a coast station by DSC techniques, ship stations should use in order of preference:

(a) a national digital selective-calling channel on which the coast station is maintaining watch;
(b) international digital selective-calling frequency 2 189.5 kHz.

The international DSC frequency 2 189.5 kHz may be assigned to any ship station. In order to reduce interference on this frequency, it may be used as a general rule by ship stations to call coast stations of another nationality.

A ship station calling another ship station by DSC techniques should use the frequency 2 177 kHz for the call. Acknowledgements of such calls should also be made on this frequency.

When making their call to ship stations by DSC techniques, coast stations should use in order of preference:

(a) a national digital selective-calling channel on which the coast station is maintaining watch;
(b) the international digital selective-calling frequency 2 177 kHz.

The international DSC frequency 2 177 kHz may be assigned to any coast station. In order to reduce interference on this frequency, it may be used by coast stations to call ships of another nationality. It may also be used in cases where it is not known on which DSC frequencies within the bands between 1 605 kHz and 4 000 kHz the ship station is maintaining watch.

The frequency to be used for transmission of an acknowledgement should normally be the frequency paired with the frequency used for the received call, as indicated in the ITU List of Coast Stations or ALRS Volume 1.

3.21.2 Bands between 1 605 kHz and 4 000 kHz: Watch

The provisions detailed below are applicable to watch-keeping by digital selective calling, except for distress, urgency and safety purposes, to which different provisions of the ITU Radio Regulations apply.

Where a coast station provides an international public correspondence service using DSC techniques it should, during its hours of service, maintain automatic DSC watch on appropriate national or international calling frequencies on the

bands between 1 605 kHz and 4000 kHz. These hours and frequencies are indicated in the ITU List of Coast Stations or ALRS Volume 1.

Ship stations equipped with DSC apparatus to work in the authorised maritime bands should, when within the coverage area of coast stations providing DSC services, maintain an automatic DSC watch on one or more appropriate DSC frequencies within these bands. Ships should take into account the DSC frequencies operated by the coast stations.

3.22 HIGH FREQUENCY (HF) DSC: BANDS BETWEEN 4 000 kHz AND 27 500 kHz

The class of emission used for digital selective calling and acknowledgement in the authorised bands in this frequency range is F1B or J2B. Transmissions of digital selective calling and acknowledgements by ships in these bands is limited to a mean power of 1.5 kW.

3.22.1 HF DSC: Call and acknowledgement

A station calling another station by DSC techniques within the authorised maritime HF bands should choose an appropriate DSC frequency, taking into account propagation characteristics.

When making their call to a coast station by DSC techniques in the authorised HF bands, ship stations should use in order of preference:

(a) a national DSC channel on which the coast station is maintaining watch;
(b) if calling cannot be made on a national DSC channel, one of the assigned international DSC frequencies may be used.

When making their call to ship stations by DSC techniques in the authorised HF bands, coast stations should use in order of preference:

(a) a national DSC channel on which the coast station is maintaining watch;
(b) one of the international DSC frequencies assigned to coast stations for routine DSC calling (not distress or safety).

3.22.2 HF DSC: Watch

The watch-keeping details below are applicable to stations using digital selective calling, except for distress, urgency and safety purposes, for which different provisions apply.

A coast station providing an international public correspondence service using DSC techniques within the maritime HF bands, should keep automatic DSC watch on the appropriate frequencies for the service, during its hours of service. Details are indicated in the ITU List of Coast Stations or ALRS Volume 1.

Ship stations equipped with DSC apparatus for working in the maritime HF bands, should maintain automatic DSC watch on appropriate DSC frequencies within these bands. Ships should take into account propagation characteristics and the calling frequencies of coast stations providing a DSC service.

3.23 VHF DSC: BANDS BETWEEN 156 MHz AND 174 MHz

The class of emission to be used for digital selective calling and acknowledgement in the authorised bands between 156 MHz and 174 MHz (the VHF band) shall be G2B.

3.23.1 VHF DSC: Call and acknowledgement

The frequency 156.525 MHz (Ch.70 VHF) is an international channel in the maritime mobile service used for distress, urgency, safety and calling using DSC techniques.

Calling by DSC techniques within the authorised maritime VHF band, for public correspondence, from ship-to-coast station, from coast station-to-ship and from ship-to-ship should, as a general rule, be made on the DSC Channel 70. Essentially, therefore ALL VHF DSC calls, whether distress, urgency, safety or routine should be made on Channel 70. Radiotelephony transmissions must NOT be used on this channel.

3.23.2 VHF DSC: Watch

Coast stations who maintain watchkeeping by automatic digital selective-calling on Ch.70 VHF are listed in the ITU List of Coast Stations or ALRS Volume 1.

Ship stations equipped with DSC apparatus to work in the authorised maritime VHF band should, while at sea, maintain an automatic DSC watch on Ch.70.

3.24 DSC CALLING FREQUENCIES

Frequency band	*Use*	*Ship transmit kHz*	*Shore transmit kHz*
MF I	International calling	458.5	455.5
MF II	Distress	2 187.5	2 187.5
	Intership calling	2 177.0	—
	International calling	2 189.5	2 177.0
HF 4 MHz	Distress	4 207.5	4 207.5
	International calling	4 208.0*	4 219.5*
		4 208.5	4 220.0
		4 209.0	4 220.5
HF 6 MHz	Distress	6 312.0	6 312.0
	International calling	6 312.5*	6 331.0*
		6 313.0	6 331.5
		6 313.5	6 332.0
HF 8 MHz	Distress	8 414.5	8 414.5
	International calling	8 415.0*	8 436.5*
		8 415.5	8 437.0
		8 416.0	8 437.5
HF 12 MHz	Distress	12 577.0	12 577.0
	International calling	12 577.5*	12 657.0*
		12 578.0	12 657.5

Frequency band	*Use*	*Ship transmit kHz*	*Shore transmit kHz*
		12 578.5	12 658.0
HF 16 MHz	Distress	16 804.5	16 804.5
	International calling	16 805.0*	16 903.0*
		16 805.5	16 903.5
		16 806.0	16 904.0
HF 18 MHz	International calling	18 898.5*	19 703.5*
		18 899.0	19 704.0
		18 899.5	19 704.5
HF 22 MHz	International calling	22 374.5*	22 444.0*
		22 375.0	22 444.5
		22 375.5	22 445.0
HF 25 MHz	International calling	25 208.5*	26 121.0*
		25 209.0	26 121.5
		25 209.5	26 122.0
VHF	Distress and international calling	Channel 70	Channel 70

* Indicates the paired frequencies (for ship/coast stations) which are the first choice international frequencies for digital selective calling.

The international DSC frequencies in the list above may be used by any ship station. In order to reduce interference on these frequencies, they shall only be used when calling cannot be made on nationally assigned frequencies. The international DSC frequencies above may be used by any coast station. In order to reduce interference on these frequencies, they shall only be used when calling cannot be made on nationally assigned frequencies. They may also be used in cases where it is not known on which DSC frequencies in the band concerned the ship station is keeping watch.

The frequency 2 177.0 kHz is available to ship stations for intership calling only.

In addition to the frequencies listed above, appropriate working frequencies in the following bands may be used for digital selective calling:

415-526.5 kHz (Regions 1 and 3)
415-525 kHz (Region 2)
1 606.5-4 000 kHz (Regions 1 and 3)
1 605-4 000 kHz (Region 2)
4 000-27 000 kHz
156–174MHz (VHF band)

3.25 DSC REGULATIONS: METHOD OF CALLING

The procedures set out in this section are applicable to the use of DSC techniques, except in cases of distress, urgency or safety, for which special provisions of the ITU Radio Regulations apply.

The call shall contain information indicating the station or stations to which the call is directed, and the identification of the calling station. The call should also contain information indicating the type of communication to be set up and may include supplementary information such as a proposed working frequency or

channel; this information shall always be included in calls from coast stations, which shall have priority for that purpose.

The duration of a routine DSC call on MF or HF is in the order of 7 seconds and about 0.45 seconds on VHF.

3.25.1 Format of routine DSC calls

The technical format of DSC calls sequence involve the use of fields of information in binary form and sent in a synchronous manner. The numerical symbols 100 to 127 are used to indicate various meanings within their respective fields. All DSC calls commence with a dot pattern and this is followed by a phasing sequence to enable receiving equipment to synchronise to the incoming transmission.

An information field known as the Format Specifier allows for the following types of calls:

Distress Call, All ships call, Individual ship call, Geographic area call, Group call and Automatic/Semiautomatic call. For routine DSC calls the next field contains information about the MMSI of the station that has to be contacted. This in turn is followed by the Category field that is used to indicate the priority of the call i.e. Distress, Urgency, Safety, Ship's business or Routine. The data stream continues with the field used for Self Identification of the sending station, this is the sending station's MMSI.

The subsequent fields are known as Telecommands 1 and 2. Telecommand 1 is selected from a list of 26 items mostly concerning the mode of emission to be used for subsequent communication. Telecommand 2 also has a 26 items available again used to convey additional information in the call. For most DSC calls however only four items are used and two of these are for use in war zones. The Ships and Aircraft indicates that the vessel is a neutral ship as defined by Resolution 18 in the ITU manual. The other item is used to denote a Medical Transport as defined by Resolution 33 in the ITU Manual for use by Maritime Mobile and Maritime Mobile-Satellite Services.

The remaining information fields allow the ship's operator to include working frequencies for subsequent transmission and reception. All DSC calls end with an End of Sequence (EOS) field and finally a field containing error check character information.

3.26 DSC REGULATIONS: ACKNOWLEDGEMENT OF CALLS

The reply to a digital selective call requesting an acknowledgement shall be made by transmitting an appropriate acknowledgement using DSC techniques, acknowledgements may be manual or automatic. Transmission of the calling signal must cease as soon as an acknowledgement has been received.

Acknowledgements shall normally be transmitted on the frequency paired with the frequency of the received call. If the same call is received on several calling channels, the most appropriate channel shall be chosen to transmit the acknowledgement.

If the call includes a proposal for a working channel or frequency which can be used immediately by the station called, the latter should transmit an acknowl-

edgement indicating this possibility. If however, the station called is unable to use the working frequency or channel proposed in the received call immediately, it should indicate this in its acknowledgement, which may also include supplementary information in that respect. Coast stations unable to comply immediately on a proposed working frequency or channel may include a proposal of an alternative working frequency or channel in the acknowledgement. If no working frequency or channel was proposed in the call, the station called should include a proposal for a working frequency or channel in its acknowledgement of the call.

Thus when an acknowledgement has been received which indicates that the called station is ready to receive traffic, the ships transmitter should be prepared for traffic as follows:

(a) Tune transmitter and receiver to the agreed working channel indicated.
(b) Initiate communication on this working channel as follows:
the nine digit MMSI or call sign or other identification of the called station
this is
the nine digit MMSI or call sign or other identification of the calling station (own ship).

Example:
352378000
THIS IS
636010525

3.27 DSC REGULATIONS: ACKNOWLEDGEMENTS, TRANSMISSION METHOD

Acknowledgements may be initiated either manually or automatically. If the ship station is unable to acknowledge a received call within a time limit of five minutes, the ship station's reply to the call should be made by transmitting a call using standard DSC calling procedure. Where automated or semi-automated systems are used, a time limit in accordance with the relevant CCIR Recommendation should apply.

3.28 DSC REGULATIONS: PREPARATION FOR EXCHANGE OF TRAFFIC

The procedures described below are applicable for manual operation. Where automated or semi-automated digital selective calling VHF or UHF systems are used, these should operate in conformity with relevant CCIR Recommendations.

After having transmitted an acknowledgement indicating that it can use the proposed working frequency or channel the station called transfers to the working frequency or channel and prepares to receive the traffic. The calling station shall prepare to transmit traffic on the working channel or frequency it has proposed. The calling station and the called station then exchange traffic on the appropriate working frequency or channel.

If it is unable to use the working frequency or channel proposed in an acknowledgement transmitted by the coast station the ship station should then transmit a new call, indicating that it is unable to comply. The coast station shall then transmit an acknowledgement indicating an alternative working frequency or channel.

On reception of the acknowledgement, the operator of the ship station transfers to the working frequency or channel and prepares to receive the traffic or transmits another call, indicating that it is unable to comply. For communication between a coast station and a ship station, the coast station shall finally decide the working frequency or channel to be used.

3.29 DSC SHORE-TO-SHIP CALLS

There are two categories of calls for commercial communications:

(a) routine calls;
(b) ship's business.

If a direct connection exists between the calling subscriber and the coast station, the coast station asks the calling subscriber the approximate position of the ship. If the ship's position cannot be indicated by the caller, the coast station operator tries to find the location in the information available at the coast station. The coast station checks to see whether the call would be more appropriate through another coast station.

Assuming a DSC call is appropriate the coast station composes the calling sequence as follows:

— selects the format specifier;
— enters the address of the ship;
— selects the category;
— selects telecommand information;
— inserts working frequency information in the message part of the sequence, if appropriate usually selects end of sequence signal "request acknowledge", (code 117). However, if the coast station knows that the ship station cannot respond, or the call is to a group of ships the frequency is omitted and the end of sequence signal should be 127 (i.e. no request acknowledge). In the latter case the following procedures relating to acknowledgement are not applicable.
— the coast station verifies the calling sequence.
— the coast station operator chooses the calling frequencies which are most suitable for the ship's location.
— after checking as far as possible that there are no calls in progress, the coast station operator initiates the transmission on one of the frequencies chosen. Transmission on any one frequency should be limited to no more than two call sequences separated by intervals of at least 45 seconds to allow for reception of an acknowledgement from the ship, or exceptionally to one "call attempt" consisting of up to five transmissions.

If appropriate, the "call attempt" may be transmitted, which may include the transmission of the same call sequence on other frequencies (if necessary with a change of working frequency information to correspond to the same band as the calling frequency) made in turn at intervals of not less than five minutes.

If an acknowledgement is received further transmission of the call sequence should not take place.

When a station called does not reply, the call attempt should not normally be repeated until after an interval of at least 30 minutes. The same call attempt

should not be repeated more than five times in every 24 hours. The aggregate of the times for which frequencies are occupied in one call attempt, should normally not exceed one minute.

3.29.1 Ship receives DSC call from a coast station

On receipt of a calling sequence at the ship station, the received message is recorded and the appropriate indication is activated as to whether the call category is "routine" or "ship's business". The category does not affect the DSC procedure on the ship.

The definition of automatic DSC operation at a ship station should be noted. This is a mode of operation employing automatic tunable transmitters and receivers, suitable for unattended operation which provide for automatic call acknowledgements upon reception of a DSC and automatic transfer to the appropriate working frequencies.

When a received call sequence contains an end of sequence signal request acknowledge (RQ), an acknowledgement sequence should be composed and transmitted. The format specifier and category information should be identical to that in the received calling sequence.

If the ship station is not equipped for automatic DSC operation, the ship's operator initiates an acknowledgement to the coast station after a delay of at least 5 seconds but no later than 4 minutes 30 seconds of receiving the calling sequence, using ship to shore calling procedure. However the transmitted sequence should contain a “BQ” end of sequence signal in place of an "RQ" signal, BQ being the reply to an RQ. If such an acknowledgement cannot be transmitted within 5 minutes of receiving the calling sequence then the ship station should instead transmit a calling sequence to the coast station using ship to shore calling procedure.

If the ship station is equipped for automatic DSC operation, the ship station automatically transmits an acknowledgement with an end of sequence signal "BQ" . The start of the transmission of the acknowledgement sequence should be within 30 seconds for HF and MF or within 3 seconds for VHF after the reception of the complete call sequence.

If the ship is able to comply immediately the acknowledgement sequence should include a telecommand signal which is identical to that received in the calling sequence indication that it is able to comply.

If the ship is not able to comply immediately the acknowledgement sequence should include a telecommand signal "unable to comply" (code 104), with a second telecommand signal giving additional information. At some later time when the ship is able to accept the traffic being offered, the ship's operator initiates a call to the coast station using ship to shore calling procedures.

If a call is acknowledged indicating ability to comply immediately and communication between coast station and ship station on the working channel agreed is established, the DSC call procedure is considered to be completed.

If the ship station transmits an acknowledgement which is not received by the coast station then this will result in the coast station repeating the call. In this event the ship station should transmit a new acknowledgement. If no repeated call is received the ship station should transmit an acknowledgement or calling sequence containing a "BQ" end of sequence signal in place of an "RQ" signal.

3.30 DSC: ROUTINE SHIP TO SHORE CALLS

The procedure outlined below should be followed both as a delayed response to a call received earlier from the coast station and to initiate traffic from the ship station.

The ship station composes the calling sequence as follows:

- — selects the format specifier;
- — enters the address;
- — selects the category;
- — selects telecommand information;
- — inserts working frequency information in the message part of the sequence, if appropriate
- — inserts the telephone number required (semi-automatic/automatic connections only)
- — selects "end of sequence" signal "RQ."

The ship

- — verifies the calling sequence.
- — selects the most appropriate calling frequency.
- — initiates the the transmission of the sequence on the frequency selected after checking as far as possible that there are no calls in progress on that frequency.

If a called station does not reply, the call sequence from the ship station should not normally be repeated until after an interval of at least 5 minutes for manual connections, or 5 seconds or 35 seconds in the case of semi-automatic/automatic VHF or MF/HF connections respectively. These repetitions may be made on alternative frequencies if appropriate. Any subsequent repetitions to the same coast station should not be made until at least 15 minutes have elapsed.

The coast station should transmit an acknowledgement sequence (after checking as far as possible that there are no calls in progress on the frequency selected), after a delay of at least 5 seconds but no later than 4 minutes 30 seconds for manual connections, or within 3 seconds for semi-automatic/automatic connections, containing the format specifier, the address of the ship, the category, the coast station self-identification (MMSI) and:

- — if able to comply immediately on the working frequency suggested, the same telecommand and frequency information as in the call request;
- — if no working frequency was suggested by the ship then the acknowledgement sequence should include channel/frequency proposal;
- — if not able to comply on the working frequency suggested but able to comply immediately on an alternative frequency, the same telecommand but an alternative working frequency;
- — if unable to comply immediately the telecommand signal code 104 (unable to comply) should be used, with a second telecommand signal giving additional information. For manual connections only, this second telecommand signal may include a queue indication.
- — the end of sequence signal BQ should also be included.

For manual connections, if a working frequency is proposed but this is not acceptable to the ship station, then the ship station should immediately transmit

a call to the coast station indicating (by the use of telecommand signals 104 or 108) that it cannot comply on that frequency. The coast station should then transmit an acknowledgement in accordance with standard DSC procedure either accepting the ship station's original suggested frequency or proposing a second alternative. Note code 108 in the second telecommand indicates unable to use proposedfrequency/channel.

On receipt of an acknowledgement which indicates ability to comply the DSC procedures are complete and both coast station and ship station should communicate on the working frequencies agreed with no further exchange of DSC calls.

If the coast station transmits an acknowledgement which is not received at the ship station then the ship station should repeat the call in accordance with standard DSC procedures.

3.31 DSC MODES OF EMISSION

F1B mode. This is a type of frequency modulation using a single channel containing quantised or digital information without the use of a modulating sub carrier. Frequency shift keying.

G2B. Phase modulation. A single channel containing quantised or digital information with the use of a modulating sub carrier.

J2B mode. This is single sideband suppressed carrier containing quantised or digital information with the use of a modulating sub carrier.

All the above modes of emission provide automatic reception.

3.32 DSC WATCHKEEPING: ROUTINE CALLS

Most modern DSC receiving equipment is provided with scanning facilities, which permit the operator to programme the receiver to scan the transmission frequencies of the desired coast station(s). As discussed previously, programming a DSC receiver to scan commercial frequencies must not prevent the monitoring of the various DSC distress frequencies by the dedicated DSC watchkeeping receiver. Generally therefore, a separate DSC receiver is required to scan ship-to-ship and commercial DSC frequencies in addition to the dedicated DSC distress watchkeeping receiver.

Under normal circumstances ship's operators should set the receiver to automatically scan the appropriate coast station frequencies of the area in which they are sailing. The frequencies selected for the scanning programme will, of course, be dependent on radio propagation conditions existing at that time. If an incoming call is received, the ship's own MMSI will be detected and the controller will sound a "received call" alarm. The operator can then manually interrogate the controller to read details of the incoming call. Controllers have a memory store in which a number of incoming calls are held for later inspection, this is called queuing. Received calls may also be displayed by being printed out by an external printer if this option is fitted.

On ships equipped for automatic DSC operation, when an individual call is received, an acknowledgement is sent automatically. For other ships the acknowledgement should be sent by the operator within five minutes of receiving the call. If unable to reply to a call within five minutes, a reply should be made later using

the normal ship to shore calling sequence. Once an acknowledgement has been made (automatic or manually), the operator should then transfer to the paired RT frequency indicated in the original call and establish communication with the coast station.

3.33 DSC ROUTINE CALLS FOR AUTOMATIC CONNECTION

These are DSC transmissions from a ship to a coast station requesting a telephone or telex channel for normal commercial communication purposes. Calls to port authorities etc. are also included in this category. Where coast stations are fitted with appropriate equipment, DSC can be used to provide automatic ship to shore telephone connection. Information contained in the ship's call automatically starts the coast station equipment searching for a free telephone line and then "dials" the subscriber. The equipment also controls the status of the line and releases it at the termination of the communication. As a result of this direct "straight through" telephone communication link, the utilisation of the channel is improved and waiting time reduced. A similar situation exists at some coast stations who can provide automatic ship to shore telex connection, this again improves channel efficiency and reduces waiting time.

3.34 DSC TESTING

DSC controllers have a facility which allows the unit to be routinely tested without its associated transmitter being activated. The unit should be tested at least once a day, without radiating signals from the transmitter to establish that the DSC facility functions correctly. When within range of coast stations fitted with DSC equipment the ship should test for the correct operation of the DSC facilities by making a "live" test call at least once a week. If the ship has been out of range of a DSC equipped coast station for more than a week, a test call must be made at the first opportunity that the ship is within range of such a coast station. It should be noted that no live test transmissions should be made on Ch. 70 VHF.

3.35 DSC: TESTING ON 2 187.5 kHz

To facilitate testing of DSC equipment, a "Test Safety" category has been introduced and HMCG Coastguard stations will respond to calls made on 2 187.5 kHz. Test calls should be transmitted by a ship station and sent to the nearest HMCG station. On reception of this call the coastguard station will acknowledge by transmitting a Test Safety of its own. Normally no further communication between the two stations is necessary. The test call procedure is as follows:

(a) check by monitoring that no transmissions are taking place on 2 187.5 kHz;
(b) tune ships transmitter to DSC distress and safety frequency 2 187.5 kHz;
(c) on the keyboard select, (or key in) the format for the test call* on the DSC controller in accordance to manufacturers instructions;
(d) key in the nine-digit MMSI of the HMCG station to be called;
(e) transmit the DSC call;
(f) await an acknowledgement.

* Note: Some ships fitted with DSC are unable to make a test call because their controllers are not programmed to make such calls. In order to meet testing requirements on UK ships, any ship without this capability must make all efforts to fit the necessary software. The ability to make test transmissions will be checked by radio surveyors and is a requirement for the issue of a Safety Radio Certificate.

3.36 DSC: EUROPEAN TESTING ARRANGEMENTS

Nine UK coast stations operated by HM Coastguard are fitted with test safety protocol on the distress and safety frequency 2 187.5 kHz and ships masters are encouraged to exchange DSC test messages. Continental stations at Ostend, Nijmegen, Blavand, Skagen and the Irish stations at Valentia and Malin Head are also prepared to exchange test messages. Test messages may also be exchanged with other ships.

No "live" test transmissions should be made on Ch. 70 VHF, however the VHF DSC equipment should be tested daily on an "internal" or loop test which checks the operation of the equipment without radiating a signal.

CHAPTER 4

Safety Related Equipment and Services

4.1 EMERGENCY POSITION INDICATING RADIOBEACON (EPIRB): INTRODUCTION

The main purpose of an EPIRB within GMDSS is as a location device, to allow the position of survivors of an emergency to be determined. Although primarily designed to be activated by survivors in liferafts and lifeboats, the device also provides a secondary method of distress alerting on board ship. In this case the activated EPIRB would indicate the position of the unit in distress. The system allows shore authorities to receive and locate the source of the transmission, search and rescue procedures can then be implemented. The system is designed to give a rapid response to distress alerts on a global scale. EPIRBs can be activated either manually or automatically when "floating free" during an emergency.

In addition to meeting the requirements of the Radio Regulations, EPIRBs also have to meet general performance standards for example:

(a) be capable of being easily activated by unskilled personnel;
(b) be fitted with adequate means to prevent accidental activation;
(c) remain watertight after immersion in water (10 metres for 5 minutes);
(d) be automatically activated after floating free;
(e) be able to be activated/deactivated manually;
(f) have some means to indicate that signals are being transmitted;
(g) withstand being dropped from a height of 20 metres into water without being damaged;
(h) be capable of being tested on board, without radiating an alert signal, to test its operation;
(i) be of highly visible yellow/orange colour and fitted with retroflective material;
(j) be equipped with a buoyant lanyard, used as a tether, this to be arranged in such a way as to prevent it being trapped by ships structure when floating free;
(k) have a low duty cycle light activated by darkness to indicate its position to survivors/rescuers;
(l) be able to float upright in calm water and have positive stability and sufficient buoyancy in all sea conditions;
(m) be capable of resisting deterioration from prolonged exposure to sunlight;
(n) not be unduly affected by seawater or oil.

The EPIRB should be able to operate in extremes of climate and environment. The battery should be of such capacity to operate the EPIRB for at least 48 hours.

Three EPIRB types are employed in the GMDSS system, these are:

(a) The COSPAS-SARSAT satellite system using two frequencies 406 MHz and 121.5 MHz as described in the following section (4.2);

(b) The satellite L band (4.6 GHz) operating with the Inmarsat communications system. This is the distress radio call system (DRCS) utilising the Inmarsat-E system;

(c) VHF EPIRBs using Ch.70 VHF.

Carriage of satellite EPIRBs by ships, became compulsory from 1st August 1993. Some EPIRBs include a search and rescue radar transponder (SART) operating on 9 GHz which is activated to assist SAR units to locate the EPIRB (see section 4.14).

4.2 COSPAS-SARSAT: INTRODUCTION

This is an international satellite aided search and rescue system designed to locate three types of distress beacons which first became operational in 1982. The system operates on two frequencies 406 MHz and 121.5 MHz, and relies on a constellation of low earth orbiting satellites (1 000 km altitude) each in a near-polar orbit, which collectively provides global coverage. This is referred to as the LEOSAR segment.

Complementing the existing COSPAS-SARSAT orbiting satellite system, are three geostationary satellites known as the GEOSAR segment. The GEOSAR segment uses existing 406 MHz beacons and have the capability to provide almost instantaneous distress alerts, but without the use of location data by Doppler effect. This can be overcome by obtaining an estimated position from an emergency point of contact if the beacon is correctly registered. New beacons with encoded position information allow the GEOSAR system to provide distress alerting and location data to receiving stations.

The satellites can communicate with a network of earth stations known as Local User Terminals (LUTs), which then pass distress alerts and location data to rescue authorities via Mission Control Centres (MCCs). The SAR authorities receiving this information may be located at a Rescue Coordination Centre (RCC) or at a Search and Rescue Point of Contact (SPOC) but in any event will have the facilities to implement rescue procedures.

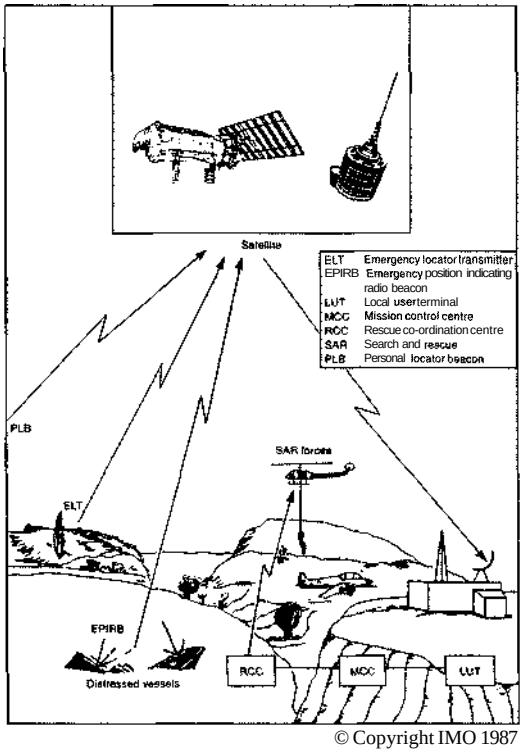
The significance of the polar orbit is that as the earth rotates, the path of the orbiting satellite will pass over a different part of the earth each time. Thus with only one satellite full earth coverage is obtained within 12 hours, as there are two separate rotations; *viz.* the satellite with its polar north /south orbit and the earths rotational spin west/east.

This unique coverage is exploited in the COSPAS-SARSAT system by having several satellites in polar orbits in different orbital planes. Thus the time between satellite passes at any fixed point on earth is decreased compared to the single satellite model. This reduces the notification time for a distress alert, i.e. the time interval between activation of the EPIRB (its first transmission), and the reception of an alert message by the appropriate RCC. The system thus provides a complete worldwide distress alert monitoring facility. The three types of beacon used are:

(a) Emergency Position Indicating Radiobeacon (EPIRB) used by ships;
(b) Emergency Locator Transmitters (ELT) used by aircraft;
(c) Personal Locator Beacons (PLB) used in various land operations.

When activated the beacons transmit a distress signal which is received by the orbiting satellite and relays the signal back to an earth station. This station, the LUT, processes the received signal and then determines the beacon's location. The LUT then alerts the RCC (via the MCC), enabling SAR operations to be implemented. The overall system has thirty-nine LEOLUTs in operation located in twenty-one countries (see Fig. 4.1 below) and nine GEOLUTs. Currently (March 2003) twenty four MCCs are operational.

IMO have decided that float-free satellite EPIRBs will be a mandatory carriage requirement under GMDSS rules. Transmissions from 406 MHz beacons contain identification codes and by employing Doppler shift measurement techniques, the LUT can determine a beacons location. Thus details of the identity and position of a beacon can be passed to a RCC.



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Fig. 4.1 Basic concept of COSPAS-SARSAT system.
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4.3 COSPAS-SARSAT: COVERAGE MODES

Two modes are used to detect and locate beacons. The first, called the local, or real-time mode, operates with beacons using the 406 MHz and 121.5 MHz systems. The second mode operates with the 406 MHz system only and is called the global coverage mode.

(a) *Local mode 406 MHz.* This system uses data processing in the following manner. When the satellite receives the distress alert transmission from the beacon, the Doppler shift is measured and the digital data recovered from the signal. The time is noted (time tagged), and processed as digital data and retransmitted to any LUT in view in real time, this data is also stored for future transmission to earth by the satellite.

(b) *Local mode 121.5 MHz.* Repeater equipment on the satellite relays the signal directly to earth. If an LUT and EPIRB are "visible" i.e. within the field of view of the satellite, the signal will be received and processed immediately.

(c) *406 MHz global coverage mode.* This system provides for the distress signals received by the satellite to be stored in the satellite's memory and subsequently rebroadcast to all LUTs as the satellite orbits the earth, in what is known as a "dumping" process. With this method each beacon can be located by all LUTs in the system. Using the combined system of LEOSAR-GEOSAR satellites improves the system by providing continuous coverage over a large area of the earth centred on the equator. This gives almost immediate alerting capability, as the GEOSAR satellites can relay transmission picked up from COSPAS-SARSAT 406 MHz beacons to earth for processing by GEOLUTs. All relevant information is then forwarded to a MRCC for evaluation as to subsequent action. Certainly the joint system reduces the mean notification time of reception of distress beacon transmissions compared with using only LEOSAR satellites.

See Figure 4.2 on page 99.

4.4 COSPAS-SARSAT: 406 MHz BEACONS

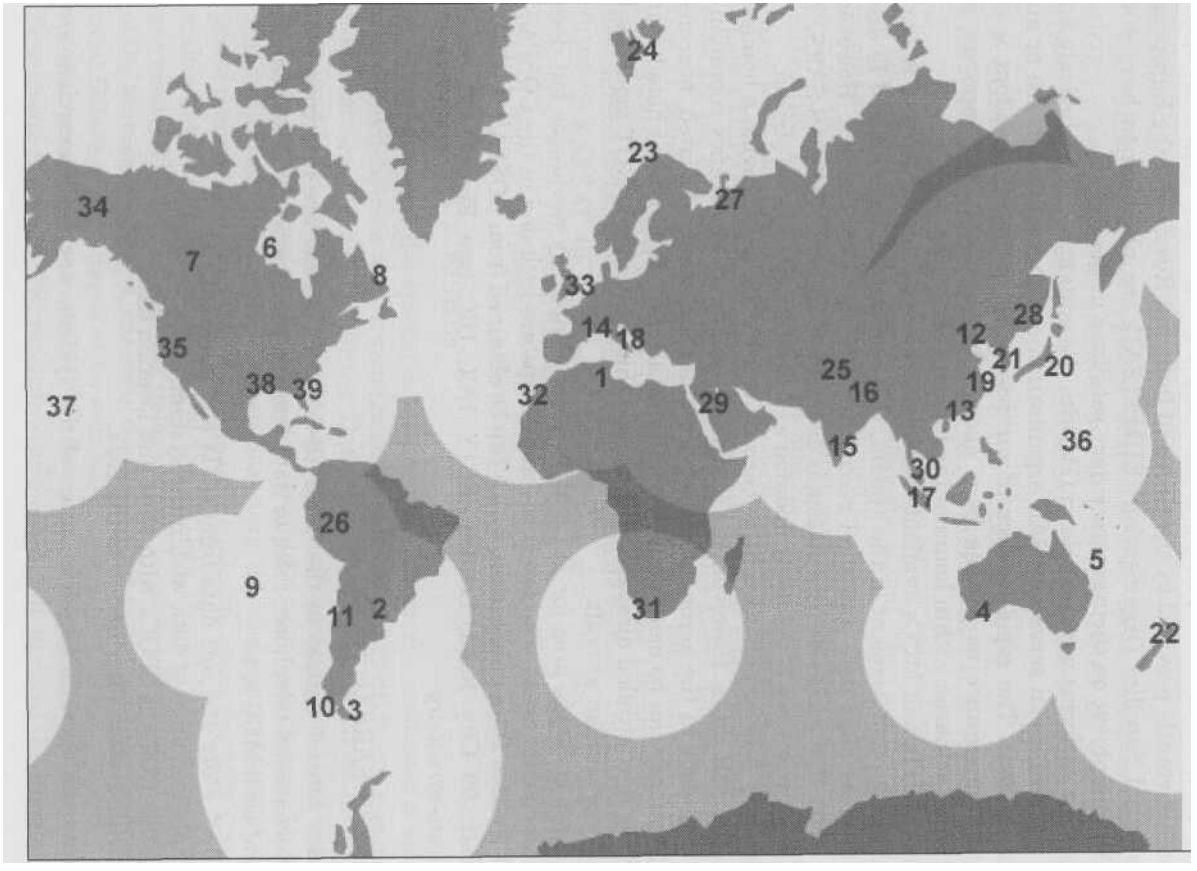
These beacons, which can be activated manually or automatically, have been specifically designed to operate with this satellite system, and have the following features. They send a 0.5 second burst of RF energy every 50 seconds on a rigidly stable frequency and which contains digitally coded information from which the MCC can rapidly determine:

(a) identity of the ship/aircraft;
(b) country of origin of the unit in distress.

An additional option allows the EPIRB message to include the ships position from information given by an on-board navigational system. The use of digitally coded identification data means that this type of beacon is more sophisticated that the 121.5 MHz type.

As discussed in the previous section, beacons operating on 406 MHz have world wide coverage in the global mode. The accuracy of location of beacons by measurement of Doppler effect by LUTs is typically better than 5 km for 406 MHz beacons, and 18 km for 121.5 MHz beacons. The overall system offers a multi-access capability where one orbiting satellite can handle more than ninety beacons operating simultaneously within its reception footprint.

The specifications for 406 MHz beacons were amended to permit the use of optionally coded position information in the transmission burst. The use of beacons with location data protocols for EPIRB, ELTs and PLBs, i.e. essentially they include integral GPS receivers have passed type approval tests and are available on the market.



NOTES

LUTs:

1	Ouargla, Algeria
2	Parana, Argentina
3	Rio Grande, Argentina
4	Albany, Australia
5	Bundaberg, Australia
6	Churchill, Canada
7	Edmonton, Canada
8	Goose Bay, Canada
9	Easter Island, Chile
10	Punta Arenas, Chile
11	Santiago, Chile
12	Beijing, China
13	Hong Kong, China
14	Toulouse, France
15	Bangalore, India
16	Lucknow, India
17	Jakarta, Indonesia
18	Bari, Italy
19	Keelung, ITDC
20	Yokohama, Japan
21	Daejeon, Korea
22	Wellington, New Zealand
23	Tromsoe, Norway
24	Spitzbergen, Norway
25	Lahore, Pakistan
26	Callao, Peru
27	Arkhangelsk, Russia
28	Nakhodka, Russia
29	Jeddah, Saudi Arabia
30	Singapore
31	Cape Town, South Africa
32	Maspalomas, Spain
33	Combe Martin, UK
34	Alaska, USA
35	California, USA
36	Guam
37	Hawaii, USA
38	Texas, USA
39	Florida, USA

Fig. 4.2 Satellite visibility area of COSPAS-SARSAT Operational LEOLUTs. Reproduced with kind permission of COSPAS-SARSAT *(www.cospas-sarsat.org). Note:* The MSG satellite is currently undergoing commissioning testing. Nevertheless, alerts detected by the MSG system are distributed in the Cospas-Sarsat system.

The number of 406 MHz beacons estimated to be in service in early 2003 was 285,000.

4.5 COSPAS-SARSAT: GENERAL INFORMATION

The COSPAS-SARSAT Programme is managed by a secretariat located at the headquarters of the International Maritime Satellite (Inmarsat) organisation in London. The system was initially established and operated by Canada, France, Russia, and the USA. Subsequently 31 other countries and organisations have joined the programme. The network of earth stations is most commonly described by the general term "ground segment", whilst the LEOSAR space segment consists of satellites put into orbit by Russia and the USA. The GEOSAR space segment is currently provided by the USA and India but Russia and the European Meteorological Satellite Organisation (EUMETSAT) are planning to have 406 MHz SAR equipment on their geostationary satellites in the future.

Although it performs a major role in GMDSS, the system is not exclusively for marine use and it can serve any SAR organisation dealing with land, sea or air rescue operations. The rapid location of survivors after a distress incident is of paramount importance as studies have shown that survival rate improves if rescue can be achieved within hours of the incident occurring.

The more LUTs and MCCs available worldwide means a higher level of redundancy in the ground segment of the system, this reduces the time needed to distribute distress alert data and send rescue units to the alert location. However there is a risk that excessive information is distributed, thus the COSPAS-SARSAT Data Distribution Plan is regularly updated as new stations come on stream.

The primary function of 121.5 MHz beacons or EPIRBs is to provide a homing function for SAR units. However, 121.5 MHz is used as an emergency communications frequency in the aeronautical service and thus, when activated, beacons on this frequency can be monitored by overflying aircraft. It should be noted that while flying in controlled airspace over Europe this frequency is NOT monitored by military or civilian aircraft.

The beacon coverage area for 121.5 MHz is not completely worldwide but does cover most coastal areas. The latest coverage maps are included in the COSPAS-SARSAT System Data publication, which can be obtained from COSPAS-SARSAT Secretariat, 99 City Road, London, EC1Y 1AX, UK. (see Figure 4.2) or at *www.cospas-sarsat.org*

4.6 COSPAS-SARSAt: MCCs

All mission control centres in the COSPAS-SARSAT system are interconnected using international telephone, telex or data transmission networks. The primary function of each MCC is to:

(a) collect, store and sort data from LUTs and other MCCs;
(b) supply exchange of data within the system;
(c) provide RCCs or SPOCs with alert and location data;
(d) provide system information to the LUT.

To ensure the operational performance of the system, worldwide exercises are held occasionally.

4.7 COSPAS-SARSAT: SATELLITE INFORMATION

The LEOSAR space segment consists of a constellation of four satellites, two provided by COSPAS (called Nadezhda) and two by SARSAT (called NOAA). All satellites are equipped with the necessary communication equipment for operating at 406 MHz and 121.5 MHz. As the satellites orbit the earth, each satellite "views" a segment of the earth over 4000 km wide, giving a field of view about the size of a continent. The time taken for each satellite to complete its orbit is approximately 100 minutes. The nominal COSPAS-SARSAT space segment operates with four satellites, but currently (2003) six satellites are in full operation with three more in the integration phase. Ten older satellites have been decommissioned.

The GEOSAR space segment currently has three satellites in operation, (GOES-E and GOES-W) operated by the USA and INSAT-2B operated by India. There are also two spare GOES satellites in orbit available for use as required.

4.8 COSPAS-SARSAT: FUTURE DEVELOPMENTS TERMINATION OF SATELLITE ALERTING SERVICE ON 121.5 MHz

The 121.5 MHz ELT beacons have relatively simple technology, a sweeping audible tone contained in an continuous analogue signal. They are low cost items predominantly used by aircraft and designed to be heard by overflying aircraft that monitor this distress frequency and these aircraft then report the details to SAR authorities. The beacon's distinctive audible tone can also be used for homing purposes by SAR units. This type of beacon does not transmit identification data which COSPAS-SARSAT can uniquely identify and more importantly, such beacons are the cause of an immense number of false alerts, leading to the degradation of some aspects of SAR efficiency. Both the International Civil Aviation Organisation (ICAO) and the International Maritime Organisation (IMO) have now agreed to terminate processing of 121.5 MHz signals. The current plan is for COSPAS-SARSAT to terminate 121.5 MHz satellite alerting services on 1st February 2009. However, the IMO recommend retaining 121.5 MHz beacons for aircraft homing purposes.

4.9 INMARSAT-E EPIRBs

The Inmarsat satellite system is designed to indicate the position of a distress situation from distress alert transmissions made by an EPIRB to any one of the four Inmarsat ocean region satellites. Data from these transmissions are relayed to dedicated Land Earth Stations in each of the four ocean regions and immediately forwarded to designated Maritime Rescue and Coordination Centres by dedicated communication links. MRCCs then deal with the distress incident in an appropriate manner. The time taken for the information to reach the MRCC is typically within two minutes of activation of the EPIRB. The EPIRB alert transmissions are received by two completely independent Inmarsat-E equipped LESs in each of the four ocean regions, ensuring duplication for each region in the event of technical failure of, or outages at an individual LES.

A feature of this type of EPIRB is that it is combined with an integrated Global Positioning System (GPS) receiver, which constantly updates the vessel's position,

accurate to within 200 metres. As a minimum requirement, the alerting transmission message will contain information as to identity and position of the EPIRB.

Shipboard Inmarsat-E EPIRB equipment can be activated in one of three ways:

(a) automatically, when floating free (sea water activated);
(b) manually, while still in its cradle;
(c) by remote control, from a remote control unit located in the wheelhouse (optional facility).

When activated an EPIRB transmits a coded distress alert which contains the following information:

(a) the identity code to uniquely identify the beacon;
(b) ships position in latitude and longitude;
(c) time of position update;
(d) nature of distress, (default condition is "unspecified");
(e) ship's course;
(f) ship's speed;
(g) time the transmission was activated.

Special transmission and reception techniques are employed in order to minimise EPIRB power consumption. Each transmission burst consists of the transmission of sixty identical frames of information, each five seconds in length. There is a short delay between EPIRB activation and the transmission of the first burst. This is to allow the integrated GPS receiver to obtain an accurate position from the satellites if no navigation interface has been used.

The start of the first transmission burst is timed as zero minutes. The second transmission burst is 45 minutes later and a third transmission burst is at 130 minutes. Transmission burst 4 is scheduled at 240 minutes (4 hours after first burst). The transmission burst is then repeated at four hourly intervals for at least 48 hours. Thus 15 transmission bursts are made within 48 hours. The EPIRB is designed to continue operating for at least 48 hours and longer if the battery capacity permits.

Digital Receiver Processor (DRP) equipment in the LES processes the distress alert and automatically forwards it to the appropriate MRCC.

Information input and remote activation from the bridge or other position are possible with certain types of float free EPIRB. Crew members can trigger an alert and include details on the nature of the situation via a keypad. The beacon contains a high intensity low duty cycle flashing light, additional features include an optional Search and Rescue Transponder (SART) and 121.5 MHz locator beacon.

Inmarsat-E EPIRBs can be fitted in place of a 406 MHz COSPAS-SARSAT EPIRB for ships sailing in areas A1, A2, and A3. However, ships sailing in area A4 must carry a 406 MHz EPIRB since the Inmarsat satellite signals do not reach vessels sailing in these high latitudes.

IMPORTANT NOTE: Since the preceding section was completed, Inmarsat announced in September 2004 that with efffect from 1st December 2006, the L-Band EPIRB system (Inmarsat-E) is being withdrawn. As Inmarsat are committed to ensuring the safety of its customers, those customers with L-Band EPIRBs will be contacted by letter and offered an alternative solution that will comply with GMDSS regulations.

4.9.1 Inmarsat-E system tests

A unique feature of the Inmarsat-E system is the method of testing the system with live transmissions. Every three hours there is a live transmission over the system by each LES using a test EPIRB to verify the overall integrity of the system. Alerts from these test transmissions are sent to the Inmarsat Network Control Centres only and not to the associated MRCCs. If a scheduled test transmission is not received at the Network Control Centre within a specified period then an alarm is raised and an investigation into what is wrong is instigated.

4.9.2 Inmarsat-E world network

Ocean Region	*Land Earth Station*	*Country*	*Associated MRCC*
Atlantic East	Goonhilly	UK	Falmouth
Atlantic East	Raisting	Germany	Bremen
Atlantic West	Goonhilly	UK	Falmouth
Atlantic West	Santa Paula	USA	USCG Alameda
Indian	Perth	Australia	Australia
Indian	Raisting	Germany	Bremen
Pacific	Perth	Australia	Australia
Pacific	Santa Paula	USA	USCG Alameda

4.10 EPIRBs IN THE VHF BAND

Vessels trading exclusively in designated GMDSS areas A1, may use EPIRBs operating on Ch.70 VHF as an alternative to either a COSPAS/SARSAT or Inmarsat-E EPIRB.

The general requirements for VHF EPIRBs are:

(a) to provide a DSC distress alert on VHF, and
(b) to provide a locating signal on 9 GHz, i.e. to be used as a SART.

The VHF EPIRB DSC signal

The signal will be transmitted on Ch. 70 VHF using class G2B mode of emission with a power level of at least 100 mW. The technical characteristics of the EPIRB DSC call are similar to the "standard" DSC distress alert calls with the following exceptions. First the "nature of distress" indication will be "EPIRB emission" while the distress coordinates and time do not need to be included. The "type of subsequent communication" will give a "no information" indication, as there is no provision on the VHF EPIRB for RT or telex operation. All the digital information is transmitted in bursts with five successive repetitions of the DSC sequences to ensure the signals are received correctly by stations within range of the EPIRB's transmission.

4.11 EPIRBs: ROUTINE MAINTENANCE AND CHECKS

The EPIRB should be routinely maintained to ensure, among other things, that the float free release mechanism is in good operational condition, and any release springs etc should be inspected, cleaned and kept greased. The EPIRB should be tested at least once per month to ensure that it is capable of proper operation. This test should be carried out using the EPIRB's "on board" test facility and the results of such tests should be recorded in the GMDSS log. "

The expiry date of the EPIRB battery should be checked and if necessary batteries should be replaced. Lanyards are often attached to an EPIRB to enable the device to be secured to a liferaft or fixed to a person in the water. It is important therefore to check that the lanyard is NOT secured to the vessel. This would prevent the EPIRB floating free in an emergency as the device would sink with the vessel and obviously no transmission could be made.

When EPIRBs have to be returned to the manufacturer to replace batteries or for routine technical testing, etc, as a precautionary measure, the EPIRB should be double wrapped in tinfoil. This will prevent the signals radiating from the device in the event of it accidentally activating during transportation.

4.11.1 406 MHz guard receivers

It should be noted that 406 MHz EPIRB guard receivers are now available commercially on the market. In addition to this device sounding an audible alarm whenever your own ships EPIRB has activated, it also has an EPIRB test facility. This feature allows the ships staff to confirm that the test transmission has been successful by displaying their own EPIRB's identity number on a LED read out. For further information see website at *www.stanguard.com*

4.12 EPIRBs: FURTHER MAINTENANCE REQUIREMENTS

Several amendments were made to the International Convention for the Safety of Life at Sea 1974 (SOLAS) that became effect from 1st July 2002. One of these amendments related to the annual testing of satellite EPIRBs and a new paragraph was added to the relevant regulation. This is quoted below.

Satellite EPIRBs shall be tested at intervals not exceeding 12 months for all aspects of operational efficiency with particular emphasis on frequency stability, signal strength and coding. The test may be conducted on board the ship or at an approved testing and servicing station.

4.12.1 EPIRBs: Guidelines for shore-based maintenance of satellite EPIRBs

The Maritime Safety Committee has approved the following guidelines for shore based maintenance of satellite EPIRBs. This maintenance must be carried out by approved shore-based maintenance provider. Shore-based maintenance must be carried out in accordance with the requirements of MSC/Circ. 1039 at an interval not exceeding five years or when battery replacement is due, such requirements are given below.

1. Introduction

1.1 The purpose of these guidelines is to establish standardised procedures and minimum levels of service for the testing and maintenance of satellite EPIRBs to ensure maximum reliability whilst minimising the risk of false distress alerting.

1.2 The guidelines are intended to be applicable both to 406 MHz EPIRBs and to L-band EPIRBs, as either type may be carried to comply with the requirements of SOLAS regulation IV/7.1.6. EPIRBs may include 121.5 MHz transmitter, or Global Navigation Satellite System (GNSS) receivers.

1.3 The guidelines also apply to service exchange EPIRBs which should be properly encoded to match the appropriate registration database.

2. Shore-based Maintenance (SBM) Provider

2.1 The SBM should:

(1) have a quality control system audited by a competent authority in respect of its servicing operation;

(2) have access to adequate calibrated test equipment and facilities to carry out the SBM in accordance with these guidelines;

(3) have access to batteries and other spare parts to the original equipment specification;

(4) have access to up-to-date technical manuals, service bulletins and the latest software versions as provided by the original equipment manufacturer;

(5) keep records of maintenance, available for inspection by the Administration as may he required;

(6) ensure that all personnel responsible for supervising and for carrying out the maintenance procedures are adequately trained and fully competent to perform their duties; and

(7) issue a shore-based maintenance report with a list of the test results and maintenance performed.

3. *Prevention of false distress alerts*

3.1 Throughout he testing and maintenance process, great care must be taken to avoid the transmission of false distress alerts. The transmissions may be picked up by aircraft as well as satellites.

3.2 A radio-frequency-screened room or enclosure should be used for all maintenance procedures involving, or likely to involve, any transmission from an EPIRB.

3.3 Provision of a 121.5 MHz monitor receiver is required; this will pick up the homing transmitter and give a warning if the EPIRB is accidentally activate outside the screened enclosure.

3.4 If a distress signal is transmitted accidentally, the local RCC should be contacted immediately and informed of the coordinates of the test site.

4. Maintenance service interval

4.1 406 MHz satellite EPIRBs should be inspected and tested in accordance with MSC/Circ. 1040. (This is quoted in the following sub section).

4.2 Shore-based maintenance of all satellite EPIRB, as defined in paragraph 1.2 should be carried out in accordance with these guidelines at intervals specified by the flag Administration and not exceeding 5 years. It is recommended that the maintenance be performed at the time when the battery is to be changed.

5. Self-test

5.1 Prior to carrying out any maintenance and, upon completion, a self-test should be performed, following the instructions on the equipment, and the results noted.

5.2 Attention is drawn to paragraph 3 on the prevention of false distress alerts. Avoidance of live transmissions is required to prevent unnecessary loading of the satellite channels.

5.3 It should be verified that the self-test mode operates properly. This check could be performed by holding the switch in the self-test mode position for 1 minute after the first self-test mode burst transmission. All transmissions should cease after releasing the self-test mode switch. Additionally, for 406 MHz satellite EPIRBs which received the COSPAS-SARSAT type approval after October 1998 (Type approval Certificates 106 or higher) the number of self-test bursts should be verified to be no more than one.

6. Battery change

6.1 The main battery should be changed in accordance with the manufacturer's recommendations, including he replacement of any other routine service parts (e.g. seals, memory battery, desiccant).

6.2 The removed batteries should be disposed of in accordance with the manufacturer's and/or national/local recommendations.

6.3 After having changed the battery, the new expiration date should be displayed on the exterior surface of the EPIRB.

7. Satellite distress transmission

7.1 The satellite EPIRB should be activated in its normal transmitting mode (i.e. not just self-test). Attention is drawn to paragraph 3 on the prevention of false distress alerts. Where seawater contacts are fitted, these should be connected together to activate the EPIRB.

7.2 The transmitted signal should be checked with a suitable test receiver to verify the signal integrity and coding.

7.3 The frequency of the transmitted signal should be recorded and verified to be within the limits required by the specification to which it is approved.

7.4 The output power of the transmitter should be check in the self-test mode. A simple method of the emission verification, such as a low sensitivity receiver placed at an unobstructed distance of at least 3 metres from the EPIRB antenna, may be used for this check. The original equipment manufacturer may suggest an appropriate method to verify the output power. Attention is drawn to paragraph 3 on the prevention of false distress alerts.

8. 121.5 MHz homing transmission

8.1 The satellite EPIRB should be activated in it's normal transmitting mode (i.e. not just self-test). Attention is drawn to paragraph 3 on the prevention of false distress alerts. Where seawater contacts are fitted, these should be connected together to activate the EPIRB.

8.2 The transmitted signal should be checked with a suitable test receiver for the characteristic swept tone modulation.

9. Global Navigational Satellite System (GNSS)

9.1 Some satellite EPIRBs are designed to transmit a position derived from a GNSS receiver, which may be internal or external to the EPIRB.

9.2 The original equipment (EPIRB) manufacturer should be consulted for a method of testing the correct operation of this function, e.g. by using a GNSS repeater/simulator or external input. This test may involve a live transmission from the EPIRB and should be performed in a screened room or enclosure in accordance with paragraph 3.2. Attention is drawn to paragraph 3 on the prevention of false distress alerts.

9.3 A test receiver should be used to verify that the signal transmitted by the satellite EPIRB contains the correctly encoded position data derived from the GNSS receiver. Attention is drawn to paragraph 3 on the prevention of false distress alerts.

10. Waterproof integrity

10.1 The satellite EPIRB should be inspected for any signs of damage or cracks to the casing, or of water ingress. Any damaged item should be replace in accordance with the manufacturer's recommended procedures.

10.2 The satellite EPIRB should be tested of the waterproof integrity at the end of the shore based maintenance. The equipment manufacturer may suggest an appropriate method to test the integrity of the EPIRB.

10.3 One method involves immersing the equipment in hot water 20-30°C above ambient) for a period of 1 minute. It can be readily seen if there are any problems with the seals, as the air inside the beacon expands and escapes as a stream of bubbles. This test should not be carried out with cool water, as the water may be drawn into the equipment without showing significant release of air bubbles.

10.4 Satellite EPIRBs equipped with seawater switches should have this function disabled during the immersion test to prevent activation, unless the complete test is

performed inside a screened room. This disabling may be achieved by immersing the EPIRB complete with a mounting bracket if the bracket includes an interlock to prevent activation before release. In some cases the EPIRB contains an inversion switch, so it will not be activated if immersed in the inverted position. The manufacturer should be consulted for specific guidance.

11. Labelling

11.1 As a minimum the equipment external labelling should be checked for the following details:

(1) manufacturer's serial number. This identifies the equipment, even if the programmed date (e.g. MMSI or callsign) is later changed;

(2) the transmitted identification code:

— for L-band EPIRBs, it will be the Inmarsat System Code; and

- for 406 MHz EPIRBs, this will be the beacon 15 Hexadecimal Identification (15 Hex ID) and other encoded identification information (MMSI/callsign) as required by the Administration. It should be verified that the label matches the information decoded from the self-test, mode transmission using the test receiver. For the COSPAS-SARSAT location protocol beacons, the 15 Hex ID should correspond to position data set to default values;

(3) the expiration date of the battery; and

(4) the date when the next shore-based maintenance is due (see paragraph 12.1).

11.2 The above checks also apply if a replacement EPIRB is provided by the SBM provider.

12. Shore-based maintenance report and other documentation

12.1 The results of shore-based maintenance should be provided in the form of a shore-based maintenance report, a copy of which is to be kept on board, and a label affixed to the exterior of the beacon detailing the name of the SBM provider and the date when the next shore-based maintenance is due.

12.2 The SBM provider may affix a tamper-proof seal or similar device on completion of the SBM.

12.3 Before returning the beacon to the owner, or when providing a replacement beacon, the SBM provider should check the registration details with the beacon registry, where practicable.

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4.12.2 EPIRBs: Guidelines on annual testing of 406 MHz satellite EPIRBs

The following information has been taken from MSC Circular 1040.

1. The annual testing of 406 MHz satellite EPIRBs is required by new SOLAS regulation IV/15.9 entering into force on 1 July 2002.
2. The testing should be carried out using suitable test equipment capable of performing all the relevant measurements required in these guidelines. All checks of electrical parameters should be performed in the self-test mode, if possible.
3. The examination of the installed 406 MHz satellite EPIRB should include:
   (1) checking position and mounting for float-free operation;
   (2) verifying the presence of a firmly attached lanyard in good condition; the lanyard should be neatly stowed, and must not be tied to the vessel or the mounting bracket;
   (3) carry out visual inspection for defects;
   (4) carrying out the self-test routine;
   (5) checking that the EPIRB identification (15 Hex ID and other required information) is clearly marked on the outside of the equipment;

(6) decoding the EPIRB 15 Hexadecimal Identification Digits (15 Hex ID) and other information from the transmitted signal, checking that the decoded information (15 Hex ID or MMSI/callsign data, as required by the Administration) is identical to the identification marked on the beacon;
(7) checking registration through documentation or through the point of contact associated with that country code;
(8) checking the battery expiry date;
(9) checking the hydrostatic release and its expiry date, as appropriate;
(10) checking the emission in the 406 MHz band using the self-test mode or an appropriate device to avoid transmission of a distress call to the satellites;
(11) if possible, checking emission on the 121.5 MHz frequency using the self-test mode or an appropriate device to avoid transmission of a distress call to the satellites;
(12) checking that the EPIRB has been maintained by an approved shore-based maintenance provider at intervals required by the administration.
(13) after the test, remounting the EPIRB in its bracket, checking that no transmission has been started; and
(14) verifying the presence of beacon operations.

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Tron UniDec EPIRB decoder (Ver. 1.00)

Beacon information

Beacon Manufacturer:	Jotron
Model:	Tron+40s
Beacon serial number:	130adl2092
Bracket type:	fbn4
Bracket serial number:	fb411837
Battery expiry:	3%2F2007

Measurements

Frequency: (MHz) (406.025±0.0025	406.024920
RF level (dB):	+53
Status:	OK

Message decoding

Hex message:	FFFEDD053F46AOA9A69AE894A4C50
Message format:	short format
Protocol:	User
Country code:	319
User type:	Maritime User
Maritime MMSI (6 digits):	036000
Specific bcn:	1
Spare:	
Aux radio device:	121.5 MHz
Emerg. Code:	Emergency Code Data Not Entered
Activation Type:	Automatic and Manual Activation
Emergency Code:	No information entered or Nationally assigned
15 Hex ID:	A7E8D41534D35D1

Monday, February 10, 2003 3:10:18 p.m.

CLOSE

Fig. 4.3 Tron UniDec EPIRB decoder (Ver. 1.00) readout.
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4.12.3 Clarification on EPIRB maintenance and testing

Some clarification of the above two sub sections is relevant. An EPIRB should and must be tested annually, but this is part of the *annual Safety Radio Survey* requirements and therefore the EPIRB will be tested to the new standards outlined in the preceding subsections. All radio survey companies carrying out surveys on behalf of flag and class authorities MUST have a new type of EPIRB tester and thus tests to the new IMO specifications and gives a printed read out. Ships staff should now always ask the Radio Surveyor to print out two copies of this test report. One copy can be retained in the ship's file so that it can be shown that the ship is in compliance for annual test with the new IMO requirements, and the other copy is with the Safety Radio Survey form which goes to flag and classification authorities. Only if a fault is found during this test is any further action required, usually the EPIRB would need to be replaced or exchanged as an EPIRB is not serviceable on board ship. A specimen of a typical read out of a modern EPIRB tester is shown in Fig. 4.3.

As outlined in para. 4.2 of section 4.12.1 all EPIRBs must be be taken ashore for stringent testing after five years service on board ship. An EPIRB's batteries will need to be replaced every few years and to do this the device must be taken ashore and then the device has to undergo extensive testing. This includes tests for water integrity tightness, it will be activated in a special "Faraday screening cage" to ensure that the device is in perfect operational condition, it will be resealed and relabelled, etc. Such tests are in addition to the annual tests mentioned above. It would therefore be convenient for ship's staff to ensure that battery replacement and the 5-year test cycle coincide to prevent the EPIRB having to be taken ashore on more than one occasion.

4.13 REGISTRATION OF EPIRBS

Internationally, major maritime states have established databases where details of EPIRB registration for that flag state will be maintained and accessible 24 hours a day. Details such as ship identity, serial numbers, owners address, etc, are kept on a database which is available to SAR authorities and is considered to be essential for the efficient use of the EPIRB system in identifying the ship in distress and to obtain other information that will assist in providing the appropriate response. Some countries have made the registration of EPIRBs mandatory, for the activation of unregistered beacons cause increased work load for RCCs in tracing details of an unregistered beacon. Investigating unregistered beacons could also, and more seriously, cause delays in processing genuine distress alerts by the RCCs, or cause the unnecessary activation of SAR units in non-distress situations. Any changes to details of the ship should be notified to the appropriate EPIRB registration database, for example change of ownership, change of name, etc.

In the UK the register of 406 MHz beacons installed on UK ships is kept by the Maritime and Coastguard Agency. The contact point for all routine EPIRB registration and database enquiries is:

The EPIRB Registry,
HM Coastguard,

MRCC Falmouth
Pendennis Point,
Castle Drive,
Falmouth,
Cornwall,
TR11 4WZ.
Telephone: + 44 (0) 1326 211569 Fax: + 44 (0) 1326 319264

A specimen registration form used by the UK EPIRB Registry is shown in Fig. 4.4.

It is essential that shipowners inform the relevant flag state registration authorities when information relating to EPIRBs changes. This will permit databases to be kept up to date and avoid unnecessary confusion during distress situations. Change of ownership, flag, vessel or change of name or loss of EPIRB through theft are examples of information which should be forwarded to the relevant authorities.

Some seventy countries have now set up 406 MHz beacon registers. For contact details of individual countries see Admiralty List of Radio Signals Volume 5.

4.13.1 Registration of Inmarsat-E EPIRBs

A separate register of Inmarsat-E EPIRBs is kept at the organisation's headquarters in London.

Owners of this type of EPIRB should register their EPIRB with Inmarsat Customer Services (Fax + 44 (0) 20 7728 1142).

The Inmarsat-E EPIRB registration form should be completed by the owners and should include all the information requested. Rescue authorities require the following information for their databases:

(a) General details about the registration and owners or manager's emergency contact details.
(b) Details of the vessel including its name, radio call sign, MMSI, gross tonnage, vessel length and number of passengers and crew on board.
(c) EPIRB details including system code in decimal form, set-up frequencies, manufacturer, model and type, approval certificate number and serial number.

After making some checks the Inmarsat Customer Services will issue an Inmarsat-E Access Authorisation Certificate, a copy of which is sent to the ship and one to the owners. This completes the registration process.

It is essential that the owners of all Inmarsat-E EPIRBs register them with Inmarsat, giving details of the vessel or craft on which they are installed, as soon as possible after installation. The reasoning behind registration is to facilitate a speedy response to people in distress, registration allows MRCCs to quickly identify the vessel to which the EPIRB has been fitted, validate the distress situation and therefore take appropriate action.

4.14 SEARCH AND RESCUE TRANSPONDERS (SARTs)

The purpose of a search and rescue transponder (SART) is to locate the position of a vessel in distress or persons in survival craft from the distressed vessel.

⊗⊗ 406 MHz EPIRB, NEW OR CHANGE OF REGISTRATION ⊗⊗

COSPAS/SARSAT No: (Please leave blank)

To be completed by Epirb owner:

Please tick the appropriate box(es):

New Registration	[ ]
Change of address	[ ]
Change of vessel ownership	[ ]
Change of vessel name	[ ]
Change of beacon	[ ]

Name of vessel:

Type of Vessel (please circle): Merchant Fishing Pleasure
Rig/Platform Aircraft Other
PLB ELT

Maximmum Number of persons carrried:

Fishing Vessel Indentity No:

MMSI:

Callsign:

Radio (please circle): VHF MF HF VHF/DSC MF/DSC HF/DSC
Inmarsat: A B C M

Beacon Manufacturer's Name:

Beacon type:

Beacon Serial No:
(from the label on the beacon)

Hex ID (15 characters):

Details of Owners:

Name(s):

Address:

Telephone/Contact No(s):

Emergency contact details:

IMPORTANT: The owner should contact the Epirb Registry if acknowledgement is not received within 2 weeks.

Contact: The Epirb Registry, HM Coastguard Southern (Falmouth), Pendennis Point, Castle Drive, Falmouth, Cornwall, TR11 4WZ
Tel: 01326 211569. Fax: 01326 319264

Fig. 4.4 UK EPIRB Registry specimen registration form.

A SAET is an easily portable device which should be taken into the lifeboat or liferaft when abandoning ship. When switched on it will transmit signals only when triggered by signals from an external marine or aircraft radar operating in the 9 GHz (3 cm) band and it should respond to interrogation from ships at 5 nautical miles range. An audible alarm or small light is incorporated into the device in order that persons in distress will be aware that a rescue ship or aircraft is within close range. Battery capacity should allow 96 hours of standby operation and the SART should be able to respond to interrogating signals even in heavy swell. SARTs may be combined with a float free EPIRB to provide a locating and homing capability.

When a SAET is activated it generates a swept frequency signal which shows up on the rescue crafts radar screen. This is a distinctive line of about 12 equally spaced blips (0.6 nautical miles between blips) extending outwards from the SARTs position along its line of bearing. The total length of *this line of blips is* approximately 9.5 nautical miles. This assists rescue craft locate and approach the scene of the incident.

4.14.1 SART signals at close range

As a rescue ship nears the SART (usually around 1 nautical mile) radar antenna sidelobe transmissions cause the dots on the radar screen to widen into concentric arcs. At even closer ranges concentric circles appear, which indicates to rescue vessels that the SART is now at very close range. The SART dotted line can be recovered by reducing the radar receiver gain.

The exact location of the SART is just inside the first narrow dot (i.e. the one closest to the radar observer).

4.14.2 Factors which increase the visibility of a SART

(1) The SART should be located as high as possible. Survivors should therefore mount the SART at least one metre above sea level, higher if possible, the aerial should be unobstructed. It is not advisable to use a radar reflector and a SART on the same survival craft. The radar reflector may obscure or reflect the SART signals.

(2) The search craft radar controls should be as follows:

(a) the preferred range is either 6 or 12 nautical miles when looking for SART transmissions;

(b) receiver gain at NORMAL setting for MAXIMUM range SART detection, i.e. a light background noise (speckle) visible;

(c) fast time constant (FTC) or anti-clutter rain (A/C rain) should be used on manual control.

(d) radar range should be selected giving a MEDIUM receiver bandwidth;

(e) receiver should be DE-TUNED in order to reduce unwanted targets such as clutter. Some radars may not permit manual detuning. Care should be taken when operating the radar in the detuned condition as vital navigational information may be lost. Retune radar as soon as possible to restore normal conditions;

(f) the anti sea-clutter control should be set to minimum for optimum range SART detection. Care should be taken that as wanted targets may be

obscured in the clutter. With heavy sea clutter conditions, the first few dots of the SAET response may not be detected, irrespective of the position of the anti sea-clutter control. Under such conditions the SART position can be estimated by measuring 9.5 nautical miles back from the most distant dot towards operators own ship.

Some radar sets are equipped with automatic/manual anti sea-clutter control features. Automatic function of such controls vary from manufacturer to manufacturer, thus it is advisable for radar operators to use manual anti sea-clutter control until the SART signal is detected. Automatic anti sea-clutter control can then be selected and its response to the SART signal can then be assessed with manual control.

Important notes:

(i) Navigators should be aware that operating the controls of radars as outlined in (a) to (f) above will produce loss of performance for normal echo returns. A radar used for locating SART signals should therefore not be used for normal radar navigational duties.

(ii) Wide bandwidth is often automatically selected on radar ranges on or below 3 nautical miles.

(iii) After first sighting of SART signals radar operators may increase radar range again if required.

4.15 SARTS: TESTS AND CHECKS

Each search and rescue transponder should be tested and checked at least once a month using the in-built test facility. Further it should be checked for security and signs of damage and results of these tests and checks noted in the GMDSS log (SART test signals may be detected on own ship's 3 cm radar). The battery expiry date should also be checked and appropriate action taken if the battery needs renewal.

4.16 MARITIME SAFETY INFORMATION (MSI)

Broadly within the GMDSS MSI provides for the dissemination of important navigational, meteorological and other valuable safety related information to shipping. This is carried out by automatic means on different frequency bands to ensure maximum coverage but generally can be described as being short range (or coastal) and long range (or oceanic). Coverage of the short range coastal service is provided by the Medium Frequency system known as NAVTEX, while the oceanic service is provided by Inmarsat SafetyNET service utilising the Enhanced Group Call (EGC) system. There is also an HF Navtex service available in area A4, the frequencies for this service are shown below:

4 210 kHz
6 314 kHz
8 416.5 kHz
12 579 kHz
16 806.5 kHz
19 680.5 kHz

22 376 kHz
26 100.5 kHz

It should be noted, however, that currently there are no dedicated HF Navtex receivers available commercially. Thus ships operators must manually tune into the frequencies shown above to receive MSI by telex on HF.

GMDSS compliant ships must, while they are at sea keep a radio watch for MSI broadcasts on appropriate frequency or frequencies for the area in which it is sailing. Any safety messages received must be logged, i.e. time of reception and identity of broadcasting station. Printed copies of any distress messages or distress traffic received must be attached to the radio log.

Seven categories of MSI are identified as follows:

(a) navigational warnings
(b) meteorological warnings.
(c) search and rescue information
(d) ice reports
(e) meteorological forecasts
(f) pilot service information
(e) electronic navigational systems updating information.

Meteorological forecasts and navigational warnings for given areas are transmitted at fixed times from various station throughout the world. Meteorological warnings and distress alerts can be sent at any time. The urgent nature of such messages cause an alarm to sound at the terminal when they are received. Both scheduled and unscheduled transmissions of MSI will be automatically printed by the appropriate equipment on board the ship.

4.17 NAVTEX

This is an international automated direct printing telex service used to promulgate navigational and meteorological warnings and also urgent information to shipping. The system uses the single frequency of 518 kHz worldwide, messages are broadcast in the English language. It is designed to offer a simple but inexpensive method of receiving Maritime Safety Information (MSI) in coastal waters as the transmission range is about 400 nautical miles.

There are three different message priorities used to effect the timing of the first broadcast of a new warning with the NAVTEX service. These are as follows:-

VITAL:for immediate broadcast, subject to avoiding interference to ongoing transmissions;
IMPORTANT: for broadcast at the next available period when the frequency in unused;
ROUTINE: for broadcast at the next scheduled transmission period.

It should be noted that the VITAL and IMPORTANT warnings will usually need to be repeated, (if still valid) at the next scheduled transmission period. To avoid unnecessary interference with scheduled services, the VITAL priority designation must be used only in extremely urgent situations, such as some distress alerts. Messages allocated this priority must be kept as brief as possible. Scheduled periods between regular transmission intervals should be incorporated into the regu-

lar transmission cycle to allow for the immediate or early transmission of VITAL messages.

In addition to the promulgation of navigational and meteorological warnings, NAVTEX provides an important method of alerting all vessels when a distress incident is in progress. It is thus a vitally important part of the GMDSS. All ships over 300 g.t. equipped for the GMDSS must carry a NAVTEX receiver and small craft and fishing vessels operating within range of NAVTEX transmissions are strongly recommended to fit such receiving equipment.

Transmitting stations are identified by a transmitter identification letter. Operators can choose to receive from one (or more) transmitters, that provide information in the sea area in which they are sailing, by programming the receiver. Internationally, transmission times are coordinated on a time sharing basis to prevent mutual interference on the single frequency.

It should be noted that the frequency 490 kHz is used exclusively for the transmission of MSI in a second or local language. In tropical zones a third frequency 4 209.5 kHz may be used exclusively for NAVTEX type transmissions of meteorological and navigational information, etc, where transmissions on 518/490 kHz may be obliterated by heavy static.

Within the NAVTEX system there are currently seventeen different message categories which can be received, as shown in the list below. The ship's operator has control of the selection of which message categories can be received by preprogramming the receiver select/reject facility. A built in micro-processor ensures that vital information, such as distress or urgency messages are always accepted, i.e. the operator cannot disable or reject these message categories.

The micro-processor control can, if desired also prevent a routine message previously received from being printed on a subsequent transmission.Thus ships coming into range of a particular NAVTEX transmitting station will receive many messages, previously broadcast for that area. Ships which are already within the transmission range and which have previously received messages, will not have them printed a second time. The updated identifying categories or subject codes for MSI broadcasts from UK station transmitting NAVTEX are :-

A = Navigational warnings, including mobile drilling rig movements. See “V” also.
B = Meteorological warnings
C = Ice reports
D = Search and rescue information, including piracy and armed robbery warnings
E = Meteorological forecasts
F = Pilot service messages
G = Available
H = LORAN messages
I = Available
J = SATNAV messages
K = Other electronic navaid messages (information relates to radio navigation services)
L = Subfact and Gunfacts warnings, brief messages with details of VHF channels and broadcast timings

V = Amplifying navigational warning information initially announced under "A"
W = Special services - trial allocation
X = Special services — trial allocation
Y = Special services — trial allocation
Z = No messages on hand

Message categories A, B, D cannot be rejected by the receiver and will always be printed. Category L should not be rejected by the receiver while the other categories can be selected or rejected as desired.

It should be noted that the indicator WZ appearing before the NAVTEX message number indicates a coastal navigational warning.

4.17.1 European NAVTEX service : **MSI Transmission Data** (North Atlantic - East)

Station	*Station Identity Letter*	*Transmission Times*
Corsen	*A*	0000 0400 0800 1200 1600 2000
Bod0	B	0010 0410 0810 1210 1610 2010
Murmansk	C	0020 0420 0820 1220 1620 2020
Grimeton	D	0030 0430 0830 1230 1630 2030
Niton	E	0040 0440 0840 1240 1640 2040
Arkhangelsk	F	0050 0450 0850 1250 1650 2050
Cullercoats	G	0100 0500 0900 1300 1700 2100
Bjurb'klubb	H	0110 0510 0910 1310 1710 2110
Gislövshammer	J	0130 0530 0930 1330 1730 2130
Niton	K	0140 0540 0940 1340 1740 2140
Rogaland	L	0150 0550 0950 1350 1750 2150
Oostende	M	0200 0600 1000 1400 1800 2200
Portpatrick	O	0220 0620 1020 1420 1820 2220
Den Helder	P	0230 0630 1030 1430 1830 2230
Malin Head	Q	0240 0640 1040 1440 1840 2240
Reykjavik	R	0250 0650 1050 1450 1850 2250
Oostende	T	0310 0710 1110 1510 1910 2310
Vard0	V	0330 0730 1130 1530 1930 2330
Valentia	W	0340 0740 1140 1540 1940 2340
In addition stations below transmit inshore local broadcasts on 490 kHz		
Portpatrick	C	0820 2020
Corsen (in French)	E	0040 0440 0840 1240 1640 2040
Niton	I	0520 1720
Cullercoats	U	0720 1920
Niton (in French)	T	0310 0710 1110 1510 1910 2310

4.18 ENHANCED GROUP CALL (EGC)

EGC is part of the Inmarsat-C system and complements the NAVTEX systems described in previous section. EGC supports two services, namely:

SafetyNETTM — for the promulgation of Maritime Safety Information (MSI), and

FleetNET™ — for transmission of fleet management and general information to particular fleets or groups of ships.

Fig. 5.1(a) (on page 125) shows the areas of the world covered by the four Inmarsat satellites each of which transmits EGC information on a designated channel. The EGC channel is optimised to enable the signal to be monitored by a small receive-only MES designed to monitor such messages. This capability can also be built into Inmarsat-A and Inmarsat-B MESs.

EGC messages can be addressed as follows:

(a) to either a geographical area (area call - see Fig. 4.6), or
(b) to groups of ships (group call — see Fig. 4.7).

Area calls will be received by all ships within the footprint of the satellite which they are currently monitoring. However, they will be printed only by those receivers which recognise the fixed area or geographical position in the message. The system enables an EGC receiver to print only those MSI messages which relate to the ships present position, intended route, or to fixed areas programmed by the operator. Certain types of message which are not relevant to a particular ship can also be suppressed.

Note: It is therefore important that operators input ship's position on a regular basis into the EGC receiver if this is not performed by automatic means such as GPS.

Messages previously received in full will be automatically suppressed. However, certain message categories such as shore-to-ship distress alerts, meteorological and navigational warnings for ships in the affected area can not be suppressed.

Information providers such as hydrographic offices, meteorological offices, RCCs, shipping companies or news services can route information to a LES for onward transmission via satellite to selected ships, ships in particular fleets, ships in given geographical areas, ships of certain flag, or all ships. As some EGC receivers may not provide wholly uninterrupted monitoring of the satellite broadcast channel, certain important unscheduled messages may be repeated six minutes after the first broadcast.

4.18.1 Setting up EGC receiver

At the start of a voyage the shipboard operator should:

(a) Select the appropriate broadcast channel. This is normally an automatic function when logging on to a particular ocean region.
(b) Input the identification code for the NAVAREAS/METAREAS and coastal MSI area for which it is desired to receive MSI.
(c) Ensure the ship's position is updated at regular intervals or confirm connection with an interfaced electronic position fixing system such as a GPS receiver.
(d) Ensure the EGC receiver is switched on whilst in port in order to receive all necessary MSI messages prior to sailing.

4.19 MSI IN UK WATERS

The main method for promulgating important information to navigators using the GMDSS in UK waters, is the terrestrial NAVTEX service on 518 kHz. Maritime Safety Information (i.e. navigation warnings etc) for the coastal waters relating to

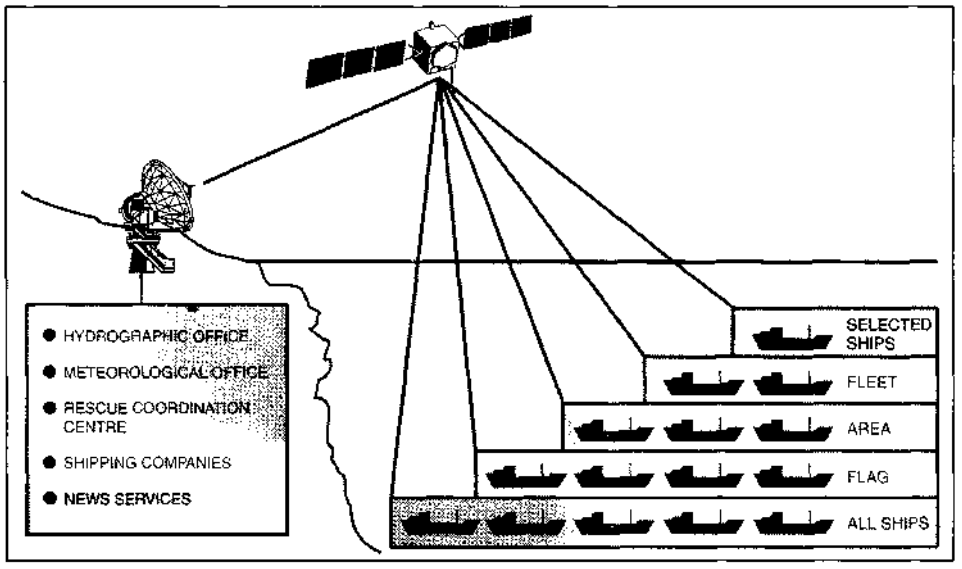


Fig. 4.5 Enhanced group call system. © Inmarsat 1996.
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the North Atlantic - East (excluding Mediterranean and Black Sea) are transmitted by this service. NAVAREA messages are transmitted using the satellite communication system, utilising the Inmarsat-C Enhanced Group Call (EGC) system.

In order to assist in the wide dissemination of maritime information ships staff are invited to co-operate with shore authorities by reporting any potential danger to navigation which they might encounter. Such messages should be transmitted to the nearest coast station and be preceded by the safety signal. In the case of meteorological messages concerning storms the service indicator = OBS = should be used, no charges will be made for these messages.

4.20 WORLD WIDE NAVIGATIONAL WARNING SERVICE (WWNWS)

This system is concerned with:

(a) long range (NAVAREA) warnings usually of a long term nature primarily concerned with routing and passage through main shipping lanes;
(b) coastal warnings, usually dealing with short term, short range information and sent by NAVTEX, MF RT or VHF;
(c) local warnings, mainly dealing with inshore waters and usually broadcast on VHF only.

WWNWS provides for the coordinated transmission of radionavigational warnings globally by dividing the world into sixteen geographical areas (see Fig. 4.8 on page 122). Known as NAVAREAS each area is identified by a Roman numeral, e.g. NAVAREA III (Spain). To provide adequate notification to ships, at least two daily transmissions are deemed necessary. Thus arrangements for this are built into transmission schedules for each NAVAREA broadcast. Messages are numbered consecutively throughout the year for each area.

Within WWNWS provision is made to meet national maritime needs by having local control and coordination for navigational warnings concerning local and coastal areas. The use of the English language for all broadcasts is mandatory but provision exists for transmitting second language broadcasts. Although the broadcasting and coordinating arrangements for NAVAREA and NAVTEX warnings are significantly different, the contents of both message classes are similar. The subject matter of such warnings include the following (but other subject items may also be included if considered necessary):

(1) reports of defective lights, fog signals, buoys, etc in main shipping lanes;
(2) the presence of dangerous wrecks in or near main shipping lanes and, if relevant, their marking;
(3) establishment of major new aids to navigation or significant changes to existing ones;
(4) the presence of unwieldy tows in congested waters;
(5) drifting mines;
(6) areas where search and rescue and anti-pollution operations are in progress or on-going (for avoidance of such areas);
(7) notification from MRCCs or MRSCs of ship/aircraft in distress, overdue or missing in open seas, seriously overdue or missing;
(8) the presence of newly discovered wrecks or natural hazards, liable to cause danger to shipping and, if relevant, their marking;
(9) unexpected suspension or alteration to established shipping routes;
(10) the existence of cable laying or underwater operations either in progress or on-going, such as towage of submerged objects (manned or unmanned), for whatever reason. Other underwater operations which potentially are a danger in or near shipping lanes;
(11) information concerning new offshore structures in or near shipping lanes;
(12) failure of, or changes to important radio-navigational aids;
(13) information relating to safety of shipping by special operations over wide areas. This could include, naval exercises, space missions, nuclear tests etc. If a degree of hazard exists, the warning should include details of the risk. Whenever possible, these warning should be originated at least five days prior to the start of the special operation, and should remain if force until the operation is concluded.

4.21 PORTABLE SURVIVAL CRAFT VHF RADIOTELEPHONES

The purpose of portable survival craft VHF radiotelephones is to provide on-scene communications to and from rescue surface craft/aircraft during SAR operations. These highly visible yellow or orange devices should be taken to the liferaft(s) or lifeboat(s) along with the EPIRB(s) and SART(s) in the event of abandoning ship.

Vessels between 300 and 500 g.t. are required to carry two portable VHF radiotelephones, vessels over 500 g.t. are required to carry three. They should be capable of working on Ch. 16, Ch. 6 and at least one other working channel. However, many such radiotelephones work on all VHF international channels and perhaps private channels as well. Although designed primarily for on-scene communications during a maritime incident, they may be used for routine on-board communications provided the rechargeable Ni-Cad batteries are employed. The

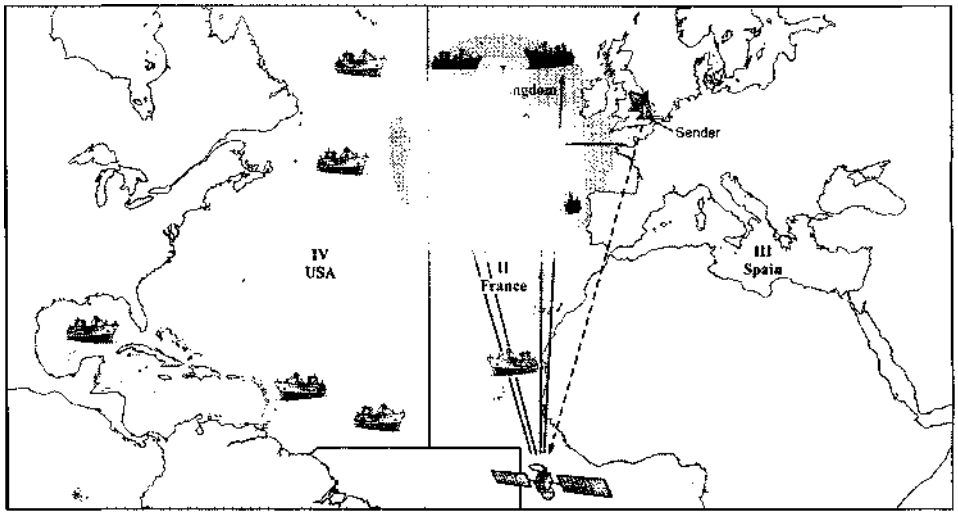


Fig. 4.6 SafetyNet call to a circular area around an emergency.
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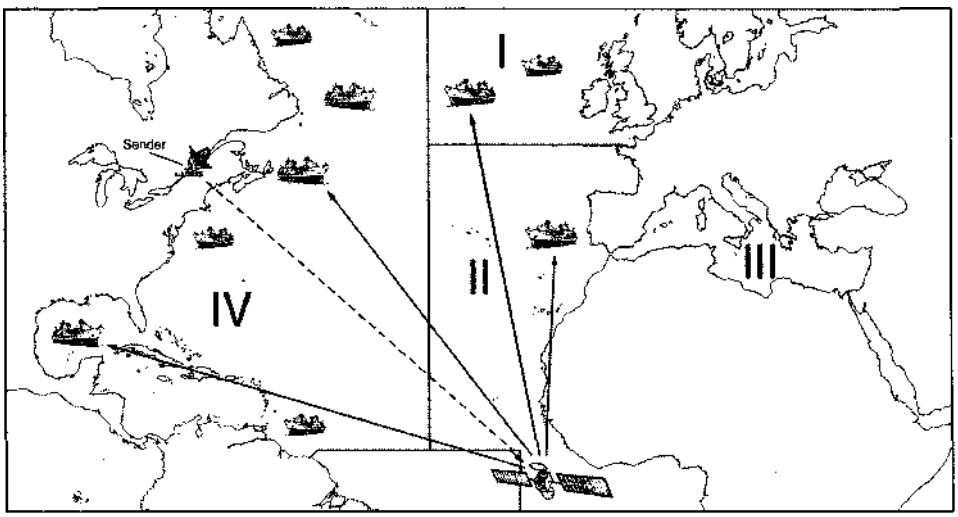


Fig. 4.7 FleetNet call to a selected group of ships.
Reproduced with kind permission of Inmarsat, London *(www.inmarsat.com).*

"Emergency" batteries are not rechargeable and are usually kept near, or at the rear of, the battery charger unit and should be taken to the liferaft/lifeboat along with the VHF transceivers. These emergency batteries have approximately five times the capacity of the rechargeable batteries. The emergency battery expiry date should be checked regularly particularly during routine tests as indicated in section 8.20.

It should be noted that portable survival craft radios may not be intrinsically safe. Using them or, in particular, changing their batteries in hazardous areas should be avoided.

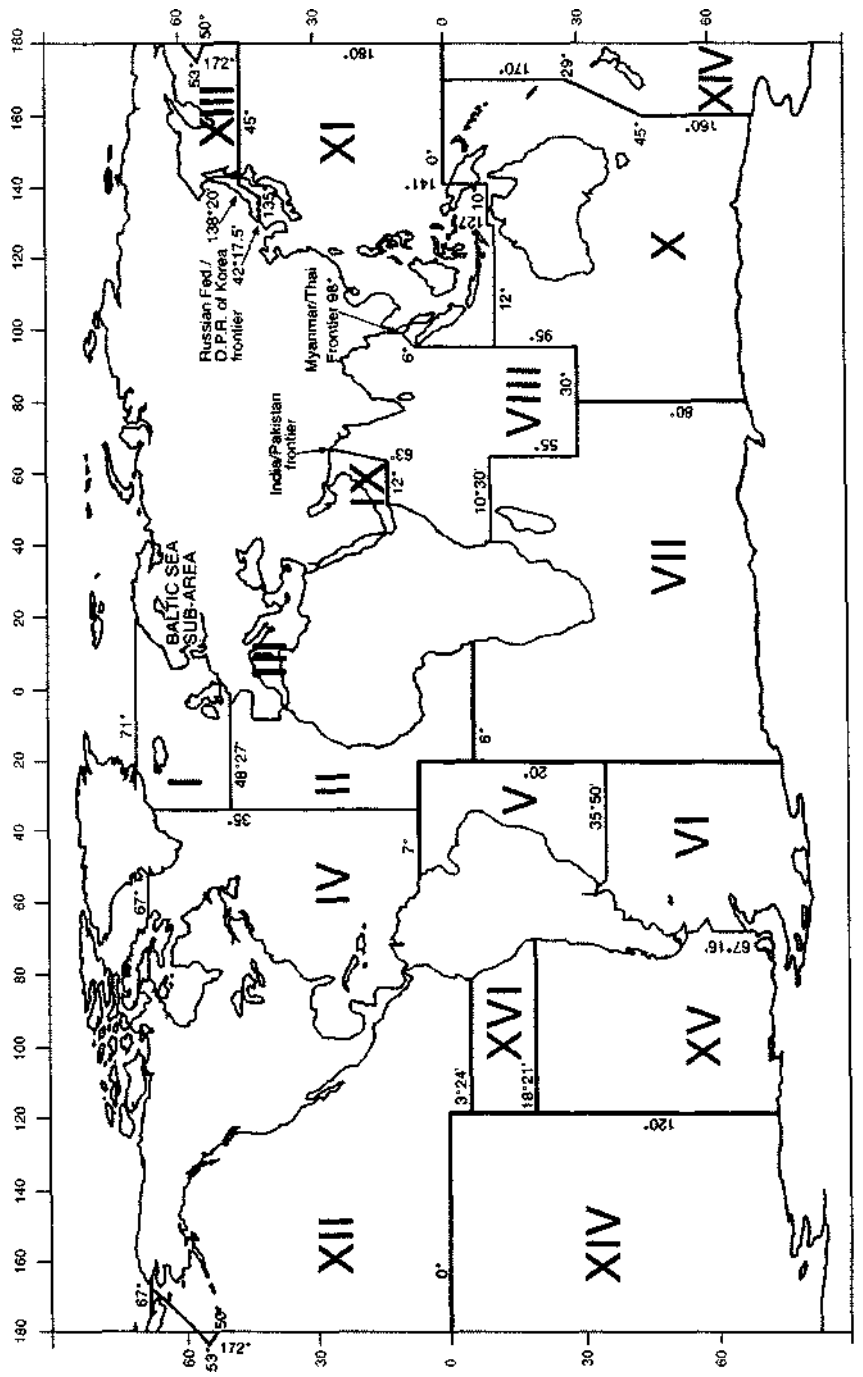


Fig. 4.8 Geographical areas for coordinating and promulgating radio-navigational warnings.

CHAPTER 5

Satellite Communications

GENERAL

The International Mobile Satellite organisation (Inmarsat) provides high quality voice, telex, data and fax circuits to and from suitably equipped ships. Similar services are also offered to land mobile and aeronautical users via the same system. These, together with distress, urgency and safety services are available using four geostationary satellites via land earth stations situated in many countries.

5.1 MASTER'S AUTHORITY

The service of a Mobile Earth Station is placed under the supreme authority of the Master or the person responsible for the ship. The person holding this responsibility shall require that each operator comply with ITU Radio Regulations and the observance of secrecy of correspondence.

5.2 THE INMARSAT ORGANISATION

Inmarsat became a private company in April 1999. Inmarsat headquarters are in London. Contact details:

Inmarsat
99 City Road
London EC1Y 1AX
United Kingdom
Telephone: +44 (0) 20 7728 1000
Fax: +44 (0) 20 7728 1044
e-mail: *customer_care@inmarsat.org*
web: *www.inmarsat.org*

To ensure that it meets its public service obligations with particular regard to GMDSS it is managed by an intergovernmental body known as the International Mobile Satellite Organisation (IMSO) at the above address but with the telephone number +44 (0) 207 728 1249.

Inmarsat has responsibility for the procurement and operation of the space segment of the global mobile satellite communications system. From 2005 the new Inmarsat 1-4 satellite system will support the Inmarsat Global Area Network providing mobile data communications at up to 432 kbit/s for internet access, mobile multimedia and other modern applications such as Inmarsat Fleet. The space segment comprises four geostationary satellites each having a particular area of coverage as follows:

Atlantic Ocean Region - East	(AOR-E)
Indian Ocean Region	(IOR)
Pacific Ocean Region	(POR)
Atlantic Ocean Region - West	(AOR-W)

Figs. 5.1(a) and (b) show the coverage area of each ocean region. Note that at latitudes greater than approximately 70° ships are not be able to maintain reliable communications with the satellites which go out of line-of-sight when this far north or south.

The complete satellite system comprises:

(a) The shipboard equipment known either as:
Mobile Earth Station (MES) or
Ship Earth Station (SES).
See also Fig. 5.2 on page 127.
(b) The four satellite regions as detailed above.
(c) Shore-based satellite stations known either as:
Land Earth Stations (LES) or
Coast Earth Stations (CES).
(d) One Network Coordinating Station (NCS) in each ocean region.
(e) Inmarsat headquarters in London which controls the entire system.

There is a range of different shipboard satellite terminal equipment available, each having its own particular features.

(a) Inmarsat-A
(b) Inmarsat-B
(c) Inmarsat-C (including mini-C)
(d) Inmarsat-M (including mini-M)
(e) Inmarsat-E
(f) Inmarsat-Fleet (F77, F55 and F33)

Each of the above systems uses the same four satellites. However, each system will require its individual receive/transmit processing at the Land Earth Stations (LES). Subsequently the LES will be designated as able to operate only on the system(s) for which it is built.

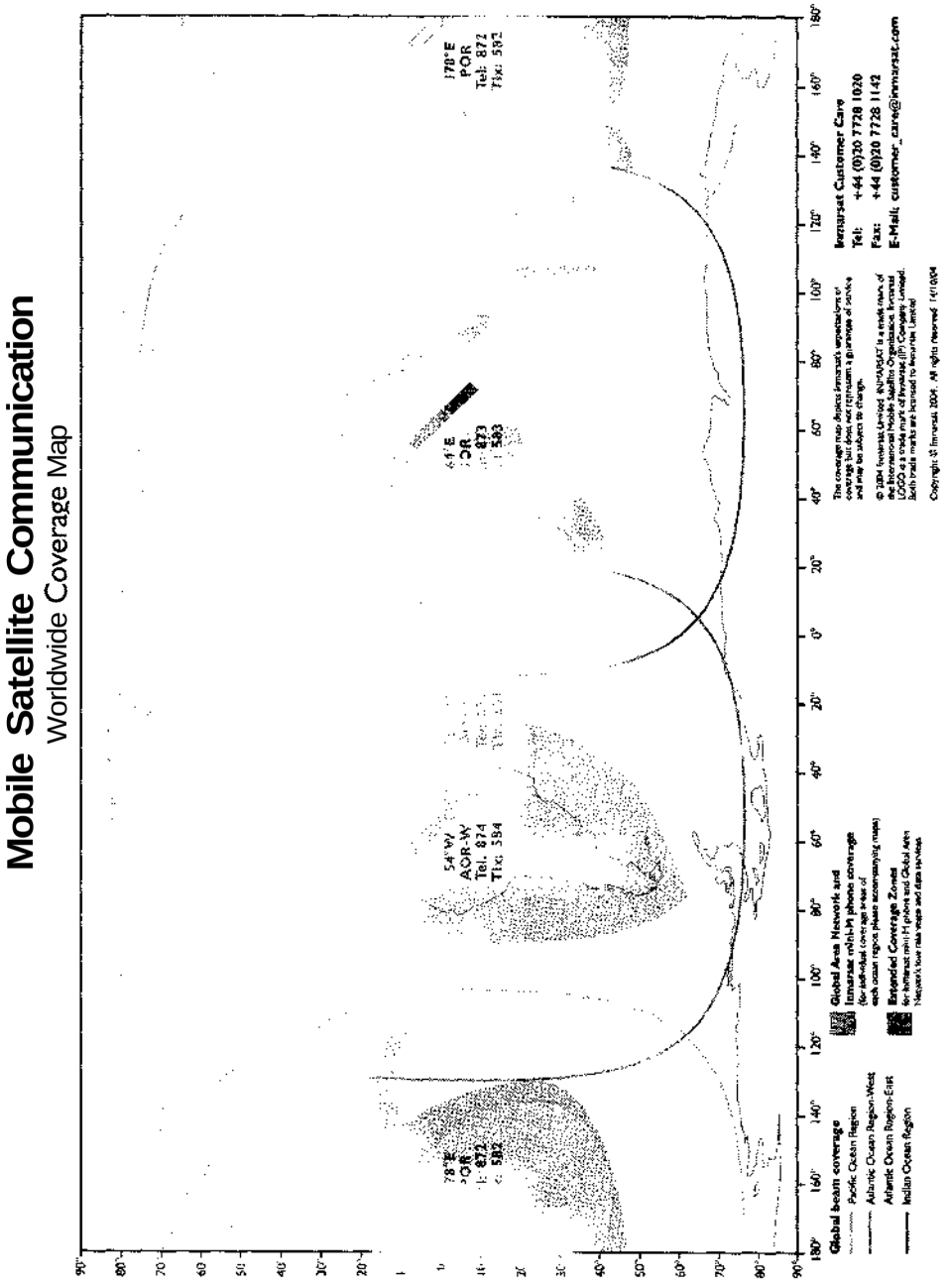
5.3 SATELLITE FREQUENCIES

Four bands of frequencies are used in the two-way communications between the MES, satellite and LES.

(a) MES to satellite uplink	1626.5-1646.5 MHz
(b) Satellite to MES downlink	1525.0-1545.0 MHz
(c) LES to satellite uplink	6425.0-6443.0 MHz
(d) Satellite to LES downlink	3600.0-3623.0 MHz

5.4 SATELLITE EARTH STATIONS

Each ocean region has one network co-ordination station (NCS) which controls the allocation of channels to MESs and LESs within its region. MESs and LESs continuously monitor one of two common signalling frequencies (TDMO or TDM1) in



(a) Inmarsat Mobile Satellite Communications Worldwide Coverage Map.
rsat 2004.
ith the kind permission of Inmarsat, London (*www.inmarsat.com*).

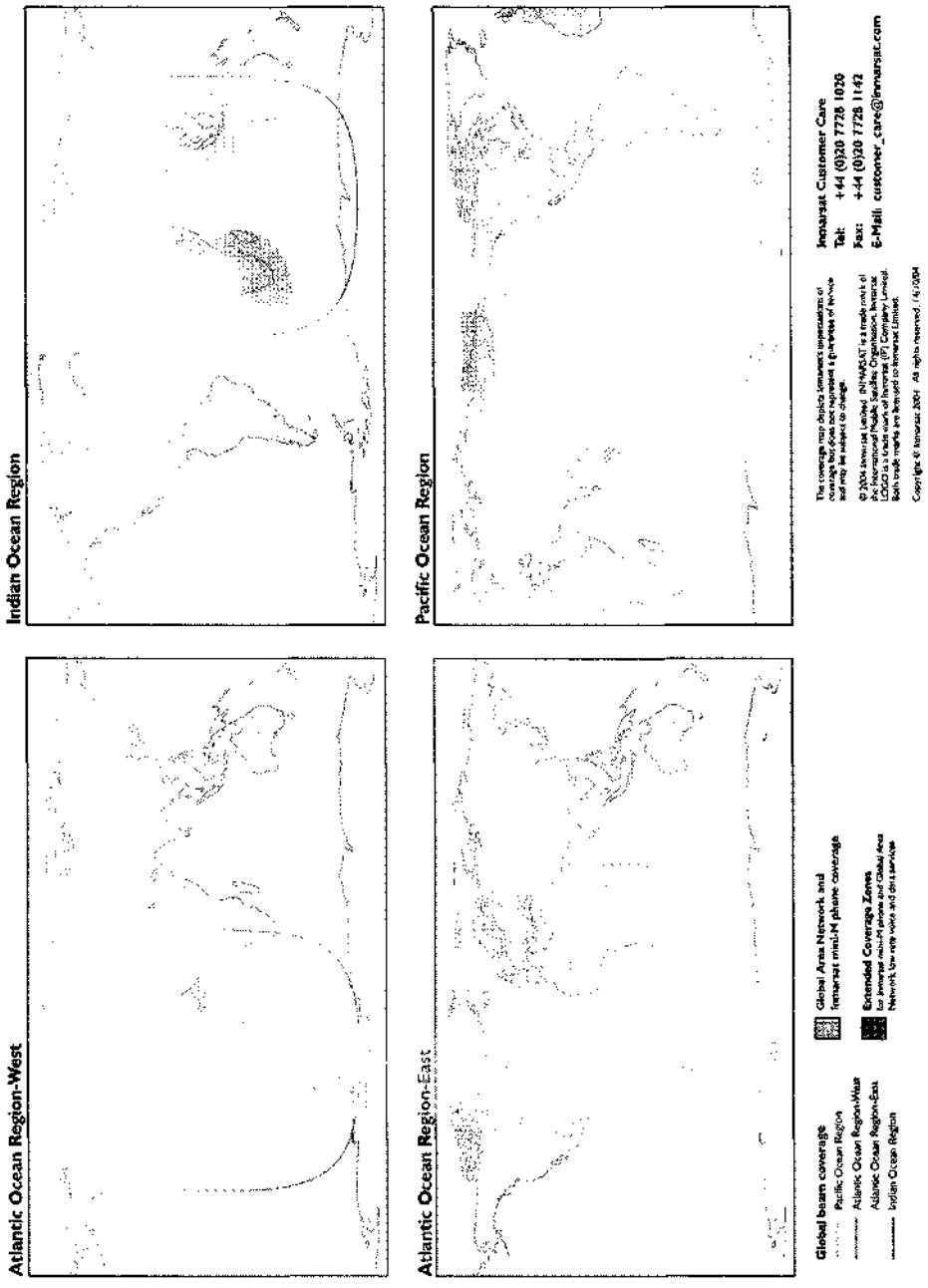


Fig. 5.1(b) Inmarsat Mobile Satellite Communications Worldwide Coverage Map. © Inmarsat 2004.
Used with the kind permission of Inmarsat, London *(www.inmarsat.com).*

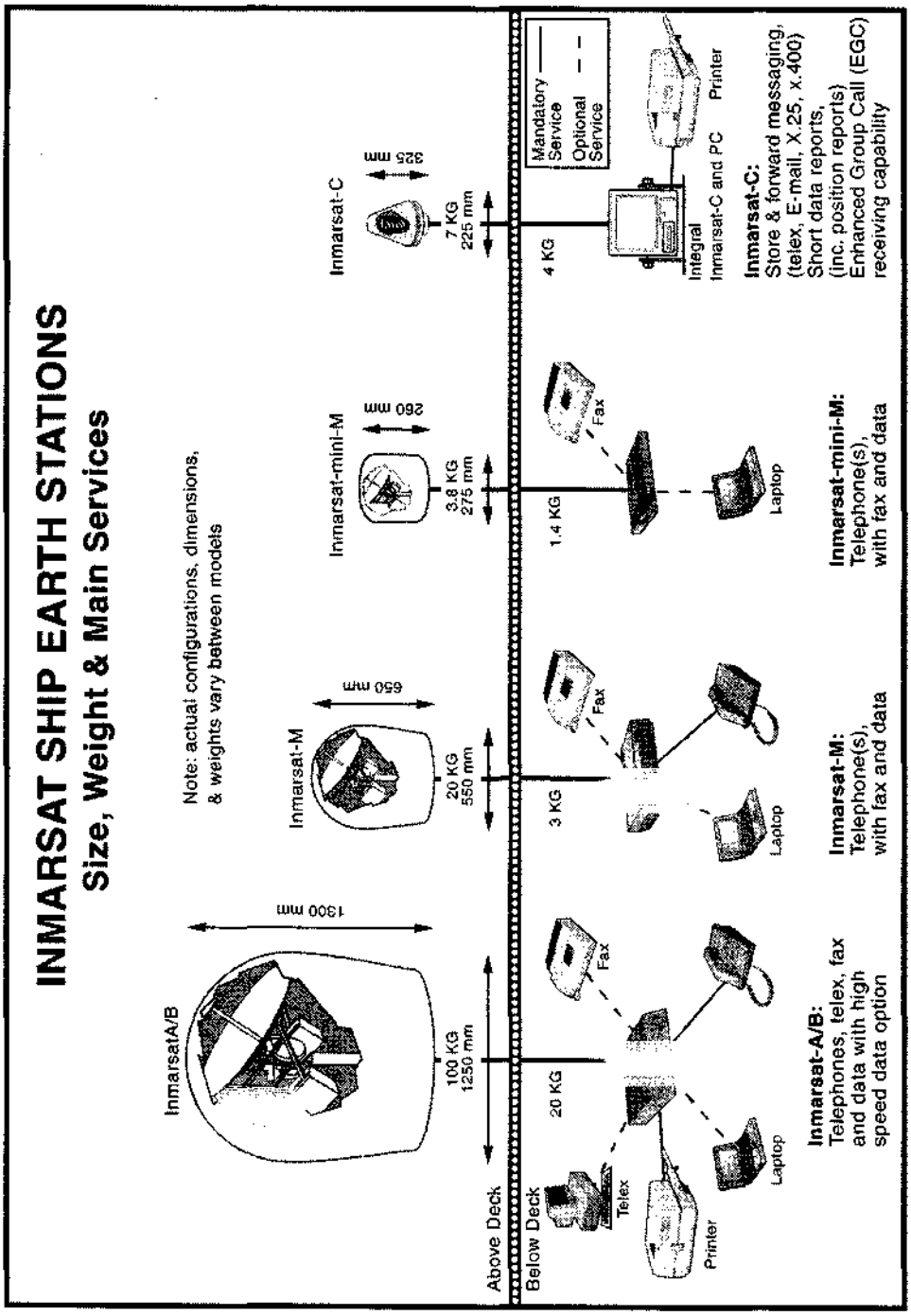


Fig. 5.2 Inmarsat Ship Earth Stations. © Inmarsat 1999.
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to

order to receive and send channel assignment commands. MESs with an even-fourth digit monitor TDMO and those with an odd fourth digit in their main Inmarsat mobile number monitor TDM1.

There are several land earth stations available in each of the four ocean regions. Their function is to connect the Inmarsat network to the national and international telecommunications systems around the world.

Note: Ship earth stations (SES) are now more commonly referred to as mobile earth stations (MES) since aircraft and land-mobile users also access the Inmarsat system.

5.5 ANTENNA ALIGNMENT

If a MES *is* within the coverage area of one of the satellites it needs to align its antenna to point directly at the satellite before communications can take place. This can be achieved with the use of information tables or azimuth and elevation maps. This relates to Inmarsat MES types A, B, F and M equipment which all have directional antennas. Inmarsat-C antennas are omnidirectional and do not require antenna alignment. Manufacturer's operating instructions indicate how to perform antenna alignment, although many modern MES's are capable of locating any available satellite if the ship position is input (usually from GPS or other electronic positioning equipment). Thereafter the antenna has to track the satellite irrespective of ship movement and once locked on, the antenna will change azimuth and elevation angles automatically in order to maintain contact. If the MES goes out of the service area of one ocean region, then the antenna will have to be realigned with the satellite within the newly entered ocean region. Some MES's are able to "search" for the new satellite when changing ocean regions.

5.6 INMARSAT MOBILE NUMBERS (IMN)

All Inmarsat Mobile Numbers (IMN) consist of nine digits apart from Inmarsat-A which has seven. Each MES is issued with a unique identification number(s). If more than one MES is fitted to a ship, each will have its own IMN. The numbering system is as follows:

Inmarsat MES	*Inmarsat IMN*
Inmarsat-A	1xxxxxx
Inmarsat-B	3xxxxxxxx
Inmarsat-C	4xxxxxxx
Inmarsat-M	6xxxxxxx
Inmarsat mini-M	76xxxxxx
Inmarsat Fleet	76xxxxxx
Inmarsat Fleet (HSD)	60xxxxxx

The Inmarsat-A IMN consists of seven digits beginning with figure 1 followed by a further 6 digits, e.g. 1234567. See section 5.7 for details of the allocation of a second IMN.

Inmarsat-B/M and mini-M follow similar patterns in that the first or second digit identify the particular Inmarsat system where 3 indicates Inmarsat-B, 6 for Inmarsat-M and 76 is Inmarsat mini-M. The last two digits of the IMN identify

the terminal on that MES such as telephone, fax, computer or telex machine.

Inmarsat-C has nine digits beginning with figure 4 followed by the country MID (see Appendix 7) and a further 5 digits e.g. 423212345. Only one IMN is allocated to each Inmarsat-C terminal and this is used for all services whether telex, data, e-mail or fax.

The ITU List of Callsigns and Numerical Identities and ITU List of Ship Stations contain details of Inmarsat IMNs. Alternatively, the Inmarsat ship directory may be accessed on *www.inmarsat.com/ships*

5.7 MULTIPLE IDENTITY OF A SINGLE MES

It is possible for an Inmarsat-A MES to have more than one IMN. It is usual for the main IMN to be available for normal telex/phone communications. The second IMN is usually used exclusively for voice-band data and/or fax, probably left in auto answer mode. Telex is not available on the second IMN. Only one IMN can be accessed at any one time.

With Inmarsat-B, F and M systems the last two digits can range from 10 to 99 to differentiate, on the same ship, between:

(a) different MES's;
(b) separate channels on a multi-channel MES; or
(c) different services such as telephone, fax, data, telex, etc.

5.8 INMARSAT PRIORITY INDICATORS

There are four levels of priority within the Inmarsat system.

Level 0	Routine Priority
Level 1	Safety Priority
Level 2	Urgency Priority
Level 3	Distress Priority

These priority levels are used particularly within the Fleet F77 service in order to comply with the new IMO GMDSS rules regarding distress, urgency and safety.

5.9 INMARSAT-F (FLEET) SYSTEM

Inmarsat-F (or Fleet as it is often known) is the latest Inmarsat service to be introduced. It provides modern voice and data services to the maritime sector. Fleet F77 will gradually replace the Inmarsat-B service. The service is based on the Inm-IV satellites scheduled to be launched in 2005. There are three Fleet versions available with Fleet 77 being the only one compliant with the GMDSS. The other two systems Fleet 55 and Fleet 33 are aimed at smaller craft users. The Inmarsat Fleet service does not support telex messaging.

Fleet 77 provides Inmarsat Mobile ISDN at 64 kbit/s enabling large volumes of data to be transferred efficiently and a mobile packet data service (MPDS) with always-on connectivity when charges are based on the volume of traffic transferred. Access to the internet, e-mail services, telemedicine, LAN and private networks are possible. Four levels of pre-emption and voice prioritisation for distress, urgency and safety applications are provided in order to comply with the latest IMO regulations. LES's currently providing Fleet 77 services are listed in section 5.11.

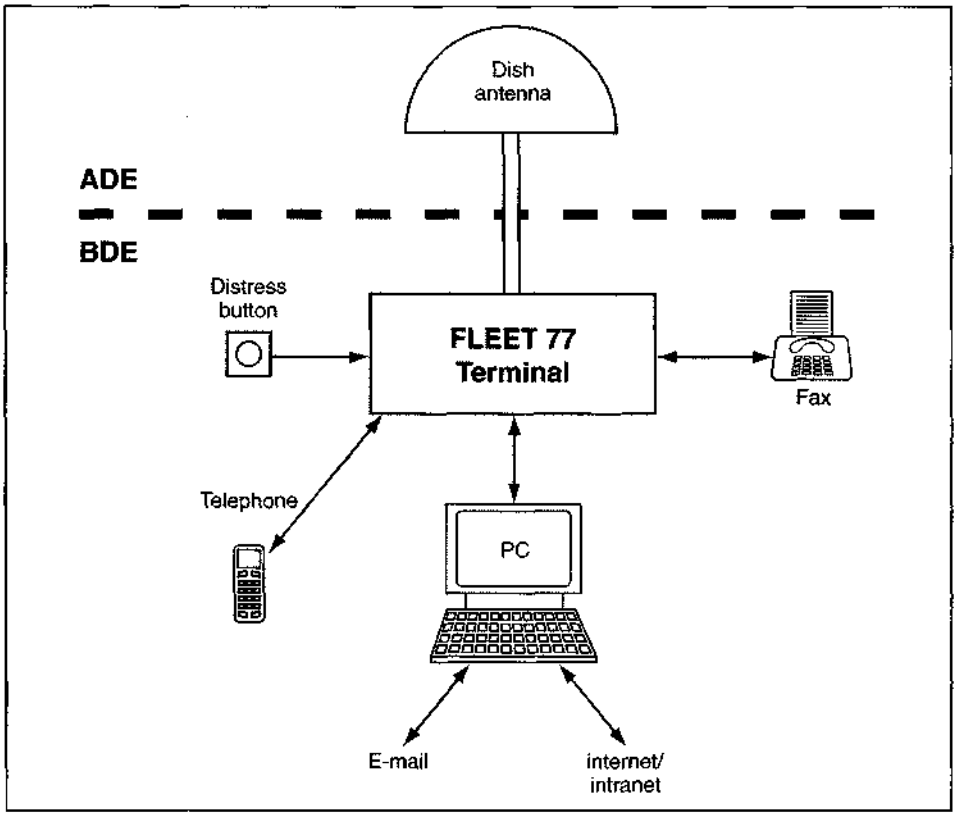


Fig. 5.3 Inmarsat Fleet F77 ship equipment layout.

5.10 FLEET 77 SERVICES

Fleet 77 services include:

(a) speech
(b) 64 kbit/s universal data
(c) 56 kbit/s universal data
(d) 3.1 kHz audio for connection to analogue fax services
(e) Mobile Packet Data Service (MPDS)
(f) mini-M voice service at 4.8 kbit/s

Each of the above services would have its own IMN.

5.11 INMARSAT FLEET F77 LAND EARTH STATION ACCESS CODES

LES Operator	*Country*	*Ocean Region*			
		AOR-E	*AOR-W*	*IOR*	*POR*
BeijingMCN	China	868	868		
France Telecom	France	011	011	011	011
KDDI	Japan	003	003	003	003

LES Operator	*Country*	*Ocean Region*			
		AOR-E	*AOR-W*	*IOR*	*POR*
Malaysia Telecom	Malaysia	060	060	060	060
O.T.E.	Greece	005	005	005	005
Singapore Telecom	Singapore	210	210	210	210
Stratos (Goonhilly)	UK	002	002	002	
Stratos (Auckland)	New Zealand	002			
Telecom Italia	Italy	555		555	
Telenor Services AS	Norway	004	004	004	004
Telenor Services Inc.	USA	001	001	001	001
Xantic	Netherlands	012	012	012	012
Xantic	Australia			022	022

5.12 INMARSAT FLEET F77 DISTRESS PROCEDURES

Four levels of priority are available in the Fleet 77 service:

Level	*Use*
0	Routine
1	Safety
2	Urgency
3	Distress

Level 3 has the highest priority and is reserved for distress. Pressing and holding the distress button for 5 seconds selects level 3. A level 3 call will pre-empt all other communications in both ship-to-shore and shore-to-ship directions hence guaranteeing a connection. A level 2 call will pre-empt safety and routine calls. A Level 1 call will pre-empt routine communications.

A Fleet 77 distress call is a quick automated method of making a voice connection with the operator at an MRCC. Once connected, the vessel position from GPS or other electronic positioning system and MES ID is automatically sent to the MRCC.

To initiate a distress priority call:

(a) Press and hold the Distress button (usually under a protective flap) for 5 seconds.

(b) Select nearest LES and press the # key. Some MES equipment will automatically select a default distress LES for each ocean region.

(c) When the MRCC operator responds give the following information:

MAYDAY
Ship's name and/or callsign
Position
Nature of distress
Assistance required
Other information

(d) The operator should be prepared to confirm own vessel's 9-digit Inmarsat IMN number and the ocean region satellite currently being used.

(e) Follow the instructions given by MRCC operator.

5.13 FLEET F55 SYSTEM AND SERVICES

The Fleet 55 system uses a smaller antenna than Fleet 77. It offers 64 kbit/s ISDN and MPDS services, but does not support the GMDSS component. F55 applications include:

(a) voice
(b) fax
(c) internet and LAN
(d) videoconferencing

Those LES's offering F55 services are listed in sections 5.15.

5.14 FLEET F33 SYSTEM AND SERVICES

Fleet 33 offers voice and data services at 9.6 kbit/s within the Inmarsat spotbeams shown on figure 5.1(b). F33 applications include:

(a) voice
(b) e-mail
(c) data transfer
(d) fax

Those LES's offering F33 services are listed in section 5.15.

5.15 INMARSAT FLEET F55/F33 LAND EARTH STATION ACCESS CODES

LES Operator	*Country*	*Ocean Region*			
		AOR-E	*AOR-W*	*IOR*	*POR*
France Telecom	France	011	011	011	
Stratos (Goonhilly)	UK	002	002	002	
Stratos (Auckland)	New Zealand				002
Telecom Italia	Italy	555		555	
Telenor Services AS	Norway	004	004	004	004
Telenor Services Inc.	USA	001	001	001	001
Xantic	Netherlands	012	012	012	012
Xantic	Australia			022	022

5.16 INMARSAT-B SYSTEM

Inmarsat-B commenced operation in 1993 and uses digital techniques exclusively. Those LES's offering Inmarsat-B services are listed in section 5.18. Two classes of Inmarsat-B ship earth stations are available:

Class 1: A MES providing the following services:

(a) Duplex telephony.
(b) Duplex 50-baud telegraphy.
(c) Reception of simplex ship-to-shore telephony (including both normal group calls and area group calls).
(d) Reception of simplex ship-to-shore 50-baud telegraphy (including both normal group calls and area group calls).
(e) Reception of Inmarsat service announcements.

Class 2: A MES providing the following services:

(a) Duplex telephony.
(b) Reception of simplex ship-to-shore telephony (including both normal group calls and area group calls).
(c) Reception of Inmarsat service announcements.

Note: Facsimile and data services are defined as being optional additions to the two basic classes.

5.17 INMARSAT-B SERVICES

(a) Immediate access for distress calls on telephony or telex.
(b) High quality telephony (duplex and simplex) at 16 kbits/s with transparency to voice-band data (including facsimile) up to 2.4 kbit/s.
(c) 50 baud telex, including shore-to-ship group calls.
(d) CCITT Group-3 Facsimile at 9.6 kbit/s over satellite links.
(e) Duplex data up to 9.6 kbit/s via PSTN and PSDN.
(f) Duplex data at 64/56 kbits/s (High speed data option).
(g) Group call services via voice, data or facsimile.
(h) Advice of call duration.
(i) Credit card charging.
(j) Enhanced privacy over voice and facsimile circuits.

Safety related features:

If an Inmarsat-B installation has telex facilities a mandatory Distress Message Generator (DMG) function is provided. See section 5.19.

5.18 INMARSAT-B and INMARSAT-M LAND EARTH STATION ACCESS CODES

LES Operator	*Country*	*Ocean Region*			
		AOR-E	*AOR-W*	*IOR*	*POR*
Beijing MCN	China	868	868	868	868
Bezeq	Israel	711	711	711	711
Thai Communications	Thailand			333	
Etisalat	U.A.E.			123	
France Telecom	France	011	011	011	011
FT MSC GmbH	France	111	111	111	111
KDDI	Japan	003	003	003	003
Korea Telecom	South Korea	006	006	006	006
Malaysia Telecom	Malaysia	060	060	060	060
Ministry of PTT	Algeria	777			
Morsviazsputnik	Russia	015	015	015	015
O.T.E.	Greece	005	005	005	005
Reach Networks Ltd.	China (Hong Kong)	118	118	118	118
Ministry of PTT	Saudi Arabia	025		025	
Singapore Telecom	Singapore	210	210	210	210
Stratos Mobile	Canada	013	013	013	013

LES Operator	*Country*	*Ocean Region*			
		AOR-E	*AOR-W*	*IOR*	*POR*
Stratos (Goonhilly)	UK	002	002	002	
Stratos (Auckland)	New Zealand				002
Telecom Italia	Italy	555	555	555	555
Telekom Polska	Poland	016		016	
Telenor Services AS	Norway	004	004	004	004
Telenor Services Inc.	USA	001	001	001	001
Vishipel	Vietnam			009	
VSNL	India	306	306	306	306
Xantic	Netherlands	012	012	012	012
Xantic	Australia	222	222	222	222

Network Coordination Stations for Inmarsat B and M are as follows:

AOR-E	Telenor Services Inc., USA
AOR-W	Telenor Services Inc., USA
IOR	OTE, Greece
POR	Telenor Services Inc., USA

5.19 INMARSAT-B DISTRESS MESSAGE GENERATOR (DMG) SYSTEM

The DMG conforms with the requirements of the IMO and consists of a pre-programmed telex distress message held in the MES memory and includes vessel ID, position and time of position update, course and speed. The DMG may be used as follows. The operator selects the identity of a LES which is held in the MES memory. By pressing and holding the "emergency" or "distress" button for six seconds a distress priority (level 3) call is activated via this LES and once the GA+ is received the stored message from the Distress Message Generator (DMG) can be sent.

5.20 INMARSAT-B DISTRESS TRANSMISSION PROCEDURE

(A) *Telephony:*

Refer to manufacturer's operating instructions and:

(a) Lift the telephone handset and wait for dial tone.
(b) Hold the "distress" button down for at least 6 seconds.
(c) Press the # key to initiate the call.
(d) When the MRCC operator answers give the following information:
MAYDAY
Ship's name and/or callsign
Position
Nature of distress
Assistance required
Other information
(e) Be prepared to give own 9-digit Inmarsat IMN number and the current ocean region satellite being used.
(f) Follow the instructions given by MRCC operator.
(g) Keep the MES clear so that the MRCC can call back when necessary.

(B) *Telex:*

Refer to manufacturer's operating instructions and:

(a) Put telex "on line".
(b) Hold the "distress" button down for at least 6 seconds.
(c) Wait for answerback and GA+ from MRCC.
(d) Then either:
  (i) Send distress message stored in DMG, or
  (ii) Type in the following information:
    MAYDAY (or SOS)
    Ship's name and/or callsign
    Position
    Nature of distress
    Assistance required
    Other information
(e) Be prepared to give own 9-digit Inmarsat IMN number and the ocean region satellite currently being used.
(f) Follow the instructions given by MRCC operator.
(g) Keep the MES clear so that the MRCC can call back when necessary.

5.21 INMARSAT-C SYSTEM

Inmarsat-C is a digital satellite communications messaging system and is an integral part of the GMDSS as it does not require a gyro input and operates easily from the 24v reserve power supply. The system does NOT handle voice communications. An Inmarsat-C terminal consists of a compact transceiver, a VDU, keyboard and printer. Many systems use a PC to handle keyboard and display operations. It is usual to interface navigational equipment such as GPS or Loran-C to an Inmarsat-C terminal and technology exists for polling of the ships position. Enhanced group call (EGC) equipment for receiving maritime safety information (MSI) is based on this system and is an integral part of all maritime Inmarsat-C equipment. (See section 4.18 for further details of EGC services.)

An Inmarsat-C MES has an omnidirectional antenna as opposed to a highly directional antenna as used in Inmarsat A, B, F and M installations.

Information can be passed to and from the ship via any one of the four ocean region satellites at a signalling rate of 600 bits per second to any Inmarsat-C equipped LES as listed in section 5.23. Messages are not transmitted in real-time, but instead are split up into data packages which are transmitted in blocks whenever the system has available capacity. Error correction codes are included in the messages and, if errors are detected, requests for retransmission are initiated and the system will repeat packages until the entire message is completely reassembled. This is called a store-and-forward system.

5.21.1 **Inmarsat mini-C**

Inmarsat mini-C was introduced in 2002 for smaller vessels such as fishing or leisure craft that require messaging, position reporting, tracking and secure communications via e-mail, telex, X-25, PSTN, fax (ship-to-shore direction only), short code messaging, data reporting and polling. It may be used for distress alerting

for non-SOLAS vessels. At present mini-C does not fully conform to the GMDSS and does not receive EGC messages. However, it may be used by merchant vessels for anti-piracy applications. Mini-C has a lightweight antenna unit containing the transceiver and GPS receiver and has a low power consumption assisted by a solar cell battery system if required.

5.22 INMARSAT-C SERVICES

Telex messages	Messages can be sent to and from any telex terminal connected to the national and international telex network.
Electronic Mail	E-mail can be sent to and from any computer terminal which is connected to the national and international PSDNs and PSTNs.
Ship-to-shore fax	Messages can be sent from your ship to a fax terminal connected to the national and international telephone networks. Fax messages can NOT be sent in the direction shore-to-ship.
Ship-to-ship	Data can be sent between Inmarsat-A/B/C/F and Inmarsat-C terminals on other ships within the four ocean regions.
Distress & safety	Distress priority messages can be sent and routed automatically to the nearest RCC which will take the necessary action to inform rescue services and other ships and aircraft in your vicinity.
EGC	All maritime Inmarsat-C MESs must have EGC receive facilities enabling the reception of broadcasts to selected ships, or areas. SafetyNETTM - Authorities can send MSI to ships within selected geographical areas. FleetNETTM — Information can be transmitted to a selected group of ships — rather like collective callsign messages. (See section 4.18 for further details of EGC.)

Note: It is mandatory for all marine Inmarsat-C LESs to offer the minimum services of store and forward telex, distress alerting, distress priority messages and EGC SafetyNET.

5.23 INMARSAT-C LAND EARTH STATION ACCESS CODES

Station Operator	*Country*	*Ocean Region*			
		AOR-E	*AOR-W*	*IOR*	*POR*
Beijing MCN	China			311	211
Bezeq	Israel	127		327	
Comms Authority	Thailand			319	
CP Radio Marconi	Portugal	118			
Embratel	Brazil	114			
France Telecom	France	121	021	321	221
FT MSC GmbH	France	115		333	
KDD	Japan	103	003	303	203
Korea Telecom	South Korea			308	

Station Operator	*Country*	*Ocean Region*			
		AOR-E	*AOR-W*	*IOR*	*POR*
Morsviazsputnik	Russia	117		317	
O.T.E.	Greece	120		305	
Polish Telecom	Poland	116		316	
Ministry of PTT	Saudi Arabia	125		325	
Singapore Telecom	Singapore			328	210
Stratos Mobile	Canada	102	002	302	202
Telecom Iran	Iran			314	
Telecom Italia	Italy	105		335	
Telenor Services AS	Norway	104	004	304	204
Telenor Services Inc.	USA	101	001		201
Vishipel	Vietnam			330	
VSNL	India			306	
Xantic	Netherlands	112	012	312	212
Xantic	Australia	122	022	322	222

Network Coordination Stations for Inmarsat C are as follows:

AOR-E	Goonhilly, UK
AOR-W	Goonhilly, UK
IOR	OTE, Greece
POR	Singapore

5.24 INMARSAT-C DISTRESS ALERTING

Note: When using the Inmarsat-C system for distress alerting ensure that the automatic scan facility is set to scan only the ocean region you are logged on to i.e. make it the preferred ocean region.

There are two methods of sending distress alerts in the Inmarsat-C system:

(a) By using the remote distress alert button(s)

(b) By keying the information into the MES terminal using the edit facilities.

(a) Using the remote distress button(s) on the MES

If distress alert buttons are fitted to the MES and there is insufficient time to use the MES keyboard an alert may be initiated in this manner. However, the information stored may be out of date unless recently updated positional information has been inserted, or it is continuously updated by an electronic navigator.

In the interests of safety of life at sea operators are advised to keep the MES logged on with up-to-date positional information in the store at all times.

Method:

1. Press the remote distress alert button(s) until an indication that the MES is in the distress alert mode. This condition should remain until an acknowledgement is received from the LES and then the nearest RCC.

2. If no acknowledgement is received from both the LES and RCC within five minutes repeat the distress alert.

3. If possible send updated information about your distress situation using the edit facilities indicated below.

(b) Using the edit facilities on the MES terminal

This method enables the operator to key in up-to-date information. However, due to a system definition change, this system will eventually be discontinued.

Method:

\. Select the distress alert menus according to the manufacturer's operating instructions.

2. When requested, complete following message:

MES *IMN*	Entered automatically by MES.
LES required	Select nearest LES within the ocean region to own ship's position.
Position	Latitude and longitude entered manually via keyboard, or automatically from electronic navigator such as GPS.
Date and time of last position update	Enter time manually or automatically from electronic navigator (all times to be quoted in UTC). Position status should read satisfactorily.
Nature of distress	One of the following: Unspecified Listing Fire/explosion Sinking Flooding Disabled and adrift Collision Abandoning ship Grounding Assistance required Piracy/armed robbery
Ship's Course (0°–359°)	Entered manually or automatically.
Ship's speed (Knots)	Entered manually or automatically.

3. Press the appropriate key(s) to send the alert.

4. If an acknowledgement is not received from both the LES and RCC within five minutes repeat the distress alert.

5.25 INMARSAT-C DISTRESS PRIORITY MESSAGE

A distress priority message should be typed into the MES using the edit facilities. The message will be routed automatically to the nearest RCC by the selected LES.

Method:

1. Ensure the MES is logged on and tuned to the common channel of the NCS appropriate to the ocean region in which the vessel is sailing.

2. Prepare the distress message using the word processing facilities of the terminal.

3. Include the information as listed in section 5.24 (b).

4. Give details of the kind of assistance required, although this may be implied by the nature of distress.

5. Select distress priority.
6. Key in the code for the nearest LES to own ship position. This LES will route the distress message to the nearest RCC.
7. Press the appropriate key(s) for sending the message.
8. Wait for an acknowledgement from the LES.

5.26 INMARSAT-C SPECIAL CODES FOR URGENCY AND SAFETY

Some Inmarsat-C LES's support 2-digit code services which can be used to access urgent and safety facilities ashore. Those available may include : -

Code 32	Medical advice
Code 38	Medical assistance
Code 39	Maritime assistance
Code 42	Navigational warnings

5.27 INMARSAT-C LOGGING ON

Follow the manufacturer's instructions on how to log on to the network coordinating station (NCS) in current ocean region. This enables messages to be sent and received at any time. As vessels move out of the service area of one ocean region they must log on to the ocean region they are entering. Vessels will only be able to communicate through LESs within the ocean region they are logged on to.

5.28 INMARSAT-C LOGGING OFF

N.B.: *When vessels wish to switch off an Inmarsat-C MES for any prolonged period of time it is important to log off first.*

This will stop the NCS in that region from sending further messages until the MES is logged on again. If this is not done, any LES with traffic for that vessel will continue to try and contact the MES. After repeated unsuccessful attempts the LES may reject the message and the ship may never receive it. Furthermore, certain LESs may charge the sender for use of satellite air time even though the message is not received.

It should be noted that certain Inmarsat-C terminals perform automatic logging functions and manual intervention may not be required by operators.

5.29INMARSAT-CCOMMUNICATIONS

Text or data messages can be sent from an Inmarsat-C MES to destinations via the telex, PSTN, PSDN and X.400 systems. Delivery to e-mail addresses is becoming increasingly popular. The terminal must be logged on to an ocean region before communications can take place. See section 5.27 for further details about logging on.

Message transmission procedure is as follows:

1. Create the message using the MES edit facilities.
2. Select "Transmit" or "Send" mode.
3. Select the message destination either from the address book or create a new address.
4. Select the LES through which you wish communications to take place (see section 5.23).

5. Select time for transmission (default is immediate).
6. Select routine priority.
7. Indicate whether confirmation of delivery is required - there is a charge for this service.
8. Enter the command to transmit the message.
9. Within a few minutes a message should be received to indicate successful transmission. This does not mean that the message has arrived at the final destination.
10. If confirmation of delivery has been requested, the LES normally issues a report within 4 to 6 minutes.
11. If the message is not delivered the LES will issue a non-delivery notification (see section 5.32).

5.30 INMARSAT-C MESSAGE ADDRESSES

Destination type	*Destination number required*
(a) Shore Telex	Telex country code followed by telex subscriber number.
(b) Ship-to-ship telex	Telex ocean region code followed by MES IMN.
(c) Ship-to-ship fax	Telephone ocean region code followed by MES IMN.
(d) Ship-to-ship data	DNIC followed by MES IMN.
(e) Ship-to-shore data	DNIC followed by X.25 data address.
(f) Shore fax	Telephone country code followed by fax number.
(g) Shore PSTN	Telephone country code followed by modem or mailbox number.
(h) X.400	Refer to X.400 service provider guide.
(i) 2-digit code	Special access codes as per section 5.31.

5.31 INMARSAT-C 2-DIGIT SERVICE CODES

Code	*Service accessed*
00	Automatically connected calls
31	Maritime enquiries
32	Request for medical advice
33	Request for technical assistance
37	Advice of time and charges for a call
38	Request for medical assistance
39	Request for maritime assistance
41	OBS messages
42	Sending weather danger and navigational reports to shore authorities
43	Position reports to shore authorities
6X	Special use such as leased lines

5.32 INMARSAT-C NON-DELIVERY NOTIFICATION CODES

Code	*Meaning*
ABS	Absent subscriber. The MES is not logged on to the ocean region.
ACB	Access barred
ADR	Addressee refuses to accept message
ANU	Deleted (if not delivered within one hour).

Code	*Meaning*
ATD	Attempting to deliver the message
BK	Message aborted. Used when a fax or PSTN connection is cleared abnormally.
BUS	Busy
CCD	Call cut or disconnected
CI	Conversation impossible
CIE	LES ran out of processing/communications capacity to process your message
CNS	Call not started
DTE	Data terminal equipment; used when an X.25 subscriber has cleared the connection during the call attempt
ERR	Error
FAU	Faulty
FMT	Format error
FSA	Fast select acceptance not subscribed
IAB	Invalid answerback from destination.
IAM	Unable to process the address information in the following message :
IDS	Invalid date from ship
IDT	Input data timeout
IFR	Invalid facility request
IMS	Message size invalid (7932 characters maximum)
IND	Incompatible destination
INH	Unable to establish the type of message from the following header :
INV	Invalid
ISR	Invalid ship request
LDE	Maximum acceptable message length or duration has been exceeded
LEF	Local equipment failure
LPE	Local procedure error
MBB	Message broken by higher priority
MCC	Message channel congestion
MCF	Message channel failure
MKO	Message killed by operator
MSO	Machine switched off
NA	Correspondence not permitted with this subscriber.
NAL	No address line present
NC	No circuits
NCH	Subscriber number changed
NDA	No delivery attempted
NFA	No final answerback
NIA	No initial answerback
NOB	Not obtainable
NOC	No connection
NP	No party. The party is not, or is no longer, a subscriber.
NTC	Network congestion
OAB	Operator aborted
OCC	Telex subscriber is occupied

Code	*Meaning*
OOO	Out of order
PAD	Packet assembler/dissembler
PRC	Premature clearing
PRF	Protocol failure
RCA	Reverse charging acceptance not subscribed
REF	Failure in the remote equipment
RLE	Resource limit exceeded
RPE	Remote procedure error
RPO	Out of order
SCC	Successfully completed call
SHE	MES hardware error
SNF	Satellite network failure
SPE	MES protocol error
SUC	Test results being delivered
TBY	Trunks busy
TGR	TDM group reset
TIM	Timeout
TMD	Too many destinations
UNK	Unknown - Used if no other failure code is suitable
WFA	Wrong final answerback
WIA	Wrong initial answerback

5.33 INMARSAT-M SYSTEM AND SERVICES

Inmarsat-M is a small scale digital communications system requiring a directional antenna. It is designed for small craft and does not comply with the requirements of the GMDSS, but can handle voice communications, low speed data and facsimile services as follows:

(a) Direct dial telephone via PSTN with good quality voice at 6.4 kbits/s (not transparent to voice band data).
(b) Distress communications which will attract priority assignment on voice channel.
(c) Direct dial group 3 facsimile at 2.4 kbits.
(d) Duplex data via PSTN, PSDN and closed user groups at 1.2 kbits or 2.4 kbits.
(e) Reception of group call services via voice, data or facsimile.
(f) Advice of call duration.
(g) Credit card charging.
(h) Enhanced privacy over voice and facsimile circuits.

Lower channel bit rate means much smaller sizes than those employed in Inmarsat-A, B and F installations. Antenna beamwidth is around 35 - 40 ° allowing low cost stabilisation systems for maritime and land-mobile applications.

5.34 INMARSAT-M LAND EARTH STATION ACCESS CODES

Inmarsat-M LESs are the same as those listed under Inmarsat-B in section 5.18. Mini-M LESs are listed in section 5.37.

5.35 INMARSAT-M DISTRESS ALERTING

Distress, urgency and safety procedures with Inmarsat-M telephony equipment are identical to those for Inmarsat-B MES's as detailed in section 5.20 and 5.44. However, there is no telex facility within the Inmarsat-M system.

5.36 INMARSAT MINI-M SYSTEM AND SERVICES

Mini-M, also known as Inmarsat-phone, is a portable version of Inmarsat-M being the size of a laptop computer. It uses the spot-beam power of the Inmarsat satellites to illuminate particular high traffic areas within the satellite footprint. Depending upon the user's equipment the following services are available via mini-M:

(a) VOICE — Direct dial phone calls, with voicemail.
(b) RECEIVE — Broadcast services such as financial and currency listings. Two-way personally addressed paging alerts and short messages. Access to e-mail and fax systems.
(c) INTERNET - Internet based e-mail, the Internet and World Wide Web.
(d) LAN — Exchanging information with databases or local area networks.
(e) VIDEO — From video conferencing to news quality video transmissions.
(f) PICTURES — Exchanging digitised pictures and diagrams.

5.37 INMARSAT MINI-M LAND EARTH STATION ACCESS CODES

LES Operator	*Country*	*Ocean Region*				
		AOR-E	*AOR-W*	*IOR*	*POR*	*IND-W*
Beijing MCN	China	868	868	868	868	
Bezeq	Israel	711	711	711	711	
France Telecom	France	011	011	011	011/402	011
FT MSC GmbH	France	111	111	111	111/403	
KDDI	Japan	003	003	003	003	
Korea Telecom	South Korea	006	006	006	006	
Malaysia Telecom	Malaysia	060	060	060	060	
Ministry of PTT	Algeria	777				
Morsviazsputnik	Russia	015	015	015	015	
O.T.E.	Greece	005	005	005	005	
Ministry of PTT	Saudi Arabia	025		025		
Singapore Telecom	Singapore	210	210	210	210	
Stratos Mobile	Canada	013	013	013	013	
Stratos (Goonhilly)	UK	002	002	002		002
Stratos (Auckland)	New Zealand				002	
Telecom Italia	Italy	555	555	555	555	
Telekom Polska	Poland	016		016		
Telenor Services AS	Norway	004	004	004	004	004
Telenor Services Inc.	USA	001	001	001	001	
VISHIPEL	Vietnam			009		
VSNL	India	306	306	306	306	
Xantic	Netherlands	012	012	012	012	012
Xantic	Australia	222	222	222/022	222/022	

5.38 INMARSAT-A SYSTEM

The Inmarsat-A system was the first maritime satellite communications system to be introduced in 1979 and is scheduled to close on 31st December 2007. It uses mainly analogue techniques to provide communications between suitably equipped ships and subscribers ashore having access to their national and international public switched telephone network (PSTN) and public switched data network (PSDN). A MES is also able to communicate with other suitably equipped MESs in any of the ocean regions. The MES requires above deck equipment (ADE) comprising a highly directional parabolic dish antenna and below deck equipment (BDE) which comprises the transceiver, telephone, telex and perhaps facsimile and data equipments. See Fig. 5.2 on page 127.

5.39 INMARSAT-A SERVICES

Inmarsat-A provides the following services:

(a) distress communications.
(b) telephone calls.
(c) telex calls.
(d) facsimile services.
(e) data communications at medium and high speeds.

Other possible services may include:

(f) maritime assistance.
(g) medical assistance.
(h) technical assistance.
(i) advice of time and duration for calls.
(j) person-to-person, collect or credit card calls.
(k) packet switching.
(l) national and international operator assistance.
(m) commissioning tests.

All of the above can be accessed by keying in the appropriate two-digit code on the telephone or telex equipments, details of these services are given in sections 5.45 and 5.55.

Those LES's offering Inmarsat-A services are listed in section 5.40.

5.40 INMARSAT-A LAND EARTH STATION ACCESS CODES

ATLANTIC OCEAN REGION - EAST (AOR-E) NCS Telenor, USA

LES Operator	*Country*	*LES Code*	
		Octal	*Decimal*
Beijing Marine	China	13-7	11-7
Embratel	Brazil	14	12
France Telecom	France	17	15
KDD	Japan	03	03
Korea Telecom	South Korea	06	06
Malaysia Telecom	Malaysia	13-2	11-2
OTE	Greece	15	13

ATLANTIC OCEAN REGION - EAST (AOR-E)—*continued*

LES Operator	*Country*	*LES Code*	
		Octal	*Decimal*
Polish Telecom	Poland	16	14
Reach Networks Ltd.	China (Hong Kong)	13-6	11-6
Singapore Telecom	Singapore	13-5	11-5
Stratos Mobile	Canada	13-1	11-1
Stratos (Goonhilly)	UK	02	02
Telecom Italia	Italy	05	05
Telenor Services AS	Norway	04	04
Telenor Services Inc.	USA	01	01
Turk Telecom	Turkey	10	08
VSNL	India	13-4	11-4
Xantic	Netherlands	12	10
Xantic	Australia	13-3	11-3

ATLANTIC OCEAN REGION - WEST (AOR-W) NCS Telenor, USA

LES Operator	*Country*	*LES Code*	
		Octal	*Decimal*
Beijing Marine	China	13-7	11-7
France Telecom	France	17	15
KDD	Japan	03	03
Korea Telecom	South Korea	13-5	11-5
Malaysia Telecom	Malaysia	13-2	11-2
OTE	Greece	07	07
Reach Networks Ltd.	China (Hong Kong)	13-6	11-6
Singapore Telecom	Singapore	10	08
Stratos (Goonhilly)	UK	02	02
Telecom Italia	Italy	05	05
Telenor Services AS	Norway	04	04
Telenor Services Inc.	USA	01	01
VSNL	India	06	06
Xantic	Netherlands	12	10
Xantic	Australia	13-3	11-3

INDIAN OCEAN REGION (IOR) NCS KDD, Japan

LES Operator	*Country*	*LES Code*	
		Octal	*Decimal*
Beijing Marine	China	11	09
France Telecom	France	17	15
KDD	Japan	03	03
Korea Telecom	South Korea	13-2	11-2
Malaysia Telecom	Malaysia	13-3	11-3
OTE	Greece	05	05
Polish Telecom	Poland	16	14
Reach Networks Ltd.	China (Hong Kong)	13-6	11-6

INDIAN OCEAN REGION (IOR) *ontinued*

LES Operator	*Country*	*LES Code*	
		Octal	*Decimal*
Singapore Telecom	Singapore	13-5	11-5
Stratos Mobile	Canada	13-1	11-1
Telecom Iran	Iran	14	12
Telecom Italia	Italy	13-4	11-4
Telenor Services	USA	01	01
Telenor Services	Norway	04	04
Turk Telecom	Turkey	10	08
VSNL	India	06	06
Xantic	Netherlands	12	10
Xantic	Australia	02	02

PACIFIC OCEAN REGION (POR) NCS KDD, Japan

LES Operator	*Country*	*LES Code*	
		Octal	*Decimal*
Beijing Marine	China	11	09
Far East Shipping	Russia	15	13
France Telecom	France	17	15
KDD	Japan	03	03
Korea Telecom	South Korea	04	04
Malaysia Telecom	Malaysia	13-3	11-3
OTE	Greece	07	07
Reach Networks Ltd.	China (Hong Kong)	13-6	11-6
Singapore Telecom	Singapore	10	08
Stratos Mobile	Canada	13-1	11-1
Stratos (Goonhilly)	Canada	05	05
Telecom Italia	Italy	13-4	11-4
Telenor Services Inc.	USA	01	01
VSNL	India	06	06
Xantic	Netherlands	12	10
Xantic	Australia	02	02

5.41 INMARSAT-A DISTRESS ALERTING

MES's have to be able to initiate a request burst having distress priority which is instantly assigned a free channel. This is automatically recognised by a LES, even though all satellite channels may be occupied at that time. A distress priority (level 3) message is initiated in some MES equipment by using an "Emergency" or "Distress" button. The MES operator then indicates an appropriate LES (usually the nearest). The call is normally routed automatically to the MRCC associated with the selected LES without having to enter further details. For example priority 3 calls made via Goonhilly LES are automatically routed to MRCC Falmouth. Many MES's have standard format Distress Message Generator (DMG) software which simplifies the transmission of distress details once connection has been established with the LES.

5.42 INMARSAT-A DISTRESS TRANSMISSION PROCEDURE

Refer to the equipment manufacturer's instructions and:

(a) Select telex or telephone operation.
(b) Select distress priority.
(c) Select access code for nearest LES (see section 5.40).
(d) Initiate the call according to manufacturer's instructions.
(e) If no response is received within 15 seconds repeat the distress alert.
(f) When contact is made with the MRCC operator send the distress message as follows:

Telephony	*Telex*
MAYDAY	SOS or MAYDAY
Vessel's name	Vessel's callsign
Position	Position
Nature of distress	Nature of distress
Assistance required	Assistance required
Additional information	Additional information

(g) Do NOT clear the connection.
(h) Await instructions from the MRCC operator and be prepared to provide details of own Inmarsat 7-digit mobile number (IMN) and the ocean region satellite in use.
(i) Expect the MRCC to call back so keep the line clear for incoming calls.

5.43 INMARSAT-E

This is an L-band satellite EPIRB system which offers instant distress alerting via any one of the four Inmarsat ocean region satellites. It can be used instead of COSPAS/SARSAT satellite EPIRB for vessels trading sea areas A1, A2 or A3. Full details of EPIRBs available within the GMDSS are given in chapter 4. *It should be noted that Inmarsat have announced that the Inmarsat-E system will be discontinued on 1st December 2006.*

5.44 INMARSAT URGENCY AND SAFETY PROCEDURES

Medical advice, medical assistance and urgent maritime assistance can be obtained via most LES's by using 2-digit codes mentioned in sections 5.45 and 5.55. MES operators should check the services provided by an LES before proceeding with the use of 2-digit codes listed below. Further details on obtaining medical services are available in sections 2.30 to 2.35.

Refer to the equipment manufacturer's instructions and:

	Telephone	*Telex*
(a)	Select telephone mode	Select telex mode
(b)	Select routine priority (level 0)	Select routine priority (level 0)
(c)	Select the nearest LES	Select the nearest LES
(d)	Initiate the call	Initiate the call
(e)	When PTS tone received select:	When GA+ received select:
For medical advice	32#	32+
Medical assistance	38#	38+
Urgent assistance	39#	39+

Notes:

(1) When making such calls via Inmarsat-B/M/F MES's once the LES is selected (or accept default) operators should merely dial the two-digit code followed by # or +.

(2) Code 32 usually connects the vessel directly to a local hospital who will give appropriate advice. Use the word MEDICO and provide the name of ship/callsign and IMN, position and condition of patient.

(3) Code 38 usually connects the vessel with nearest MRCC to selected LES and should be used when medical evacuation of a patient is required. Be prepared to provide the name of ship/callsign and IMN, position and condition of patient.

(4) Code 39 usually connects the vessel with nearest MRCC to selected LES and should be used only when immediate assistance is required concerning the safety of person(s) or the vessel, e.g. man overboard, steering gear failure, not under command, or oil pollution. Be prepared to provide the name of ship/callsign and IMN, position of the ship and details of the incident.

(5) There is no telex facility in the Inmarsat-M and Fleet services.

5.45 INMARSAT TELEPHONY 2-DIGIT CODES

An international telephone service from suitably equipped ships is available via Inmarsat-A, Inmarsat-B, Inmarsat-F and Inmarsat-M (including mini-M). The MES operator can obtain fully automated telephone facilities by keying in the appropriate two-digit code.

Code	*Service*	*Notes*
00	Automatic dialling	Use this code to make automatic telephone, facsimile and voice band data calls using International Direct Dial (IDD) codes.
11	International operator	Use this code to obtain information from the international operator of the country where the LES is situated.
12	International information	Use this code to obtain information about subscribers located in countries other than that in which the LES is situated.
13	National Operator	Use this code to obtain assistance to connect to telephone subscribers within the country in which the LES is situated. In some countries which do not have an international operator, use this code instead of code 11.
14	National information	Use this code to obtain information about subscribers located in the country in which the LES is situated.
17	Telephone bookings	This code may be used via some LESs to book telephone calls, although normally this code is used via the telex service.
20	Access to a Maritime PAD	This code is used when using a voice band data modem to access a Maritime Packet Assembly/ Disassembly (PAD) facility in the packet switched public data network. The PAD is accessed via telephone circuits and the prefix 20 should be followed by two additional digits indicating the required data rate.

Code	*Service*	*Notes*
23	Abbreviated dialling	This code is to be used by some LESs to allow (short code selection) Inmarsat-A equipped subscribers to utilise abbreviated dialling codes for their regularly dialled numbers.
28	Internet access	This code is used by some LESs to allow access to the internet. Terminals must be registered with the LES before using this service.
31	Maritime enquiries	This code may be used for special enquiries e.g. ship location, authorisation etc.
32	Medical advice	Use this code to obtain medical advice. Some LESs have direct connections to local hospitals when this code is used.
33	Technical assistance	Use this code if you are having technical problems with your Inmarsat terminal. Technical staff at LESs would normally be able to give assistance.
34	Person to person call	Use this code to contact the operator for a person to person call
35	Collect calls	Use this code to contact the operator for a collect call (Charges payable by the recipient of the call).
36	Credit card call	Use this code to charge a telephone call to a credit card or charge card.
37	Time and duration	This code should be used at the start of a call instead of the code 00 for an auto telephone call. This service will enable the time and duration for the call being set up to be advised to the Inmarsat terminal operator. This will be either by a call back from the LES operator on the telephone or a short telex message at the end of the connection containing the duration and time relating to the call. Code 37 can not work with a second IMN on Inmarsat-A, Inmarsat-F or an Inmarsat-M MES as there is no associated telex line.
38	**Medical assistance**	**This code should be used if the condition of an ill or injured person on board the vessel requires urgent evacuation ashore or the services of a doctor aboard the vessel. This code will ensure that the call is routed to the appropriate agency/ authority ashore to deal with the situation.**
39	**Maritime assistance**	**This code should be used to obtain maritime assistance if the vessel requires help such as a tow, oil pollution etc.**
41	Meteorological reports	This code should be used by weather observing vessels to send their weather observations (OBS). In most cases where this service is available the service is free of charge to the vessel, the national weather authority paying the relevant charges.
42	**Navigational hazards and warnings**	**This code provides a connection to a navigational office for transmission of information from the vessel on any hazards which could endanger safety of navigation (e.g. wrecks, derelicts, floating obstructions, defective radiobeacons or light vessels, icebergs, floating mines, etc.).**

Code	*Service*	*Notes*
43	Ship position reports	This code provides a connection to an appropriate national or international centre collecting ship movement information for search and rescue purposes, e.g. AMVER, AUSREP etc.
6(x)	Admin. Specialised use	For use by administrations for specialised use. Often used for leased lines etc. The 'x' digit is allocated on a national basis and would not normally be used for the same service/leased line for more than one LES.
70	Databases	This code will normally be used by the LES to allow automatic access to its information retrieval database, if available.
91	Automatic line test	This code should be used to obtain a test levels and tones when setting up a modem or voice band data equipment.
92	Commissioning tests	This code is used when a vessel is ready to commence its Inmarsat-A commissioning tests. The code should only be used for this purpose and then only used via the LES through which the commissioning has been arranged.

5.46 TELEPHONE CALLS VIA INMARSAT

Establishing a telephone call

Note: These instructions are of a general nature and are for guidance only. Operators should refer to manufacturers operating instructions for specific details on making telephone calls from your particular MES.

Making a telephone call is divided into two separate stages:

1. Establishing a link between your MES via a satellite to your chosen LES

(a) Select telephone mode (normally by simply lifting telephone handset).

(b) Select routine priority and channel type 01, (normally available by default). Channel type 02 is an uncompanded channel and should always be used when transmitting some type of voice band data or fax).

(c) Select the appropriate LES access code in your ocean region through which the call is to be routed.

(d) Initiate the telephone channel request burst in accordance with the manufacturers instructions for your MES.

(e) Within approximately 12 seconds the MES should have made contact with the LES and the LES header should appear on the printer and/or VDU followed by GA+, the proceed to select (PTS) tone should be heard indicating successful connection to the LES.

Note: If no indication is received within 12 seconds the request burst should be repeated.

2. Establishing the link between the LES and the telephone subscriber

(a) When GA+ and the PTS tone is received from the LES key in the appropriate 2-digit code for the telephone service you require (see section 5.45).

(b) Key in the access code for the country of destination (see Appendix 5).

(c) Key in the number of the telephone subscriber required followed by #

Example: *00441512521111#*

where 00	is a request for an automatic telephone call
44	is the telephone country code (in this case, the UK)
151	is the area code minus the 1st "0" (in this case, Liverpool)
252 1111	is the telephone subscriber's number
#	is the end of calling sequence

(d) Within approximately 15 seconds the ringing tone of the called subscriber should be heard. When the subscriber answers this means that the telephone link has been successfully established and the charging period begins. The charging period will end from the time the link to the subscriber is disconnected.

Notes: The end of number selection symbol # must always be used to signify the end of the calling sequence.

(e) If another call is required key in the code for the chosen LES and thereafter repeat the process.

(f) When no further traffic is on hand place the telephone "off-line". This breaks the connection with the satellite.

5.47 TELEPHONE CALLS TO ANOTHER MES

(a) Follow the instructions above to select a LES and obtain a channel.

(b) On hearing the tone, select 00 for automatic connection.

(c) Key in the code for the satellite region in which the selected MES is sailing (see section 5.48).

(d) Key in the seven or nine digit IMN for the required MES, followed by #
e.g. 00872323212345#
for a call to an Inmarsat-B LES in the Pacific Ocean Region.

(e) Replace the handset at the end of the call.

5.48 INMARSAT TELEPHONE OCEAN REGION CODES

Atlantic Ocean Region - East	871
Pacific Ocean Region	872
Indian Ocean Region	873
Atlantic Ocean Region - West	874
Inmarsat Ocean Region Code	870

The 870 ocean region code automatically routes the caller regardless of the ocean region that the called vessel may be in. However, if the Inmarsat IMN begins with a '1' or '8' the appropriate individual ocean region code must be used.

5.49 TELEPHONE CALLS TO SHIPS FITTED WITH VHF/MF/HF RT EQUIPMENT

Telephone calls from a ship with a MES to ships fitted with VHF, MF or HF radiotelephones may be booked using the telex two-digit code 31#.

5.50 CHARGES FOR INMARSAT TELEPHONE CALLS

(a) Automatically connected calls

Automatically connected calls via the Inmarsat system are usually charged on the basis of six second minimum charge with six second incremental steps.

For example: an auto connected call of 45 seconds is charged as for 48 seconds.
an auto connected call of 2 minutes 33 seconds is charged as for 2 minutes 36 seconds.

However, it should be noted that some LES operators may employ 1 second increments or other timing systems.

(b) Operator connected calls

Operator connected calls via the Inmarsat system are charged on the basis of three minutes minimum charge with one minute incremental steps.

For example: an operator connected call of 45 seconds is charged as for three minutes.
an operator connected call of 3 minutes 30 seconds is charged as for four minutes.

5.51 DATA COMMUNICATIONS VIA INMARSAT

The MES needs to be connected to voice band data or facsimile equipment. A normal telephony channel is used for the communications link between the MES and the distant data or facsimile terminal. Inmarsat also offers High Speed Data (HSD) and Duplex High Speed Data (DHSD) services. Use of data communications gives access to a wide range of information databases such as weather bulletins, marine chart corrections, technical maintenance information etc. These are usually only available on a subscription basis. Remote monitoring of ships position, course, speed, fuel consumption, cargo condition, engine room parameters etc. is also possible.

5.52 DATA NETWORK IDENTIFICATION CODES (DNICs)

Atlantic Ocean Region - East	1111
Pacific Ocean Region	1112
Indian Ocean Region	1113
Atlantic Ocean Region — West	1114

5.53 FACSIMILE VIA INMARSAT

Suitably equipped ships can send facsimile messages via the national and international telephone networks and to other suitably equipped MESs. The procedures for establishing a facsimile call are the same as those for a telephone call as detailed in section 5.46. However, best results will be achieved if an uncompanded (type 02) channel is selected. When tones from the remote facsimile equipment are received press the 'start' button to commence transmission. If someone answers the telephone instead of hearing tones operators should inform the

person to transfer to their facsimile equipment. When the tones are received press the 'start' button.

It is important not to start the facsimile transmission until the tones from the distant equipment are heard — otherwise the circuit may fail.

5.54 TELEX VIA INMARSAT

International telex service to and from suitably equipped ships is available via Inmarsat-A, Inmarsat-B and Inmarsat-C systems. The MES operator can obtain fully automated telex facilities by keying in the appropriate two-digit code.

5.55 INMARSAT TELEX 2-DIGIT CODES

Some or all of these telex services may be offered by Land Earth Stations operating in the Inmarsat-A, B and C networks.

Code	*Service*	*Notes*
00	Automatic dialling	Use this code to make automatic telex calls using international telex country codes.
11	International operator	Use this code to obtain information from the international operator of the country within which the LES is situated.
12	International information	Use this code to obtain information about subscribers located in countries other than that in which the LES is situated.
13	National operator	Use this code to obtain assistance to connect to telex subscribers within the country in which the LES is situated. In some countries which do not have an international operator, use this code instead of code 11.
14	National information	Use this code to obtain information about subscribers located in the country in which the LES is situated.
15	Radiotelegram service	This code will connect the caller to the radiotelegram service position for the transmission of radiotelegrams originated via telex.
17	Telephone bookings	This code may be used via some LESs to book telephone calls.
21	Store-and-Forward	This code is used to gain access to a Store and Forward (international) Unit (SFU) for international calls.
22	Store-and-Forward	This code is used to gain access to a Store and Forward (national) Unit (SFU) for national calls.
24	Telex letter service	This code is used for directly transmitting a message originated from a MES to a selected telegraph office for delivery by mail or appropriate means (Inmarsat-C only).

31	Maritime enquiries	This code may be used for special enquiries e.g. ship location, authorisation etc.
32	Medical advice	Use this code to obtain medical advice. Some LESs have direct connections to local hospitals when this code is used.
33	Technical assistance	Use this code if you are having technical problems with your Inmarsat-A terminal. Technical staff at LESs would normally be able to give assistance.
36	Credit card call	Use this code to charge a telex call to a credit card or charge card.
37	Time and duration	This code should be used at the start of a call instead of the code 00 for an auto telex call. This service will enable the time and duration for the call being set up to be advised to the Inmarsat operator. This will be normally a short telex message at the end of the connection containing the duration and time relating to the call. It is more normal to terminate a telex call by using five full stops when the time and duration for the call will be automatically returned.
38	Medical assistance	This code should be used if the condition of an ill or injured person on board the vessel requires urgent evacuation ashore or the services of a doctor aboard the vessel. This code will ensure that the call is routed to the appropriate agency/authority ashore to deal with the situation.
39	Maritime assistance	This code should be used to obtain maritime assistance if the vessel requires help such as a tow, oil pollution etc.
41	Meteorological Reports	This code should be used by weather observing vessels to send their weather observations (OBS). In most cases where this service is available the service is free of charge to the vessel, the national weather authority paying the relevant charges.
42	Navigational hazards	This code provides a connection to a and warnings navigational office for transmission of information from the vessel on any hazards which could endanger safety of navigation (e.g. wrecks, derelicts, floating obstructions, defective radiobeacons or light vessels, icebergs, floating mines etc.).
43	Ship position reports	This code provides a connection to an appropriate national or international centre collecting ship movement information for search and rescue purposes e.g. AMVER, AUSREP, etc.

51	Meteorological forecasts	This code is used for retrieving meteorological forecasts.
52	Navigational warnings	This code is used for retrieving navigational warnings.
6(x)	Admin. Specialised use	For use by administrations for specialised use. Often used for leased lines etc. The 'x' digit is allocated on a national basis and would not normally be used for the same service/leased line for more than one LES.
70	Databases	This code will normally be used by the LES to allow automatic access to its information retrieval database, if available.
91	Automatic telex test	This code should be used to obtain a telex receiver check. The LES normally transmits as follows: THE QUICK BROWN FOX JUMPS OVER THE LAZY DOG 1234567890
92	Commissioning tests	This code is used when a vessel is ready to commence its Inmarsat-A commissioning tests. The code should only be used for this purpose and then only via the LES through which the commissioning has been arranged.

5.56 TELEX ANSWERBACKS

A MES telex answerback consists of the vessels IMN followed by four letters (usually its international call sign) and then the letter "X" to indicate a maritime mobile station.

e.g. 1234567 ABCD X

Each MES answerback is unique to a particular ship and it serves as a positive identification when an external telex terminal calls the ship up. The answerback is automatically sent by the ship when it receives a "Who are you?" (WRU) request.

5.57 TELEX CALLS VIA INMARSAT-A AND INMARSAT-B

Before operators make any telex call they should ensure that their MES is properly set up for the required ocean region.

If possible, prepare the message in advance by typing into the memory with the telex in "local" mode which allows editing of the message prior to transmission.

The telex message format should generally include the following information:

(a) The destination company and/or name of addressee.
(b) The message originator's name and/or title.
(c) A message reference number and/or subject of the call.
(d) The ocean region satellite through which your vessel can be contacted if a reply is required.
(e) The text of the message.

Establishing a telex link

Note: These instructions are of a general nature and are for guidance only. Operators should refer to manufacturer's operating instructions for specific details on making telex calls from your particular MES.

Making a telex call is divided into two separate stages:

1. Establishing a link between the MES via a satellite to a chosen LES

(a) Select telex mode.
(b) Select routine priority (normally available by default).
(c) Select a suitable LES in vessel ocean region through which the call is to be routed.
(d) Initiate the telex channel request burst in accordance with the manufacturer's instructions for the MES.
(e) Within approximately 10 seconds the MES should have made contact with the LES and the LES header should appear on the printer and/or VDU followed by GA+ indicating successful connection to the LES.

Note: If an indication is not received within 10 seconds retransmit the request burst.

2. Establishing the telex link between the LES and the telex subscriber

(a) When GA+ is received from the LES key in the appropriate 2-digit code for the telex service required (see section 5.55).
(b) Key in the access code for the country of destination (see Appendix 6).
(c) Key in the telex number of the telex subscriber to be called followed by +

Example:	*005112345+*
where 00	is a request for an automatic telex call
51	is the telex country code (in this case, the UK)
12345	is the telex subscriber number
+	is the end of calling sequence

(d) Within approximately 15 seconds the answerback of the telex subscriber should be received. This indicates that the telex link has been successfully established.

Notes: The end of number selection symbol + must always be used to signify the end of the calling sequence. If the call is unsuccessful a fault code may be indicated. See sections 5.32 and 5.65 for the meaning of such codes.

(f) When the call is finished key five full stops (.....) at the beginning of a new line.
(g) A date and time group followed by the duration of the call will be received
e.g. 01/JUN/05 15:30 3.5 MINS (or 210 seconds)
(h) If another call is required key in the code for the chosen LES and thereafter repeat the process.
(i) When no further traffic is on hand place the telex "off-line". This breaks the connection with the satellite.

5.58 TELEX DIALLING

The required telex number should be keyed without long pauses between digits. It may take up to 30 seconds for the call to be connected and the subscriber's answerback to be received. If the answerback is not received, try to obtain the answerback by sending the "who are you?" request. If the answerback is still not received it can be presumed that the call has failed. If there is no connection but a service code received, take the following action:

Code received:	*Action:*
NC or OCC	Clear call , wait at least one minute then try again.
NA, NCH or NP	Check telex number before trying again.
ABS, CI, DER, INF or JFE	Contact directory enquiries.

Automatically dialled services:

(i) *Urgent assistance required*
Select priority 0
Key 39+

(ii) *Medical assistance required*
Select priority 0
Key 38+

(iii) *Example of routine auto-dialled call to UK*
Select priority 0
Key 005154321+

(iv) *Example of call to another ship in POR*
Select priority 0
Key 005827654321+

5.59 FORMAT OF RADIOTELEGRAMS, AMVER AND MEDICAL MESSAGES

Telegrams to shore addresses can be made from a MES. It should be noted that the telegram service in the UK has been replaced by the "Telemessage" which arrives by first class post.

Note: LS = letter shift; LF = line feed; CR = carriage return; e.g. 10 x LS means 10 letter shifts

(a) Contact the selected LES via the satellite within your ocean region as previously described.
(b) Select a duplex channel.
(c) Select routine priority.
(d) Receive the LES answerback followed by the "who are you?" symbol.
(e) The MES will automatically reply with your ID number and answerback.
(f) GA+ will be received from the LES.
(g) Key in the two-digit code for the telegram service (15+).
(h) Send 12 x LS followed by 10 x LF.
(i) Receive answerback.
(j) Send 1 x LS, then send MES answerback code.
(k) Send 1 x CR, then 3 x LF.

(l) Send ship's name and callsign, MES ID, date, time and AAIC.
(m) Send service instructions (if any).
(n) 1 x CR, then 3 x LF.
(o) Send paid service instructions (if any).
(p) Send 1 x CR, then 1 x LF.
(q) Send address as follows:
1st line: name of addressee, then 1 x CR, then 1 x LF
2nd line: house name or number, street, then 1 x CR, 1 x LF
3rd line: town of destination as per telegraphic address list.
(r) Send 1 x CR, then 3 x LF.
(s) Send telegram text placing 1 x CR and 1 x LF at end of each line.
(t) Send signature then 1 x CR, then 3 x LF.
(u) Send collation (if required) followed by 1 x CR, then 3 x LF.
(v) Request answerback by sending "who are you?" symbol, then 1 x LS.
(w) Send own MES answerback, then 1 x CR and 10 x LF.

Note: The AAIC is NOT required for medical advice messages as these will not normally be charged to the ship. However, if the facilities accessed are abused the public authority responsible for meeting the charges reserve the right to reject the bill which will then become payable by the ship. Codes 32, 38 and 39 are only to be used for urgency and safety.

Since the introduction of GMDSS certain authorities now charge for onward transmission of AMVER messages to New York.

5.60 TELEX AND GREETINGS TELEX LETTERS

Some administrations offer telex letter and telex greetings card services. Messages must have full postal address (with postcode) and are sent by first class mail. A range of greetings card designs may be available to suit appropriate occasions.

(a) Follow instructions for contacting LES and exchanging answerbacks.
(b) When GA+ is received immediately key in either:
24+ for telex letter service or
25+ for greetings card service.
(c) LES answerback will be received.
(d) Send answerback followed by ship's name, ID, date and AAIC, then 1 x CR and 6 x LF.
(e) Send either:
RTL for telex letter or
GTLXX for greetings card (where XX refers to the type of card required).
(f) Send 1 x CR, the 1 x LF followed by the address as follows : -
1st line: Name of addressee, then 1 x CR, 1 x LF
2nd line: House number or name, 1 x CR, 1 x LF
3rd line: Town of destination plus postcode, 1 x CR, 1 x LF
4th line: Country of destination, then 1 x CR, then 6 x LF.
(g) Send text and signature of message.
(h) Send 1 x CR, the 1 x LF to receive answerback.
(i) Send own answerback.

Notes: (1) There is no need to separate each line with 1 x CR and 1 x LF
(2) If the address and text exceeds 50 words, it should be marked off every 50 words as follows:

Send (50), 1 x CR, then 3 x LF

5.61 TERMINATING INMARSAT TELEX CALLS

Five full stops clears the call by disconnecting the link between the LES and the telex subscriber and also the link between the satellite and the LES. The date, time and duration of the call to the nearest decimal point will be received in the following form:

01/MAY/05 13:15 2.5 MINS (or 150 seconds)

Telex follow-on calls may be obtained with some LESs by sending five commas. This breaks the circuit with the previous subscriber but retains the connection with the LES. A date and time group will be issued followed by advice of duration of last call. GA+ will be received after which details of the next call or service required should be entered.

5.62 INMARSAT TELEX CALLS TO ANOTHER MES

Follow the same process for making connection with the desired LES. After the LES header followed by GA+ is received:

(a) Key in 00 for automatic connection.
(b) Key in the satellite region code in which the ship being called is situated (see section 5.63 for details).
(c) Key in the IMN for the receiving ship followed by + sign

e.g. 005831234567+ to contact a ship in the Indian Ocean.

(d) The answerback of the called ship is received if connection is successful.
(e) Send your telex message.
(f) Clear by sending five full stops (.....) at the beginning of a new line, which will give date and time group followed by duration of the call.

5.63 INMARSAT TELEX OCEAN REGION CODES

Atlantic Ocean Region — East	581
Pacific Ocean Region	582
Indian Ocean Region	583
Atlantic Ocean Region — West	584

5.64 TELEX CALLS TO SHIPS FITTED WITH MF/HF EQUIPMENT

Contact the LES as described in section 5.57 and exchange answerbacks.
Key in code 31+ for the maritime enquiry service.
Send full details of the ship's name, callsign and position (if known).
When the ship has been contacted on a terrestrial radio circuit operators will be instructed to pass their telex traffic.

5.65 TELEX SERVICE FAULT CODES

Fault code	*Meaning*
ABS	Subscriber absent — received if shore telex subscriber is switched off or faulty.
DER	Indicates that the path to the telex subscriber is faulty and the telex terminal does not respond to the WRU signal.
NC	Indicates that no circuits are available in the land-based system.
NP	No party. The called number is no longer connected.
NA	Connection not admitted.
OCC	Telex subscriber terminal engaged.

CHAPTER 6

Radio Telex

GENERAL

The information in this chapter relates to the terrestrial radio telex services in the MF and HF marine bands. There is no telex facility in the maritime mobile VHF band. For information regarding the format of telex traffic refer to Chapter 9. For examples of distress procedures using telex refer to Chapter 2.

Marine telex is also referred to as "Narrow Band Direct Printing" (NBDP) or sometimes Radio Teletype (RTT).

Telex is also available using satellite communications which is covered in Chapter 5 of this book.

6.1 AVOIDANCE OF INTERFERENCE

Before transmitting stations should take precautions to ensure that their emissions will not interfere with transmissions already in progress. If such interference is likely, the station shall wait for an appropriate break in the communications in progress. This obligation does not apply to stations where unattended operation is possible through automatic means.

6.2 TELEX MODES OF EMISSION

6.2.1 Band between 415 kHz and 535 kHz

All ships equipped with telex equipment in this band shall be able to:

(a) send and receive class F1B or J2B emissions on the working frequencies necessary to carry out their service.

(b) receive class F1B emissions on 518 kHz if complying with GMDSS.

6.2.2 Bands between 1 605 kHz and 4 000 kHz and 4 000 kHz and 27 500 kHz

All ships equipped with telex to work in these bands shall be able to send and receive class F1B or J2B emissions on working frequencies necessary to carry out their service.

6.3 TELEX COMMUNICATION BETWEEN TWO STATIONS

For communication between only two stations the ARQ mode should be used when available.

6.4 TELEX BROADCASTS TO MORE THAN ONE RECEIVING STATION

For transmissions from a coast station or ship station to two or more other stations Forward Error Correction (FEC) mode should be used when available.

N.B.: FEC is therefore the mode of telex operation normally used when sending and receiving distress, urgency and safety information.

6.5 TELEX PUBLIC CORRESPONDENCE

The services provided by each coast station open for public correspondence shall be indicated in the ITU List of Coast Stations and details of ships open for public correspondence should be indicated in the ITU List of Ship Stations. Information on charging should be available in both cases.

6.6 SELCALL NUMBERS

Each ship station and coast radio station having radio telex facilities is assigned a unique selcall number in addition to its international callsign. The first number(s) of the selcall indicates the country to which a particular station belongs.

Ship selcall numbers have five digits e.g. 47579
Coast station selcall numbers have four digits e.g. 3220

UK allocation of selcall numbers are as follows:

Ship stations 45500 to 50499
Coast stations 3200 to 3259

Ships selcall numbers will be found in ITU List of Ship Stations. Coast station selcall numbers will be found in ALRS Volume 1 and ITU List of Coast Stations.

Some administrations may introduce the use of the 9-digit MMSIs for radiotelex (see section 8.26) in place of the 5-digit selcall numbers previously described.

6.7 MANUAL TELEX CALLING PROCEDURES

When using telex in the maritime mobile frequency bands the call may, by prior arrangement, be made on a working frequency available for such systems.

6.7.1 Ship to shore telex calling

The operator of the ship station establishes communication by telephony, or by other means using normal calling procedures. The ships operator requests telex and exchanges working frequency arrangements to be used and gives the ship's telex selcall number. The coast station then establishes communication on the agreed frequencies. Alternatively, the ship operator calls the coast station on a predetermined telex frequency using the coast station telex selcall number. The coast station should reply on the appropriate paired transmit frequency.

6.7.2 Shore to ship telex calling

The coast station operator calls the ship using telephony, or other means using normal calling procedures. The ship then applies the ship to shore procedures detailed in the preceding section.

6.7.3 Telex intership communications

The operator of the calling ship establishes communication by telephony, or other means using normal calling procedures. The operator requests telex and supplies information regarding frequencies to be used and gives own ships telex selcall number. The operator of the called ship then establishes communication on the agreed frequency using the appropriate selcall of the calling ship.

6.8 AUTOMATIC TELEX CALLING PROCEDURES

6.8.1 Ship to shore automatic calling

The ship station calls the coast station on a predetermined frequency using telex equipment and the selcall number of the coast station. The coast station telex equipment detects the call and responds on the appropriate paired frequency either automatically or manually.

6.8.2 Shore to ship automatic calling

The coast station calls the ship station on a predetermined transmit frequency using telex equipment and the ship's selcall number. If the ship's equipment detects the call the reply is given in either of the following ways:

(a) the ship station replies either immediately on the corresponding paired transmit frequency, or at a later stage using the procedures above; or
(b) the ship station's transmitter is automatically started on the corresponding transmit frequency and sends appropriate control signals to indicate readiness to receive traffic automatically.

6.9 TELEX TRANSMISSION FORMAT

Where the appropriate facilities are provided by the coast station, traffic may be exchanged with the telex network either:

(a) in conversational mode where the stations are connected directly, either automatically or manually; or
(b) in a store and forward mode where traffic is stored at the coast station until the circuit to the called subscriber can be set up, either automatically or under manual control.

6.10 TELEX MESSAGE FORMAT

In the direction shore to ship the message format should conform to normal telex network practice. In the direction ship to shore, the message format should conform to the operational procedures specified in the relevant CCIR Recommendations. Section 9.26 has further details.

6.11 FEC OPERATION MODE

Messages in forward error correction (FEC) mode may be sent, by prior arrangement, from a coast station or ship station to one or more ship stations in the following cases:

(a) Where a receiving ship station is not able to use its transmitter or is not permitted to do so;
(b) Where the message is intended for more than one ship;
(c) Where unattended reception of the message in the FEC mode is necessary and automatic acknowledgement is not required.

N.B.: All messages in the FEC mode should be preceded by at least one carriage return (CR) and at least one line feed (LF) signal. With modern telex equipment one CR followed by one LF are often inserted by operating the "Enter" or "Return" key as individual carriage return and line feed keys may not be available on the telex keyboard.

6.12 ACKNOWLEDGEMENT OF FEC MESSAGES

Ship stations may acknowledge receipt of messages in FEC mode by telephony or other means.

6.13 TELEX FREQUENCIES

General: All radio telex frequencies quoted in this book are the assigned frequencies. Operators should refer to equipment manufacturer's operating instructions to determine if an offset of 1.5 kHz, 1.7 kHz or 1.9 kHz should be subtracted from transmitter and receiver assigned frequencies before attempting to receive or send telex signals. More modern microprocessor controlled transceivers automatically apply this offset when "telex" mode is selected.

Note; The modem offset frequency in most common use is 1.7 kHz.

490 kHz

490 kHz will be used exclusively for the transmission by coast stations of meteorological and navigational warnings and urgent information to ships by means of telex. Broadcasts on this frequency will normally be in a local language.

518 kHz

In the MF maritime mobile service 518 kHz is used exclusively for the transmission by coast stations of meteorological and navigational warnings and urgent information to ships, by telex (international NAVTEX system). See section 4.17 for further details.

2 174.5 kHz

Exclusively reserved for distress and safety traffic using telex in the MF band. It should be used for ship-to-ship on-scene communications in FEC mode. See chapter 2 for further details.

4 209.5 kHz

In the HF maritime mobile service 4 209.5 kHz is used exclusively for the transmission by coast stations of meteorological and navigational warnings and urgent information to ships, by NAVTEX type transmission.

4 177.5 kHz; 6 268 kHz; 8 376.5 kHz; 12 520 kHz; 16 695 kHz

In the HF maritime mobile bands these frequencies are reserved exclusively for distress and safety using telex.

4 210 kHz; 6 314 kHz; 8 416.5 kHz; 12 579 kHz; 16 806.5 kHz; 22 376 kHz; 26 100.5 kHz

In the HF maritime mobile bands these frequencies are reserved exclusively for the transmission by coast stations of maritime safety information (MSI) by telex in FEC mode.

6.14 TELEX DISTRESS FREQUENCIES

All ship stations using telex shall be able to send and receive on the frequency designated for distress and safety traffic in the band in which they are operating. Appendix 8 lists all distress frequencies.

6.15 PROTECTION OF FREQUENCIES

Any emission capable of causing harmful interference to distress, urgency or safety communications on any of the telex frequencies listed in the preceding sections is prohibited.

6.16 TEST TRANSMISSIONS

Test transmissions should be kept to a minimum on the distress and safety frequencies identified in the preceding sections and should, wherever practicable, be carried out on artificial antennas or with reduced power. Further details on test transmissions see section 8.19.

6.17 TELEX ANSWERBACK

To ensure that an operator has been connected to the correct coast radio station, ship station, or telex subscriber ashore it is normal to exchange answerbacks at the commencement and termination of a telex link. Each telex installation has a unique answerback to identify itself and this is programmed into the equipment. When, for example, the automatic radio telex facility controlled by Hong Kong radio is accessed, the following answerback will be received: "1480 HKRDO HX".

Where	1480	is Hong Kong radio's selcall number
	HKRDO	identifies the radio station
	HX	indicates the country — in this case Hong Kong, China.

A ship's answerback might be: "47579 GFCV X"

Where	47579	indicates the ship's selcall number
	GFCV	is the ship's international call sign and
	X	indicates that it is a maritime mobile station.

Telex subscribers ashore have answerbacks which include their telex number followed by a short word or group of letters indicating the name of the company or organisation and finally the country identifier, e.g. 987321 LLOYDS G.

6.18 MODES OF TELEX OPERATION

Radio circuits are liable to interference, fading or bursts of static which can mutilate sections of a telex message. An effective means of error detection and correction is therefore needed. There are two main modes of operation namely ARQ and FEC.

6.18.1 ARQ means Automatic Repetition reQuest and provides error detection and error correction. However, it requires both communicating stations to have their respective transmitters and receivers active simultaneously. This mode is normally restricted in use between two stations e.g. a ship and a coast station or, perhaps, two ships.

6.18.2 FEC means Forward Error Correction and provides error detection only. If any doubtful letters occur in the text, a gap (or sometimes an asterisk) will appear. With FEC the receiving station(s) do not need an active transmitter. This mode of operation is therefore ideal for broadcasting information to numerous stations simultaneously and it is used for sending traffic lists, weather and navigational warnings etc. It is therefore sometimes referred to as the "broadcast mode" and is the preferred telex mode for distress, urgency and safety messages.

N.B.: It is most important when sending in FEC mode to allow the initial call to last at least 10 seconds and then to send at least one carriage return (CR) followed by at least one line feed (LF). This is nowadays achieved by using the keyboard "Enter" or "Return" key. If this is not done receiving equipments will not respond to the broadcast.

6.18.3 SELFEC

Selective FEC is a derivation of FEC and is called SELFEC. Again the recipients transmitter does not have to be active and it is similar to FEC in all respects except that the transmission is addressed to a particular receiving station (i.e. by addressing the call to the appropriate selcall number). It is an ideal mode of transmission for ships to receive telex messages whilst in a port where the use of transmitters may be restricted or forbidden.

6.18.4 Direct

Some telex installations may have DIRECT mode in which case there is no error detection or error correction. For that reason it is seldom used for marine radio telex circuits.

6.19 PROCEDURE FOR INITIATING A RADIO TELEX CIRCUIT TO A COAST STATION

(a) Refer to ITU List of Coast stations or ALRS Volume 1 for particulars of coast station telex frequencies.
(b) Decide which telex channel to use by tuning the receiver to the coast station transmit frequencies. (Some coast stations emit channel free signals). If strong signals are heard the operator can presume that the coast station will hear the call.

(c) If certain that the channel is free — key in the selcall number of the receiving station, tune the transmitter to the appropriate paired ship station transmit frequency and start the call in ARQ mode.
(d) If the call is heard a response will be received, e.g. when contacting Hong Kong radio 1480 HKRDO HX will appear on the VDU and/or printer.
(e) Key in the code for the service required, e.g. DIRTLX12345+ (see section 6.21 for list of more commonly used radio telex codes).
(f) When connected to the distant subscriber, exchange answerback codes, then proceed to exchange traffic.
(g) When finished with that subscriber send KKKK which disconnects the circuit. A date and time group will be issued, followed by duration of the call and the invitation to proceed with the next call or telex facility (GA+). Note, this does not break the radio link with the coast station, allowing a follow-on call.
(h) When all traffic has been completed sending BRK+ breaks the radio link with the coast station and the telex equipment should be switched back to the "Standby" condition immediately, otherwise that telex channel will be blocked and prevent other users from accessing it.

6.20 TRAFFIC LISTS ON TELEX

Most coast radio stations transmit lists of vessels for whom they hold telex traffic. Traffic lists are transmitted at regular intervals, details of which may be found in ITU List of Coast Stations and ALRS Volume 1. Vessels should make sure that their telex receiver is tuned to the correct frequency. Alternatively, vessels with scanning receivers should ensure that stations from whom they regularly receive traffic are included in their scan list.

6.21 TELEX COMMANDS AS USED IN THE AUTOMATIC RADIO TELEX SYSTEM

Note: All commands should be terminated with a + sign which means "execute". There should be no spaces in a telex command:

i.e. DIRTLX12345+ is correct, whereas
DIR TLX 12345 + is not acceptable

AMEND+	Change previous watchkeeping instructions given in a FREQ+ message.
AMV+	Ship's position report which is routed to USCG, New York.
BRK+	Break (terminate) radio circuit.
CANCEL+	Used to cancel previous watchkeeping instructions in a FREQ+ message.
DATAx	Requests a message to be forwarded using data facilities, where x indicates the subscriber number.
DIRTLXx+	Request for direct connection to a specific telex subscriber. (Where x indicates the telex subscriber number). All telex calls outside the coast station country are to be preceded with 0 followed by the telex country code and then the subscribers number, e.g. DIRTLX02312345+ where 0 requests international call, 23 requests USA country code and 12345 is the subscriber's number.
ERROR+	Incorrect watchkeeping instruction received within a FREQ+ message.
FAX....+	Telex messages can be delivered to a fax number by keying in FAX followed by telephone country code (for overseas calls) and/or area code and

fax number terminated with +. This is available only in the direction ship-to-shore.

FREQ+ To advise radio station of your ships watchkeeping arrangements.

GA+ Invitation to go ahead.

HELP+ Ship's help file. Radio station sends operator information and guidance. At the end you will receive GA+ which invites you to select a telex facility command.

INF+ Request for databank information held at the coast station. INF+ returns a directory listing and a subsequent facility code selects the desired information.

MAN+ A message to be stored and forwarded manually to a country which can not he accessed automatically.

MED+ Instantly connects you with manual assistance point when you require medical assistance at sea.
For less urgent assistance you should use OPR+

MSG+ Indicates that the ship station needs to immediately receive any messages held for it at the coast station. If the MSG+ command is not given by the ship the traffic will not be transferred.

MULFAX..+ For delivering telex to more than one fax number. Each number should be separated by / and last number terminated with +

MULTLX+ For sending the same text to multiple addresses. Each telex number should be separated by an oblique stroke (/) and overseas numbers should be prefixed with a 0 and then the appropriate country code, e.g.

MULTLX54321/02312345/040522233+

First telex is charged at full rate and subsequent numbers at half normal rate.

MULTLA+ Request for advice of delivery of multi-address facility.

NAV+ Request for navigational warnings.

OBS+ Ship's weather report which is automatically routed to local Meteorological Office.

OK Indicates FREQ+ instructions accepted by computer.

OPR+ Connects ship to a manual operator at the radio station for assistance of a general nature.

POS+ Voluntary reporting of ship's position which is stored and used to improve safety of life at sea. It can also be used to assist in the subsequent automatic transmission and reception of messages by determining OTF and/or directional antennas.

RDL+ Redials last DIRTLX number.

RPTx Requests repetition of a message, using ARQ mode, previously transmitted using FEC mode, where x is used to identify the message number.

RTL+ Radio Telex Letter which is received at the coast station by radio telex and then sent to the destination by post.
A RTL should always be in the following format:

(a) Ship's name/callsign, date and time of handing in
(b) RTL
(c) Name of addressee
(d) Full Postal Address including post code and country if necessary
(e) Text
(f) KKKK

STA+ Request for status report on all store and forward messages for which advice of delivery has not been received.

STSx+	Ship-to-ship telex facility. where x is the 5 or 9 digit identity number of the addressed ship.
SVC+	To send unpaid service information to radio station e.g. to acknowledge receipt of a SELFEC broadcast.
TELx	Requests a telex message to be telephoned by the coasts station to the telephone number x.
TGM+	Indicates a from ship telegram.
TLX....+	Request for store and forward facility where the subscriber's number should be inserted in place of the dots. Own ships telex message will be electronically stored at radio station. At some later time it will be sent to the telex subscriber's number. This facility is useful if the distant telex machine is unmanned or temporarily not available (e.g. engaged). If the message cannot be sent within 24 hours a service message will be received cancelling the original telex message.
TLXA....+	Store and forward when advice of delivery is requested.
TRAFFIC HELD	Telex traffic on hand for own ship at radio station. (To obtain own ships traffic MSG+ should be keyed in)
TRF+	indicates that the ship needs to receive information, automatically transmitted, on tariffs currently applicable to the coast station.
TST+	Indicates that the ship needs to receive an automatically transmitted test text (e.g. "the quick brown fox....”).
URG+	Instantly connects ship with manual assistance point at radio station where an audible alarm will be activated. To be used when urgent assistance regarding safety of life at sea is required. See Chapter 2 for full details.
VBTLXx+	Telex voicebank facility where the coast station dictates the message to a telephone voicebank number x for subsequent retrieval by a telephone subscriber.
WX+	Request for weather information.
xx	Watchkeeping stop time (in FREQ+ instructions).
+?	Means over. It places the other station in the transmit mode and puts own station in receive mode.

6.22 TELEX ADVICE CODES AS USED IN THE AUTOMATIC RADIO TELEX SYSTEM

Code	*Meaning*
ABS	Absent subscriber/Office closed.
ADD	Please input your international telex number.
ANUL	Delete.
BCT	Broadcast call.
BK	I cut off.
BMC	No end of message or end of transmission received, therefore message cancelled.
CFM	I confirm/Please confirm.
CI	Conversation impossible.
COL	I collate/Collation please.
CRV	Do you receive well?/I receive well.
DER	Out of order.
DF	You are in communication with the called subscriber.
EXM	Connection cleared due to exhaustion of text recording medium at either end of called or calling terminal.
FMT	Format error.

Code	*Meaning*
GA	Go ahead you may transmit or may I transmit?
IAB	Invalid answerback from destination.
IMA	Input message acknowledgement.
INF	Subscriber temporarily unobtainable, call the information service.
ITD	Input transaction accepted for delivery.
ITL	I transmit later.
JFE	Office closed because of holiday.
LDE	Maximum acceptable message length or duration has been exceeded.
MNS	Minutes.
MOM	Wait/waiting.
MUT	Mutilated.
NA	Correspondence with this subscriber is not admitted.
NC	No circuits.
NCH	Subscriber's number has been changed.
NDN	Non-delivery notification.
NI	No line identification available.
NP	The called party is not, or is no longer, a subscriber.
NR	Indicates your call number/My call number is ...
OCC	Subscriber is engaged.
OK	Agreed/Do you agree?
PPR	Paper.
R	Received.
RAP	I shall call you back.
RDI	Redirected call.
REF	Reference of the message delivered to the telex side from a conversation facility for telex/teletex interworking.
REI	Address validation failure/non-compliant answerback received.
RPT	Repeat/I repeat.
RSBA	Retransmission still being attempted.
SSSS	Change of alphabet.
SVP	Please.
T	Stop your transmission. This is to be repeated until the transmission is brought to a stop.
TAX	What is the charge/The charge is ...
TEST MSG	Please send a test message.
THRU	You are in communication with a telex position.
TMA	Maximum number of addresses exceeded.
TPR	Teleprinter.
TTX	Designation of telex facility (CF) for telex/teletex interworking.
VAL	Validation response.
W	Words.
WRU	Who are you/Who is there?
XXXXX	Error.

6.23 CHARGES FOR TELEX CALLS

Charges depend upon three factors:

(a) Duration of the telex circuit. (See note on next page.)
(b) Location of distant subscriber from coast radio station. (land line charge).
(c) Frequency band used. It is sometimes more expensive using HF than MF.

However, some administrations may offer a standard telex charge to any destination worldwide in which case the call is charged on a time basis only.

Note: Automatically connected calls are normally based on a minimum charge of 6 seconds with 6 second incremental steps. Manually connected calls are based on a minimum charge of 3 minutes with one minute incremental steps.

When subscribers are connected an automatic timer records time used. However, if poor conditions exist, the timer will stop whilst the equipment is handling repetitions of corrupted text. Sending KKKK or BRK+ at the end of a telex circuit prompts the automatic equipment to quote duration of the call.

CHAPTER 7

Radiotelephony (RT)

PART I - MARITIME RT BANDS: USE OF FREQUENCIES

7.1 RT: INTRODUCTION

7.1.1 Nomenclature of frequency bands used in radiocommunication

The radio spectrum is subdivided into nine frequency bands as shown below. The unit of frequency is the Hertz (Hz). Frequencies are expressed in kilohertz (kHz), up to and including 3 000 kHz, in megahertz (MHz), above 3 MHz, up to and including 3 000 MHz and in gigahertz (GHz), above 3 GHz, up to and including 3 000 GHz.

Frequency Range	*Name / Symbol*
3 to 30 kHz	Very Low Frequency (VLF)
30 to 300 kHz	Low Frequency (LF)
300 to 3 000 kHz	Medium Frequency (MF)
3 to 30 MHz	High Frequency (HF)
30 to 300 MHz	Very High Frequency (VHF)
300 to 3 000 MHz	Ultra High Frequency (UHF)
3 to 30 GHz	Super High Frequency (SHF)
30 to 300 GHz	Extremely High Frequency (EHF)
300 to 3 000 GHz	Unnamed

7.1.2 SSB designation

When designating frequencies for single-sideband radiotelephony the carrier frequency is always to be designated. The assigned frequency shall be 1.4 kHz higher than the carrier frequency.

Single-sideband apparatus in radiotelephone stations of the maritime mobile service operating in the bands allocated to this service between 1 605 kHz and 4 000 kHz and in the bands allocated exclusively to this service between 4 000 kHz and 27 500 kHz shall satisfy the technical and operational conditions specified in the relevant ITU Radio Regulations.

7.2 BANDS BETWEEN 1 605 kHz AND 4 000 kHz

7.2.1 Mode of emission of stations

Unless otherwise specified in the present ITU Radio Regulations notably the use of different modes of emissions on specified distress frequencies, the class of emission to be used in the bands between 1 605 kHz and 4 000 kHz shall be J3E.

The normal mode of operation for each coast station shall be indicated in the ITU List of Coast Stations. Transmissions in the bands 2 170-2 173.5 kHz and 2 190.5-2 194 kHz with the carrier frequency 2 170.5 kHz and the carrier frequency 2 191 kHz respectively are limited to class J3E emissions and are limited to a peak envelope power of 400 W. However, on the frequency 2 170.5 kHz and with the same power limit, coast stations may also use class H2B emissions when using the selective calling system and exceptionally, in Regions 1 and 3 and in Greenland, may also use class H3E for safety messages.

7.2.2 Use of 2 182 kHz

The frequency 2 182 kHz is used for GMDSS radiotelephony distress and safety traffic. It is used for radiotelephony distress calls and traffic, and for urgency calls and messages and for calls preceded by the safety signal. Safety messages should be transmitted where practicable on a working frequency. The class of emission to be used shall be J3E.

The class of emission J3E is used for the exchange of distress traffic on 2 182 kHz following the acknowledgement of a distress alert call using DSC techniques. However operators must take into account other shipping in the vicinity which may not be able to receive this traffic (mode of emission).

Where administrations provide at their coast stations a watch on 2 182 kHz for receiving class J3E emissions as well as class A3E and H3E emissions, ship stations may call those coast stations for safety purposes using class H3E or J3E emissions.

The frequency 2 182 kHz may also be used:

(a) for call and reply;
(b) by coast stations to announce the transmission, on another frequency, of traffic lists.

In addition, an administration may assign to its stations other frequencies for call and reply. To facilitate use of the frequency 2 182 kHz for distress purposes, all transmissions on 2 182 kHz shall be kept to a minimum. Before transmitting on the carrier frequency 2 182 kHz, a station shall listen on this frequency for a reasonable period to make sure that no distress traffic is being sent, this provision does not however apply to stations in distress.

Coast stations which use 2 182 kHz for calling shall be able to use at least one other frequency in the authorised bands between 1 605 kHz and 2 850 kHz. Coast stations authorised to use radiotelephony on one or more frequencies other than 2 182 kHz in the authorised bands between 1 605 kHz and 2 850 kHz shall use class J3E emissions on those frequencies.

7.3 MF RT WORKING FREQUENCIES IN REGION 1

The peak envelope power of ship radiotelephone stations operating in the authorised bands between 1 605 kHz and 2 850 kHz shall not exceed 400 W.

7.3.1 MF ship to shore and intership working frequencies

All stations on ships making international voyages should, in addition to their normal working frequencies, be able to use:

(a) the ship to shore working frequency, if required by their service, carrier frequency 2 045 kHz for class J3E emissions;
(b) The intership frequency, if required by their service, carrier 2 048 kHz for class J3E emissions.

This frequency may be used as an additional ship to shore frequency. This frequency shall not be used for working between stations of the same nationality.

Ships frequently exchanging correspondence with a coast station of a nationality other than their own may use the same frequencies as ships of the nationality of the coast station;

where mutually agreed by the administrations concerned; or

where the facility is open to ships of all nationalities by virtue of a note against the frequencies concerned in the ITU List of Coast Stations.

The following ship to shore frequencies may be assigned to coast stations as receiving frequencies.

carrier frequency 2 051 kHz
carrier frequency 2 054 kHz
carrier frequency 2 057 kHz

7.3.2 MF intership working frequencies

Also in Region *1,* frequencies assigned to stations operating in the bands between 1 850 kHz and 3 800 kHz should, whenever possible be in accordance with the following sub divisions:

2 262.5-2 498 kHz intership SSB RT
3 340-3 400 kHz intership SSB RT
3 500-3 800 kHz intership SSB RT

7.4 BANDS BETWEEN 4 000 kHz AND 27 500 kHz

7.4.1 Mode of emission of stations

The class of emission to be used for RT transmissions in the bands between 4 000 kHz and 27 500 kHz shall be J3E.

Ship RT stations employing J3E emissions in the bands between 4 000 kHz and 27 500 kHz shall at no time use a peak envelope power in excess of 1.5 kW per channel.

7.4.2 Call and reply

Ship stations may use the following carrier frequencies for calling on RT.

Ship frequency	*ITU channel number*	*Coast frequency*
4 125 kHz	421	4 417 kHz
6 215 kHz	606	6 516 kHz
8 255 kHz	821	8 779 kHz
12 290 kHz	1221	13 137 kHz
16 420 kHz	1621	17 302 kHz
18 795 kHz	1806	19 770 kHz
22 060 kHz	2221	22 756 kHz
25 097 kHz	2510	26 172 kHz

7.4 *Radiotelephony (RT)*

Before transmitting on the carrier frequencies 4 125 kHz, 6 215 kHz , 8 291 kHz, 12 290 kHz and 16 420 kHz, stations should listen on the frequency for a short period to ensure no distress traffic is being sent. However this provision does not apply to stations in distress. The ship frequencies above are paired with associated coast station frequencies, thus coast stations may use the carrier frequencies for calling and replying to ships on RT.

7.4.3 Traffic

When using duplex telephony, the transmitting frequencies of the coast station and the corresponding ship station will be in pairs as indicated by Appendix 16 of the ITU Radio Regulations. These frequencies may be used by ship stations of any category according to traffic requirements.

7.5 VHF BANDS BETWEEN 156 MHz AND 174 MHz

7.5.1 Call and reply

Channel 16 (156.8 MHz) is the international channel for distress traffic and for calling on RT on the authorised maritime mobile service VHF bands. The class of emission used for RT on this channel must be G3E. Ch.16 VHF may also be used:

(a) by coast and ship stations for call and reply;
(b) by coast stations to announce the transmission of traffic lists and important maritime information on another channel;

Any of the channels designated for public correspondence in Appendix 18 of the ITU Radio Regulations may be used as a calling channel if an administration so desires. Coast and ship stations in the public correspondence service may use a working frequency for calling purposes.

To assist in the reception of distress calls and distress traffic, all transmissions on Ch.16 VHF shall be kept to a minimum and shall not exceed one minute. Also, before transmitting, a station should listen on Ch.16 VHF for a short period to ensure that no distress traffic is being transmitted on this channel. This provision does not apply to stations in distress.

7.5.2 Watch

Until 1st February 2005, ship stations should, where practicable, maintain watch on Ch.16 VHF when within the service area of a coast station providing international maritime mobile RT service in the band 156–174 MHz. Ships fitted only with RT equipment operating in the authorised maritime mobile service VHF bands should maintain watch on Ch.16 VHF when at sea.

A coast station in the port operation service in an area where Ch.16 VHF is being used for distress, urgency or safety shall during its hours of service keep additional watch on Ch.12 VHF or some other port operations channel.

A coast station in the ship movement service in an area where Ch 16 is being used for distress, urgency or safety shall during its hours of service keep additional watch on the ship movement channels indicated in heavy type in the ITU List of Coast Stations.

The method of working (simplex or duplex) as specified in Appendix 18 of the ITU Radio Regulations for each channel should be used in the international service. The carrier power of ship station transmitters shall not exceed 25 W.

7.5.3 **VHF Ch.13** Watch

Ships complying with the provisions of the Radio regulations should, where practicable, maintain watch on Ch.13 VHF for communications related to safety of navigation.

PART II - GENERAL RT PROCEDURE IN THE MARITIME MOBILE SERVICE

7.6 RT: GENERAL PROCEDURES

The procedures detailed in the following sections are applicable to radiotelephone stations, except in cases of distress, urgency or safety, to which special provisions of the ITU Radio Regulations are applicable. The service of ship radiotelephone stations shall be performed or controlled by an operator holding an appropriate operating certificate recognised by a competent administration.

The IMO has expressed concern at the widespread misuse of VHF channels at sea. This is particularly so on the distress and safety channels (Ch.16 and Ch.70), but also on channels set aside for port operations, ship reporting and ship movement services. All users should be aware that they must comply with the ITU Radio Regulations and operate the equipment in accordance with these regulations. Operators are reminded to pay particular attention to the following points:

(i) Ch. 16 VHF should only be used for distress, urgency and brief safety communications. It may be used to establish other communications, which should then be transferred to a suitable working channel.

(ii) Ch.70 VHF is used exclusively for DSC and no speech communication should be used on this channel.

(iii) Port operations, ship reporting and ship movement channels. Only messages relating to the operational handling of such services, and to the safety of persons are permitted on these channels.

(iv) Stations must ALWAYS identify themselves by using the vessel's name or callsign every time they transmit "on air".

(v) The VHF RT equipment on board a ship must be under the control of an operator holding appropriate qualifications recognised by the administration of the ship's country of registration.

7.7 RT: STATION IDENTIFICATION

How RT stations identify themselves is given in section 8.24, briefly identification is as follows:

7.7.1 Coast stations

(a) by use of a call sign, or

(b) by use of the geographical place name as it appears in the List of Coast Stations, followed preferably, by the word RADIO, or some other appropriate indication.

7.7.2 Ship stations

(a) by use of a call sign, or
(b) by the official name of the ship preceded, if necessary by the name of the owner on condition that there is no possible confusion with distress, urgency and safety signals, or
(c) by its selective call number or MMSI .

The radiotelephone public correspondence service provided on ships should, if possible, be operated on a duplex basis. Devices providing for the emission of a signal to indicate that a call is in progress on a channel may be used in this service on a non-interference basis to the service provided by coast stations. The use of devices for continuous or repetitive calling or identification in a manually operated radiotelephony service is not permitted. A station may not transmit identical information simultaneously on two or more frequencies when communicating with only one other station.

A station shall not emit any carrier wave between calls. However, stations using an automatically operated radiotelephone system may emit marking signals on request by a ship station to establish an RT call. This tuning signal duration should be no longer than 10 seconds.

Radiotelephone stations should, as far as possible, be equipped with devices for instantaneous switching from transmission to reception and vice versa. This equipment is necessary for all stations participating in communication between ships and subscribers of the land telephone system.

Stations equipped for RT may transmit and receive radiotelegrams by means of radiotelephony. Coast stations providing such service and open for public correspondence shall be indicated in the ITU List of Coast Stations.

When it is necessary to spell out certain expressions, difficult words, service abbreviations or figures, the phonetic alphabet given in Appendix 4 should be used.

7.8 SIMPLEX AND DUPLEX

Telephony which is a form of telecommunication set up for the transmission of speech or, in some cases, other sounds. In the maritime mobile service radiotelephone calls are generally conducted on duplex channels. Duplex operation is an operating method in which transmission is possible simultaneously in both directions on a telecommunication channel. Duplex therefore requires the use of two frequencies, one for the ship to shore direction and the other shore to ship.

Simplex operation on the other hand is an operating method in which transmission is made possible alternately in each direction of a telecommunication channel, for example, by means of manual control. This is sometimes referred to as a single frequency, or unidirectional, method of operation.

7.9 PRELIMINARY RT OPERATIONS

Before transmitting, a station shall take precautions to ensure that its emissions will not interfere with transmissions already in progress; if such interference is likely, the station shall await for an appropriate break in the working. If, after taking these precautions, the emissions of the station should nevertheless interfere with a transmission already in progress, the following rules shall be applied:

(a) the ship station whose emission causes interference to the communication of a mobile station with a coast station shall cease sending at the first request of the coast station;

(b) the ship station whose emission causes interference to communications already in progress between mobile stations shall cease sending at the first request of one of the other stations;

(c) the station which requests this cessation shall indicate the approximate waiting time imposed on the station whose emission it suspends.

7.10 RT CALLS FROM SHIPS

As a general rule, it rests with the ship station to establish communication with the coast station. For this purpose the ship station may call the coast station only when it comes within the coast station's service area. However, a coast station having traffic for a ship station may call this station if it has reason to believe that the ship station is keeping watch and is within its service area.

7.11 RT CALLS FROM COAST STATIONS

Coast stations shall, so far as practicable, transmit its calls in the form of "traffic lists" consisting of the call signs or other identification in alphabetical order of all ship stations for which it has traffic on hand. These calls shall be made at specified times fixed by agreement between the administrations concerned and at intervals of not less than two hours and not more than four hours during the working hours of the coast station. Coast stations shall transmit their traffic lists on their normal working frequencies in the appropriate bands. The transmission shall be preceded by a general call to all stations.

The general call to all stations announcing the traffic lists may be sent on a calling frequency in the following form:

"Hello all ships" or CQ (spoken as CHARLIE QUEBEC) not more than three times;

the words THIS IS (or DE spoken as DELTA ECHO in case of language difficulties);

“ . . . Radio" not more than three times;

"Listen for my traffic list on . . . kHz".

In no case may this preamble be repeated.

However, in the VHF band when the conditions for establishing contact are good, the call described above may be replaced by:

"Hello all ships" or CQ (spoken as CHARLIE QUEBEC), once;

the words THIS IS (or DE spoken as DELTA ECHO in case of language difficulties);

" . . . Radio", twice;
"Listen for my traffic list on channel . . .

In no case may this preamble be repeated.

The calls outlined above are obligatory when 2 182 kHz or Ch.16 VHF are used.

7.12 RT TRAFFIC LISTS

The hours at which coast stations transmit their traffic lists and the frequencies and classes of emission which they use for this purpose are listed in the ITU List of Coast Stations and ALRS Volume 1.

Ship stations should as far as possible listen to the traffic lists transmitted by coast stations. On hearing their call sign or other identification in such a list they must reply as soon as possible. When the traffic cannot be sent immediately, the coast station shall inform each ship station concerned of the probable time at which working can begin, and also, if necessary, the frequency and class of emission which will be used.

7.13 ORDER OF WORKING

When a coast station receives calls from several ship stations at practically the same time, it decides the order in which these stations may transmit their traffic. Its decision shall be based on the priority of the radiotelegrams or radiotelephone calls that the ship stations have on hand and on the need for allowing each calling station to clear the greatest possible number of communications.

7.14 REPETITION OF CALLS

When a station called does not reply to a call sent three times at intervals of two minutes, the calling shall cease. However, when a station called does not reply, the call may be repeated at three-minute intervals. Ship stations shall not radiate a carrier wave between calls.

In areas where reliable VHF communication with a called coast station is practicable, the calling ship station may repeat the call as soon as it is ascertained that traffic has been terminated at the coast station.

7.15 RT CALL AND REPLY

The RT call consists of:

- the call sign or other identification of the station called, not more than three times;
- the words THIS IS (or DE spoken as DELTA ECHO in case of language difficulties);
- the call sign or other identification of the calling station, not more than three times.

Example:

KINGSTONRADIOKINGSTONRADIOKINGSTONRADIO
THIS IS
ALKAID ALKAID ALKAID

However, in the VHF band when the conditions for establishing contact are good, the call described above may be replaced by:

the call sign of the station called, once;
the words THIS IS (or DE spoken as DELTA ECHO in case of language difficulties);
the call sign or other identification of the calling station, twice.

Example:

KINGSTON RADIO
THIS IS
ALKAID ALKAID

When calling a VHF coast station operating on more than one channel, a ship station calling on a working channel should include the number of that channel in the call.

When contact is established, the call sign or other identification may thereafter be transmitted *once only*.

When the coast station is fitted with equipment for selective calling and the ship station is fitted with equipment for receiving such selective calls, the coast station shall call the ship by transmitting the appropriate DSC signals. The ship station shall then call the coast station by speech using the standard RT calling procedure described above.

The reply to an RT call consists of:

the call sign or other identification of the calling station, not more than three times;
the words THIS IS (or DE spoken as DELTA ECHO in case of language difficulties);
the call sign or other identification of the called station, not more than three times.

Example:

ALKAID ALKAID ALKAID
THIS IS
KINGSTON RADIO KINGSTON RADIO KINGSTON RADIO

7.16 FREQUENCY TO BE USED FOR CALL AND REPLY

7.16.1 Bands between 1 605 kHz and 4 000 kHz

A radiotelephone ship station calling a coast station should use for the call, in order of preference:

(a) a working frequency on which the coast station is keeping watch;
(b) the carrier frequency 2 182 kHz;
(c) in Regions 1 and 3 and in Greenland, the carrier frequency 2 191 kHz when a carrier frequency of 2 182 kHz is being used for distress;
(d) in Region 2 except for Greenland, the carrier frequency 2 191 kHz as a supplementary calling frequency in those areas of heavy usage of 2 182 kHz.

A radiotelephone ship station calling another ship station should use for the call:

(a) the carrier frequency 2 182 kHz;
(b) an intership frequency, whenever and wherever traffic density is high and prior arrangements can be made.

Coast stations shall, in accordance with the requirements of their own country, call ship stations of their own nationality either on a working frequency or, when calls to individual ships are made, on the carrier frequency 2 182 kHz.

However, a ship station which keeps watch simultaneously on the carrier frequency 2 182 kHz and a working frequency should be called on the working frequency. As a general rule, coast stations should call radiotelephone ship stations of another nationality on the carrier frequency 2 182 kHz.

Coast stations may call ship stations equipped to receive selective calls in accordance with the provisions of the ITU Radio Regulations.

When a ship is called on the carrier frequency 2 182 kHz, it should reply on the same carrier frequency unless another frequency is indicated by the calling station. When a ship is called by selective calling techniques, it shall reply on a frequency on which the coast station keeps watch. When a ship station is called on a working frequency by a coast station of the same nationality, it shall reply on the working frequency normally associated with the frequency used by the coast station for the call.

When calling a coast station or another ship station, a ship station shall indicate the frequency on which a reply is required if this frequency is not the normal one associated with the frequency used for the call. A ship station which frequently exchanges traffic with a coast station of another nationality may use the same procedure for reply as ships of the coast stations nationality, where this has been agreed by the administrations concerned.

As a general rule a coast station shall reply:

(a) on the carrier frequency of 2 182 kHz to calls made on the carrier frequency 2 182 kHz unless another frequency is indicated by the calling station;
(b) on a working frequency to calls made on a working frequency;
(c) on a working frequency to calls made in Region 1 and 3 and in Greenland on the carrier frequency 2 191 kHz.

7.16.2 Bands between 4 000 kHz and 27 500 kHz

A ship station calling a coast station by radiotelephony shall use either one of the HF calling frequencies listed earlier in section 7.4.2. Alternatively a ship may use the working frequency associated with that of the coast station, that is SSB transmitting frequencies for duplex (two frequency) operation.

A coast station calling a ship station by radiotelephony shall use:

(a) one of the HF calling frequencies listed earlier in section 7.4.2
(b) one of its working frequencies shown in the ITU List of Coast Stations or,
(c) the carrier frequency 4 125 kHz or 6 215 kHz.

The provisions above do not apply to communications between ship stations and coast stations using SSB transmitting frequencies for simplex operation or for intership cross-band operation. A ship station called by a coast station shall reply

either on one of the ship calling frequencies listed in section 7.4.2 or on one of the working frequencies associated with that coast station.

A coast station called by a ship station shall reply either on one of the calling frequencies listed in section 7.4.2 or on one of its working frequencies shown in the ITU List of Coast Stations. The provisions above do not apply to communications between ship stations and coast stations using SSB transmitting frequencies for simplex operation or for intership cross-band operation. When a station is called on the carrier frequency 4 125 kHz or 6 215 kHz, it should reply on the same frequency unless another frequency is indicated for that purpose by the calling station.

7.16.3 VHF band (156–174 MHz)

In the VHF band as a general rule intership calling and coast station to ship calling should be made on Ch.16. However coast station to ship calling may be conducted on a working channel or on a two frequency calling channel designated by administrations for that purpose. Except for distress, urgency or safety communications, when Ch.16 VHF should be used, ship to coast station calling should whenever possible, be made on a working channel or on a two frequency calling channel designated by administrations for that purpose.

When a station is called on Ch.16 VHF it should reply on the same channel unless another channel is indicated by the calling station.

When a coast station open to public correspondence calls a ship either by speech or by selective calling, using a duplex channel the ship station shall reply by speech on the frequency associated with that of the coast station. Conversely, a coast station shall reply to a call from a ship station on the channel associated with the ship station.

7.17 INDICATION OF THE FREQUENCY TO BE USED FOR TRAFFIC

7.17.1 Bands between 1 605 kHz and 4 000 kHz

If contact is established on the carrier frequency 2 182 kHz, coast and ship stations shall transfer to working frequencies for the exchange of traffic.

7.17.2 Band between 4 000 kHz and 27 500 kHz

After a ship station has established contact with a coast station, or another ship station, on the calling frequency of the selected band, traffic shall be exchanged on their respective working frequencies.

7.17.3 VHF band (156–174 MHz)

Whenever contact has been established between a coast station in the public correspondence service and a ship station either on Ch.16 VHF or a two frequency calling channel, both stations shall transfer to one of their normal pairs of working channels for exchange of traffic. The calling station should indicate the channel to which it is proposed to transfer by reference (preferably) to its channel designator.

A ship station when it has established contact with another ship on Ch.16 VHF, should indicate the intership channel to which it proposes to transfer by reference (preferably) to its channel designator.

However, a brief exchange of traffic (not to exceed one minute), concerning the safety of navigation need not be transmitted on a working frequency when it is important that all ships within range receive the transmission. Stations hearing a transmission concerning the safety of navigation shall listen to the message until they are satisfied that the message is of no concern to them. They shall not make any transmission likely to interfere with the message.

7.18 AGREEING THE FREQUENCY TO BE USED FOR TRAFFIC

If the station called is in agreement with the calling station, it shall transmit:

(a) an indication that from that moment onwards it will listen on the working frequency or channel announced by the calling station;

(b) an indication that it is ready to receive traffic of the calling station.

Where the station called is not in agreement with the calling station on the working frequency or channel to be used it shall transmit an indication of the working frequency or channel proposed.

For communications between a coast station and a ship station the coast station shall finally decide the frequency or channel to be used. When agreement is reached regarding the working frequency or channel which the calling station shall use for its traffic, the station called shall indicate that it is ready to receive the traffic.

7.18.1 **Indication of traffic on hand**

When the calling station wishes to exchange more than one radiotelephone call, it should indicate this when contact is established with the station called.

7.19 RECEPTION DIFFICULTIES

If the called station is unable to receive traffic immediately, it should reply in the normal manner but followed by "Wait ... minutes" (or AS spoken as ALFA SIERRA ... (minutes) in case of language difficulties), indicating the probable duration of waiting time in minutes.

Example:

BENGOUR
THIS IS
PENANG RADIO
WAIT FIVE MINUTES
OVER

If the probable waiting time exceeds 10 minutes the reason for the delay shall be given. Alternatively the station called may indicate, by any appropriate means, that it is not ready to receive traffic immediately.

When a station receives a call without being certain that such a call is intended for it, it shall not reply until the call has been repeated and understood.

When a station receives a call which is intended for it, but is uncertain of the identification of the calling station, it shall reply immediately asking for a repetition of the call sign or other identification of the calling station.

Example:

STATION CALLING STARDANIA
THIS IS STARDANIA,
REPEAT YOUR CALL
OVER.

7.20 SENDING TRAFFIC

Every station should transmit its traffic (radiotelephone calls or radiotelegrams) on one of its working frequencies in the band in which the call has been made. In addition to its normal working frequency, printed in heavy type in the ITU List of Coast Stations, a coast station may use one or more supplementary frequencies in the same band.

The use of frequencies reserved for calling shall be forbidden for traffic, except distress traffic. After contact has been established on the frequency to be used for traffic, the transmission of a radiotelegram or radiotelephone call shall be preceded by:

the call sign or other identification of the station called;
the words THIS IS (or DE spoken as DELTA ECHO in case of language difficulties);
the call sign or other identification of the calling station .

The call sign or other identification need not be sent more than once.

7.21 ESTABLISHMENT OF RT CALLS

In setting up a radiotelephone call, the coast station should establish connection with the telephone network as quickly as possible. In the meantime, the ship station shall maintain watch on the appropriate working frequency as indicated by the coast station. However, if the connection cannot be quickly established, the coast station shall inform the ship station accordingly. The latter station shall then either:

(a) maintain watch on the appropriate frequency until an effective circuit can be established; or
(b) contact the coast station later at a mutually agreed time.

When a radiotelephone call has been completed, the end of work signal "OUT" should be sent unless further calls are on hand at either station.

7.22 TRANSMISSION OF RADIOTELEGRAMS BY RT

The transmission of a radiotelegram by RT should be made using the standard format, i.e. preamble, address, text and signature. The preamble of the radiotelegram allows it to be identified by the telegraph service. The standard format of a preamble is as follows; the office of origin (name of ship, etc.), number of the telegram, the number of words, date, time (time telegram was handed in aboard ship), service indicator (usually AAIC).

7.22 *Radiotelephony (RT)*

Radiotelegrams sent from ships to individual coast stations must be numbered in sequence and on a daily basis, with numbering commencing at 0001 hours UTC. Each radiotelegram should be transmitted once only by the sending station. However, it may, when necessary, be repeated in full or in part by the receiving or the sending station.

In transmitting groups of figures by RT, each figure shall be spoken separately and the transmission of each group or series of groups shall be preceded by the words "in figures".

Numbers written in letters shall be spoken as they are written, their transmission being preceded by the words "in letters". See Chapter 9 for further details concerning radiotelegrams.

7.23 SIGNALS USED FOR CLARIFICATION

Occasionally, because of interference or static, the receiving station will have to seek clarification about certain words or groups in the radiotelegram. This is to prevent errors or confusion and to assist in ensuring that the correct information is received. To do this, some standard words or signal procedures are used. These are listed below.

ALL AFTER; ALL BEFORE; CORRECT; CORRECTION; IN FIGURES; IN LETTERS; I SAY AGAIN; I SPELL; OUT; OVER; RADIO CHECK; READ BACK; RECEIVED; SAY AGAIN; STATION CALLING; TEXT; TRAFFIC; THIS IS; WAIT; WORD AFTER; WORD BEFORE; WRONG.

Word	*Meaning*	*Example*
ALL AFTER	Everything after a particular word	Repeat, all after Rotterdam
ALL BEFORE	Everything before a particular word	Repeat, all before Pobol
CORRECTION	Used to correct an error	Six, correction sixty
CORRECT	The information is accurate	Correct
IN FIGURES	The numbers are written in figures	In figures, 60675
IN LETTERS	The numbers are written in letters	In letters, six hundred
I SAY AGAIN	I repeat the information	Rothenburg, I say again Rothenburg
I SPELL	Used to clarify difficult names or groups	CALMAC, I spell, Charlie Alpha Lima Mike Alpha Charlie
OUT	End of work	This is Finnamore, out
OVER	End of own transmission and an invitation to receiving station to transmit	....message ends, over
RADIO CHECK	Used for making test call	Penang Radio this is Finnamore, radio check, over
READ BACK	Repeat by reading back to me, the message I have sent you	Read back message number three
RECEIVED	Acknowledgement of reception, i.e. the message has been correctly received	Received message number two
SAY AGAIN	A request for repetition, or an indication of repetition	Cannon, I say again, Cannon
TEXT	Used to indicate the main text of a radiotelegram	Repeat text, over

Word	*Meaning*	*Example*
TRAFFIC	Used to indicate a telegram or radiotelephone call on hand at a coast station	.....Radio, I have traffic for the following ships.....
WORD AFTER	The word after a particular word in the text	Repeat, word after Rome
WORD BEFORE	The word before a particular word in the text	Repeat, word before New York
WRONG	The information is incorrect or wrong.	Wrong, I say again.....

7.24 ACKNOWLEDGEMENT OF RECEIPT

The acknowledgement of receipt of a radiotelegram or a series of radiotelegrams shall be given by the receiving station in the following manner:

the call sign or other identification of the sending station;
the words THIS IS (or DE spoken as DELTA ECHO in case of language difficulties);
the call sign or other identification of the receiving station;
"Your number.... received, over" (or R spoken as ROMEO ... (number), K spoken as KILO in case of language difficulties); or
"Your number.... to number.... received, over" (or R spoken as ROMEO... (numbers), K spoken as KILO in case of language difficulties).

Example:

BENGOUR
THIS IS
PENANG RADIO
YOUR NUMBER THREE RECEIVED
OVER

The radiotelegram, or series of radiotelegrams, shall not be considered as cleared until this acknowledgement has been received.

The end of work between two stations shall be indicated by each of them by means of the word "Out" (or VA spoken as VICTOR ALFA in case of language difficulties).

7.25 DURATION AND CONTROL OF WORKING

Calling, and signals preparatory to traffic, shall not exceed one minute when made on the carrier frequency 2 182 kHz or on Ch.16 VHF, except in cases of distress, urgency or safety.

In communications between coast stations and ship stations, the ship station shall comply with the instructions given by the coast station in all questions relating to the order and time of transmission, to the choice of frequency, and to the duration and suspension of work.

In communications between ship stations, the station called controls the working, ie frequency to be used, the duration and suspension of work. However, if a coast station finds it necessary to intervene, the ship stations shall comply with the instructions given by the coast station.

7.26 RT TESTS

When it is necessary for a ship station to send signals for testing or adjustments which are liable to interfere with the working of neighbouring coast stations, the consent of these stations shall be obtained before such signals are sent.

When it is necessary for a station to make test signals, either for the adjustment of a transmitter before making a call or for the adjustment of a receiver, such signals shall not be continued for more than ten seconds, and shall include the call sign or other identification of the station emitting the test signals. This call sign or other identification shall be spoken slowly and distinctly.

Any signals sent for testing shall be kept to a minimum, particularly on the frequencies to the maritime mobile and maritime mobile-satellite services for distress and safety purposes.

7.27 ESTABLISHING RT CALLS USING DSC

Calling by digital selective calling techniques and acknowledgement of a received DSC call shall be carried out in accordance with the provisions of the relevant sections of the ITU Radio Regulations. An appropriate digital selective calling channel shall be used for the call (see Chapter 3).

When agreement regarding the RT working frequency or channel to be used for the exchange of traffic has been reached, the two stations then transfer to the working frequency or channel agreed for the exchange of traffic. The forwarding of traffic and the control of working shall be carried out in accordance with standard RT procedures.

7.28 ON BOARD COMMUNICATIONS

Calls for internal communications on board ship when in territorial waters shall consist of:

(a) From the master station:

the name of the ship followed by a single letter (ALFA, BRAVO, CHARLIE, etc.) indicating the sub-station not more than three times;
the words THIS IS;
the name of the ship followed by the word CONTROL.

Example:

TENAKA ALPHA TENAKA ALPHA TENAKA ALPHA
THIS IS
TENAKA CONTROL TENAKA CONTROL TENAKA CONTROL

(b) From the sub-station:

the name of the ship followed by the word CONTROL not more than three times;
the words THIS IS;
the name of the ship followed by a single letter (ALFA, BRAVO, CHARLIE, etc.) indicating the sub-station.

7.29 PILOT SERVICE : CALLING PROCEDURE

A ship station calling stations offering a pilot service should use for the call, in order of preference:

(a) an appropriate channel in the maritime VHF band;
(b) a working frequency in the bands between 1 605 and 4 000 kHz;
(c) the carrier frequency 2 182 kHz and then only to determine the working frequency to be used.

7.30 PORT OPERATIONS AND SHIP MOVEMENT SERVICES

Communications in the port operations service shall be restricted to those relating to operational handling, the movement and the safety of ships and, in emergency, to the safety of persons. Messages of a public correspondence nature are excluded from this service.

Communications in the ship movement service shall be restricted to those relating to the movement of ships. Messages of a public correspondence nature are excluded from this service.

Ships wishing to use the port operations service or ship movement service should call on the appropriate working channel for that service, indicated in heavy type in the ITU List of Coast Stations. When Ch.16 VHF is being used for distress, urgency or safety communications, ships wishing to use the port operations service may establish contact on Ch.12 VHF, or other port operations channel indicated in heavy type in the ITU List of Coast Stations.

When contact on Ch.16 VHF has been established between a coast station in the port operations service and a ship station, the ship station should indicate the particular service required. For example, navigational information, docking instructions etc. The port station shall then indicate the channel to be used for exchange of traffic by reference (preferably) to its channel designator.

When contact on Ch.16 VHF has been established between a coast station in the ship movement service and a ship station, the coast station shall then indicate the channel to be used for exchange of traffic by reference (preferably) to its channel designator.

7.31 AUTOLINK RT SERVICE

Autolink is a means of self-dialling link calls in the MF, HF and VHF bands without engaging the services of a coast radio station operator. It is available via the coast radio stations in many countries throughout the world, see Admiralty List of Radio Signals Volume 1 for details of stations.

An additional item of equipment which is interfaced with the ship's RT transceiver permits the operator to direct dial telephone numbers on a national or international basis. Connection can only be effected by entering the correct PIN number prior to dialling the required telephone number. Each vessel can be allocated up to 99 PIN numbers each of which can be billed separately. Autolink offers the added advantage that RT calls may be scrambled over the radio path hence increasing the security of telephone conversations.

7.32 GUIDANCE ON THE USE OF VHF AT SEA

The following information is extracted from IMO Resolution A.474 (XII). Masters, skippers and owners are urged to ensure that VHF channels are used in accordance with these guidelines.

(1) Preparation

Before transmitting, think about the subject which have to be communicated and, if necessary, prepare written notes to avoid unnecessary interruptions and ensure that no valuable time is wasted on a busy channel

(2) Listening

Listen before commencing to transmit to make certain that the channel is not already in use. This will avoid unnecessary and irritating interference.

(3) Discipline

VHF equipment should be used correctly and in accordance with the Radio Regulations. The following in particular should be avoided:

(a) calling on Channel 16 for purposes other than distress, urgency and very brief safety communications when another calling channel is available;
(b) communication on Channel 70 other than for Digital Selective Calling;
(c) communication not related to safety and navigation on port operation channels;
(d) non-essential transmissions, e.g. needless and superfluous signals and correspondence;
(e) transmitting without correct identification;
(f) occupation on one particular channel under poor conditions;
(g) use of offensive language.

(4) Repetition

Repetition of words and phrases should be avoided unless specifically requested by the receiving station.

(5) Power reduction

When possible, the lowest transmitter power necessary for satisfactory communication should be used.

(6) Communications with shore stations

Instructions given on communication matters by shore stations should be obeyed.

Communications should be carried out on the channel indicated by the shore station.

When a change of channel is requested, this should be acknowledged by the ship.

On receiving instructions from a shore station to stop transmitting, no further communications should be made until otherwise notified (the shore station may be receiving distress or safety messages and any other transmissions could cause interference).

(7) Communications with other ships

During ship-to-ship communications the ship called should indicate the channel on which further transmissions should take place. The calling ship should acknowledge acceptance before changing channel.

The listening procedure outlined above should be followed before communications are commenced on the chosen channel.

(8) Distress communications

Distress calls/messages have absolute priority over all other communications. When hearing them all other transmissions should cease and a listening watch should be kept.

Any distress calls/messages should be recorded in the ship's log and passed to the master.

On receipt of a distress message, if in the vicinity, immediately acknowledge receipt. If not in the vicinity, allow a short interval of time to elapse before acknowledging receipt of the message in order to permit ships nearer the distress to do so.

(9) Calling

Whenever possible, a working frequency should be used. If a working frequency is not available, Channel 16 may be used, provided it is not occupied by a distress call/message.

In case of difficulty to establish contact with a ship or shore station, allow adequate time before repeating the call. Do not occupy the channel unnecessarily and try another channel.

(10) Changing channels

If communications on a channel are unsatisfactory, indicate change of channel and await confirmation.

(11) Spelling

If spelling becomes necessary (e.g. descriptive names, callsigns, words which could be misunderstood) use the spelling table contained in the International Code of Signals and the Radio Regulations.

(12) Addressing

The words "I" and "You" should be used prudently. Indicate to whom they refer.

Example:

"Seaship, this is Port Radar, Port Radar, do you have a pilot?"
"Port Radar, this is Seaship, I do have a pilot."

(13) Watchkeeping

Ships fitted with VHF equipment should maintain a listening watch on Channel 16 and, where practicable, Channel 13 when at sea.

In certain cases Governments may require ships to keep watch on other channels.

CHAPTER 8

General Regulations

8.1 RADIO REGULATIONS

Radio regulations are required to ensure that those who need (especially in emergency situations) or wish to use radio, can do so without causing and suffering from undue interference. The regulations regarding use of maritime radio are set out in the International Radio Regulations, as published by the International Telecommunications Union (ITU), and the Wireless Telegraphy Act of 1949 (WT Act), as amended.

8.2 THE OFFICE OF COMMUNICATIONS (OFCOM)

Ofcom replaced the former Radiocommunications Agency in December 2003. It is responsible for managing most of the civil radio spectrum in the UK and for implementing the Radio Regulations. Ofcom ensures that the appropriate kind of spectrum is available to those who need it and that it is used efficiently whilst causing minimal interference. Together with the ITU, Ofcom agrees maritime frequency bands. It offers advice on the radio service, investigates complaints of radio interference and takes the necessary action to those who may contravene their licence terms and conditions. Its staff carry out routine spot checks to ensure maritime users hold the appropriate radio licences and operator's certificates.

Contact details:

Ofcom
2a Southwark Bridge Road
London SE1 9HA
Telephone: +44 (0) 20 7981 3000 (switchboard)
Facsimile: +44 (0) 20 7981 3333
Local call numbers within the UK: telephone: 0845 456 3000
facsimile: 0845 456 3333

To download free information leaflets and other Ofcom publications contact the website at:

www.ofcom.org.uk

Printed versions can be ordered by e-mail from the contact centre at:

contact@ofcom.org. uk

For information about the International Radio Regulations consult the ITU — contact details given in section 8.14.

8.3 MARITIME AND COASTGUARD AGENCY (MCA)

The Maritime and Coastguard Agency (MCA) was established in April 1998 by merging the former Maritime Safety Agency (MSA) and the Coastguard Agency. The MCA is an executive agency of the Department of the Environment, Transport and the Regions (DETR) and has seven main areas of responsibility : -

(a) Search and Rescue
(b) Ship Safety
(c) Seafarer's Standards
(d) Counter Pollution
(e) Survey and Certification of Ships
(f) Registration of Ships and Seafarers

Included in the above is responsibility for type approval of marine radio and also administration of Marine Radio Operator Certification as detailed in Chapter 12.

Contact details:

Maritime and Coastguard Agency
Tutt Head
Mumbles
Swansea
West Glamorgan
SA3 4HW

The MCA infoline may be contacted as follows:

Telephone: 0870 6006505
e-mail: *infoline@mcga.gov.uk*

For information about the GMDSS or on matters concerning the approval of maritime radio equipment contact:

MCA
Spring Place
105 Commercial Road
Southampton
Hampshire
SO15 1EG
Tel: +44 (0) 2380 329100
Fax: +44 (0) 2380 329252
Web: *www.mcga.gov.uk*

8.4 BREACH OF RADIO REGULATIONS

Where a radio operator or a radio officer infringes any provisions of the Radio Regulations then they are guilty of an offence and liable on summary conviction to a fine. Should the Radio Regulations be contravened in any other respect in relation to the ship, the owner or master shall each be guilty of an offence and liable on summary conviction to a fine or, on conviction or indictment to imprisonment for a term not exceeding two years and a fine. Additionally, the Authority to Operate (ATO) attached to a certificate of competence may be suspended or

revoked thus prohibiting the holder from operating the radio station on a British ship. It shall be a defence in proceeding for an offence under the regulations to prove that all reasonable steps had been taken to ensure that the ship complied with the regulations.

8.5 INFRINGEMENTS OF THE RADIO REGULATIONS

It is the duty of radio operators to report to Ofcom through their employers any infringements of the Radio Regulations which they may detect.

8.6 AVOIDANCE OF INTERFERENCE

All stations are forbidden to carry out:

(a) unnecessary transmissions ;
(b) the transmission of superfluous signals and correspondence ;
(c) the transmission of false or misleading signals ;
(d) the transmission of signals without identification.

Any emission capable of causing harmful interference to distress, urgency or safety communications on the frequencies: 518 kHz, 2 174.5 kHz, 2 182 kHz, 2 187.5 kHz, 4 125 kHz, 4 177.5 kHz, 4 207.5 kHz, 4 209.5 kHz, 6 215 kHz, 6 268 kHz, 6 312 kHz, 8 291 kHz, 8 376.5 kHz, 8 414.5 kHz, 12 290 kHz, 12 520 kHz, 12 577 kHz, 16 420 kHz, 16 695 kHz, 16 804.5 kHz, 121.5 MHz, 156.525 MHz (Ch. 70), 156.8 MHz (Ch. 16), or in the frequency bands 406–406.1 MHz, 1 544-1 545 MHz and 1 645.5-1 646.5 MHz is prohibited.

8.7 MASTER'S AUTHORITY

The radio service of a ship is placed under the supreme authority of the master or person responsible for the ship or other vessel carrying the radio station.

The person holding this authority shall require that each operator comply with the international Radio Regulations and that the ship station or mobile earth station for which the operator is responsible is used, at all times, in accordance with those regulations.

8.8 RADIO SECRECY

The master or person responsible, as well as all persons who may have knowledge of the text or even of the existence of a radiotelegram, or of any other information whatever obtained by means of the radiocommunication service, are placed under the obligation of observing and ensuring the secrecy of correspondence. These provisions shall also apply to personnel who operate mobile earth stations.

By international regulation administrations agree to take necessary steps to prohibit and prevent:

(a) the unauthorised interception of radiocommunications not intended for general use of the public.
(b) the divulgence of the contents, simple disclosure of the existence, publication or any use whatever, without authorisation of information of any

nature obtained by the interception of the radiocommunication mentioned in (a) above.

On UK ships, communications staff must therefore take all possible steps to preserve the confidentiality of international radio communications as outlined in Section 11 of the Post Office (Protection) Act of 1884, and Part 1, Section 5 of the Wireless Telegraphy Act 1949 and subsequent legislation. A copy of the 1884 Act shall be exhibited near to the radiocommunications equipment.

To protect correspondence it is forbidden to intercept radiocommunications except for those for which the ship radio station is authorised. Where public correspondence is inadvertently received its existence must be denied, it must not be reproduced, disclosed to other parties or used in any other way.

8.9 SHIP RADIO LICENCE

The International Radio Regulations requires that any vessel carrying maritime radio transmission equipment must be licensed. This regulation is implemented in the UK by Ofcom (see section 8.2). A ship Radio Licence is required if you have maritime radio equipment available for use on board a vessel. It is an offence to install or use any radio equipment on board a UK registered ship, craft, hovercraft, lightship, oil rig or any other ship predominantly moored in UK waters without a ship radio licence. This includes vessels registered in the Channel Islands and the Isle of Man. A licence is required even if the transmitting equipment is not in constant use, or if it is used only for distress purposes. For example even if EPIRBs are the only type of radio equipment a ship radio licence will still be required.

The Ship Radio Licence allows personnel to install and use any combination of maritime radio equipment and covers the following:

(a) Terrestrial radio equipment in MF, HF and VHF maritime bands.
(b) Mobile earth stations (Inmarsat A, B, C, F and M).
(c) Marine radar in 3 cm (X) and 10 cm (S) bands.
(d) Low powered on board portable communications equipment.
(e) On board repeater stations.
(f) Marine EPIRBs.

The licence does not permit the use of these items on land.

Where VHF and UHF transportables are not dedicated to a particular vessel they can be used on a number of different vessels. In this case a separate licence will be required to cover the transportable only in which case a callsign will not be issued, instead it is issued with a T reference. It is usual to use the vessel name as identity. Calls through coast stations are not permitted. Other items of equipment that can be used with a Portable Maritime Licence are: Portable VHF VHF DSC Radio, 121.5 MHz Personal Locator Beacon, and 406 MHz/121.5 MHz Personal Locator Beacon. A separate fee is charged for each transportable.

The Licence comprises:

(a) Licence document.
(b) Licence terms booklet.
(c) Ship Licence disc.

The disc shows the date of expiry and a licence number and is renewed annually on payment of the appropriate fee. The fee payable depends upon whether the radio equipment is fitted on a vessel used solely for pleasure purposes or other (standard) purposes.

The Licence shows the following information:

(a) name of the ship (or registration).
(b) callsign or other identification and owner of the ship.
(c) the public correspondence category.
(d) Maritime Mobile Service Identity (MMSI).
(e) Selective Calling Number (Selcall).

Where possible the ship radio licence disc should be kept with the terms booklet and on display in a prominent position on the port side of the vessel. The licence document and any notices of variation should be kept adjacent to the ships' radio communications equipment.

The licence must be kept in such a way that it can be produced upon request for inspection by the competent authorities in ports at which the ship calls.

For all enquiries contact:

Ship Radio Licensing Centre
PO Box 1495
Bristol
BS99 3QS
Telephone: 0870 243 4433
Facsimile: +44 (0) 117 975 8911
Web: *www.radiolicencecentre.co.uk*

8.10 INSPECTION AND SURVEY OF SHIP RADIO AND SHIP EARTH STATIONS

All ship radio stations and ship earth stations are subject to inspection by officers appointed for that purpose by the Secretary of State for Trade and Industry in order to ascertain that the conditions imposed by the Ship Radio Licence are being met. Ship radio stations and ship earth stations which are required by the Merchant Shipping Acts and the Merchant Shipping Regulations made thereunder to be equipped with a radio installation are surveyed annually by a radio surveyor appointed by the Department of Transport for the purpose of renewal of the appropriate Safety Radio Certificate.

Inspections to ensure compliance with the Ship Radio Licence under the Wireless Telegraphy Act 1949 are conducted separately from those undertaken by the Radio Surveyors who check that the installations meet the safety requirements of the Merchant Shipping Acts and Regulations. Ship Radio Licence inspections are made randomly, nationwide, although in general ships' Masters can expect no more than one such inspection per year.

It is the duty of the master or person responsible for the ship to permit any person acting on behalf of the Secretary of State to have access to the ship station at all reasonable times for the purpose of inspection and/or survey and testing of the radio apparatus. The inspectors shall have in their possession an identity badge or card issued by the competent authority, which they shall show on request of the

master or person responsible for the ship or other vessel carrying the ship station or ship earth station.

The competent authorities of any port state where a ship calls may require the production of the Safety Radio Certificate and the certificates of competence of the radio personnel. However, proof of professional knowledge may not be demanded. On failure to produce these documents or where manifest irregularities are observed, the authorities can inspect the radio apparatus in order to satisfy themselves that it conforms to the requirements of the International Radio Regulations and, if appropriate, the International Safety of Life at Sea Convention then in force. A responsible representative of the ship owner who can operate the equipment should be in attendance, if possible, during an inspection. This ensures that equipment is not accidentally damaged and that personnel are not put at risk. If no representative is available the surveyor will not proceed until he has the agreement of the Master and considers it safe to do so.

Before leaving, the inspector shall report the result of his survey to the Master, or person responsible for the ship. If any breach of the Regulations is observed, the inspector shall make this report in writing to the owner or his representative.

The Department of Transport (Maritime and Coastguard Agency), has appointed Marconi Selenia Communications Limited to conduct surveys and certification of radio installations on UK registered vessels in UK ports from the 1st November 2001. Marconi Selenia will also, on request, survey UK registered vessels overseas and assist in Port State Control inspections of foreign ships as required by the local Marine Office. The same company will also conduct surveys and certification of radio installations on non-UK registered vessels in UK ports at the request of the Flag Administration.

To assist with applications for a radio survey for ships within ports of the UK, Marconi Selenia have established dedicated central administrative centre at Chelmsford to coordinate bookings of surveys, issue of certificates and fee collection. Contact details:

Marconi Selenia Communications Ltd.
Marconi House
New Street
Chelmsford
Essex
CM1 1PL
United Kingdom
Telephone: +44 (0) 1245 275888
Facsimile: +44 (0) 1245 275500
e-mail: *marine-surveys@marconiselenia.com*
Web: *www.marconi-marine.com*

Further details of the conduct and booking of radio surveys in the UK are give in Maritime and Coastguard Agency's Marine Guidance Note, MGN 206 (M + F).

8.11 RADIO DOCUMENTATION

Ships complying with the ITU Radio Regulations should carry the following documentation:

(a) The Safety Radio Certificate, record of equipment or radio facilities or report of inspection — radio, as appropriate.
(b) the Ship Radio Licence.
(c) the certificate(s) of the operator(s).
(d) the radio log book(s).
(e) details of the radio operating procedures to be used in the maritime mobile bands for which the vessel is equipped, together with detailed explanations of distress and safety communication procedures, and
(f) documents required by the ITU Radio Regulations appropriate to the class of ship.

Each GMDSS ship should keep a radio log in which details as specified in the Radio Regulations and details of equipment and reserve power supply checks are recorded.

8.12 DOCUMENTS TO BE CARRIED ON UK SHIPS

UK registered ships must carry documents according to radio installation as follows:

(a) The Ship Radio Licence.
(b) The radio certificate(s) of the operator(s).
(c) GMDSS radio logbook.
(d) ITU List of Callsigns and/or Numerical Table of Identities of Stations used by the Maritime Mobile and Maritime Mobile-Satellite Services.
(e) Particulars of coast stations and land earth stations participating in GMDSS; a list of coast stations and land earth stations with which communications are likely to be established, showing watchkeeping hours, frequencies and charges; and a list of coast stations and land earth stations providing navigational and meteorological warnings and other urgent information for ships.
(f) ITU List of Ship Stations.
(g) ITU Manual for use by the Maritime Mobile and Maritime Mobile-Satellite Services.
(h) Admiralty List of Radio Signals as detailed in section 8.13.

8.13 ADMIRALTY LISTS OF RADIO SIGNALS (ALRS)

The Hydrographer of the Navy publishes information of interest to users of the maritime radio service in six volumes. Weekly corrections are issued free of charge to keep information updated and may be obtained from agents for the sale of Admiralty Charts.

ALRS Volume 1: Coast Radio Stations (Parts 1 and 2)

This volume is produced annually in two parts giving worldwide coverage and contains the following:

(a) Maritime Public Correspondence Stations listings.
(b) Details on how to use Inmarsat satellite services.

(c) Ship Reporting Systems including AMVER.
(d) Piracy and Armed Robbery Reporting Procedures.
(e) Station identity by DSC MMSI numbers.

ALRS Volume 2: Radio Aids to Navigational, Satellite Navigation Systems, Legal Time, Radio Time Signals and Electronic Position Fixing Systems

This volume is revised annually and includes the following:

(a) Listings of all Radio Direction-Finding Stations.
(b) Listings of all Radar Beacon services.
(c) Major Satellite Navigation Systems.
(d) Listings of DGPS transmitting beacons.
(e) International Standard and Daylight Saving Times and Dates.
(f) International Time Signal Broadcast details.
(g) Electronic Position Fixing System details.

ALRS Volume 3: Maritime Safety Information Services (Parts 1 and 2)

This volume is produced in two parts giving worldwide coverage of the following:

(a) Maritime Radio Weather Services.
(b) Maritime Safety Information (MSI) broadcasts.
(c) Listings of NAVTEX stations.
(d) Submarine and Gunnery Warning details (Subfacts and Gunfacts).
(e) Listings of Radio Facsimile stations.

ALRS Volume 4: Meteorological Observation Stations

This volume is revised every 18 months and gives details of the following:

(a) All Meteorological Observations Stations

ALRS Volume 5: Global Maritime Distress and Safety System (GMDSS)

This volume is revised annually and contains:

(a) Worldwide communication requirements for distress, search and rescue.
(b) A SOLAS guide for GMDSS.
(c) A chart showing world coverage of all Maritime Search and Rescue Regions (SRR).

ALRS Volume 6: Pilot Services, Vessel Traffic Services and Port Operations (Parts 1, 2, 3, 4, and 5)

This five part volume is revised annually and contains the following:

(a) Pilot Information.
(b) Vessel Traffic Service Information.
(c) Diagrams depicting Reporting Points, Pilot Boarding Positions and Ports.
(d) Listings of Harbour VHF facilities.

8.14 ITU DOCUMENTS

The International Telecommunications Union based in Geneva publish several documents of interest to users of the maritime mobile and maritime mobile-satellite services, some of which are compulsorily carried on ships. (See sections 8.11 and 8.12). The ITU contact details:

ITU
Place des Nations
CH 1211
Geneva 20
Switzerland
website: *www.itu.int*

ITU List of Coast Stations

This document is published every two years and kept up to date by three semi-annual supplements. It contains particulars of coast stations and coast earth stations providing a public correspondence service with their charges and accounting authorities. An annex is included giving details of different categories of stations participating in the GMDSS.

ITU List of Ship Stations

This document is published every year. It is available in either paper or CD-ROM format as follows:

Part A (Paper format): This contains particulars of those ships for which a MMSI number has been notified to the Radiocommunication Bureau, with the exception of yacht pleasure ships (i.e. ship classification of PL-YAT in column 5 of the List) unless advised otherwise by the responsible Administration. This part is kept up to date by means of two quarterly supplements in addition to a half-yearly recapitulative supplement.

Part B (CD-ROM format): This part contains particulars of all ship stations notified to the Radiocommunication Bureau, with or without a MMSI number, including those installed on board yacht pleasure ships. The information is presented in pdf format. The CD-ROM also contains all ship stations in a database format with associated software interface. This part is kept up to date by means of quarterly publications on CD-ROM which will contain recapitulative information on the particulars of all ship stations at the time of printing the supplement in pdf format as well as all ship stations in a database format.

ITU List of Callsigns and Numerical Identities of Stations Used by the Maritime Mobile and Maritime Mobile-Satellite Services

This document is published every two years and kept up to date by recapitulative supplements every three months. It contains callsigns and other alphanumeric identities (coast station identification numbers, ship station selective call numbers, MMSI's, national system telex numbers and Inmarsat IMNs) used in the emission of maritime radio beacons and stations of the Maritime Mobile and Maritime Mobile-Satellite Services. Ship stations and ship earth stations con-

tained in this List are those for which a MMSI number has been notified, with the exception of pleasure ships with the individual classification of yacht (PL-YAT) in the ITU List of Ship Stations.

ITU List of Radiodetermination and Special Service Stations

This document is published at intervals determined by the Secretary-General of the ITU, normally about every three and a half years. It is kept up to date by recapitulative supplements issued every six months and contains particulars of radio direction-finding stations and radiobeacon stations of the maritime radionavigation service and the particulars of radiodetermination-satellite systems available for maritime use, ocean station vessels, direction-finder calibration stations as well as stations transmitting standard frequency and time signals, regular meteorological bulletins, notices to navigators, medical advice and ursigrams.

ITU Manual for Use by the Maritime Mobile and Maritime Mobile-Satellite Services

This document is revised as needed, especially after administrative conferences and Plenary Assemblies of the CCITT and/or the CCIR. New editions are published at intervals to be determined by the Secretary-General. The Manual is available in printed format in English, French and Spanish or in CD-ROM (containing all three languages) and contains extracts from:

(a) the Constitution of the International Telecommunications Union.
(b) the Radio Regulations in force.
(c) the ITU-R Recommendations.
(d) the International Telecommunications Regulations.
(e) the ITU-T Recommendations.
(f) the Preface to List VIIA.

8.15 RADIO WATCH

Every GMDSS fitted ship while at sea shall maintain a continuous watch

(a) on VHF DSC distress, safety and calling channel 70, if the ship is fitted with a VHF radio installation as required by the radio regulations;
(b) on the distress and safety frequency of 2 187.5 kHz if the ship is required by the radio regulations to carry an MF DSC installation;
(c) on the distress and safety DSC frequencies of 2 187.5 kHz, and 8 414.5 kHz and also at least one other HF DSC distress and safety frequency from the following; 4 207.5 kHz, 6 312 kHz, 12 577 kHz or 16 804.5 kHz, appropriate to the time of day and the geographical position of the ship, if the ship is required by the regulations to carry an MF/HF radio installation ; this watch can be kept by means of a scanning receiver;
(d) for satellite shore-to-ship distress alerts, if the ship is fitted with an Inmarsat mobile earth station.

All GMDSS fitted ships while at sea will maintain a radio watch for broadcasts of maritime safety information (MSI) on the appropriate frequency or frequencies on which such information is broadcast for the area in which the ship is sailing.

Until 1st February 2005 every ship while at sea shall maintain, when practicable, a continuous listening watch on the bridge on Ch. 16 VHF; such watch shall be kept at the position from which the ship is normally navigated.

All GMDSS vessels are required to maintain watch on the bridge to bridge safety of navigation frequency Ch. 13 VHF when practicable whilst at sea.

8.16 RADIO LOG BOOKS

A radio logbook should be maintained on all vessels carrying compulsorarily fitted radio.

The GMDSS Radio Logbook incorporates instructions for its completion to meet the legal requirements. It must be kept on the navigating bridge convenient to the radio installation and should be made available for inspection by a radio surveyor appointed by Marconi Selenia or any officer authorised by the Department of Transport. The logbook comprises three sections and an annex as follows:

Section A: Particulars of ship

1 (a) Ship's name.
(b) International callsign .
(c) MMSI number.
(d) Port of Registry.
(e) IMO number.
(!) Gross Tonnage.
(g) Date keel was laid.
(h) Sea area(s) in which ship is certified to operate.
(i) Date of expiry of current Safety Radio Certificate.

2 (a) Method(s) used to ensure availability of radio facilities:
(i) Duplication of equipment (DOE).
(ii) Shore-based maintenance giving details of name and address of service company (SBM).
(iii) At-sea maintenance capability (ASM).
(b) Name and address of owner, managing owner or agent.

Section B: Qualified Personnel

(1) Name(s).
(2) Dates on board or dates of attachment.
(3) Class(es) of Certificate and date(s) of issue.
(4) Number(s) of Certificate(s) .
(5) Name and rank, or muster number, of designated person(s) with responsibility for radio communications during emergencies.
(6) Name of person(s) nominated to carry out appropriate tests and checks and log entries (if different from 5 above).

Section C

(1) Period of log: From.....To.....
(2) Details of disposal of log to Superintendent of marine office together with his signature and that of the master.

Section 3 of the GMDSS radio logbook comprises the diary record of the operation of the radio installation. Entries in this part are to be prepared in duplicate. The summary column must include the following:

(1) A summary of communications relating to distress, urgency and safety traffic. The summary must include dates and times, details of the vessels involved and their positions.

(2) A record of important incidents connected with the radio service. For example:
   (a) a breakdown or serious malfunction of the equipment.
   (b) a breakdown of communications with coast stations, coast earth stations or satellites.
   (c) adverse propagation conditions, i.e. ionospheric, static, atmospheric noise, general interference.
   (d) serious breaches of radio procedures by other stations.
   (e) any significant incidents concerning the exchange of commercial traffic such as disagreements over charges, the non-receipt of messages and so on.

(3) The position of the ship at least once a day. The position may be given relative to a geographical point, if appropriate, as an alternative to its latitude and longitude.

(4) The details of the daily, weekly and monthly tests carried out as identified in the Annex.

The Annex

This contains details of the daily, weekly and monthly tests to be carried out on GMDSS radio equipment — see section 8.20.

Notes on keeping the log

The Master must nominate one or more crew members, normally the person(s) qualified for distress and safety radio communications to maintain the log and to carry out the tests. Distress and distress-related communications received as hard copy via Navtex, EGC or telex should be noted in the log and filed in date order at the rear of the log.

If any radio equipment is found not to be in full working order the nominated person must notify the Master and record details of the deficiencies in the log.

The master must inspect and sign each day's entries in the GMDSS radio logbook.

The duplicates of Section C (carbon copy perforated sheets) must be detached and carefully fastened together in the correct order to form the record of the operation of the radio installation. They should finally be disposed of in the manner directed by the operating company or the shipowner as the case may be.

The Master shall then deliver the completed Log to the appropriate Superintendent of a Marine Office, or if abroad, to the appropriate officer, at the same time as the official log book; that is within 48 hours of the last person discharged from the crew agreement, or in any other case, when the ship first calls at a port more than 6 months after the first entry in the official log book.

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8.17 COORDINATED UNIVERSAL TIME (UTC) AND DATES USED FOR RADIOCOMMUNICATIONS

Any date used in relation to radiocommunication shall be according to the Gregorian Calendar. If in a date the month is not indicated either in full or in an abbreviated form, it shall be expressed in an all-numeric form with the fixed sequence of figures, two of each representing the day, month and year.

Whenever a date is used in connection with Coordinated Universal Time (UTC), this date shall be that of the prime meridian at the appropriate time, the prime meridian corresponding to zero degrees geographical longitude.

Whenever a time is used in international radiocommunication activities, UTC shall be applied, unless otherwise indicated, and shall be presented as a four digit group (0000-2359). The abbreviation UTC shall be used in all languages.

Coordinated Universal Time reckoned from 0000 to 2359 h beginning at midnight, shall be used for all entries in the radiocommunication service log and in all similar documents of ships compulsorily equipped with radiocommunications apparatus in compliance with an international agreement. This same provision will apply, as far as possible, to other ships. Every station in the maritime mobile and maritime mobile-satellite service shall have an accurate clock correctly regulated to UTC.

8.18 RADIO TIME SIGNALS

These are transmitted by various coast stations worldwide. The signals originate from Observatories, National Physical Laboratories or Hydrographic Institutes etc. of the host nation. The standards used conform to those set by the International Time Bureau (Bureau International de l' Heure BIH). Using an atomic resonant frequency based on caesium standards they provide a very precise unit of time interval known as International Atomic Time (TAI).

Most time signal transmissions are organised to give Coordinated Universal Time (UTC), this meets the requirements of navigators and others who require an extremely accurate scale of time interval directly related to the earths rotation. However many time signal transmissions contain a code to allow the difference between Astronomical Time (UTI) and UTC to be identified. This is known as Difference Universal Time International (DUTI) and can be regarded as a correction to the broadcast UTC signal. This difference which can amount to a maximum of 0.7s a year is indicated by a series of double pulses within the time signal sequence.

It should be noted that within the sophisticated time signal sequence, provision is made for identifying the correct minute and the correct second. There may be breaks within the sequence to include station identification signals using either Morse code or normal voice transmission.

Time signals are transmitted on various high frequencies but of particular use are those which are sent out on "standard frequencies". Signals transmitted on the Standard Frequency and Time Signal service provide a radiocommunication service for scientific, technical and other purposes. They offer transmissions on specified frequencies, time signals, or both for general reception. These frequencies are very highly stable being derived from atomic sources, thus the accuracy

of the ship's receiving apparatus can be monitored. Such signals can be received on 2.5, 5, 10, 15 and 20 MHz.

In addition to the stations referred to above, many national and local broadcasting authorities transmit time signals at specified times. Details of stations sending time signals, together with information about their frequencies and sequences can be obtained from the ITU List of Radiodetermination and Special Service Stations and Admiralty List of Radio Signals Volume 2. (See sections 8.13 and 8.14).

8.19 TEST TRANSMISSIONS

Test transmissions shall be kept to a minimum on the distress and safety frequencies identified in section 8.6 and should, wherever practicable, be carried out on artificial antennas with reduced power. When it is necessary for a ship to send signals for testing or adjustments which are liable to interfere with the working of neighbouring coast stations, the consent of these stations shall be obtained before such signals are sent.

When it is necessary for a station to make test signals, either for the adjustment of a transmitter before making a call or for the adjustment of a receiver, such signals shall not be continued for more than ten seconds, and shall include the call sign or other identification of the station emitting the test signals. This call sign or other identification shall be spoken slowly and distinctly.

8.20 TESTING OF RADIO EQUIPMENT

The testing of compulsorily fitted radio equipments is governed by regulation. Under Schedule 2 to S.I. No. 2070 of 1998 (The Merchant Shipping (Radio Installations) Regulations 1998) the following tests should be carried out on GMDSS fitted ships:

Daily tests:

(a) The proper functioning of the DSC facilities shall be tested at least once per day, without radiation of signals, by use of the means provided by the equipment.

(b) Batteries providing a source of energy for any part of the radio installations shall be tested daily and, where necessary, brought up to the fully charged condition.

Weekly tests:

(a) The proper operation of the MF/HF DSC facilities shall be tested at least once per week by means of a test call, when within communication range of a coast station fitted with DSC equipment. Where a ship has been out of communication range of a coast station fitted with DSC equipment for a period longer than a week, a test call shall be made on the first opportunity that the ship is in communication range of such a coast station.

Note: Live tests should not be made on VHF DSC equipment.

(b) Where the reserve source of energy is not a battery (for example, a motor generator), the reserve source of energy shall be tested weekly.

Monthly tests:

(a) Each EPIRB and satellite EPIRB shall be examined at least once a month to determine its capability to operate properly; particularly its ability to float free (where required to do so) in the event of the ship sinking, its security and for signs of damage. See also sections 4.11 and 4.12.

(b) Each search and rescue radar transponder (SART) shall be checked at least once per month for security and signs of damage (see section 4.15).

(c) Each survival craft two-way VHF equipment shall be tested at least once per month on a frequency other than Ch. 16 VHF.

(d) A check shall be made at least once per month on the security and condition of all batteries providing a source of energy for any part of a radio installation. The battery connections and compartment shall also be checked.

The above tests are a legal requirement. However, the Annex to the UK GMDSS radio logbook as published by the MCA recommends the following additional tests:

(a) Printer(s) should be checked daily to ensure there is an adequate supply of paper.

(b) A check should be made at least once per month on the condition of all aerials and insulators.

Annual tests:

For details of annual tests to EPIRBs see section 4.12.

8.21 CONTROL OF WORKING

In communication between coast stations and ship stations, the ship station shall comply with the instructions given by the coast station, in all questions relating to the order and time of transmission, to the choice of frequency and class of emission, and to the duration and suspension of work.

In communication between ship stations, the station called shall control the working. However, if a coast station finds it necessary to intervene, the ship stations shall comply with the instructions given by the coast station.

Note: the provisions of this section are not applicable in the cases of distress, urgency or safety.

8.22 ORDER OF PRIORITY OF COMMUNICATIONS IN THE MARITIME MOBILE AND MARITIME MOBILE-SATELLITE SERVICES

The order of priority for communications in the maritime mobile and maritime mobile-satellite service shall be as follows, except where impracticable in a fully automated system in which, nevertheless, distress calls, messages and traffic shall receive priority as follows:

(1) Distress calls, distress messages and distress traffic.
(2) Communications preceded by the urgency signal.
(3) Communications preceded by the safety signal.
(4) Communications relating to radio direction-finding.

(5) Communications relating to the navigation and safe movement of aircraft engaged in search and rescue operations.

(6) Communications relating to the navigation, movements and needs of ships and aircraft, and weather observation messages (OBS) destined for an official meteorological service

(7) ETATPRIORITE - Radiotelegrams relating to the application of the United Nations charter.

(8) ETATPRIORITE - Government radiotelegrams with priority and government calls for which priority has been expressly requested.

(9) Ordinary private radiotelegrams and Red Cross Telegrams for which priority has been requested.

(10) Service communications relating to the working of the telecommunications service or to communications previously exchanged.

(11) Government communications other than those shown in *8* above, ordinary private communications and Red Cross Telegrams.

(12) Radiomaritime letters.

8.23 CLOSURE OF SHIP RADIO STATIONS

Any ship station arriving in port, and whose service is therefore about to close, shall:

(a) notify accordingly the nearest coast station and, if appropriate, the other coast stations with which it generally communicates.

(b) not close until after the disposal of traffic on hand, unless this conflicts with the regulations in force in the country of the port of call.

On departure from port the ship station shall notify the coast station or stations concerned that its service is reopening as soon as such reopening is permitted by the regulations in force in the country of the port of departure. However, a ship station not having hours of service fixed by the ITU Radio Regulations may defer such notification until the station first reopens its service after departure from port.

8.24 IDENTIFICATION OF STATIONS

All transmissions from ships shall be capable of being identified. All transmissions with false or misleading identification are prohibited. Such identification may be one or more of the following:

(a) Land and fixed station

— a callsign of two characters and one letter or,

— two characters and one letter followed by not more than three digits (other than the digits 0 and 1 in cases where they immediately follow a letter).

Stations using telephony shall be identified as above; or by

- the geographical name of the place as it appears in the ITU List of Coast Stations, followed preferably by the word "RADIO" or by any other appropriate indication.

(b) Ship stations

— two characters and two letters, or
- two characters, two letters and one digit (other than the digits 0 and 1).

Stations using telephony shall be identified as above; or by

- the official name of the ship preceded, if necessary, by the name of the owner on condition that there is no possible confusion with distress, urgency and safety signals;
  or by
— its selective call number or signal; or
— two characters (provided that the second is a letter) followed by four digits (other than 0 or 1 in cases where they immediately follow a letter); or
— two characters and one letter followed by four digits (other than 0 or 1 in cases where they immediately follow a letter).

Stations using digital selective calling techniques:

— the ships MMSI number (see section 8.26).

Ship earth stations:

- the ships 7-digit Inmarsat-A IMN or 9-digit Inmarsat-B, C, F or M IMN (see section 5.6).

(c) Aircraft stations

— two characters and three letters.

Stations using telephony shall be identified as above; or by

— its callsign, which may be preceded by a word designating the owner or the type of aircraft; or
— a combination of characters corresponding to the official registration mark assigned to the aircraft; or
— a word designating the airline, followed by the flight identification number.

(d) Ship's survival craft stations

— the callsign of the parent ship followed by two digits (other than 0 and 1 in cases where they immediately follow a letter).

Stations using telephony shall be identified as above; or by

— a signal or identification consisting of the name of the parent ship followed by two digits.

(e) EPIRB stations

— the name and/or callsign of the parent ship to which the radiobeacon belongs; or

For satellite EPIRBs

- the 3-digit MID to identify the country in which the vessel is registered followed by either:
  — the final 6 digits of the vessels MMSI number; or
  — a unique four-digit code recorded with EPIRB manufacturer and the MCA, or
  — the ship's callsign.

(f) Aircraft survival craft stations

— the complete callsign of the parent aircraft, followed by a single digit other than 0 or 1.

(g) Land mobile stations

— two characters (provided that the second is a letter) followed by four digits (other than the digits 0 and 1 in cases where they immediately follow a letter), or

— two characters and one or two letters followed by four digits (other than the digits 0 or 1 in cases where they immediately follow a letter).

Stations using telephony shall be identified as above; or by

- the identity of the vehicle or any other appropriate indication.

(h) Amateur and experimental stations

— one character and a single digit (other than 0 or 1) followed by a group of not more than three letters, or

— two characters and a single digit (other than 0 or 1), followed by a group of not more than three letters.

Note: The prohibition of the use of the digits 0 and 1 does not apply to amateur stations.

(i) Stations in the space service

When callsigns for stations in the space service are employed, it is recommended they consist of:

— two characters followed by two or three digits (other than the digits 0 or 1 in cases where they immediately follow a letter).

(j) Base stations

— a callsign; or

— the geographical name of the place followed, if necessary, by any other appropriate indication.

(k) Aeronautical stations

— the name of the airport or geographical name of the place followed, if necessary, by a suitable word indicating the function of the station.

8.25 SELECTIVE CALL NUMBERS IN THE MARITIME MOBILE TELEX SERVICE

(a) Coast station selcall numbers:

- four digits

(b) Ship station selcall numbers

— five digits

(c) Predetermined groups of ship stations

— five digits consisting of:
  — the same digit repeated five times; or
  — two different digits repeated alternately.

8.26 MARITIME MOBILE SERVICE IDENTITIES (MMSIs) IN THE MARITIME MOBILE AND MARITIME MOBILE-SATELLITE SERVICES

(a) Ship station identities

a 9-digit code formed as follows:

M I D X X X X X X

— where M I D consists of three digits representing the country (see Appendix 7), (i.e. the UK allocation is 232, 233, 234 and 235),
— and X X X X X X consists of six digits indicating the particular ship.

(b) Group ship station identities

Group ship station identities are used for calling simultaneously more than one ship and are a 9-digit code formed as follows:

O M I D X X X X X

— where the first figure is zero.
— M I D represents a three-digit country code (see Appendix 7)
— X X X X X identifies the particular group of ships.

Note: The particular MID represents only the country assigning the group ship station call identity and so does not prevent group calls to fleets containing more than one ship nationality.

(c) Coast station identities

a 9-digit code formed as follows:

O O M I D X X X X

— where the first two figures are zeros.
— M I D represents the coast station country code (see Appendix 7)
— X X X X represents four digits to identify the particular coast station.

(d) Group coast station call identities

Group coast station identities are for calling simultaneously more than one coast station and are a 9-digit code formed as follows:

O O M I D X X X X

— where the first two figures are zeros.
– M I D represents the coast station country code (See Appendix 7)
— X X X X represents four digits to identify the particular group of coast stations.

8.27 CLASSES OF EMISSION

The basic characteristics are:

(1) The first symbol — type of modulation of the main carrier.
(2) The second symbol — nature of signal(s) modulating the carrier.
(3) The third symbol — type of information to be transmitted.

(1) The first symbol

N Unmodulated carrier.

The following are forms of amplitude modulation:

A Double sideband.
H Single sideband full carrier.
R Single sideband reduced carrier.
J Single sideband suppressed carrier.
B Independent sidebands.
C Vestigial sidebands.

The following are where the main carrier is angle-modulated:

F Frequency modulation.
G Phase modulation.
D Amplitude and angle modulation either simultaneously or in a preestablished sequence.

The following are forms of pulse modulation:

P Unmodulated pulses.
K Pulse amplitude.
L Pulse width/duration.
M Pulse position/phase.
Q Angle modulation during the period of the pulse.
V Combination of foregoing or produced by other means.

Special types of modulation:

W Modulation in a combination of two or more modes.
X Cases of modulation not otherwise covered.

(2) The second symbol

0 No modulating signal.
1 Single channel containing quantised or digital information without the use of a modulating sub-carrier.
2 Single channel containing quantised or digital information with the use of a modulating sub-carrier.
3 Single channel containing analogue information.
7 Two or more channels containing quantised or digital information.
8 Two or more channels containing analogue information.
9 Composite system with one or more channels containing quantised or digital information, together with one or more channels containing analogue information.
X Cases not otherwise covered.

(3) The third symbol

N No information transmitted.
A Telegraphy — for aural reception (i.e. Morse).
B Telegraphy for automatic reception (i.e. Telex, D SC, etc.).
C Facsimile.
D Data transmission, telemetry, telecommand.
E Telephony (including sound broadcasting).
F Television (video).

W Combination of the above.
X Cases not otherwise covered.

There are optional additional characters (fourth and fifth symbols) to represent details of signal(s) and nature of multiplexing respectively.

Following are examples of classes of emission commonly used in marine radiocommunications:

(1) Radiotelephony

J3E Single sideband suppressed carrier.
H3E Single sideband full carrier (permitted on 2 182 kHz only).
F3E Frequency modulation.
G3E Phase modulation.

(2) Radiotelex and DSC

F1B Frequency shift keying of carrier with error correction.
J2B Frequency shift keying of sub-carrier with error correction.
G2B Phase modulation. A single channel containing quantised or digital information with the use of a modulating sub-carrier.

8.28 RADIO TRANSMISSIONS BY SHIPS ON UK INLAND WATERWAYS AND IN UK HARBOURS

The use of radio apparatus licensed by Ofcom and installed on board ships is permitted in the harbours and estuaries of the UK, subject to the conditions of the ship radio licence and for the following purposes only: -

(a) For the exchange of communications through coast stations.
(b) For radiodetermination and for the reception of messages sent from authorised broadcasting stations for general reception.
(c) for the exchange of communications in a port operations service, or in a private mobile service licensed by Ofcom for a specific purpose.

On inland waterways, radio apparatus may be used as above and also for inter-ship communication when the ships communicating are under way. Communication between a ship underway and a ship at a berth or anchorage is not permitted other than through a coast radio station. Lock stations which enter into radiocommunication with ships must themselves be stations of a licensed port operations service or private mobile service.

8.29 BROADCASTING FROM MOBILE STATIONS

The establishment, use and operation of broadcasting stations (sound and television) on board ships, aircraft or any other floating or airborne objects outside national territories is prohibited.

8.30 AMATEUR RADIO STATIONS ABOARD UK SHIPS

The establishment and use of amateur sending and receiving stations on United Kingdom (registered) ships is permitted under *the* Amateur Radio Licence, granted by the Secretary of State for Trade and Industry. On a vessel the licensee shall:

(a) Install, use or make changes to the station only with the written permission of the vessel's master ; and
(b) observe radio silence on the advice of the vessel's master.

When a vessel is in international waters, the Licensee shall use only those frequency bands which, in accordance with the Radio regulations, have an allocation to the amateur service in the ITU Region being visited.

The full terms, provisions and limitations with which the licensee shall comply are contained in booklet BR68 which is available from the Ofcom's amateur and citizens' band radio unit contact details are indicated in section 8.2. Individual licence enquiries and correspondence should be sent to the Radio Licence Centre contact details in section 8.9.

There are severe penalties for not keeping within the laws regarding use of radio equipment at sea. Penalties for misuse include fines, imprisonment and the forfeiture of anything used within the offence. Anyone convicted may also face revocation of their licence and the suspension of their Authority to Operate (ATO).

8.31 HOURS OF SERVICE OF COAST STATIONS AND LAND EARTH STATIONS

The service of coast stations and land earth stations (LESs) are, as far as possible, continuous (day and night). Certain coast stations, however, may have a service of limited duration. The hours of service of coast stations are published in the ITU List of Coast Stations.

Coast stations whose service is not continuous shall not close before:

(a) finishing all operations resulting from a distress call or from an urgency or safety signal.
(b) exchanging all traffic originating in or destined for ship stations which are situated within their service area and have indicated their presence before the actual cessation of work.
(c) making a general call to all stations announcing the closing down of the service and advising the time of reopening, if other than their normal hours of service.

8.32 TECHNICAL REQUIREMENTS OF RADIO APPARATUS

The radio apparatus on UK ships shall comply with the requirements of the Ship Radio Licence and the relevant Ofcom published Performance Specification - (these Specifications are being gradually replaced by European Technical Standards Institution (ETSI) Specifications). On compulsorily-fitted ships of 300 g.t. and over the ship radio installation shall, in addition, comply with the Merchant Shipping (Radio Installations) Regulations currently in force.

8.33 SOURCES OF ENERGY FOR RADIO EQUIPMENT

(a) Main power supply:

There shall be available at all times while the ship is at sea a supply of electrical energy sufficient to operate the radio installations and to charge any batteries used as part of a reserve source or sources of energy for the radio installations.

(b) Reserve power supply:

A reserve source or sources of energy shall be provided on every ship, to supply radio installations, used for the purposes of conducting distress and safety radiocommunications, in the event of a failure of the ship's main and emergency sources of electrical power. This supply shall be capable of simultaneously operating all those distress and safety equipments operating in the bands as appropriate to the sea area(s) for which the ship is equipped as determined by the Merchant Shipping (Radio Installations) Regulations 1998 including any additional loads such as the emergency electrical lighting to illuminate radio controls and those navigational equipments which are required to supply an uninterrupted input of information to a radio installation for a period of at least:

(a) one hour on ships provided with an emergency source of electrical power; or
(b) six hours on ships not provided with an emergency source of electrical power.

The reserve source or sources of energy need not be capable of supplying independent HF and MF radio installations simultaneously. Such sources of energy shall be independent of the propelling power of the ship's electrical system.

8.34 BATTERY SUPPLIES

Where the reserve source of energy consists of rechargeable batteries a means of automatically charging such batteries shall be provided which shall be capable of charging them to minimum capacity requirements within 10 hours and the capacity of such batteries shall be checked, using an appropriate method, at intervals not exceeding 12 months when the ship is not at sea.

The siting and installation of such batteries should ensure:

(a) the highest degree of service
(b) a reasonable lifetime
(c) reasonable safety
(d) that battery temperatures remain with the manufacturer's specifications whether under charge or idle
(e) that when fully charged the batteries will provide at least the minimum required hours of operation under all weather conditions.

8.35 CAPACITY OF RESERVE SOURCES OF ENERGY

To calculate the required capacity of the reserve source of energy, the total current used in calculations shall be equal to the highest sum of all the radio installations which simultaneously can be connected to the source of energy based on the following:

(a) the current consumption of the VHF receiver;
(b) one fifth of the current consumption of the VHF transmitter;
(c) the current consumption of a MF or MF/HF receiver and of the transmitter when it is in condition that operation of the "press to transmit" switch will make it ready for immediate transmission;
(d) one third of the current which may be drawn by a MF or MF/HF transmitter for speech transmission on the frequency at which the current consumption of the transmitter is at a maximum;

(e) the current consumption of an Inmarsat ship earth station when it is receiving transmissions;
(f) one quarter of the current which may be drawn by and Inmarsat ship earth station when it is transmitting in the mode at which the current consumption is at a maximum; and
(g) the total current consumption of all additional loads to which the reserve source may supply energy in times of distress or emergency.

8.36 ELECTRICAL INTERFERENCE

Electrical interference or mechanical noise produced by the radio installation or any other equipment on board the vessel can seriously degrade the performance of radio systems or electronic navigational equipment. Extraneous noise may be of such intensity that effective reception of signals is lost. Thus radio surveyors finding any excessive interference or mechanical noise during an inspection will notify the owner, or his representative, and the master.

Methods of preventing interference are addressed in British Standard BS 5260, and interested parties should be aware of this standard particularly in the early design stages of a ship. For example the screening of radio equipment rooms offers many advantages in reducing interference. Another standard, British Standard Specification No. 1597 deals with suppression of radio interference on marine installations, measures to be taken in respect of antennas, rigging electrical wiring etc. and the standards of components which should be used for suppression purposes.

Ultimate responsibility for locating and clearing interference lies with the owner of the vessel.

CHAPTER 9

Radio Traffic

GENERAL

The information in this chapter relates to traffic transmitted and/or received in the radiotelephone and radiotelex services. Additional information concerning traffic exchanged exclusively in the Inmarsat system may be found in Chapter 5.

9.1 RADIOTELEPHONE TRAFFIC

Suitably equipped ships may use the international maritime radiotelephone service to connect with telephone subscribers on shore. This is achieved by using coast radio stations to access the local or international switched telephone networks.

The ITU List of Coast Stations and ALRS Volume 1 contain information on which coast stations offer a radiotelephony service, frequencies to be used and hours of service etc.

In the international telephone service, by agreement among the administrations concerned, the following additional facilities may be granted:

(a) Personal calls.
(b) Data Calls.
(c) Collect calls.
(d) Credit-card/Charge card calls.
(e) Conference calls.
(f) Fax calls.
(g) Home country direct calls.

These additional facilities are available in the direction ship-to-shore but particular services may be subject to special restrictions or conditions applied by individual administrations.

In the shore-to-ship direction such additional facilities are permitted for reception only if accepted by the ship station's operating agency. Where the additional facilities services are requested the sender should be advised that there may a supplementary charge to pay.

A personal call: is one between the number of a caller who may give his/her name (or the number of an extension) and some specific person (or extension); the person required must be adequately described (by name, position, address, etc.). It should be noted that all RT calls from shore-to-ship are considered as personal calls and such calls do not incur any extra charge.

A data call: is one requested for the purpose of exchanging data of any kind between telephone stations specially equipped to transmit and receive such data.

A collect call: is a call for which the caller, when requesting the call, specifies that he/she wishes the charge to be paid by the called party or, in some cases, by a third party.

A credit-card/chargecard call: is a call for which the caller, when requesting the call, specifies that he/she wishes it to be charged to his/her charge card/credit-card number.

A conference call: is a call established between three or more stations.

In the UK there are several ways of paying for radiotelephone calls:

(a) Credit card.
(b) Yacht Telephone Debit (YTD).
(c) Transfer charge.
(d) By quoting the vessels AAIC.

Stations having an Autolink RT unit on board may settle the accounts for link calls, etc., once registered with British Telecom by:

(a) having charges appear on the bills of a nominated UK telephone number.
(b) using a British Telecom Chargecard.
(c) paying via ship's Accounting Authority by quoting AAIC.

Home country direct or International Operator direct calling (IODC): is when the user, whilst travelling abroad, can dial a special number to place a direct call to an operator in his/her own country, requesting a call to a subscriber in the home country, which is chargeable either as a collect call or to a ITU-T-type international charge card/credit card or to national cards of the home country. The called subscriber may be notified of the call he will have to pay. In certain countries the caller may be charged to access the service. Enquiry services may be excluded in some countries.

9.2 TELEPHONE CREDIT CARD CALLS

The following particulars should be included in a credit-card call requests:

(a) number of the credit-card.
(b) if necessary, name of the caller.

The operator of the outgoing exchange must check that the characteristics of the credit-card indicated to him/her by the user are valid. Administrations may, however, provide for the validity of a credit-card number to be decided by the operator at the incoming exchange, and for this incoming operator to time and charge the call.

If the characteristics of the card are not correct, the operator must inform the caller thereof without indicating the inaccurate details. The operator shall not agree to set up the call with such a credit-card.

For approximately one month before the issuing period begins, and for one month after it ends, the credit-card symbol or pattern from either period shall be accepted. Beyond that time the caller shall be advised to obtain a new card.

9.3 AUTOLINK RT SERVICE

Autolink RT is available via several coast stations in the MF, HF and VHF bands. Full details are given in section 7.31 on page 189.

9.4 PRIORITY OF RADIOTELEPHONE CALLS

Radiotelephone calls are subject to the order of priority of communications as indicated in section 8.22.

9.5 CHARGES FOR RADIOCOMMUNICATIONS USING THE TERRESTRIAL SERVICES

The charges for radiocommunications consists of:

(a) The landline charges;
(b) The land station charges;
(c) The mobile station charges (if any);
(d) Any charges for special services for telegrams that have to be considered in the accounting; and
(e) Any special charges for special facilities.

The landline charge for national telecommunication channels applicable to radiocommunications between a mobile station and a country of the land station is notified either in special drawing rights (SDRs) or in gold francs to the ITU General Secretariat by the land station administration.

When a single land station is used as an intermediary between mobile stations, two land station charges are collected. If the land station charge applicable to traffic with the original mobile station is different from that applicable to traffic with the destination mobile station, the sum of these two charges is collected.

When it is necessary to use two land stations as intermediaries between two mobile stations, the land station charge for each station is collected and also the landline charge between the two land stations.

9.5.1 Special charging arrangements

Radiotelegrams of immediate special interest

No charge for radio transmission in the Maritime Mobile Service is made for radiotelegrams of immediate special interest, provided they are:

(a) distress messages or replies thereto;
(b) radiotelegrams originating in mobile stations notifying the presence of icebergs, derelicts, mines and other dangers to navigation, or announcing cyclones and storms;
(c) radiotelegrams originating in mobile stations announcing unexpected phenomena threatening air navigation or the sudden occurrence of obstacles at airports;
(d) radiotelegrams originating in mobile stations notifying sudden changes in the position of buoys, the working of lighthouses, devices connected with buoyage, etc., or
(e) service radiotelegrams relating to the Maritime Mobile Service.

Radiotelegrams relating to medical advice

No charge for radio transmission is made for radiotelegrams relating to medical advice, provided that:

(a) they are exchanged directly between mobile stations and land stations that are shown in the ITU List of Radiodetermination and Special Service Stations as providing such a service and,
(b) they are addressed in accordance with the conditions indicated in that List.

Meteorological radiotelegrams

Land stations charges applicable to meteorological radiotelegrams should be reduced by 50% in all relations.

9.6 CHARGES FOR RADIOTELEPHONE CALLS

The information regarding charges for RT calls is divided into those conducted via terrestrial circuits and those over satellite links.

If no uniform charges apply in respect of the land stations of a country, different land station charges shall be fixed for the MF, HF and VHF bands and for the Maritime Mobile-Satellite Service.

Automatic calls should be charged by one of the following two methods:

(a) charging by minute,
(b) charging by periodic pulses of the type used in the national automatic service.

The unit charge is the charge for an ordinary private call of one-minute duration except for automatic operation calls when periodic pulses are used.

The minimum charge for manual, single-operator or semi-automatic call is three units.

When handled through a land station the chargeable duration of a manual or single operator call will be fixed at the end of the call by the land station; if two land stations are participating in the handling of the call, the opinion of the land station that accepted the call from the originating mobile will prevail.

Where two mobile stations are engaged in direct RT communications, the chargeable duration of a radiotelephone call will be set by the mobile station which originated the call.

9.6.1 Operator connected calls

The basis for calculating operator connected charges is a three minute minimum charge plus one minute increments.

For example: a 1-minute 30-second call is charged as for three minutes.
a 4-minute 30-second call is charged as for five minutes.

9.6.2 Special charges

The special charges for personal RT calls (from mobile stations to land) for either credit card or collect calls, if admitted, shall be applied with manual or semi-automatic operation.

No special charges for personal RT calls to mobile stations handled by manual or single-operator service shall be applied in the Maritime Mobile Service, with the possible exception of the Maritime Mobile-Satellite Service.

The charge of special call facilities shall be calculated on the basis used for an international call, applied to either:

(a) the landline charge only, or
(b) all components of the call charge.

Unless special arrangements between the administrations or the recognised private operating agencies are in effect, supplementary charges for personal calls (in the direction ship-to-shore) and collect calls (if admitted) shall be applied.

When the booking of a radiotelephone call which is liable to the payment of a special charge (for example, a collect call) is accompanied by a booking of a personal call, only one special charge shall be collected.

9.6.3 Autolink RT calls

Charges for Autolink RT calls are based on a one minute minimum with one minute increments.

For example : a 30-second Autolink call is charged as for one minute.
a 2-minute 45-second Autolink call is charged as for three minutes.

9.6.4 Refunds for radiotelephone and radiotelex calls

When, through any fault of the service, the booking of a call is not followed by the calling and called stations being placed in communication, no charge shall be payable. If the amount of the charge has been paid, it shall be refunded.

In order to simplify operating and accounting procedures, administrations may decide that no charge shall be payable when a requested connection has not been set up whatever the reason.

However, administrations may decide to collect charges in cases where there is no fault of service. In that case the basis of charging shall be notified to the ITU General Secretariat for inclusion in the ITU List of Coast stations.

When, through any fault in the service, difficulty is experienced in the course of a call, the chargeable duration of the call shall be reduced to the total time during which transmissions conditions have been satisfactory, taking into account CCITT Recommendations.

9.7 CHARGES FOR RT CALLS USING THE INMARSAT SYSTEM

For information regarding charges for telephone calls via the Inmarsat system refer to Chapter 5 section 5.50.

9.8 RADIOTELEGRAM SERVICE

Radiotelegrams are accepted for onward transmission to many countries. These may be dictated by RT or transmitted via radiotelex from the ship to coast radio or land earth stations for onward transmission to the addressee by appropriate land line. It should be noted that the UK telegram service has ceased and been replaced by the Telemessage service. Radiotelegrams received at UK coast **radio** stations are forwarded to addresses in the UK by means of first class post.

9.9 RADIOTELEGRAM CHARACTERS

Radiotelegrams have to be written in the characters normally used in the country of origin and must contain an adequate address so that it can be delivered without requests for further information. Radiotelegrams must have a text and may contain a signature.

Everything which the sender requests to be transmitted is chargeable; however, route indicators are included free of charge. When a radiotelegram cannot be delivered to the addressee, a service advice will be sent from the office of destination to the office of origin indicating the reason for non delivery.

The radiotelegram must be written in characters which have an equivalent in telegraph signals. They include:

(a) the letters A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
(b) the figures 1 2 3 4 5 6 7 8 9 0
(c) the following punctuation marks and signs:

Full stop .
Comma ,
Colon or division sign :
Question mark ?
Apostrophe '
Cross or addition sign +
Hyphen, dash or subtraction sign -
Fraction bar or division sign /
Double hyphen =
Left-hand bracket (parenthesis) (
Right-hand bracket (parenthesis))

9.10 CLASSES OF RADIOTELEGRAMS ADMITTED

Only the following classes of radiotelegrams can be accepted in the international maritime mobile service:

Class	*Indicator*
Telegrams relating to the safety of life at sea	SVH
Telegrams relative to the application of the United Nations Charter	ETATPRIORITE
Government telegrams	ETAT or ETATPRIORITE
Meteorological telegrams	OBS
Telegrams concerning persons protected in times of war by the Geneva Conventions of 12th August 1949	RCT
Ordinary private telegrams	P (or no indicator used)
Service telegrams or advices	A

Administrations have the option of admitting the classes of telegrams listed below; however if they do not admit such telegrams in their own service they must let them pass in transit.

Optional telegrams

Class	*Indicator*
Postal financial services telegrams	POSTFIN
Letter telegrams	LT
Government letter telegrams	LTF
Franking privilege telegrams (ITU)	CONFERENCE

Special services

Class	*Indicator*
Urgent transmission and delivery	URGENT
De luxe form	LX
De luxe form of condolence	LXDEUIL
Period of retention of radiotelegrams at land stations	Jx (x = number of days)
Telephone delivery	TFx (x = telephone number)
Facsimile delivery	FAXx (x = facsimile number)

Note: Urgent radiotelegrams and telegrams having telephonic addresses are not accepted by UK coast stations.

9.11 RADIOTELEGRAM ADDRESSES

The following categories of address of a radiotelegram from a ship are permitted. However, all telegrams are subject to the telegram services available in the country of delivery:

(a) full postal address.
(b) registered address.
(c) telephonic address.
(d) telex address.
(e) teletext address.
(f) facsimile address.
(g) "Poste restante" or "Telegraphe restant" address.
(h) post office box address.

The address must contain all the particulars necessary to ensure delivery of the telegram to the addressee without enquiries or requests for information. In all cases of insufficient address the telegrams shall be accepted only at the risk of the sender, if the sender insists on sending it, the sender also has to bear the consequences.

Note: The inland telegram service in the UK has been replaced by the "Telemessage" service. Telegrams arriving for delivery to addresses in the UK are forwarded by first class post.

9.12 URGENT RADIOTELEGRAMS

The sender of an ordinary private radiotelegram may obtain priority in transmission and delivery by requesting the special urgent transmission and delivery (URGENT). The service indicator URGENT shall be shown before the address. In such cases the landline charge between coast station and town of destination is

doubled in addition to normal coast station charges. The minimum number of chargeable words is seven. URGENT telegrams are not accepted for addresses in the UK or the Republic of Ireland.

9.13 POST RESTANTE, OR TELEGRAPHE RESTANT ADDRESS

These are radiotelegrams to be called for. The address of telegrams intended to be delivered to a POSTE RESTANTE or TELEGRAPHE RESTANT must be composed of:

(a) the name of the addressee including, where possible, the forename(s) or initials,
(b) the words POSTE RESTANTE, TELEGRAPHE RESTANT (or the equivalent in a language of the country of destination), and
(c) the name of the telegraph office of destination.

Initials alone, figures, given (forename) name only, fictitious names or arbitrary signs of any kind shall not be allowed in the address.

Note: The UK does not admit telegrams addressed to post restante, or telegraphe restant.

9.14 PAID SERVICE INDICATORS

A paid service indicator appears in a radiotelegram between the preamble and the address and is itself counted as a chargeable word. This indicator identifies a special class of telegram as in the examples below:

= LX = is a request by the sender to have the telegram delivered on a special de luxe form and/or in a special envelope,
= Jx = (where x represents the number of days) is a request by the sender to be advised of non-delivery of a radiotelegram before the expiry of the standard period of five days.

9.15 COUNTING WORDS IN RADIOTELEGRAMS

When counting words in a radiotelegram each separate word or group of characters will be counted as one actual word. Also a distinction has to be made between the actual number of words and the chargeable number of words. In a radiotelegram, if the chargeable and the actual number of words differ they are both sent but separated by a fraction bar. For example 24/21 indicates 24 chargeable words and 21 actual words. If chargeable and actual number of words are the same they need only be shown as a single number.

Everything which the sender asks to be transmitted is chargeable, including any service indicators used in the telegram. Such service indicators shall be counted as actual words, for example the De Luxe telegram indicator = LX = counts as one chargeable word. However route indicators added to the service instructions of a radiotelegram for the benefit of the telegraph service are included free.

Radiotelegrams are charged at 10 characters to the word and where a word exceeds ten characters they shall be counted at the rate of one chargeable word for each ten characters or part thereof.

Example of word counts

Example Words	*Number of characters*	*Number of words*
RP6.50	6	1
DA4270PSA	9	1
141630z	7	1
23/3/92	7	1
CFC-113(C2F3CL3)	16	2
Chlorofluorocarbons	19	2
Jose Maria Monfort		3
Josemaria Monfort		2
Saint Lukes Square		3
SaintLukesSquare	16	2
(Documentation carried)*		3
San Francisco Calif.		3
SanFranciscoCalif.	18	2

Each bracket forms part of the nearest word or group, thus in the example shown "(Documentation" has 14 characters and this counts as two words, and "carried)" has 8 characters and counts as one word.

The word count of the office of origin is decisive in the case of radiotelegrams for mobile stations, and that of the controlling operator is decisive in the case of telegrams originating in mobile stations.

9.16 RADIOTELEGRAM FORMAT

A radiotelegram is set out in a standard format and consists of a preamble, address, text and signature. The preamble consists of various components in the following sequence:

Prefix (if any)
Name of the ship or office of origin
Serial number of the radiotelegram
Number of words
Date
Time
Service instructions.

Notes:

The prefix is an indicator (one, two or three letters) used to identify the type of message being sent for example, P indicates a private radiotelegram.

In the office of origin where confusion could occur if two stations have the same name, each station should identify itself by its name followed by its call sign.

Radiotelegrams sent from ships to individual coast stations must be numbered in sequence and on a daily basis. Numbering commenced at 0001 hours UTC and the same sequence is used irrespective of the service used ie RT or radiotelex

The date and time of handing in at the station of origin are given in the preamble of the radiotelegram. The day of the month is given using figures (1 to 31), the time of handing is also given by figures (0000 to 2359) indicating hours and minutes, UTC is used for the latter.

Service instructions are included where necessary to give routing instructions for the telegraph service or more commonly for indicating the AAIC of the station of origin.

9.17 MARITIME RADIO ACCOUNTING

Accounting Authorities are permitted to operate under the terms of the ITU Convention as set down in Article 66 of the Radio Regulations and Appendix 2 of the International Telecommunication Regulations. They are used to facilitate the effective collection and distribution of charges accruing to radiocommunication between ships and authorised stations ashore and effectively act as an intermediary between the service providers, network operators and the subscriber on the mobile.

The accounting authorities must have the facilities, experience and financial resources to handle complex billing for an international market and be able to provide a continuous worldwide service to their mobile customers. These authorities are therefore responsible for the collection of charges from ship stations based on accounts submitted from service providers or network operators, of terrestrial of satellite systems. They are also responsible for payment of charges to service providers or network operators for the ship station licensees or shipowners for which they have accounting responsibility.

To ensure accurate information exchange, accounting authorities have to maintain a high level of quality control. They are therefore obliged to keep accurate records and notify service providers in the event of change of subscribers, change of flag state of change of ownership etc. They must provide service providers with updated mobile subscriber list on a regular basis. New lists with all changes clearly identified must be issued at least every ninety days.

National administrations are responsible for registers of accounting authorities located within their territory and for allocating each accounting authority with its AAIC.

9.17.1 Accounting Authority Identification Code (AAIC)

This is an indicator which should be inserted into the service instructions of a telegram by the ships radio operator as a matter of routine and without waiting on a request from the coast station to do so. It allows the receiving operator to determine the discrete identification of the accounting authority responsible for settling maritime accounts for a particular station. These accounting authorities may be state administrations or private operating agencies licensed by the national administration of their respective countries.

The AAIC is in two parts:

(a) the first part consists of one or two letters representing the country in which the accounting authority is based. For example the letters NO indicates Norway as the accounting country.

(b) the second part consists of two numbers which identifies the particular accounting authority.

For example the indicator N001 indicates the accounting authority as the Norwegian Telecommunications Administration.

Details of a ships AAIC are contained in the ITU List of Ship Stations, specifically under column 11 in Part II of the List. The names and addresses of particular accounting authorities can then be found in Section 3, Part IV of the same list.

9.17.2 Providing AAIC information to coast stations

When the name and address of the administration or private operating agency controlling a ship station are not given in the appropriate list of stations or are no longer in agreement with the particulars given therein, it is the duty of the ship station to furnish as a matter of regular procedure, to the coast station to which it transmits traffic, all the necessary information in this respect.

9.18 SERVICE MESSAGES

Service telegrams are defined as the exchange of telecommunications between:

(a) administrations;
(b) recognised private operating agencies;
(c) administrations and recognised private operating agencies;
(d) administrations and recognised private operating agencies on the one hand and the Secretary-General of the ITU on the other ; which concern the public international telecommunication network.

Service Advices are exchanged between telegraph offices and relate to the details of service or the working of circuits and the transmission of traffic (see also section 9.21).

9.19 NON-DELIVERY OF TELEGRAMS

When for any reason a radiotelegram originating in a mobile station and destined for a place on land cannot be delivered to the addressee, an advice of non delivery is addressed to the land station or the telegraph office that received the telegram. After checking the address the land station forwards the advice, when possible, to the mobile station where the ships operator compares the address quoted in the advice against the address in the original telegram. If any discrepancy is found this can be corrected by transmitting a service advice via the original coast station wherever possible. If no error is found the sender of the original telegram is informed of the non-delivery and the reason. The sender may then wish change or add to the address information of the original telegram using a paid service advice.

When a telegram received at a mobile station cannot be delivered, that station must inform the land office or mobile station of origin by a service advice. In respect of a telegram originating on land this service advice is sent whenever possible, to the land station through which the original telegram passed, or if necessary to the land station of the same country, or of a neighbouring country, as far as existing conditions or special arrangements permit. In such cases the name or call sign of the station from which the telegram was received is quoted. A separate service message should be sent for each undelivered telegram and the date quoted in the non-delivery advice should be that on which the original telegram was handed in.

9.20 CANCELLING A TELEGRAM AT SENDER'S REQUEST

The sender of a telegram or his authorised representative may, on establishing their status and identity, cancel their telegram only if it has not been transmitted by the office of origin. After transmission has taken place, the addressee can only be notified of the cancellation of the telegram by means of another telegram from the sender.

9.21 SERVICE ADVICES

Service advices must be used only where essential and must be worded as briefly as possible, they are transmitted free of charge. They are preceded by the service abbreviation A, but otherwise are similar to an ordinary telegram containing a preamble etc. The address must contain the name of the office of destination. The text of such advices relate to a telegram previously transmitted and this is indicated by reference to:

(a) its local or international serial number and the date separated by a fraction bar,
(b) the service indications (if any),
(c) the name of the addressee.
(d) the address (excluding the office of destination),
(e) the signature (if any).

Examples:

(a) Non-delivery on shore

GLASGOW 31 8 4 1109 = (Preamble)
A GRANDPRINCESS = (Address)
2/3RD COLEMAN GIBRALTAR ROW
ADDRESSEE UNKNOWN (Text)

The number 2 of 2/3RD represents the serial number of the original telegram and 3rd is the date it was handed in, while Coleman is the name of the addressee. Addressee unknown is reason for non-delivery, other reasons for non-delivery are:

addressee left, addressee deceased, addressee not arrived, addressee not registered, addressee no longer registered, refused.

When a ship station receives a non-delivery service advice it may correct a discrepancy in the address by transmitting a service advice to the original destination, as in the example below.

GRANDPRINCESS 1 10 4 1425 = (Preamble)
A GLASGOW = (Address)
2/3RD COLEMAN GIBRALTAR ROW
DELIVER GOLDMAN GIBRALTAR ROAD (Text)

(b) Non-delivery on board

KATRINEMAERSK 9 12 23 1830 = (Preamble)
A MANCHESTER (Address)
17/22 CRINYON KATRINEMAERSK SIGNED MORRIS
ADDRESSEE NO LONGER ON BOARD (Text)

Additional reasons for non-delivery on board ships include:
addressee not on board, addressee unknown, refused.

9.22 METEOROLOGICAL (OBS) MESSAGES

Certain ships are designated to make regular meteorological observations and transmit such information ashore as weather reports in order to assist in weather forecasting. Such weather messages are given the prefix OBS and if sent via the UK are routed directly to the Meteorological Office at Bracknell. OBS messages can be sent using RT or Telex in the terrestrial or satellite services.

9.23 RADIOMARITIME LETTERS

Radiomaritime letters may be accepted taking into account CCITT recommendations relating to letter telegrams, if the telegram service is used to convey radiomaritime letters.

The total charge shall include the postal charge (by ordinary letter or airmail letter) due for delivery in the land station country.

An additional charge may be collected where delivery is to be made to a country other than that of the land station.

An additional charge may be collected where applicable:

(a) charges due for special services;
(b) the landline charge when transmission on the land section is exceptionally by telegraph.

The retransmission of radiomaritime letters is not permitted in the mobile service. Radiomaritime letters bear the service indication SLT. This precedes the address.

The address must enable delivery to be effected without enquiry or request for information. Registered or abbreviated address are admitted when, exceptionally radiomaritime letters are forwarded telegraphically on the land section.

Radiomaritime letters are *NOT* accepted for addresses in the UK.

9.24 RADIOTELEXOGRAM

A radiotelexogram is a message sent by telex direct from a subscriber to a foreign land station for transmission to a mobile station or a message sent from a mobile station to a land station for transmission by telex direct to a foreign subscriber.

Note: A radiotelexogram is different from a radiotelex call. In particular, a radiotelexogram is normally transmitted between the mobile station and the land station as a radiotelegram by radiotelephony.

9.24.1 Radiotelexogram: Information supplied to the land station by the calling party.

(a) Radiotelexogram to a mobile station

(a) telex number and/or answerback code of the calling subscriber.
(b) the national telex network to which the subscriber belongs.

(c) the date and time of origin.
(d) the word RADIOTELEXOGRAM.
(e) name or designation of the addressee with supplementary particulars if necessary.
(f) the name of the mobile station followed, when necessary, by its call sign or where this is not known, the particulars of the passage made by the mobile station.
(g) any specific delivery instructions.

(b) Radiotelexogram from a mobile station

(a) name and/or callsign of the mobile station.
(b) identification of the accounting authority.
(c) the date and time of origin.
(d) the word RADIOTELEXOGRAM.
(e) destination country and/or network.
(f) called subscriber's telex number and answerback code.

9.25 RADIOTELEX SERVICE

Suitably equipped ships may use the international maritime radiotelex service to connect with telex subscribers on shore. This is achieved by using coast radio stations to access the local or international switched telex networks. The ITU List of Coast Stations details information on which coast stations offer this service, frequencies to be used and hours of service etc.

9.26 TELEX MESSAGE LAYOUT

In order to bring some conformity into the telex service users are advised to follow certain recommendations:

9.26.1 General

Where the recipient may be in doubt of the identity of the caller, it is recommended that the calling subscriber indicate:

(a) name and place of the sender, preceded by the word *FROM*
(b) name and place of the addressee, preceded by the word *TO*
(c) if required, name and place of information addressee(s), preceded by the word *COPY*

Observing these recommendations will often save additional work for the recipient when messages have to be distributed for action.

9.26.2 Telex message format

After the exchange of answerbacks the calling subscriber can transmit his message for which the following format is recommended:

(a) start a new line and mention own reference, if any, and the date of dispatch.
(b) start a new line and indicate the priority of the message, if desirable, such as *URGENT, VERY URGENT,* etc.

(c) start a new line and indicate the subject if appropriate and/or the name of the person or department for whose attention the message is intended.
(d) start a new line and mention any references, such as *REF YOUR TELEX 987 OF 8th JAN*, etc.
(e) start a new line and transmit the text of the message.
(f) after completing the message start a new line and transmit a plus sign (+) indicating the end of the message.
(g) obtain the answerback of the called subscriber, check it in order to be sure that the connection is still in good order and generate your own answerback.
(h) if there are any further messages, they should be separated from each other by at least 8 line feeds, after the exchange of answerbacks as mentioned in (g) above.
(i) after transmission of the last message and the exchange of answerbacks send at least 8 line-feeds and give the clearing signal.

9.26.3 Sending fractions

When a group, or part of a group, is composed of a whole number and an ordinary fraction, the fraction should be separated from the number by means of a dash without a space.

For example: For two and one half send: *2-1/2*

9.26.4 Sending percentages

In order to avoid misunderstanding, a whole number, a fractional number, or a number followed by a % sign should be transmitted by joining them up to the % sign, or transmitted in full as appropriate.

For example: For 10% transmit *10-0/0* or *10 PER CENT*

9.26.5 Repetition of important groups

When important figures or words appear in the text it is desirable to repeat them immediately after the group followed by a space either in brackets or preceded by the word REPEAT

For example: *2300 (2300)*
2300 REPEAT 2300
CANCEL REPEAT CANCEL

9.26.6 Starting a new line

To pass to the beginning of the next line, i.e. to start a new line, first press carriage-return (CR) and then line-feed (LF). On modern telex terminals one LF followed by one CR is achieved by pressing the "ENTER" or "RETURN" key once.

9.26.7 Correction of errors

An error is corrected in the following manner:

(a) In manual transmission, by the sequence XXXXX (letter X repeated five times and followed by a space. (Note that the sequence E E E is also in use but is not preferred.)

Example: ETA NAB TOGXXXXX TOWER

(b) In automatic transmission, when preparing perforated tape, by backspacing (if necessary by counting the number of characters to be erased, including spaces and shifts, and by backspacing by that number) to the erroneous character and then operating the letter-shift key to erase all the characters up to and including the last punched character. Then start again with the character to be sent immediately after the last correctly punched character.

(c) If the procedure mentioned in b) above for any reason cannot be followed, an error should be corrected as in a) above.

(d) If an error is detected after the transmission of a message but before the exchange of answerbacks it should be corrected by clearly indicating under the text of the message what change is required :

Examples: *CORRECT 5TH WORD 3RD LINE TO READ 2330 REPEAT 2330*
DELETE 5TH WORD 3RD LINE
INSERT THE WORD 2330 BETWEEN THE 3RD AND 4TH WORD OF 1ST LINE

9.26.8 Cancelling a telex message

If, for any reason, a message has to be cancelled during transmission this should be clearly indicated on a new line by transmitting the words ANUL ANUL ANUL

9.26.9 Precautions to be observed when preparing perforated tape

In preparing a perforated tape for automatic transmission, care should be taken that:

(a) the signal *who are you?* (figure case D) does not appear on the tape, in order to prevent the text from being garbled by the returned answerback of the other party,

(b) the procedure in section 9.26.6 above is observed,

(c) the tape is perforated to the end with a series of letter-shifts.

9.26.10 Non-standardised telex characters

Since figure case signs or letters coupled with the letters F, G and H (e.g. £, $, etc.) are not universally standardised, they must not be used in international communications, but should be transmitted in full, e.g. *POUNDS STERLING, DOLLARS,* etc.

9.27 RADIO TELEX LETTERS (RTL)

A radio telex letter (RTL) is sent by radiotelex to the coast station and forwarded by first class post to addresses locally or by airmail to addresses overseas. The service is available in the direction ship to shore only.

Coast stations cannot amend or correct incomplete addresses of an RTL. They are posted to the addresses as furnished by the sender who is responsible for the sufficiency of the address which should include the post code. Officers accepting RTLs on ships should advise senders that they are responsible for the accuracy of the address and that inaccuracy or insufficiency may entail delay or non-delivery.

Charges for RTLs are based on time plus a fixed handling charge. There is a one minute minimum charge. One minute of radiotelex time allows approximately 45 words in an RTL. The access code is RTL+ and when the coast station issues a reference number it will request commencement of the RTL by sending MSG+?

Thereafter operators should key in the following:

(a) Ship's name/callsign; Date; Time
(b) RTL
(c) At least 4 line-feeds (4 x LF)
(d) Name and full postal address including post code if appropriate
(e) At least 4 line-feeds (4 x LF)
(f) Text
(g) KKKK

It is important that each individual RTL is terminated with KKKK, after which the time duration will be received for charging purposes. If the radio circuit fails before KKKK is received the transmission is disregarded and the entire message should be resubmitted.

9.28 FURNISHING TR INFORMATION TO COAST STATIONS

The coast station may, by means of the abbreviation TR (spoken as TANGO ROMEO), ask the ship station to furnish it with the following information:

(a) position and, whenever possible, course and speed;
(b) next port of call.

TR information should however be furnished by ship stations, whenever this seems appropriate, without prior request from the coast station. The provision of this information is authorised only by the master or the person responsible for the ship. Sending a TR also assists with traffic routing and ships should send voyage particulars to coast stations using a message prefixed by the abbreviation TR. There is no charge for this service.

TRs should be sent to coast stations without waiting for a request to do so and as a general rule should be sent under the following circumstances:

(a) When leaving or arriving at port.
(b) When changing from one coast station area into another coast station area.

Examples:

TR LaChacra/GKDU 51° 56´ North 08° 15´West course 190° speed 16 knots leaving Cork bound Lisbon.

TR Hayling/MNWN 15 miles SSW St Catherines Point speed 10 knots from Ipswich bound Fowey.

TR Viking Trader/GXIS from Jersey now entering Plymouth closing station.

TR Dendbula/GHQL leaving Hull bound Iceland speed 15 knots.

9.29 MONETARY UNIT

In the absence of special arrangements concluded between administrations or recognised private operating agency(ies), the monetary unit to be used in the composition of accounting rates for international telecommunication services and in the establishment of international accounts shall be either the monetary unit of the International Monetary Fund (IMF), currently the Special Drawing Right (SDR) as defined by that organisation or the gold franc, equivalent to 1/3.061 SDR.

9.30 SHIP REPORTING SYSTEMS

Many maritime countries operate ship reporting systems, some of which are voluntary, whilst others are mandatory they are useful in the event of a Search and rescue incident. Individual systems vary in that some countries make reporting mandatory only for vessels of their own flag. The purpose of such schemes is to provide an information databank about shipping present in a particular area for use during search and rescue incidents. Vessels would normally report their position to shore authorities through coast radio stations. The most well known of these systems is AMVER, operated by the US Coast Guard as detailed in the next section (9.31).

The UK operates a voluntary reporting system for vessels in the English Channel called MAREP, based on the use of VHF to report to appropriate coast stations.

For further details of schemes relating to traffic management, for example national reporting systems for deep draught vessels etc. ALRS Volume 6 should be consulted.

9.31 AMVER MESSAGES

The Automatic Mutual-assistance Vessel Rescue (AMVER) service is operated by the US Coast Guard (USCG) and is available to any merchant vessel of greater than 1,000 gt on a voyage greater than 24 hours. Ships registered with AMVER can send messages via many overseas coast stations as well as to USCG radio stations. AMVER messages may also be forwarded over the Inmarsat system. These messages can be transmitted using RT or telex for onward transmission to USCG New York where the data is stored on computer. If a marine incident occurs the computer will produce a picture of vessels in the area (SURPIC) which may be in a position to render assistance.

All AMVER messages should be addressed as follows: AMVER followed by the name of the participating radio station. For example *AMVER NEW YORK RADIO*. However, messages forwarded via German coast stations should be addressed to *AMVER FRANKFURT AM MAIN*.

Previously AMVER messages via UK coast radio station or Inmarsat land earth stations, were accepted free of charge. From November 1993 this situation has changed and such AMVER messages via UK Inmarsat land earth stations are now charged to the ship concerned.

Note: In emergencies all distress messages should be routed to the nearest MRCC and not to the AMVER Centre.

9.32 VESSEL MONITORING SYSTEM FOR FISHING VESSELS

Under a European directive which is likely to be mandatory by 1st January 2000 certain fishing vessels will be required to participate in a Vessel Monitoring System (VMS). The system will be satellite based, using Inmarsat-C, and its purpose is to monitor the position of EC fishing vessels. Polling requests from ashore will be sent at regular intervals to fishing vessels in a given geographical area which will respond automatically by returning a message containing position derived from a navigational aid such as GPS which has to be interfaced with the Inmarsat-C equipment.

VMS will relate to new fishing vessels built after 1st January 1998 of 12 metres and up to 24 metres and for existing fishing vessels of over 12 metres and up to 45 metres.

9.33 RADIOCOMMUNICATION BETWEEN BRITISH MERCHANT SHIPS AND HM SHIPS

As a rule, radiotelegrams to HM ships from British Merchant ships are sent to a coast radio station for onward transmission over Ministry of Defence Communications Networks.

When at sea, and as required by the GMDSS Distress and Safety Regulations dependent upon which sea area the vessel is operating in (A1, A2, A3 or A4) HM ships maintain a constant watch on DSC VHF Ch. 70 and RT VHF Ch. 16. When outside VHF range of shore stations fitted with VHF DSC, HM ships will also keep a listening watch on the DSC frequencies of 2 187.5 kHz and 8 414.5 kHz plus one other of the HF DSC Distress and Safety frequencies.

CHAPTER 10

Meteorological Services

10.1 INTRODUCTION

All shipping with suitable receiving equipment may listen to meteorological transmissions which are initiated by the appropriate maritime authority in their respective countries. Many coast stations worldwide transmit meteorological information to shipping for particular areas on a regular basis. In addition to these services, national broadcasting stations also transmit weather reports at fixed times which may be of interest to mariners.

Such forecasts can be for coastal or inland waters, the high seas, fishing areas, surface analysis and ice reports. Actual weather conditions, shipping bulletins and weather reports received from shipping or aircraft may also be broadcast. Forecasts are usually based on the conditions expected within next 12 or 24 hours. The mode of transmission varies but includes RT, Fax, Telex, NAVTEX and enhanced group call (EGC).

In addition to the scheduled meteorological broadcasts, gale and storm warnings will be transmitted immediately they are received from the appropriate meteorological authorities. Such warnings will be announced on the appropriate calling frequencies, e.g. 2 182 kHz or Ch.16 VHF together with an indication of the broadcast channel to be used. These messages are preceded by the safety signal (SECURITE sent three times). In addition to the forecasts outlined above ships may call coast stations and request current weather information from them, however a charge may be made for this by some administrations.

10.2 NAVTEX

As previously described in section 4.17, NAVTEX is an automated direct printing telex service used to promulgate navigational and meteorological information to shipping on 518 kHz. By using this frequency, a reception range of about 400 nautical miles is possible and it is therefore ideal for broadcasting weather reports concerning coastal waters. Information transmitted by NAVTEX under the Maritime Safety Information (MSI) including weather reports and warnings are broadcast in the English language. Provision exists for second language broadcasts of essential meteorological information to ships by NAVTEX on 490 kHz.

Routine area weather forecasts are transmitted by UK stations using the NAVTEX service on 518 kHz. Letters are used to identify subject code or message category on a ship's NAVTEX receiver. For example, the letter B indicates meteorological warnings while E indicates meteorological forecasts.

Ship's staff should programme Navtex equipment to receive weather information from coast radio stations appropriate to the area in which the ship is sailing or is about to enter.

Full details of stations broadcasting weather reports and forecasts by NAVTEX can be found in the ITU List of Radiodetermination and Special Services published by the ITU or in Admiralty List of Radio Signals Volume 3.

10.3 WEATHER INFORMATION VIA ENHANCED GROUP CALL (EGC)

The EGC facility on board ship should be logged on to the correct Inmarsat satellite to receive Maritime Safety Information from SafetyNet broadcasts appropriate to the area the ship is navigating or about to enter. Users should be aware that this may entail monitoring broadcasts from different satellites. See section 4.18.1 for information about setting up the EGC receiver.

10.4 NORTH ATLANTIC - METAREA 1 BROADCASTS

Under the SOLAS Convention the United Kingdom has to provide maritime safety information concerning warnings about gales and weather bulletins suitable for shipping. The area has been named Metarea 1, and includes the waters around the British Isles, northwest Europe and the high seas of the North Atlantic, north of 48° 27' North and east of 35° West. Such information can be received by vessels equipped with EGC receivers as outlined in the preceding section. GMDSS equipped vessels in coastal water receive this information via NAVTEX on 518 kHz. It should be noted that provision of this NAVTEX service has been extended out to 20° West due to the coverage areas of the Republic of Ireland's stations at Valentia and Malin Head.

Forecasts for and warnings for METAREA 1 are broadcast by Goonhilly LES through the AOR (E) satellite. Warnings are also broadcast via the AOR (W) satellite as soon as possible after receipt. The forecasts are also broadcast via the SafetyNet Service of Inmarsat-C as EGC messages within GMDSS.

10.5 METEOROLOGICAL INFORMATION FROM HMCG STATIONS

The six Coastguard MRCCs and fifteen MRSCs located around the UK are continuously manned. When requested for information concerning actual weather conditions in the immediate vicinity of their station such stations may offer present weather conditions subject to no SAR communications being in progress. The information is for the local area only, it will not include other areas or forecasts.

Most, but not all, stations transmit strong wind warnings (Force 6 and above) for its local area every two hours, at scheduled times. Such warnings will be issued on receipt from the Meteorological Office and retransmitted at designated broadcast times until cancelled. They cover up to five miles from the coastline to cater for the inshore seafarer. They are issued when winds of force 6 or more are expected within the next twelve hours and when no equivalent gale warning is in operation for the adjoining sea area. Such warnings are also broadcast by the BBC, NAVTEX service and EGC SafetyNET.

Coastguard stations will announce their scheduled transmissions on Ch.16 VHF before transmitting the broadcast on Ch.67 VHF. Weather message broadcasts, giving the local area forecast, are transmitted by all Coastguard stations at four hourly intervals and are announced and broadcast in a similar manner to strong wind warnings.

For details of transmission times for strong wind warnings, local weather messages and local navigational warnings from individual Coastguard stations, consult *Admiralty List of Radio Signals Volume 3 (1).* It should be noted that the scheduled broadcasts referred to above may be disrupted or suspended during times when SAR communications are in progress at individual MRCCs or MRSCs.

MRCC are located at Clyde, Swansea, Falmouth, Dover, Yarmouth and Aberdeen. Existing MRSCs are located at Shetland, Pentland, Forth, Tyne/Tees, Humber, Thames, Solent, Portland, Brixham, Milford Haven, Holyhead, Liverpool, Oban, Stornoway and Belfast.

10.6 GALE WARNINGS

Gale warnings are broadcast by coast stations appropriate to the area in which the gale is expected, Force 8 or above. The RT transmission is made at conversational speed then repeated at dictation speed. Warnings are issued when mean wind speeds are expected to reach or exceed 34 knots, Force 8 and above, or gusts expected to reach or exceed knots, (other than isolated thunderstorms). For the high seas of the North Atlantic, storm warnings are issued when mean wind speeds are expected to reach or exceed 48 knots, Force 10 or above.

The BBC broadcasts a daily weather bulletin for shipping on Radio 4 on 198 kHz (1 515m), for other regional transmitter frequencies. These bulletins include a summary of any gale warnings in force, a general synopsis, sea area forecasts and coastal station reports, the latter at 0048 and 0536 local time only. These broadcasts are made at the following local times:

0048 0536 1201 1754

Gale warnings are issued as separate messages and are broadcast on Radio 4 at the first available programme junction following receipt by the BBC. If this junction does not coincide with a news bulletin, the warning is repeated after the next news bulletin. The cancellation of warnings is not broadcast by the BBC since each of the four shipping forecast bulletins includes a summary of warnings that are in force.

10.7 WINTER FORECASTS FOR FISHING FLEETS

This service is operated between 1st October and 31st March and provides a three day wind direction and force forecast. Where appropriate, moderate to severe icing will also be included. Such broadcasts are for fishing fleets in the North Sea and South Western approaches.

These broadcasts are made by HM Coastguard stations at Aberdeen, Humber and Falmouth on 2 226 kHz as follows:

Northern North Sea

Transmitted by MRCC Aberdeen at 0820 UTC with repetition at 2020 UTC.
Covers sea areas: Viking, Forties, Cromarty, Forth, Fair Isle and Fisher.



Fig. 10.1 United Kingdom Shipping Forecast Areas.
Used with the kind permission of the Meteorological Office *(www.metoffice.gov.uk).*

Southern North Sea

Transmitted by MRSC Humber at 0910 UTC with repetition at 2140 UTC
Covers sea areas: German Bight, Dogger, Tyne, Fisher, Humber and Thames.

South Western Approaches

Transmitted by MRCC Falmouth at 0950 UTC with repetition at 2150 UTC
Covers sea areas: Plymouth, Fastnet, Lundy, Sole and Fitzroy

These forecasts are also available, on request, from neighbouring Maritime Rescue Sub-Centres.

10.8 MARINE CALL AND METFAX

This dial-up telephone weather information service is provided by the Meteorological Office and Marinecall. Users can access the system by telephone, fax SMS and email, or through prepaid subscription services. Mariners wishing to obtain the latest inshore weather information for anywhere in the UK, simply call the premium-rate MARINECALL number for the area required. This is a 24 hour service, with two day inshore forecasts covering coastal waters out to 12 miles offshore plus certain Irish Sea and English Channel sea crossings. Forecasts are updated twice a day, at 7 am and 7 pm. A similar system called MetFAX is also available.

For the telephone service dial 09068 500 followed by the area number . For the Marinecall fax forecasts dial 09060 100 followed by the area number (standard service) and 09065 300 followed by the area number (advance service)

Area location	*Area number*
3 to 5 day national inshore	450
Cape Wrath to Rattray Head including Orkney	451
Rattray Head to Berwick	452
Berwick to Whitby	453
Whitby to The Wash	454
The Wash to North Foreland	455
North Foreland to Selsey Bill	456
Selsey Bill to Lyme Regis	457
Lyme Regis to Land's End to Hartland Point	458
Hartland Point to St. David's Head	459
St. David's Head to Colwyn Bay	460
Colwyn Bay to Mull of Galloway including Isle of Man	461
Mull of Galloway to Mull of Kintyre and North Channel	462
Mull of Kintyre to Ardnamurchan	463
Ardnamurchan to Cape Wrath and Western Isles	464
Lough Foyle to Carlingford Lough	465

Marinecall Fax forecasts

At the time of going to press the charges for these fax forecasts started at £1 per minute for the standard service and £1.50 per minute for the advance.

Marinecall Advance. Inshore Coastal

This three-page forecasting service offers:

— A 48 hour report which includes a summary of gale and strong wind warnings.
— The general situation, wind speed and direction, probability and strength of gusts, developing weather conditions, visibility and sea state anticipated over the next 48 hours.
- A coastal location forecast in tabular format plotting the changing weather picture for the current hour and next five hours for four key sailing points within each inshore/coastal area.

— A synoptic chart covering today and tomorrow.

Marinecall Advance. Offshore

This three-page forecasting service offers:
— 2 to 5 day planning forecast.
— A synoptic chart covering days 2 to 5.
— A significant Wave Height Contour Graph.

Marinecall Standard. Inshore coastal

This two-page forecasting service offers:
— A 48 hour forecast.
— A synoptic chart covering today and tomorrow.

Marinecall Standard. Offshore

This two-page forecasting service offers:
— 2 to 5 day Planning forecast.
— A synoptic chart covering days 2 to 5.

Offshore areas	*Standard* (09060 100)	*Advanced* (09065 300)
North-west Scotland	468	275
Northern North Sea	469	276
Biscay	470	274
English Channel	471	270
Southern North Sea	472	271
Irish Sea	473	273

To use this service simply dial the number required and setting your fax machine to "poll" mode. Press the start button on your fax when prompted to begin transmission. A slight delay might be experienced between dialling the number and receiving your fax due to data being downloaded to your fax equipment.

The Meteorological Office offers a number of other services that includes the following:

MetWEB. A subscription service for weather services on line including:
surface pressure charts
weather reports
coastal reports
gale warnings shipping forecasts
satellite pictures
2 day inshore forecasts
3 to 5 day inshore outlooks
2 to 5 day planning

To set up a MetWEB account call 0845 300 0300.

Mobile marine services. A range of services are available including the ability to obtain a weather forecast for your exact location. Inshore waters forecasts, shipping forecasts, gale warning for sea areas and actual coastal reports are also available. To set up a registration for these services view the website at *www.metoffice.com* then click on Mobile Marine Services.

10.9 WEATHER ROUTEING SERVICES

Various commercial organisations offer weather routeing information and forecasts for the maritime industry. For details consult the Admiralty List of Radio Signals Volume 3.

10.10 REPORTING WEATHER INFORMATION

In order to assist in the dissemination of maritime weather information, ships staff are invited to co-operate with shore authorities by reporting any excessive weather conditions. Such messages should be transmitted to the nearest coast station, or land earth station if using the satellite service, and be preceded by the safety signal. In the case of meteorological messages concerning storms the service indicator = OBS — should be used, no charges will be made for these messages.

10.11 VOLUNTARY OBSERVING SHIP PROGRAMME

This is a voluntary meteorological observation by ships run under the auspices of the world Meteorological Organisation (WMO) and the Intergovernmental Oceanographic Commission (IOC). The observations made by ships are transmitted ashore using GMDSS equipment and systems to the relevant meteorological authorities where they are used to supplement the data obtained by weather satellites. No charges are levied on the communication relating to the observations for ships participating in the voluntary service.

CHAPTER 11

Electronic Position Fixing Systems

GENERAL

Electronic navigational aids have undergone some significant changes in recent times. The UK MF radiobeacon service closed in February 1999 and the Decca Navigator system closed on 31st March 2000 — an updated Loran-C system taking its place. A European global satellite navigation system known as Galileo is currently being developed. With effect from 1st July 2002 SOLAS regulations require that vessels carry a Global Navigation Satellite System receiver. This chapter aims to give an overview of some of the more recent satellite and terrestrial systems.

Note: Users should be aware that positions derived from such systems are not infallible and should be treated with caution.

11.1 UPDATING POSITION IN GMDSS RADIO EQUIPMENT

Changes in the requirements to SOLAS Chapter IV: Radio Communications as indicated in MSN 1779 (M+F) states that "all two-way communication equipment carried on board ship which is capable of automatically including the ship's position in the distress alert shall be automatically provided with this information from an external or internal navigation receiver, if either is installed. If such a receiver is not installed, the ship's position and the time at which the position was determined, shall be manually updated at intervals not exceeding 4 hours, while the ship is under way, so that it is always ready for transmission by the equipment."

11.2 GLOBAL POSITIONING SYSTEM (GPS)

GPS is also known as “Navstar” and is operated by the US Department of Defence. GPS provides a worldwide continuous positional fix capability with a constellation of 24 operational satellites (four in each of six orbital planes) at altitudes of approximately 20,000 km providing two-dimensional fixes (latitude and longitude) for marine users and three-dimensional fixes (latitude, longitude and altitude) for aeronautical, survey users, etc.

A GPS position fix is achieved by measuring the range from a number of selected satellites to the receiver. Range is determined by measuring the propagation time of received signals, but since this would require highly accurate synchronisation between satellites and receivers, the user presumes his measurements to contain a fixed clock error. Using microprocessor technology this clock error can

be resolved provided at least three satellites are in view for a two-dimensional fix. At least four satellites are needed for a three-dimensional fix.

Satellites transmit orbital information on two frequencies in the L-band:

L1 = 1 575.42 MHz Precision (P) code and Coarse /Acquire (C/A) code
L2 = 1 227.60 MHz P code only

Users should be aware that the United States Department of Defence has stated that the GPS satellites will be used primarily for military purposes and are subject to change without prior notice. The use of GPS satellites is therefore at the user's risk. The USCG has established a GPS information centre in Alexandria, Virginia which provides voice and on-line computer information services. Contact details:

USCG Navigation Center
7323 Telegraph Road
Alexandria
Va 22315-3998
USA
Tel: +1 703 313 5900
Fax: +1 703 313 5920
e-mail: *nisws@navcen.uscg.mil*
web: *www.navcen.uscg.mil*
Voice recording : +1 703 313 5907
Fax on demand : +1 703 313 5931 or 5932

11.3 GLOBAL NAVIGATION SATELLITE SYSTEM (GLONASS)

When fully operational GLONASS will have a space sector consisting of 24 orbiting satellites (eight in each of three orbital planes) at altitudes of around 19,000 km. Each satellite broadcasts its precise position and, to lesser accuracy, the position of other satellites in the constellation. Shipboard receivers calculate position, velocity and time from measurements made by satellite broadcasts. Accuracies in the region of 15 to 20 metres are expected.

GLONASS satellite status and other information can be obtained from:
Coordinational Scientific Information Centre
Russian Space Forces
PO Box 14
Moscow 117279
Russia
Tel: +7 095 333 7200
Fax: +7 095 333 8133
E-mail: *sfcsic@space.ru*
Web: *www.rssi.ru/SFCSIC/SFCSIC_main.html*

11.4 GALILEO NAVIGATION SYSTEM

Galileo is a joint European Commission and European Space Agency initiative to provide a non-military global satellite position fixing system with an expected accuracy of a few metres. Close co-operation with the IMO and ICAO will ensure that the Galileo system will be certified for SOLAS applications. It will have a constel-

lation of 30 satellites orbiting at an altitude of 24,000 km in three planes. Within each orbital plane a spare satellite will be available to replace any that might fail. Although independent of other systems Galileo will have compatibility with the present military GPS and GLONASS signals to ensure that users will be able to use GPS and Galileo signals within a single receiver. Development and validation phases are scheduled between 2002 and 2005 with operational satellites being deployed from 2006. Hopefully the system will offer a continuous worldwide service from 2008.

Determination of position works by measuring very accurately the time interval between the precise instant when signals are emitted by a satellite and the time it arrives at the receiver hence giving a distance measurement. When distance measurements from receiver to at least four satellites are known the receiver can accurately calculate its latitude, longitude and altitude.

Control of the satellites and their atomic clocks will be performed by two control centres located in Europe. A global network of uplink stations will transfer data to the satellites. A network of global sensor stations will be provided to monitor the integrity of each satellite's signals. Integrity information will be transmitted globally together with each satellite's navigation signal.

An open service will be available free of charge to mass-market users. Higher performance commercial services will be available on payment of a fee. The search and rescue service provided by Galileo will form part of Europe's contribution to the worldwide SAR network.

11.5 INTEGRATED GPS AND GLONASS

The 55° inclination of the GPS orbits favours accuracy in mid latitudes, whilst the 65° inclination of GLONASS orbits favour high latitude fixes. A receiver capable of receiving both systems takes advantage of the best features of each individual system. Additionally more satellites will be in view at any given geographical location giving a much faster acquisition time from cold start. Also improved overall system integrity is achieved with double the satellites in view since data from a bad satellite is easier to detect without the need for over-complicated receiver processing.

11.6 IMPROVING THE ACCURACY OF SATELLITE NAVIGATION

Ionospheric refraction of the satellite signals and receiver clock drift produce errors affecting the accuracy of position fixes. IMO performance standards from 1st July 2003 require higher levels of accuracy. Two systems to augment the accuracy of GPS and GLONASS are currently available (DGPS and WAAS). Other satellite-based augmentation systems (SBAS) are under development. GPS without augmentation gives an expected accuracy of 10 m (95%), DGPS 5 m (95%) and WAAS 3 m (95%). GPS navigators designed to receive both DGPS and WAAS differential signals are currently available to maritime users.

11.6.1 Differential GPS (DGPS)

To improve GPS accuracy correction data is transmitted over MF radio in order to reduce local inaccuracies. Differential GPS (DGPS) works on the principal of a fixed receiving station knowing its exact position. That station derives a GPS



Fig. 11.1 UK and Irish Stations that offer DGPS services.

position from the satellite system and if any error exists between the true and GPS positions then correction data is calculated and broadcast to vessels at sea. By applying this correction data to own GPS position a ship can determine position more accurately.

Ships may require an additional receiver operating in the marine MF band (283.5–325 kHz) in order to receive DGPS signals. However, modern GPS receivers have DGPS receivers built in. Typical range of DGPS transmissions is up to around 200 nautical miles. The use of these signals sometimes incurs a fee, although the service around UK waters is free of charge.

In the UK and Ireland DGPS services are offered by the following stations which are depicted in Fig. 11.1:

UK DGPS station	*Frequency*	*Range*
Point Lynas Light	297.5 kHz	150 miles
Nash Point	309.5 kHz	100 miles
Lizard Light	306.0 kHz	150 miles
St. Catherine's Point	307.5 kHz	100 miles
North Foreland	299.5 kHz	100 miles
Wormleighton (inland)	291.0 kHz	170 miles
Flamborough Head Light	290.5 kHz	150 miles
Stirling	285.5 kHz	200 miles
Girdle Ness Light	297.0kHz	150 miles
Sumburgh Head Light	291.5 kHz	200 miles
Butt of Lewis Light	295.5 kHz	200 miles
Irish DGPS station	*Frequency*	*Range*
Tory Island	288.5 kHz	200 miles
Loop Head	293.0 kHz	150 miles
Mizen Head	284.0 kHz	150 miles

Messages transmitted by each of the above station includes differential GPS corrections for all satellites, reference stations parameters and health of the satellites.

Other countries with DGPS stations with operational or planned services include Alaska, Algeria, Argentina, Australia, Bahrain, Belgium, Bermuda, Brazil, Bulgaria, Canada, China, Denmark, Egypt, Estonia, Faeroes, Finland, France, Germany, Hawaii, Hong Kong Iceland, India, Japan, Korea, Kuwait, Latvia, Netherlands, Norway, Poland, Portugal, Puerto Rico, Russia, Saudi Arabia, Singapore, South Africa, Spain, Sri Lanka, Svalbard, Sweden, United Arab Emirates, and USA. Full details are given in *ALRS Volume 2*.

The NELS Loran-C stations detailed in section 11.7 are being used to send DGPS signals on an experimental basis called Eurofix. This would have an extended range for DGPS signals (around 1000 km) with an expected accuracy of better than 5 metres.

11.6.2 Wide Area Augmentation System (WAAS)

The wide area augmentation system is a differential system consisting of a number of earth reference stations which constantly monitor the GPS satellites. Any differential corrections are transmitted via Inmarsat geostationary satellites. The AOR-W and POR satellites are being used at present (July 2004). A third satellite will eventually be employed to give almost global coverage.

11.6.3 Satellite augmentations systems being developed

The European Space Agency, together with its partners, is currently developing a system to augment GPS and GLONASS allowing users to achieve a position accuracy to within 5 m. Known as the European Geostationary Navigation Overlay System (EGNOS) it is scheduled to become fully operational in 2004 broadcasting differential corrections via three geostationary satellites. Japan is developing a similar system.

11.7 LORAN-C

Loran-C is a pulsed hyperbolic position fixing system operating at 100 kHz. Stations are arranged into groups comprising a master and up to four secondaries given identification letters W, X, Y and Z. Reliable range of a Loran-C chain is in the order of 1000 nautical miles from the master station, although this can be considerably extended using less accurate skywave reception.

Transmission from each station is delayed by specific time intervals known as secondary coding delay. The interval between successive transmissions from the master is unique to each chain and this is termed the group repetition interval (GRI). Receiving equipments are able to identify transmissions from stations within a particular chain from the chain's GRI.

A line of position (LOP) is derived by measuring the difference in arrival time of signals from the master and secondary stations. Two such LOPs are required for a position fix.

Earlier Loran-C receivers displayed LOP information as time delays in microseconds. Fixed propagation error corrections (Found in Publication No. 221 issued by the Defence Mapping Agency Hydrographic/Topographic Centre, Washington, DC) should be applied to these raw readings before being plotted on a Loran-C lattice chart. However, some US and Admiralty charts do have these corrections incorporated in the lattice lines and would be marked "ASF corrected" in which case corrections should not be applied.

Modern Loran-C receivers convert raw LOPs into latitude and longitude readings for direct transfer on to non-lattice charts.

Within Northwest Europe six countries have adopted Loran-C as the terrestrial complement to satellite systems. This consortium, consisting of Denmark, France, Germany, Ireland, The Netherlands and Norway laid the foundation for a regional Loran-C system called the Northwest European Loran-C Systems (NELS). NELS results in extended coverage around Northwest European waters as indicated on the predicted coverage diagram Fig. 11.2.

Norway has accepted the leading role of NELS and may be contacted at:

Norwegian Defence Communications and Data Services Administration
Operation & Maintenance Division
Oslo mil/Akershus
NO-0015 OSLO
Norway
Telephone: +47 23 09 24 00
Fax: +47 23 09 25 30

In addition a coordinating agency has been set up in Norway. Contact details:

NELS Coordinating Agency Office
Langkaial
N-1050 OSLO
Norway
Telephone: +47 23 09 24 76
Fax: +47 23 09 23 91
WWW: *http://odin.dep.no/fid/publ/nels*

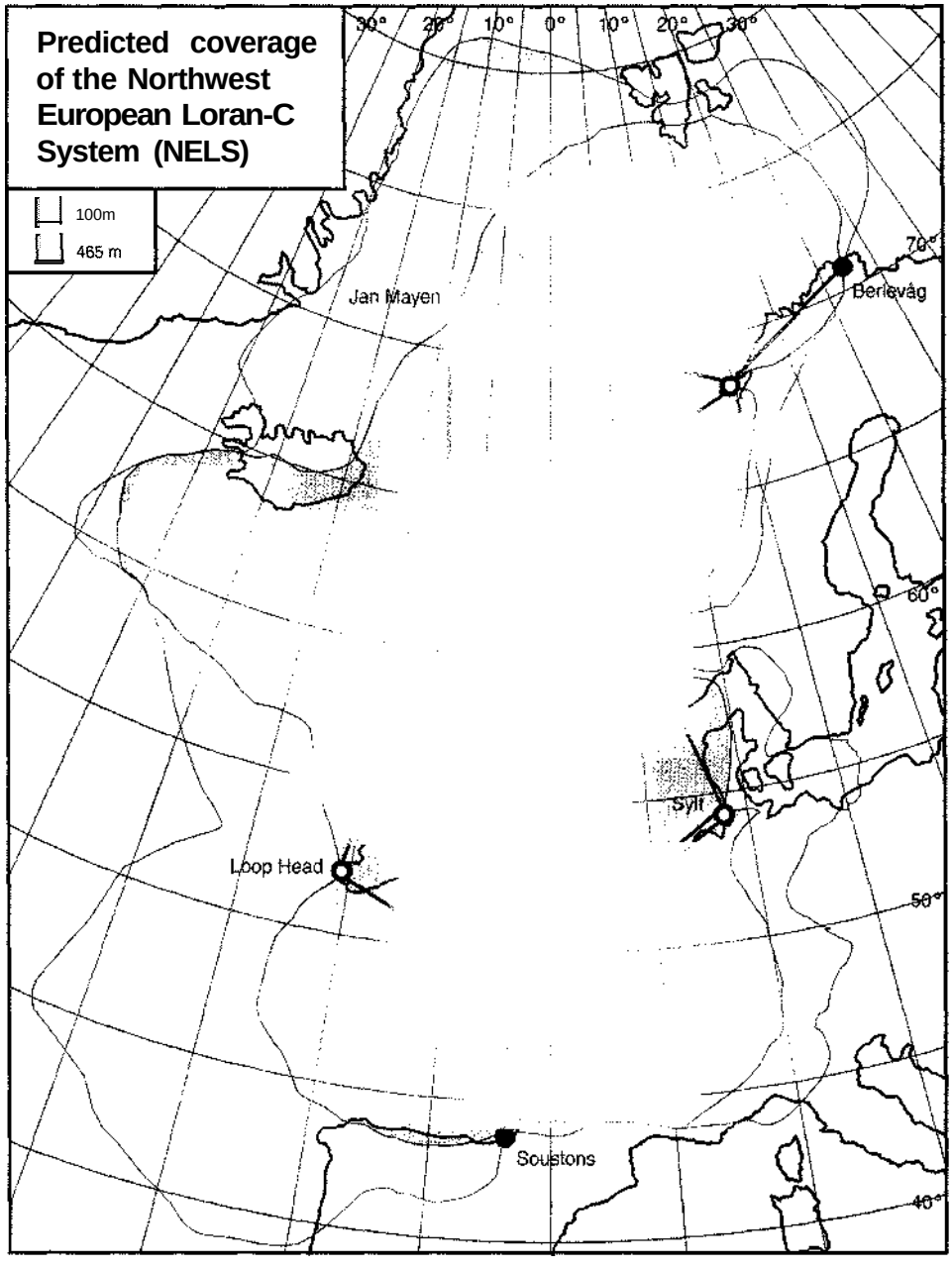


Fig. 11.2 Predicted coverage of the Northwest European LORAN-C System. Reproduced with the kind permission of NODECA, Norway.

The NELS layout consists of four Loran-C chains as follows:

(1) B0 chain GRI 7001
Master B0 and secondaries at
Jan Mayen (X) and
Berlevåg (Y)

(2) Ejde chain GRI 9007
Master Ejde and secondaries at
Jan Mayen (W)
Bø (X)
Værlandet (Y) and
Loop Head (Z) (Not transmitting at present - May 2004)

(3) Sylt chain GRI 7499
Master Sylt and secondaries at
Værlandet (Y) and
Lessay (X)

(4) Lessay chain GRI 6731
Master Lessay and secondaries at
Loop Head (Y) (Not transmitting at present — May 2004)
Sylt (Z) and
Soustons (X)

Location of above Loran-C transmitting stations is as follows:

Berlevag, Norway	70° 50' 43.014" N	29° 12' 15.980" E
Jan Mayen, Norway	70° 54' 51.478" N	08° 43' 56.525" W
B0, Norway	68° 38' 06.216" N	14° 27' 47.350" E
Ejde, Faeroe Islands	62° 17' 59.837" N	07° 04' 26.079" W
Værlandet, Norway	61° 17' 49.435" N	04° 41' 46.618" E
Sylt, Germany	54° 48' 29.975" N	08° 17' 36.856" E
Lessay, France	49° 08' 55.224" N	01° 30' 17.029" W
Soustons, France	43° 44' 23.099" N	01° 22' 49.584" W
Loop Head, W. Ireland	52° 35' 01.576" N	09° 49' 07.969" W

The NELS Loran-C system is controlled from a control centre at Brest, France. Other Loran-c chains around the world include:

Chain name	*GRI*
Newfoundland East coast	7270
Canadian East Coast	5930
Northeast USA	9960
Southeast USA	7980
Great Lakes	8970
Gulf of Alaska	7960
North Pacific	9990
Russian-American	5980
Russian	7950
Northwest Pacific	8930
Canadian West coast	5990
West coast USA	9940
Korean	9930

Chain name	*GRI*
North China Sea	7430
East China Sea	8390
South China Sea	6780
North Saudi Arabian	8830
India (Bombay)	6042
India (Calcutta)	5543

11.8 INTEGRATED LORAN-C AND SATELLITE NAVIGATION SYSTEMS

Loran-C as a system to complement GPS or GLONASS has the advantage that signals and transmission paths from terrestrial and satellite systems are totally different. Hence the weaknesses of one system can be compensated for by the strength of the other. Using the common time reference of UTC can also be used to advantage by using both systems to compute position. Integrating Loran-C with a satellite system will improve repeatable accuracy, availability, integrity and redundancy to meet the most stringent performance requirements. NELS is working towards addressing the development of low cost hybrid Loran-C/GPS receivers for use around Europe.

11.9 VHF DIRECTION-FINDING SERVICE

UK Coastguard Maritime Rescue Co-ordination Centres (MRCC) or Maritime Rescue Sub-Centres (MRSC) remotely operate the VHF direction-finding service around UK shores. Such shore station receiving antenna sites are denoted on charts with the symbol RG. This service is to be used in *EMERGENCIES* only.

Radio watch is maintained continuously on Ch. 16 VHF by MRCCs and MRSCs. If a vessel is in distress it will be requested to transmit on Ch. 16 VHF and shore stations will take DF bearings. The bearings from these stations will be communicated to the vessel and by plotting these bearings onto a chart the position of the vessel can be determined.

Vessels not in distress should not use Ch. 16 VHF but instead transmit on Ch. 67 VHF. The following list shows the precise location of VHF receiving aerials. Figure 11.3 shows the approximate location of these stations.

Great Britain

MRSC Liverpool controls :-

Snaefell	54° 15.84' N 4° 27.66' W
Walney Island	54° 06.61' N 3° 16.00' W

MRSC Holyhead controls

Great Ormes Head	53° 19.96' N 3° 21.25' W
Rhiw	52° 50.00' N 4° 37.82' W

MRSC Milford Haven controls

St Ann's Head	51° 40.97' N 5° 10.52' W

MRCC Swansea controls

Hartland	51° 01.22' N 4° 31.40' W

MRCC Falmouth controls

Lands End	50° 08.13' N 5° 38.19' W
Lizard	49° 57.60' N 5° 12.06' W
St. Mary's, Scilly Islands	49° 55.73' N 6° 18.25' W
Trevose Head	50° 32.91' N 5° 01.99' W

MRSC Brixham controls

Berry Head	50° 23.97' N 3° 29.05' W
East Prawle	50° 13.10' N 3° 42.50' W
Rame Head	50° 19.03' N 4° 13.20' W

MRSC Portland controls

Grove Point	50° 32.93' N 2° 25.20' W
Hengistbury Head	50° 42.95' N 1° 45.64' W

MRSC Solent controls

Boniface	50° 36.21' N 1° 12.03' W
Newhaven	50° 46.93' N 0° 03.01' E
Selsey	50° 43.80' N 0° 48.22' W

MRCC Dover controls

Fairlight	50° 52.19' N 0° 38.74' E
Langdon Battery	51° 07.97' N 1° 20.59' E
North Foreland	51° 22.53' N 1° 26.72' E

MRSC Thames controls

Bawdsey	51° 59.60' N 1° 25.00' E
Shoeburyness	51° 31.38' N 0° 46.50' E

MRCC Yarmouth controls

Lowestoft	52° 28.60' N 1° 42.20' E
Skegness	53° 09.00' N 0° 21.00' E
Trimingham	52° 54.57' N 1° 20.60' E

MRSC Humber controls

Cullercoats	55° 04.00' N 1° 28.00' W
Easington	53° 39.13' N 0° 05.90' E
Flamborough	54° 07.08' N 0° 05.21' W
Hartlepool	54° 41.79' N 1° 10.57' W
Newton	55° 31.01' N 1° 37.10' W
Tynemouth	55° 01.07' N 1° 24.99' W
Whitby	54 29.40' N 0° 36.30' W

MRSC *Forth controls*

Cross Law	55° 54.48' N 2° 12.31' W
Fife Ness	56° 16.70' N 2° 35.30' W
Inverbervie	56° 51.10' N 2° 15.65' W

MRCC Aberdeen controls

Dunnet Head	58° 40.31' N 3° 22.52' W
Noss Head	58° 28.80' N 3° 03.00' W
Windyhead	57° 38.90' N 2° 14.50' W

MRSC Shetland controls

Compass Head	59° 52.05' N 1° 16.30' W
Wideford Hill	58° 59.29' N 3° 01.40' W

MRSC Stornoway controls

Barra	57° 00.81' N 7° 30.42' W
Rodel	57° 44.90' N 6° 57.41' W
Sandwick	58° 12.65' N 6° 21.27' W

MRCC Clyde controls

Law Hill	55° 41.76' N 4° 50.46' W
Tiree	56° 30.62' N 6° 57.68' W
Kilchiaran	55° 45.90' N 6° 27.19' W

Channel Islands

This service is to be used in *EMERGENCIES* only.

Continuous watch is maintained on Ch. 16 VHF by Guernsey and Jersey coast guard stations. Ships should transmit on Ch. 16 VHF when in distress otherwise on Ch. 67 VHF for Guernsey or Ch. 82 VHF for Jersey. Ship's bearing from the station will be transmitted by the Coastguard on the frequency used for determining the DF bearing.

Coastguard Guernsey	49° 26.27' N 2° 35.77' W
Coastguard Jersey	49° 10.85' N 2° 14.30' W

Northern Ireland

Radio watch is maintained continuously on Ch. 16 VHF by MRCCs and MRSCs. If a vessel is in distress it will be requested to transmit on Ch. 16 VHF and shore stations will take DF bearings. The bearings from these stations will be communicated to the vessel and by plotting these bearings onto a chart the position of the vessel can be determined.

Vessels not in distress should not use Ch. 16 VHF but instead transmit on Ch. 67 VHF.

MRSC Belfast controls

Orlock Point	54° 40.41' N 5° 34.97' W
West Torr	55° 11.70' N 6° 05.20' W

11.10 RADAR BEACONS (RACONS AND RAMARKS)

Radar beacons enable navigators with radar displays to determine position with a greater degree of accuracy. They come in two forms known as Racons and Ramarks, either of which transmit microwave signals which appear on the radar screen as distinctive markings.

Racons transmit only when triggered by signals from a ship's radar usually in the 3 cm (X-band), although more racons now also respond to signals in the 10 cm

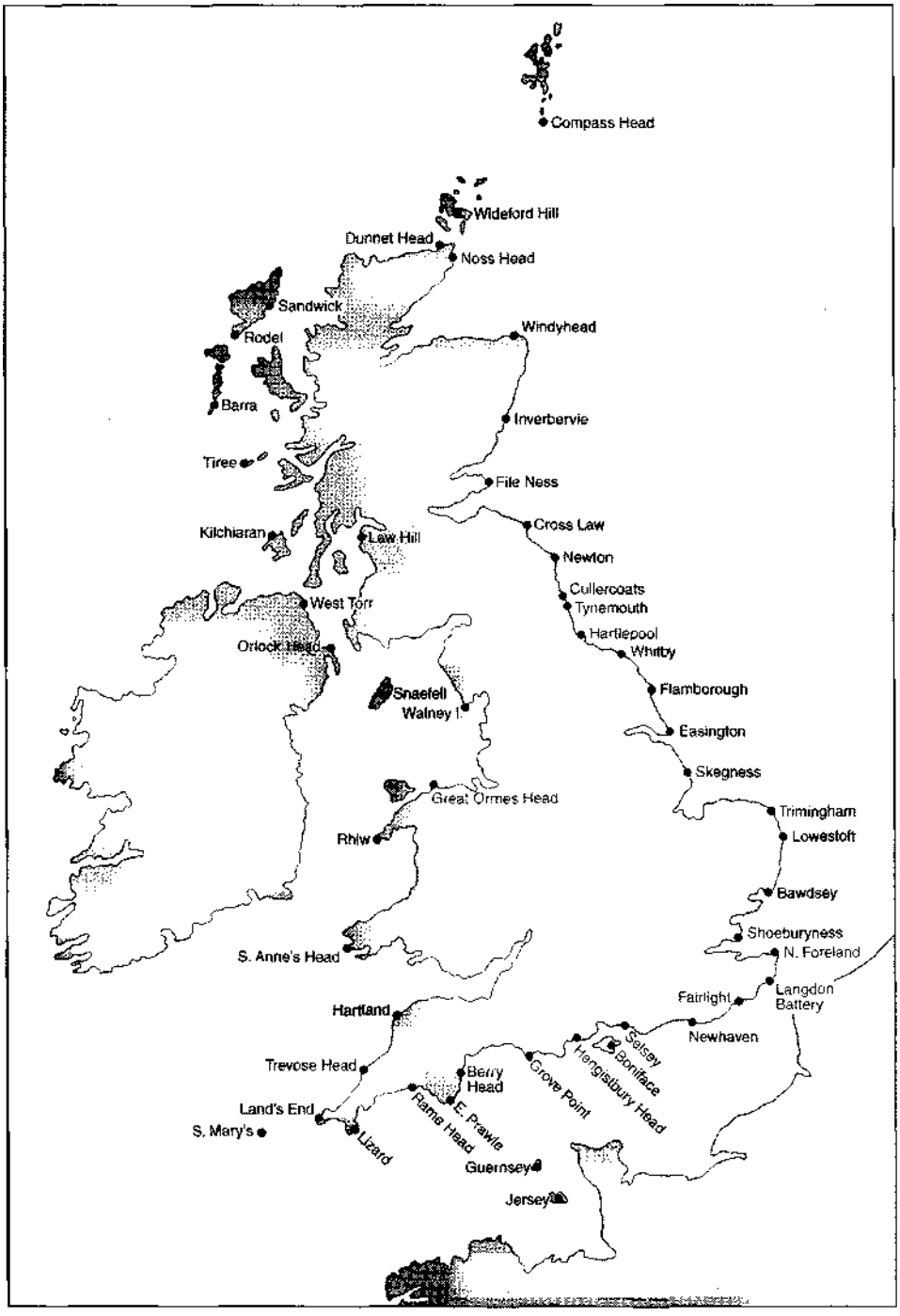


Fig. 11.3 UK VHF Direction-Finding Stations.

(S-band). The site of the racon is indicated on the user's radar screen by a single line or narrow sector which starts slightly beyond its position and then extends radially away from ship's own position (i.e. towards the edge of the radar screen). There are approximately 100 racons around UK and Irish waters with operational ranges between 3 and 25 nautical miles many of them operating on both X and S bands.

Ramarks operate without having to be triggered by external radar signals. Their appearance is similar to that of Racons except that they extend from the ship's position towards the edge of the radar screen and consequently give no indication of range.

Details of radar beacons can be found in *ALRS Volume 2.*

11.11 AUTOMATIC IDENTIFICATION SYSTEM (AIS)

The Automatic Identification System (AIS) is a radio transponder system that automatically transmits short bursts of data in the marine VHF band for reception by other ships and shore stations within range. IMO require all vessels be fitted with AIS equipment on a rolling basis. MSN 1780(M) published by the MCA in 2004 gives the revised carriage requirements for AIS which applies to all ships of 300 gt. and upwards on international voyages or calling at a port of a member state of the EU and all passenger ships, including high speed craft, irrespective of size or of 300 gt. and upwards if engaged in domestic trade as follows:

Vessel Type	*Date by which must be fitted*
1. New ships constructed on or after 1st July 2002.	Date of build.
2. Ships constructed before 1st July 2002	
(a) Passenger ships	1st July 2003
(b) Tankers	1st safety equipment survey on or after 1st July 2003
(c) Ships other than tankers or passenger ships of 50,000 gt. or more	1st July 2004
(d) Ships other than tankers or passenger ships over 300 gt. but less than 50,000 gt. engaged on international voyages.	1st safety equipment survey after 1st July 2004 or by 31st December 2004, whichever occurs earlier
(e) Ships other than tankers or passenger ships 10,000 to 49,999 gt. not engaged on international voyages.	1st July 2005
(f) Ships other than tankers or passenger ships 300 to 9,999 gt. not engaged on international voyages.	1st July 2006
(g) Ships other than tankers or passenger ships 300 to 2,999 gt. not engaged on international voyages.	1st July 2007

AIS works primarily on two dedicated frequencies VHF Ch. 87B and Ch. 88B with AIS receivers monitoring both channels.The system can also operate on VHF DSC channel 70. In some areas (such as the coasts of the USA) other channels may be used. When under the control of a vessel traffic system (VTS) the AIS can be retuned remotely to other suitable channels. AIS has three main functions:

(a) Collision avoidance when in ship-to-ship mode.
(b) Vessel monitoring when in ship-to-shore mode.
(c) Traffic management when integrated with a VTS.

Vessel information provided by AIS includes:

(a) static data programmed during installation including IMO number, vessel callsign and name, MMSI, type of vessel, length and beam, location of GPS antenna (aft of bow and port/starboard of centreline)
(b) dynamic data derived from sensors such as GPS, gyro compass, speed log etc. including position, UTC time, course and speed over ground, heading, status (at anchor, not under command etc.) and rate of turn.
(c) Voyage data entered manually by ship personnel including draught, type of cargo, destination and ETA (Master's discretion).
(d) Safety data entered manually by ship personnel at any time as needed.

The AIS equipment may have its own display unit or may be interfaced with radar/ECDIS equipment from which other vessel data may be monitored. Pilots can build up details of other vessel's movements in the immediate area. Shore authorities can monitor ship movements and can 'poll' passing ships for any of their stored data. Shore stations are able to transmit important information such as tidal and weather data. AIS is also useful during SAR operations as it allows shore authorities to monitor the movement of rescue craft. A GPS/DGPS receiver is integrated into the ship's AIS equipment in order to provide continuously updated position and accurate UTC time which is vital for system operation.

11.12 UK AUTOMATIC SHIP IDENTIFICATION AND REPORTING SYSTEM (AIRS)

The UK has developed an AIS based system for ship identification and reporting known as AIRS. The entire UK coastline is covered utilising around 20 existing MRCCs and MRSCs. It is intended that this system will eventually replace the RT ship reporting and monitoring system currently used around UK shores. Using the VHF channel 70 DSC receiver built in to a vessels AIS receiver, UK Coastguard stations can interrogate ships within VHF range in order to request vessel position, course, speed, type of vessel, length, draught, any draught or manoeuvrability restrictions, next port of call etc.

These AIS requests will be addressed to all vessels within a specified geographical area and ships equipment will respond automatically. Telecommands in the DSC request can request vessels to transfer to a suitable VHF channel in order to conduct RT communications and/or to automatically transmit their position at regular intervals (usually 30 minutes) for vessel tracking purposes.

CHAPTER 12

Certification

12.1 CERTIFICATION FOR PERSONNEL OF SHIP STATIONS AND MESs

The ITU's World Administrative Radio Conference (WARC) of 1987 modified the provisions of Article 55 of the Radio Regulations to introduce four new certificates for the GMDSS. The service of every ship RT station or MES must be controlled by an operator holding a certificate issued or recognised by the government to which the station is subject.

The service of automatic communication devices (teleprinters, data transfer systems etc) fitted to a ship station must be controlled by an operator holding a certificate issued or recognised by the government to which the station is subject. Other personnel on board will be able to operate the equipment under the supervision of the certificate holder.

In the case of complete unavailability of the operator in the course of a sea passage and solely as a temporary measure, the master or person responsible for the station may authorise an operator holding a certificate issued by another ITU member government to perform the radiocommunication service. In such circumstances the operators duties are limited solely to signals of:

(a) distress, distress alerting, urgency and safety;
(b) messages relating to (a) above;
(c) messages relating directly to safety of life at sea;
(d) urgent messages relating to the movement of the ship.

Temporary operators are bound to preserve the secrecy of correspondence and such operators must be replaced as soon as possible with an operator holding the prescribed national certificate.

In the maritime mobile service all certificates must carry the following information in one of the working languages of the ITU:

(a) the name and date of birth of the holder;
(b) the title of the certificate and its date of issue;
(c) if applicable, the number and period of validity of the certificate;
(d) the issuing administration.

All operators are under the obligation to preserve the secrecy of public correspondence. See section 8.8

12.2 CATEGORIES OF CERTIFICATES FOR SHIP STATION AND MES OPERATORS ON GMDSS SHIPS

There are four categories of certificates for personnel of GMDSS ship stations and MESs using the frequencies and techniques prescribed for those stations and for public correspondence:

(a) the first-class Radio Electronic Certificate
(b) the second-class Radio Electronic Certificate
(c) the General Operator's Certificate
(d) the Restricted Operator's Certificate

12.3 CONDITION FOR THE ISSUE OF OPERATORS CERTIFICATES

Administrations issuing certificates may, before authorising an operator to carry out their duties on board a ship, require other conditions to be fulfilled. For example, experience with automatic communication devices, further technical and professional knowledge, physical fitness etc. Administrations can take whatever steps they consider necessary to ensure the continued proficiency of operators after prolonged absences from operational duties. In the maritime mobile service administrations should take whatever steps they consider necessary to ensure the continued proficiency of operators while in service.

12.4 CERTIFICATES OF COMPETENCY AND SERVICE QUALIFICATIONS OF RADIO PERSONNEL

1.0 Introduction

1.1 The Merchant shipping (Training and Certification) Regulations 1997 (the Regulations), implemented in the United Kingdom some of the requirements of the International Convention on Standards of Training, Certification and Watchkeeping for Seafarers, 1978, as amended in 1995 (STCW 95).

1.2 Regulation 9 of the Regulations prescribes the mandatory minimum requirements for certification of radio personnel appointed for distress and safety radiocommunication purposes in accordance with regulation 19 of the Merchant shipping (Radio Installations) Regulations 1998 (hereafter known as the Radio Regulations). The Radio regulations implement in the United Kingdom, Chapter IV of the International Convention for the Safety of Life at Sea 1974 (SOLAS).

2.0 Certification Requirements

2.1 All radio personnel serving on United Kingdom registered ships, in accordance with the Radio Regulations, must reach the required vocational and academic standards as specified in STCW Code A-IV/2 and hold one on the following suitably endorsed radio certificates of competency.

1. a Restricted Operator's Certificate (ROC); or
2. a General Operator's Certificate (GOC).

2.2 In addition, all radio personnel and Electro Technical Officers (ETOs) must undertake some ancillary training, as specified in STCW Code A-VI and listed below in paragraph 3.2.

2.3 In accordance with Regulation IV/1 of STCW 95, with effect from 1st February 2002 every officer in charge of a navigational watch must be in possession of an appropriate radio operator's certificate i.e. at least a ROC. For operation outside Sea Area A1, every vessel is required to carry at least one GOC holder.

2.4 A United Kingdom GOC or ROC is obtained following examination by the MCA appointed examination body, the Association of Marine Electronics and

Radio Colleges (AMERC). Further information on the examinations is available from:

AMERC Ltd
National Administration Centre
C/o Wray Castle Limited
Bridge Mills
Stramongate
Kendal
LA9 2UB
United Kingdom
Tel: + 44 (0) 1539 742745
Fax: + 44 (0) 1539 742746
Email: *amerc_nac@wraycastle.com*

3.0 STCW 95 Endorsement

3.1 In order to comply with the requirements of STCW 95, all radio operators' certificates must be endorsed by the MCA. Applications for endorsement should be made to:

Seafaring Training and Certification Branch
Maritime and Coastguard Agency
Spring Place
105 Commercial Road
Southampton
SO15 1EG
Tel: + 44 (0) 23 8032 9136
Fax: + 44 (0) 23 8032 9252

3.2 Any application for a STCW 95 endorsement must include the following:

(1) the original GOC or ROC;
(2) a valid medical certificate;
(3) evidence that the ancillary requirements listed below have been met.

Ancillary Training Requirements	STCW 95 Regulation VI/I	STCW Code Ref.
Personal survival Techniques	Yes	A-VI/1-1
Fire Prevention and Fire Fighting	Yes	A-VI/1-2
Elementary First Aid	Yes	A-VI/1-3
Personal Safety and Social Responsibility	Yes	A-VI/1-4

3.3 Holders of deck and engineer certificates of competency will not be required to produce evidence of the ancillary requirements listed above by virtue of the training undertaken to obtain their certificates of competency.

3.4 Electro Technical Officers must hold a valid medical fitness certificate and evidence that the ancillary requirements listed in 3.2 above have been met.

4.0 Revalidation

4.1 An endorsement given for a United Kingdom GOC or ROC must be revalidated by the MCA at interval not exceeding 5 years, in accordance with Reg I/11 of STCW 95, to qualify for seagoing service.

4.2 Requests for revalidations should be sent to the MCA address given in paragraph 3.1 above, together with the following:

(1) the original GOC or ROC; and
(2) proof of service performing functions appropriate to the certificate held, for a period of at least one year in total during the previous five years.
(3) a valid Medical Certificate

5.0 Training and Maintenance of GMDSS installations

5.1 To ensure the availability of radio equipment, as required by Chapter IV of SOLAS 74, Regulation 18 of the Radio Regulations and Merchant Shipping Notice MSN 1692 (M) make provision for the use of "at sea electronic maintenance capability".

5.2 The person designated to perform at sea electronic maintenance is required to hold an appropriate certificate as specified by the International Telecommunications Union (ITU) Radio Regulations, and as referred to in STCW Code B-I/VI, or have equivalent at sea electronic maintenance qualifications. Further information and details on suitable training courses may be obtained from AMERC at the address given in paragraph 2.4 above.

6.0 Further Advice

6.1 Candidates who need further information are advised to contact the MCA at the address given below:

Seafaring Training and Certification Branch
Maritime and Coastguard Agency
Spring Place
105 Commercial Road
Southampton
SO15 1EG
Tel: + 44 (0) 23 8032 9136
Fax: + 44 (0) 23 8032 9252

12.5 ADDITIONAL KNOWLEDGE/TRAINING REQUIREMENTS FOR RT OPERATORS AND RADIO OFFICERS

(a) The provision of radio services in emergencies including:
  (i) abandon ship;
  (ii) fire aboard ship;
  (iii) partial or full breakdown of the radio station.
(b) The operation of lifeboats, liferafts, buoyant apparatus and their equipment, with special reference to portable and fixed lifeboat radio apparatus and EPIRBs.
(c) Survival at sea.
(d) First aid.
(e) Fire prevention and fire-fighting with particular reference to the radio installation.
(f) Preventive measures for the safety of ship and personnel in connection with hazards related to radio equipment including electrical, radiation, chemical and mechanical hazards.

(g) The use of the International Aeronautical and Maritime Search and Rescue Manual (IAMSAR) (including any document amending the Manual which is considered by the Secretary of State to be relevant from time to time and is specified in a Merchant Shipping Notice) with particular reference to radiocommunications.

(h) Ship position-reporting systems and procedures.

(i) The use of the International Code of Signals and the Standard Marine Navigational Vocabulary.

(j) Radio medical systems and procedures.

12.6 THE ASSOCIATION OF MARINE ELECTRONIC AND RADIO COLLEGES (AMERC)

AMERC is an international organisation that has wide experience in running a range of vocational courses principally to meet the needs of the maritime and related industries. In addition to this function the organisation has for many years been responsible for setting and marking examinations for nationally recognised qualifications. An agreement was drawn up between the Radiocommunications Agency of the Department of Trade and Industry (now Ofcom) and AMERC in 1991 to conduct Radio Operator Examinations, to administer the system and to distribute Certificates of Competence and Authorities to Operate on its behalf.

Currently, the responsibility for all UK Radio Operator Examinations has been transferred to the Maritime Coastguard Agency (MCA) an executive agency for the Department of the Environment, Transport and the Regions. However AMERC continues to provide an examination service for the following examinations.

- CEPT GMDSS General Operator's Certificate of Competence (GOC)
- CEPT GMDSS Restricted Operator's Certificate of Competence (ROC)
- CEPT GMDSS Long Range Radiotelephone Operator's Certificate of Competence (LRC) including the Satellite Module, DSC Module and replacement of W/T, R/T and GMDSS Certificates.

The Executive Committee of AMERC has the ultimate responsibility for maintaining examination standards and ensuring efficient operation of the examination system. In order to maintain a uniformly high standard throughout their home and overseas centres, quality assurance procedures have been introduced into the scheme. The Executive Council of AMERC appoints a Chief Examiner and an Examination Panel to moderate examination papers as well as ensuring that statistical analysis of results is maintained by the National Administration Centre (NAC). Policy changes that may effect the examination system have to be agreed between the MCA and the Executive Committee. A Maritime Board appointed by the Executive Committee is responsible for overseeing that the conduct and quality assurance procedures are carried out to the appropriate standards.

The National Administration Centre of AMERC was set up to act as central coordinating administration for the entire examination system. Among other duties, the NAC is responsible for dealing with general enquiries, receiving candidates examination fees and issuing Certificates to successful candidates.

The address for the National Administration Centre is given in section 12.4

12.6.1 Regional Examination Centres (RECs)

The bulk of the examination service is provided by a number of Regional Examination Centres (RECs) both in the UK and overseas which have the necessary equipment and staff to conduct the relevant courses and examinations. All RECs must have course approval certificates from the MCA before they can offer courses or examination services.

Overseas RECs operate in exactly the same manner as UK RECs apart from an initial induction period. During this period the conduct of the SOLAS and regulations papers is supervised by the British Council and the Chief Examiner. It should be noted that all GMDSS related examination are conducted in the English language at all overseas RECs.

12.7 GMDSS EXAMINATION STRUCTURE IN THE UK

Candidates for the GOC examination are required by the MCA to undertake an approved course of study. Each candidate for examination must make application to a Regional Examination Centre and pay the appropriate examination fee. Candidates must also submit two recent passport type photographs of themselves and provide a valid course completion certificate before sitting the examination. There is an age restriction on candidates. While those under the age of 18 may complete a course and successfully pass the examination, certificates cannot be issued to any candidate until they have attained 18 years of age. For GOC certificates issued by the UK administration there are no nationality requirements.

The GMDSS General Operator's Certificate examination consists of the following four elements.

(a) A written paper relating to Safety of Life at Sea, where a time limit of 15 minutes is allowed. The pass mark for this element is 30/40 or 75%.

(b) A multiple choice paper of 15 minutes duration relating to questions on Regulations. The pass mark for this element is 9/15 or 60%.

(c) An RT practical examination of simulated distress operating procedures. The pass mark for this element is 36/48 or 75%. The examination time for this element is typically 30 minutes for six candidates.

(d) An Operational Performance Test carried out on a range of GMDSS equipment. The pass mark for this element is 85/120 or 70% with a maximum examination time of 90 minutes per candidate.

12.7.1 GMDSS examination appeals procedure

As described earlier, AMERC administers examinations for the GMDSS General Operator's Certificate through approved Regional Examination Centres (RECs). Candidates for this certificate who are not satisfied with the services they have received from an REC, should in the first instance appeal in writing to that REC. If a satisfactory outcome is not achieved from this action the candidate may then appeal by writing to the AMERC NAC (GMDSS) at Kendal requesting that they consider the appeal. The situation is also investigated by the Chief Examiner who then makes a report to AMERC's Maritime Board.

12.8 CEPT GMDSS GENERAL OPERATOR'S CERTIFICATE: OUTLINE EXAMINATION SYLLABUS

A harmonised GMDSS General Operators Certificate, recognised by those European countries belonging to the Conference of European Postal and Telecommunications Administrations (CEPT) has been agreed, Recommendation T/R 31-03 E (Bonn 1993). Individual CEPT administrations will have a national examination for this certificate for the maritime mobile service.

The national examinations which qualify the candidate for the GMDSS General Operators Certificate shall cover the subjects necessary for safe and efficient use of a ship station and survival craft radio equipment. The examination shall consist of theoretical and practical tests and shall include at least:

A. Knowledge of the general principles and basic features of:

(a) The Maritime Mobile Service.
(b) The Maritime Mobile Satellite-Service.

B. Detailed practical knowledge and ability to use the basic equipment of a ship station.

(a) Use in practice the basic equipment of a ship station.
(b) Digital Selective Calling (DSC).
(c) General principles of NBDP and TOR systems. Ability to use maritime NBDP and TOR equipment in practice.
(d) Usage of Inmarsat systems. Inmarsat equipment, or simulator in practice.
(e) Fault locating.

C. Operational procedures and detailed practical operation of GMDSS system and sub-systems.

(a) Global Maritime Distress and Safety System (GMDSS).
(b) Inmarsat.
(c) NAVTEX.
(d) Emergency Position Indicating Radio Beacons (EPIRBs).
(e) Search and Rescue Transponder (SART).
(f) Distress, urgency and safety communication procedures in the GMDSS.
(g) Distress, urgency and safety communication with non-SOLAS ships which use RT
(h) Search and rescue operation (SAR).

D. Miscellaneous skills and operational procedures for general communications.

(a) Ability to use English language, both written and spoken, for the satisfactory exchange of communications relevant to the safety of life at sea.
(b) Obligatory procedures and practices.
(c) Practical and theoretical knowledge of general communication procedures.

12.9 CEPT GMDSS GENERAL OPERATOR'S CERTIFICATE: DETAILED EXAMINATION SYLLABUS

A. Knowledge of the basic features of the maritime mobile service and the maritime mobile-satellite service.

A1.1 The general principles and basic features of the maritime mobile service.

(a) Types of communication in the maritime mobile service.

Distress, urgency and safety communications
public correspondence
port operations service
ship movement service
intership communication
on-board communications.

(b) Types of station in the maritime mobile service.
Ship stations
Coast stations
Maritime rescue co-ordination centres (MRCC)
Pilot Stations, port stations etc.
Aircraft stations.

(c) Elementary knowledge of frequencies and frequency bands.
The concept of frequency
The equivalence between frequency and wavelength
The units of frequency. Hz, kHz, MHz and GHz
The subdivision of the most significant part of the radio spectrum: MF, HF, VHF, UHF, SHF.

(d) Characteristics of frequencies.
Different propagation mechanisms: propagation in free space, ground wave, ionospheric propagation
Propagation of MF frequencies
Propagation of different HF frequency bands
Propagation of VHF and UHF.

(e) Knowledge of the role of the various modes of communication.
DSC
Radiotelephony
Radiotelex (NBDP)
Facsimile
Data
Morse telegraphy.

(f) Knowledge of the different types of modulation and classes of emission.
Classes of emission
Carrier frequency and assigned frequency
Bandwidth of different emissions
Official designations of emissions (eg F1B, J3E, A3E A1A etc)
Unofficial designations of emissions (eg TLX, SSB, AM, CW etc).

(g) Frequencies allocated to the maritime mobile service.
The usage of MF, HF, VHF, UHF and SHF frequencies in the maritime mobile service
The concept of radio channel. Simplex, semi-duplex and duplex. Paired and unpaired frequencies
Frequency plans and channelling systems
HF telephony and VHF telephony (Relevant appendix of the ITU Radio Regulations)
HF telex (Relevant appendixes of the ITU Radio Regulations)
MF telephony and telex for Region 1 (Geneva 85 plan)
Distress and safety frequencies of the pre-GMDSS system

GMDSS distress and safety frequencies
Calling frequencies.

A1.2 The general principles and basic features of the maritime mobile-satellite service.

(a) Basic knowledge of satellite communications.
Inmarsat space segment
Modes of communication:
Telex services
Telephone services
Data and facsimile communications
Store and forward operation
Distress and safety communications
Inmarsat-A communications services
Inmarsat-B communications services
Inmarsat-C communications services
Inmarsat-M communications services
Inmarsat Enhanced Group Call (EGC) system.

(b) Types of stations in the maritime mobile-satellite service.
Land earth stations (LES)
Network Co-ordination Station (NCS)
Mobile Earth Stations (MES).

B. Detailed practical knowledge and ability to use the basic equipment of a ship station.

B1.1 Knowledge of, and ability to use in practice, the basic equipment of a ship station.

(a) Watchkeeping receivers.
The controls and usage of a VHF DSC watch receiver
The controls and usage of a MF DSC watch receiver and MF/HF DSC watch receiver.

(b) VHF radio installation.
Channels
Controls
Usage
DSC.

(c) MF/HF radio installation.
Frequencies
Typical controls and usage, eg connecting the power, selecting receiver frequency, selecting transmitter/receiver frequency, selecting ITU channel number, tuning the transmitter, selecting the class of emission, using volume and squelch controls, using clarifier or receiver fine tuning, controlling RF gain, using automatic gain control (AGC), using 2 182 kHz instant selector

(d) Antennas.
Isolators
VHF whip antennas
MF/HF whip antennas
MF/HF wire antennas
Satellite antennas

(g) Batteries.
Different kinds of batteries and their characteristics
Charging
Maintenance of batteries
UPS systems

(h) Survival craft radio equipment.
Portable two-way VHF radiotelephone apparatus
SART
EPIRB.

B2.1 Digital Selective Calling (DSC).

(a) Call format specifier:
distress call
all ships call
call to individual station
geographic area call
group call
automatic/semi-automatic service.

(b) Call address selection with the MMSI number system :
the nationality identification
group calling numbers
coast station numbers
ship station numbers

(c) Call categorisation:
distress
urgency
safety
ship's business
routine.

(d) Call telecommand and traffic information:
distress alerts
other calls
working frequency information.

B2.2 Knowledge of the general principles of NBDP and TOR systems. Ability to use maritime NBDP and TOR equipment in practice.

(a) NBDP systems.
Automatic systems
Semi-automatic systems
Manual systems
ARQ mode
FEC mode
ISS/IRS arrangement
Master and slave
Radiotelex number
Answerback
Numbering of the SSFC selective calling system.

(b) Telex Over Radio (TOR) equipment.
Controls and indicators
Keyboard operation.

B2.3 Knowledge of the usage of Inmarsat systems. Ability to use Inmarsat equipment, or simulator in practice.

(a) Inmarsat-A/B MES:
Satellite acquisition
telex services
telephone services
data and facsimile communications.

(b) Inmarsat-C MES:
Components of an Inmarsat-C MES terminal
entering/updating position,
usage of an Inmarsat-C MES
sending and receiving test messages.

(c) Inmarsat EGC Receiver:
Programming a MES for EGC message reception
Selecting operating mode for EGC reception.

B2.4 Fault locating:

Proficiency in elementary fault localisation by means of built in measuring instruments or software in accordance with equipment manuals. Elementary fault repair such as replacement of fuses and indicator lamps and the like.

C. *Operational procedures and detailed practical operation of GMDSS system and sub-systems.*

C1.1 Global Maritime Distress and Safety system (GMDSS).

(a) Sea areas and GMDSS master plan.
(b) Watchkeeping on distress frequencies.
(b) Functional requirements of ship stations.
(d) Carriage requirements of ship stations.
(e) Sources of energy of ship stations.
(f) Means of ensuring availability of ship station equipment.
(g) Licences, radio safety certificates, inspections and surveys.

C1.2 Inmarsat usage in the GMDSS.

(a) The Inmarsat-A/B Mobile Earth Station.
Distress communications
use of the distress facility
satellite acquisition
telex and telephony distress calls
procedures for distress calls
MRCCS associated with the Land Earth Stations.

(b) Inmarsat-C MES:
Distress and safety services:
sending a distress alert
sending a distress priority message
Inmarsat-C safety services
two-digit code safety services.

(c) Inmarsat EGC:
Purpose of EGC system
all-ships messages and Inmarsat system messages
Classes of Inmarsat-C MES and their EGC reception.

C1.3 NAVTEX.
(a) The NAVTEX system:
Purpose of NAVTEX
NAVTEX frequencies
reception range
message format (transmitter identity, message type, message number).
(b) NAVTEX receiver:
Selection of transmitters
selection of message type
messages which cannot be rejected
use of subsidiary controls and changing paper.

C1.4 Emergency Position Indicating Radio Beacons (EPIRBs).
(a) Satellite EPIRBs:
Basic characteristics of operation on 406 MHz
basic characteristics of operation on 1.6 GHz
121.5 MHz including homing functions
information contents of a distress alert,
manual usage
float-free function.
Routine maintenance:
testing
checking battery expiry date
checking the hydrostatic release mechanism expiry date
(b) VHF DSC EPIRB:
Basic characteristics of operation on Ch.70

C1.5 Search and Rescue Transponder (SART).
(a) Search and Rescue Transponder SART:
The main technical characteristics
operation
range of a SART transmitter
Routine maintenance: checking battery expiry date.

C1.6 Distress, Urgency and safety communication procedures in the GMDSS
(a) Distress Communications.
DSC distress alert :
The definition of distress alert
transmission of a distress alert
transmission of a shore to ship distress alert relay
transmission of a distress alert by a station not itself in distress.
Receipt and acknowledgement of DSC distress alert:
Acknowledgement procedure by RT
acknowledgement procedure by telex (NBDP)
receipt and acknowledgement by a coast station
receipt and acknowledgement by a ship station.

Handling of distress alerts:
Preparations for handling distress traffic
distress traffic terminology
Testing DSC distress and safety calls
On-scene communications
SAE operations.

(b) Urgency and safety communications:
The meaning of urgency and safety communications
procedures for DSC urgency and safety calls
urgency communications
radio medical services
medical transports
safety communications.

(c) Radiotelephony communication with non-GMDSS stations.
distress signal
distress call
distress message
acknowledgement of a distress message
distress traffic terminology
transmissions of a distress message by a station not itself in distress
medicaladvice.

(d) Reception ofMSI.
Reception by NAVTEX
reception by Inmarsat EGC
reception by HF telex
the navigational warning signal of the of the old distress and safety system
the navigational warnings transmitted by RT.

(e) Protection of distress frequencies.
Guard bands
tests of the distress frequencies
transmissions during distress traffic
avoiding harmful interference
prevention of unauthorised transmissions.

(f) Distress urgency and safety communications with non-SOLAS ships which only use RT distress signal
distress call
distress message
acknowledgement of a distress message
distress traffic terminology
transmission of a distress message by a station not itself in distress
urgency signal
medical advice
safety signal

C1.7 Search and rescue operation (SAR).

(a) The role of RCCs.

(b) International Aeronautical and Maritime Search and Rescue Manual (IAMSAR).

(c) Maritime rescue organisations.
(d) Ship reporting systems.

D. Miscellaneous skills and operational procedures for general communications.

D1.1 Ability to use English language, both written and spoken, for the satisfactory exchange of communications relevant to the safety of life at sea.

(a) Use of the International Code of Signals and IMO Standard Marine Navigational Vocabulary/Seaspeak.
(b) Recognised standard abbreviations and commonly used service codes.
(c) Use of the international phonetic alphabet.

D1.2 Obligatory procedures and practices.

(a) Effective use of obligatory documents and publications.
(b) Radio record keeping.
(c) Knowledge of the regulations and agreements governing the maritime mobile service and the maritime mobile-satellite service.

D1.3 Practical and theoretical knowledge of general communication procedures.

(a) Selection of appropriate communication methods in different situations.
(b) Traffic lists.
(c) Radiotelephone call.
Method of calling a coast station by RT
ordering for a manually switched link call
ending the call
special facilities of calls
method of calling a coast station by DSC
selecting an automatic RT call.
(d) The radiotelegram.
The parts of a radiotelegram *viz,* preamble, service instructions and indications, address, text, signature
Addresses *viz,* full address, registered address, telephonic address, telex address.
Counting of words
Transmission of a telegram by RT
Transmission of a telegram by telex.
(e) Traffic charges.
International charging system
Inmarsat communication charging system
AAIC code
the meaning of land line charge
coast station charge and ship charge
currencies used in international charging.
(e) Practical traffic routines.
(f) World geography, especially the principal shipping routes and related communication routes.

12.10 ROC EXAMINATION STRUCTURE

Candidates for the ROC examination are required by the MCA to undertake an approved course of study. Each candidate for examination must make application

to a Regional Examination Centre and pay the appropriate examination fee. Candidates must also submit two recent passport type photographs of themselves and provide a valid course completion certificate before sitting the examination. There is an age restriction on candidates. While those under the age of 18 may complete a course and successfully pass the examination, certificates cannot be issued to any candidate until they have attained 18 years of age. For ROC certificates issued by the UK administration there are no nationality requirements.

The examination for the CEPT GMDSS Restricted Operator's Certificate (ROC) consists of the following three elements:

(a) Safety of Life at Sea (SOLAS), which may be given as a written or oral examination, the duration of which is subject to the examiners discretion. This is in two sections:

Section 1. ONE question which carries 12 marks.
Section 2. EIGHT questions each carrying 4 marks.

The candidate has to achieve at least 9 marks in Section 1, and a total of 33/44 marks (75%) overall.

(b) Radio Regulations/Operational Performance Test on a range of marine radio equipment, the duration of which is typically 45 minutes per candidate. The pass mark for this element *is* 54/72 (75%).

(c) A radiotelephony distress communication working of simulated SOLAS operating procedures, the duration of which is typically 30 minutes per group of six candidates. The pass mark for this element is 36/48 (75%).

12.10.1 Conduct of ROC GMDSS examination and appeals procedure

As with the GOC, Candidates for the ROC certificate who are not satisfied with the services they have received from from an AMERC approved REC, should in the first instance appeal in writing to the Coordinator of that REC. If a satisfactory outcome is not achieved from this action the candidate may then appeal by writing to the AMERC NAC (GMDSS) at Kendal. The matter is then investigated by the AMERC Chief Examiner who subsequently makes a report directly to the AMERC Maritime Board.

12.11 CEPT GMDSS RESTRICTED OPERATOR'S CERTIFICATE: OUTLINE EXAMINATION SYLLABUS

The examination should consist of theoretical and practical tests and shall include at least:

A. Knowledge of the basic features of the maritime mobile service.
B. Detailed practical knowledge and ability to use of the basic equipment of a ship station.
B1. Use in practice the basic equipment of a ship station.
B2. Digital Selective Calling (DSC).
C. Operational procedures and detailed practical operation of GMDSS system and sub-systems.
C1. Global Maritime Distress and Safety System (GMDSS).
C2. NAVTEX.

C3. Emergency Position Indicating Radio Beacons (EPIRBs).
C4. Search and Rescue Transponder (SART)
C5. Distress, urgency and safety communication procedures in the GMDSS.
C6. Distress, urgency and safety communications with non-SOLAS ship which only use RT
C.7 Search and rescue operation (SAR).
D. Miscellaneous skills and operational procedures for general communications.
D1. Ability to use English language, both written and spoken, for the satisfactory exchange of communications relevant to the safety of life at sea.
D2. Obligatory procedures and practices.
D3. Practical and theoretical knowledge of general communication procedures.

12.12 CEPT GMDSS RESTRICTED OPERATOR'S CERTIFICATE: DETAILED EXAMINATION SYLLABUS

A. Knowledge of the basic features of the maritime mobile service

1.1 Types of communication in the maritime mobile service.
- Distress, urgency and safety communications
- public correspondence
- port operations service
- ship movement service
- intership communication
- on-board communications.

1.2 Types of station in the maritime mobile service.
- Ship stations
- Coast stations
- Maritime rescue co-ordination centres (MRCC)
- Pilot Stations, port stations etc
- Aircraft stations.

1.3 Elementary knowledge of frequencies and frequency bands.
- The concept of frequency.

1.4 Characteristics of frequencies.
- Propagation of VHF and UHF.

1.5 Frequencies allocated to the maritime mobile service.
- The usage of VHF and UHF frequencies in the maritime mobile service
- The concept of radio channel
- Simplex, semi-duplex and duplex
- Paired and unpaired frequencies
- Frequency plans for VHF telephony (Relevant appendix of the ITU Radio Regulations)
- GMDSS distress and safety frequencies
- Calling frequencies.

B. Detailed practical knowledge and ability to use the basic equipment of a ship station.

Bl. Knowledge of and ability to use in practice, the basic equipment of a ship station.

1.1 VHF radio installation
- Channels
- Controls
- Usage
- DSC.

1.2 Antennas
- VHF antennas
- Antennas for the NAVTEX system.

1.3 Batteries
- Different kinds of batteries and their characteristics :
- Charging
- Maintenance of batteries
- UPS systems.

1.4 Survival craft radio equipment
- Portable two-way VHF radiotelephone apparatus
- SART
- EPIRB.

B2. Digital Selective Calling (DSC).

2.1 Call format specifier:
- distress call
- all ships call
- call to individual station
- geographic area call
- group call
- automatic/semi-automatic service.

2.2 Call address selection with the MMSI number system:
- the nationality identification
- group calling numbers
- coast station numbers
- ship station numbers.

2.3 Call categorisation:
- distress
- urgency
- safety
- ship's business
- routine.

2.4 Call telecommand and traffic information:
- distress alerts
- other calls
- working frequency information.

2.5 Usage of VHF Ch.70.

C. Operational procedures and detailed practical operation of GMDSS system and sub-systems.

C1. Global Maritime Distress and Safety system (GMDSS).

1.1 Sea areas and GMDSS master plan.

1.2 Watchkeeping on VHF distress frequencies.

1.3 Functional requirements of ship stations sailing within the limits of sea area A1.

1.4 Carriage requirements of ship stations sailing within the limits of sea area A1.
1.5 Sources of energy of ship stations.
1.6 Means of ensuring availability of ship station equipment.
1.7 Licences, radio safety certificates, inspections and surveys.

C2. NAVTEX.
2.1 The NAVTEX system :
Purpose of NAVTEX
NAVTEX frequencies
reception range
message format (transmitter identity, message type, message number).
2.2 NAVTEX receiver:
Selection of transmitters
selection of message type
messages which cannot be rejected
use of subsidiary controls and changing paper.

C3. Emergency Position Indicating Radio Beacons (EPIRBs).
3.1 Satellite EPIRBs:
Basic characteristics of operation on 406 MHz
basic characteristics of operation on 1.6 GHz
121.5 MHz including homing functions
information contents of a distress alert
manual usage
float-free function.
Routine maintenance:
testing
checking battery expiry date
cleaning of the float-free release mechanism.
3.2 VHF DSC EPIRB.
Basic characteristics of operation on Ch.70.

C4. Search and Rescue Transponder (SART).
4.1 Search and Rescue Transponder (SART) :
The main technical characteristics
operation
range of a SART transmitter.
Routine maintenance of a SART:
checking battery expiry date.

C5. Distress, urgency and safety communication procedures in the GMDSS.
5.1 Distress Communications.
DSC distress alert:
The definition of distress alert
transmission of a distress alert
transmission of a shore-to-ship distress alert relay
transmission of a distress alert by a station not itself in distress.
Receipt and acknowledgement of DSC distress alert :
Acknowledgement procedure
receipt and acknowledgement by a coast station

receipt and acknowledgement by a ship station.
Handling of distress alerts:
Preparations for handling distress traffic
distress traffic terminology.
Testing DSC distress and safety calls .
On-scene communications
SAR operation.
5.2 Urgency and safety communications via DSC equipment:
The meaning of urgency and safety communications
procedures for DSC urgency and safety calls
urgency communications
radio medical services
medical transport
safety communications.
5.3 Reception of MSI.
Reception by NAVTEX
the navigational warnings transmitted by RT.
5.4 Protection of distress frequencies.
Guard bands
tests of the distress frequencies
transmissions during distress traffic
avoiding harmful interference
prevention of unauthorised transmissions.

C6. Distress urgency and safety communications with non-SOLAS ships which only use RT
distress signal
distress call
distress message
acknowledgement of a distress message
distress traffic terminology
transmission of a distress message by a station not itself in distress
urgency signal
medical advice
safety signal

C7. Search and rescue operation (SAR).
6.1 The role of RCCs.
6.2 International Aeronautical and Maritime Search and Rescue Manual (IAMSAR).
6.3 Maritime rescue organisations.
6.4 Ship reporting systems.

D. Miscellaneous skills and operational procedures for general communications.

Dl. Ability to use English language, both written and spoken, for the satisfactory exchange of communications relevant to the safety of life at sea.
1.1 Use of the International Code of Signals and IMO Standard Marine Navigational Vocabulary/Seaspeak.
1.2 Recognised standard abbreviations and commonly used service codes.

1.3 Use of the international phonetic alphabet.

D2. Obligatory procedures and practices.

2.1 Effective use of obligatory documents and publications.

2.2 Radio record keeping.

2.3 Knowledge of the regulations and agreements governing the maritime mobile service.

D3. Practical and theoretical knowledge of general communication procedures.

3.1 Traffic lists.

3.2 Radiotelephone call.
Method of calling a coast station by RT
ordering for a manually switched link call
ending the call
special facilities of calls,
method of calling a coast station by DSC
selecting an automatic RT call.

3.3 Traffic charges.
International charging system
AAIC code
currencies used in international charging
the meaning of land line charge, coast station charge and ship charge.

3.4 Practical traffic routines.

3.5 Principal shipping routes and related communication routes *appropriate* for ships sailing within the limits of sea area A1.

12.13 RADIO CERTIFICATION FOR NON-SOLAS VESSELS : GENERAL INFORMATION

The introduction of GMDSS in February 1992 made it necessary to harmonise examination requirements of professional radio personnel, based on the syllabi of Article 55 of the Radio Regulations. The GMDSS GOC is a certificate which allows the holder to operate radio equipment on GMDSS ships in all four sea areas.

For compulsory fitted vessels sailing exclusively in GMDSS sea area A1, the Restricted Operators Certificate (ROC) is available for professional radio personnel. Courses and examinations leading up to the issue of ROCs to successful candidates became available in the UK from 1st January 1998.

Considerable advantages exist for non SOLAS vessels which voluntary install GMDSS radio equipment, using some but not all of the GMDSS frequencies. However radio personnel on such craft would not need the same level of certification as radio personnel on compulsory fitted ships. Thus for the non-compulsory craft, two certificates have been made available called the Long Range Certificate (LRC) and the Short Range Certificate (SRC).

12.13.1 The Long Range Certificate (LRC)

The Long Range Certificate has sufficient flexibility in its depth of study and the range of knowledge in its syllabus, that makes it appropriate to the needs of radio personnel sailing beyond the range of VHF DSC coast stations. This syllabus also provides for the certification in the use of satellite equipment where appropriate.

The LRC replaces the Restricted Radiotelephone Operator's Certificate (VHF, MF, HF), and examinations for the LRC commenced nationally from the 1st September 1995. Successful candidates are issued with a CEPT Long Range Radiotelephone Operator's Certificate of Competence plus an Authority to Operate granted by the Secretary of State for Trade and Industry.

12.13.2 The Short Range Certificate (SRC)

As stated previously relating to the LRC, considerable advantages exist for non SOLAS vessels which voluntary install GMDSS radio equipment, using some but not all of the GMDSS frequencies. For non-compulsory SOLAS fitted vessels sailing exclusively in GMDSS sea area A1, the CEPT GMDSS Short Range Certificate (SRC), which deals with VHF topics only, replaces the Restricted (VHF) Radiotelephone Operator's Certificate. Once again, a certificate has been made available, with sufficient flexibility in its depth of study and range knowledge in its syllabus, which is appropriate to the needs of radio personnel sailing within range of VHF DSC coast stations. Courses and examinations leading up to the issue of SRCs to successful candidates became available in the UK from 1st January 1998.

Note: Existing Restricted Radiotelephone Operator's Certificates remain valid for the lifetime of their holders for use in those areas of radiocommunication covered by their original examination. These DO NOT include GMDSS.

12.14 LRC: EXAMINATION STRUCTURE

There are no nationality requirements in respect of LRC certificates issued by the UK Administration, certificates cannot be issued to candidates who are under 16 years of age. Candidates are strongly recommended to undertake a course of study prior to the examination.

The examination consists of the following four elements:

(a) A SOLAS paper consisting of 12 questions, which may be written or oral. The pass mark is 36/48 or 75%. Duration is left to examiners discretion.

(b) A Radio Regulations paper consisting of 10 multiple choice questions, which may be written or oral. The pass mark is 6/10 or 60%. Duration is left to examiners discretion.

(c) Operational performance test on marine radio equipment. The pass mark is 56/80 or 70%. Duration, typically 45 minutes per candidate.

(d) RT distress communications working of simulated SOLAS procedures. The pass mark is 36/48 or 75%. Duration, typically 30 minutes for a group of six candidates.

12.14.1 LRC satellite module: Examination structure

The examination for this module can be taken while undertaking the examination for the LRC at no extra cost to the candidate. However, if taken separately from the LRC examination, a separate fee will be payable. This module may also be taken by holders of an existing Restricted RT Certificate, on payment of an examination fee. Successful candidates who pass this option will have to return their Restricted RT Certificate (or LRC) to the AMERC NAC for endorsement.

The examination consists of the following three elements:

(a) A SOLAS paper consisting of 5 questions, which may be written or oral. The pass mark is 12/20 or 60%. Duration is left to examiners discretion.

(b) A Radio Regulations paper consisting of 10 multiple choice questions, which may be written or oral. The pass mark is 6/10 or 60%. Duration is left to examiners discretion.

(c) Operational performance test on Inmarsat-C equipment. The pass mark is 10/14 or 70%. Duration, typically 15 minutes per candidate.

12.14.2 LRC DSC module: Examination structure

Holders of an existing Restricted RT Certificate who pass examination of this module may apply to upgrade their Certificate to the CEPT LRC on payment of an examination fee. Successful candidates will have to return their certificates to the NAC for endorsement.

The examination consists of the following three elements:

(a) A SOLAS paper consisting of 5 questions, which may be written or oral. The pass mark is 12/20 or 60%. Duration is left to examiners discretion.

(b) A Radio Regulations paper consisting of 5 multiple choice questions, which may be written or oral. The pass mark is 3/5 or 60%. Duration is left to examiners discretion.

(c) Operational performance test on DSC and NAVTEX equipment and aspects on general ability. The pass mark is 54/72 or 70%. Duration, typically 20 minutes per candidate.

12.14.3 LRC GMDSS examination appeals procedure

As with the two other GMDSS radio qualifications, there is a two stage appeals procedure for those LRC candidates dissatisfied with the services they have received from an AMERC approved REC. Initially the dissatisfied candidate should appeal in writing to the Coordinator of the REC where the examination took place. If still dissatisfied as a result of this appeal, the candidate may then further appeal by writing to the AMERC NAC (GMDSS) at Kendal. The matter is then investigated by the AMERC Chief Examiner who subsequently makes a report directly to the AMERC Maritime Board for further consideration.

12.15 CEPT LONG RANGE CERTIFICATE (LRC) OUTLINE EXAMINATION SYLLABUS

Note: This certificate is for vessels not subject to compulsory fit under SOLAS convention. Syllabus extracted from Annex 1 and 2 of the text proposed by the "Radio Regulatory" (RR) Working Group of the Harmonised Examination Procedures for the General Operator's Certificate (GOC) — Recommendation T/R 31-03 E (Bonn 1993), 1st May 1993 Edition (a) CEPT Long Range Certificate (LRC).

The examination should consist of theoretical and practical tests and should include at least:

(A) General knowledge of radiocommunications in the maritime mobile service.
(Al) The general principles and basic features of the maritime mobile service.
(B) Detailed practical knowledge and ability to use radio equipment.
(B1) The VHF radio installation. Use VHF equipment in practice.
(B2) The MF/HF radio installation. Use MF/HF equipment in practice.
(B3) Purpose and use of Digital Selective Calling (DSC) facilities.
(C) Operational procedures of the GMDSS and detailed practical operation of GMDSS subsystems and equipment appropriate to non SOLAS vessels.
(C1) Basic introduction to Global Maritime Distress and Safety System (GMDSS) procedures.
(C2) Distress, urgency and safety communication procedures in the GMDSS.
(C3) Distress, urgency and safety communication procedures by RT in the old distress and safety system.
(C4) Protection of distress frequencies.
(C5) Maritime Safety Information (MSI) in the GMDSS.
(C6) Alerting and locating signals in the GMDSS.
(D) Miscellaneous skills and operational procedures for RT communications.
(D1) Ability to exchange communications relevant to the safety of life at sea.
(D2) Regulations, obligatory procedures and practices.
(D3) Practical and theoretical knowledge of radiotelephone procedures.

12.16 CEPT LONG RANGE CERTIFICATE (LRC): DETAILED EXAMINATION SYLLABUS

(A) General knowledge of radiocommunications and basic features of the maritime mobile service.

(Al) The general principles and basic features of the maritime mobile service.

1.1 Types of communication in the maritime mobile service.
Distress, urgency and safety communications; public correspondence; port operations service; ship movement service; intership communications: on-board communications.

1.2 Types of stations in the maritime mobile service.
Ship stations; coast stations; pilot stations, port stations etc.; aircraft stations; Rescue Coordination Centres (RCC).

1.3 Elementary knowledge of radio frequencies and frequency bands.
Frequency and wavelength: the unit of frequency ie Hz, kHz, MHz, GHz: the subdivision of the most significant part of the radio spectrum ie MF, HF, VHF, UHF and SHF.
Different propagation mechanisms and typical ranges.
Propagation on MF, HF frequency bands, VHF and UHF.

1.4 Frequencies allocated to the maritime mobile service.
The usage of MF, HF, VHF, UHF and SHF frequencies in the maritime mobile service.
Modes of communication (Radiotelephony, DSC, NBDP, Facsimile) and classes of emission.
Bandwidth of different emissions, carrier frequency and assigned frequency.
Official designations of emission (e.g. F1B, J3E, A3E, F4, etc.).
Unofficial designations of emission (e.g. TLX, SSB, AM, FM, etc.).

The concept of radio channel: simplex semi-duplex and duplex, paired and unpaired channels

Frequency plans and channelling systems in the MF, HF and VHF maritime mobile bands including allocations for the GMDSS.

Distress and safety frequencies; small craft safety; intership communications; port operations ; ship movement ; calling frequencies.

1.5 Maintaining the functionality of ship station equipment.

Sources of energy of ship stations.

Different kinds of batteries and their characteristics; charging; maintenance of batteries.

(B) Detailed practical knowledge and ability to use radio equipment.

(El) The VHF radio installation.

1.1 Radiotelephone.

Channel selection and controls; dual watch facilities.

1.2 Basic controls and usage, for example:-

Connecting the power, press to transmit switch, high/low power output switch, volume control, squelch control, dimmer.

1.3 Portable two-way VHF radiotelephone apparatus.

1.4 Maritime VHF antennas and their maintenance.

(B2) The MF/HF radio installation.

2.1 Frequencies/channels and selection criteria.

2.2 Typical controls and usage, for example:-

Connecting the power, selecting receiver frequency, selecting transmitter frequency, selecting ITU channel number, tuning the transmitter, selecting the class of emission, using volume and squelch controls, using clarifier or receiver fine tuning, controlling RF gain, using automatic gain control (AGC), using 2 182 kHz instant selector, testing the alarm generator, using the alarm generator.

(B3) Purpose and use of Digital Selective Calling (DSC) facilities.

3.1 The general principles and basic features of DSC.

DSC messages.

DSC attempt: single frequency call attempt, multi-frequency call attempt.

Call acknowledgement; call relay.

3.2 Types of call:

Distress call, all ships call, call to individual station, geographic area call, group call, call to individual station using automatic/semiautomatic service.

3.3 The Maritime Mobile Service Identity (MMSI) number system:

The MMSI number system.

The nationality identification, Maritime Identification Digits (MID), ship station numbers, group calling numbers, coast station numbers.

3.4 Call categorisation and priority:

Distress, urgency, safety, ship business, routine.

3.5 Call telecommand and traffic information:

Distress alerts, designated distress message, undesignated distress message, distress co-ordinates, time and validity of distress co-ordinates.

Other calls and messages.

Working frequency and channel information.

3.6 DSC facilities and usage:
The channel 70 instant alert selector.
The channel 2 187.5 kHz instant alert selector.
Manual settings J2B and F1B modes,
e.g. 2 187.5 kHz/2 185.8 kHz and 8 414.5 kHz/8 412.8 kHz.
DSC data entry and display, updating vessel position, entering preset message, entering traffic information, reviewing received messages, DSC watchkeeping functions and controls.

3.7 Testing of DSC.
Internal self testing procedures; live transmission testing.

(C) Operational procedures of the GMDSS and detailed practical operation of GMDSS subsystems and equipment appropriate to non SOLAS vessels.

(C1) Search and rescue (SAE) procedures in the Global Maritime Distress and Safety System (GMDSS).

1.1 Sea areas, the GMDSS master plan and access to GMDSS facilities.
1.2 The role of the RCCs.
1.3 Organisation of search and rescue.

(C2) Distress, urgency and safety communication procedures in the GMDSS.

2.1 Distress communications via DSC equipment:
DSC distress alert, the definition of distress alert, transmission of a distress alert, transmission of a shore to ship distress alert relay, transmission of a distress alert by a station not itself in distress.
Receipt and acknowledgement of DSC distress alert: acknowledgement procedure, receipt and acknowledgement by a coast station, receipt and acknowledgement by a ship station.
Handling of distress alerts : preparations for handling distress traffic, distress traffic terminology.
On-scene communications.
SAR operations.

2.2 Urgency and safety communications via DSC equipment: the meaning of urgency and safety communications.
procedures for DSC urgency and safety calls.
urgency communications.
safety communications.

(C3) Distress, urgency and safety communication procedures by RT in the old distress and safety system.

3.1 Distress communications:
The RT alarm signal, format of the alarm signal and purpose.
Distress signal: the correct use and meaning of the signal MAYDAY.
Distress call.
Distress message.
Acknowledgement of a distress message: obligation to acknowledge a distress message, correct form of acknowledgement, action to be taken following acknowledgement.
Control of distress traffic:
The correct use and meanings of the signals:

SEELONCE MAYDAY, SEELONCE DISTRESS, PRU-DONCE, SEELONCE FEENEE.
Transmissions of a distress message by a station not itself in distress.
The correct use and meanings of the signal MAYDAY RELAY

3.2 Urgency communications.
The correct use and meanings of the signal PAN-PAN.
Urgency message.
Obtaining urgent medical advice through a coast radio station.

3.3 Safety communications.
Safety signal.
The correct use and meanings of the signal SECURITE.
Safety message.
Special procedures for communication with appropriate national organisations on matters affecting safety.

(C4) Protection of distress frequencies.

4.1 Avoiding harmful interference.
4.2 Transmissions during distress traffic.
4.3 Prevention of unauthorised transmissions.
4.4 Test protocols and procedures: testing DSC equipment, RT test procedures.
4.5 Guard bands.
4.6 Procedures to follow when a false distress alert is transmitted.

(C5) Maritime Safety Information (MSI) in the GMDSS.

5.1 Safety information transmitted by VHF/MF/HF RT.

5.2 The NAVTEX system: purpose of NAVTEX, NAVTEX frequencies, antenna for NAVTEX receiver, reception range.
Message format (transmitter identity, message type, message number).
The NAVTEX receiver : Selection of transmitters, selection of message type, messages which cannot be rejected, use of subsidiary controls, ensuring the integrity of message output.

(C6) Alerting and locating signals in the GMDSS.

6.1 Purpose and definition.

6.2 Emergency Position Indicating Radio Beacons (EPIRBs).
Registration and coding.
Information contents of a distress alert.
Operation including automatic (float-free) and manual activation and avoidance of false alerts.
COSPAS-SARSAT 406 MHz EPIRB.
Inmarsat-E 1.6 GHz EPIRB.
VHF DSC EPIRB.
121.5 MHz homing function.
Mounting considerations.
Routine maintenance : testing, checking battery expiry date, checking the float-free release mechanism.

6.3 Search and Rescue Transponder (SART).
The main technical characteristics.
Operation: operating height, effect of radar reflector.

Range of a SART transmitter.
Routine maintenance of a SART: checking battery expiry date.

(D) Miscellaneous skills and operational procedures for RT communications.

(D1) Ability to exchange communications relevant to the safety of life at sea.

1.1 Awareness of the existence and use of the IMO Standard Marine Navigational Vocabulary and knowledge of the following basic signals:

ALL AFTER; ALL BEFORE; CORRECT; CORRECTION; IN FIGURES; IN LETTERS; I SAY AGAIN; I SPELL; OUT; OVER; RADIO CHECK; READ BACK; RECEIVED; SAY AGAIN; STATION CALLING; TEXT; TRAFFIC; THIS IS; WAIT; WORD AFTER; WORD BEFORE; WRONG.

1.2 Recognise standard abbreviations and commonly used service abbreviations.

1.3 Use of the international phonetic alphabet.

(D2) Regulations, obligatory procedures and practices.

2.1 Awareness of international documentation.

Publications of the International Telecommunication Union (ITU): List of Ship Stations, The Annex to the List of Coast Stations which contains particulars of coast stations participating in the GMDSS, The Alphabetical List of Call Signs, The Manual for use by the Maritime Mobile and Maritime Mobile-satellite Services, List of Coast Stations with which communications are likely to be established (watchkeeping hours, frequencies and charges), List of Coast Stations providing navigational and meteorological warnings and other urgent information to ships.

2.2 Availability and knowledge of national documentation.

2.3 Knowledge of the regulations and agreements governing the maritime mobile service

Requirement for ship radio licence.
Regulations concerning control of the operation of radio equipment by the holder of the appropriate certificate of competence.
Regulations concerning the RT log.
Preservation of the secrecy of correspondence.

(D3) Practical and theoretical knowledge of radiotelephone procedures.

3.1 Practical traffic routines using RT.

Preliminary operations.
Correct use of call signs.
Procedure for establishing communication: intership calling, public correspondence, small craft safety, port operation and ship movement channels, calling a coast station providing pilot service.
Control of communications and the role of coast stations:
Selection of frequencies to be used for traffic, reply procedure, duration of calls, traffic lists transmitted by coast stations, traffic reports transmitted by ship stations.
Procedure for unanswered calls and garbled calls.
Difficulties in reception and inability to accept traffic.
Types of call and types of message which are prohibited.

3.2 Practical traffic routines using DSC.

Calling a coast station or a ship station by DSC.
Acknowledging receipt of a call using DSC.

Subsequent handling of traffic.

3.3 Public correspondence procedures.
Establishing an RT link call via a coast station: ordering for a manually switched link call, ending the call, calls to ships from coast radio stations, special facilities calls.
Traffic charges:
the international charging and accounting system,
Accounting Authority Identification Code (AAIC).

12.17 LRC, SATELLITE MODULE: SYLLABUS

The examination relates to vessels not subject to compulsory fit under the SOLAS Convention. The examination should consist of theoretical and practical tests and should include at least:

1. The general principles and basic features of the maritime mobile-satellite service relevant to non-SOLAS vessels.

1.1 Maritime satellite communications via Inmarsat systems.
Inmarsat space segment.
Ocean areas and satellite acquisition.
Communication services:
telephone, telex, facsimile and data services.

1.2 Types of stations in the maritime mobile-satellite service.
Coast Earth Stations (CES), Land Earth Stations (LES).
Network Co-ordination Station (NCS).
Ship Earth Stations (SES), Mobile Earth Stations (MES).
Inmarsat-A/-B/-C/-M systems.

2. Operational procedures and detailed practical operation of ship earth stations in the GMDSS appropriate to non SOLAS vessels.

2.1 Inmarsat-C Mobile Earth Station:
components of Inmarsat-C terminal,
entering/updating position,
usage of an Inmarsat-C MES,
sending and receiving text messages,
distress and safety services: sending a distress alert, sending a distress priority message, 2-digit code safety services;
avoiding of initiating a false distress alert: procedure to follow when a false distress alert is transmitted.

2.2 Inmarsat Enhanced Group Call (EGC) system:
purpose of the EGC system,
programming a MES for EGC reception,
updating position,
selecting navigation and meteorological areas.

12.18 LRC: DSC MODULE

The examination will be on the DSC and NAVTEX components contained in the GMDSS GOC syllabus and the oral or written questions should cover SOLAS aspects, see sections 12.8 and 12.9.

PASS MARKS FOR GMDSS CERTIFICATES OF COMPETENCY

Certificate	SOLAS	*REGS*	*RT*	*OPT*
GOC	30/40	9/15	36/48	85/120
	75%	60%	75%	70%
ROC	33/44	N/A	36/48	54/72
	75%	N/A	75%	75%
LRC	36/48	6/10	36/48	56/80
	75%	60%	75%	70%

Certificate	*SOLAS*	*OPT/REGS*
LRC SAT MOD	12/20	17/24
	60%	70.8%
LRC DSC MOD	12/20	25/36
	60%	69%

Fig. 12.1 Pass marks for GMDSS Certificates of Competency.

12.19 CEPT SHORT RANGE CERTIFICATE (SRC): OUTLINE EXAMINATION SYLLABUS

The examination should consist of theoretical tests, supplemented by practical tests and/or assessed practical training, and should include at least:

(A) General knowledge of VHF radiotelephone communications in the maritime mobile service.

(A1) The general principles and basic features of the maritime mobile service relevant to vessels not subject to a compulsory fit under the SOLAS convention.

(B) Detailed working knowledge of radio equipment.

(B1) The VHF radio installation.

(B2) Purpose and use of Digital Selective Calling (DSC) facilities.

(C) Operational procedures of the GMDSS and detailed practical operation of GMDSS subsystems and equipment.

(C1) Search and rescue (SAR) procedures in the Global Maritime Distress and Safety System (GMDSS).

(C2) Distress, urgency and safety communication procedures in the GMDSS.

(C3) Protection of distress frequencies.

(C4) Maritime Safety Information (MSI).

(C5) Alerting and locating signals.

(D) Operational procedures and regulations for VHF radiotelephone communications.

(D1) Ability to exchange communications relevant to the safety of life at sea.

(D2) Regulations, obligatory procedures and practices.

(D3) Practical and theoretical knowledge of radiotelephone procedures.

12.20 CEPT SHORT RANGE CERTIFICATE (SRC): DETAILED EXAMINATION SYLLABUS

(A) General knowledge of VHF radiotelephone communications in the maritime mobile service.

(A1) The general principles and basic features of the maritime mobile service relevant to vessels not subject to a compulsory fit under the SOLAS convention.

1.1 Types of communication in the maritime mobile service.
Distress, urgency and safety communications; public correspondence; port operations service; ship movement service; intership communications: onboard communications.

1.2 Types of stations in the maritime mobile service:
Ship stations, coast stations, pilot stations, port stations etc, aircraft stations, Rescue Co-ordination Centres (RCC).

1.3 Elementary knowledge of radio frequencies and channels appropriate to the VHF maritime mobile band.
The concept of frequency.
Propagation on VHF frequencies.
Range for voice communications, range for DSC transmissions.
The usage of VHF frequencies in the maritime mobile service.
The concept of radio channel: simplex, semi-duplex and duplex.
Channel plan for VHF, including allocations for the GMDSS.
Distress and safety channels; national channels for small craft safety; intership communications; port operations ; ship movement ; calling channels; public correspondence channels.

1.4 Functionality of ship station equipment.
Sources of energy of ship stations.
Batteries.
Different kinds of batteries and their characteristics; charging, maintenance of batteries.

(B) Detailed working knowledge of radio equipment.

(B1) The VHF radio installation.

1.1 Radiotelephone channels.
Channel selection and controls; dual watch facilities and controls.

1.2 Basic controls and usage, e.g.:
Connecting the power; press to transmit switch; high/low power output switch; volume control; squelch control; dimmer.

1.3 Portable two-way VHF radiotelephone apparatus.

1.4 Maritime VHF antennas.

(B2) Purpose and use of Digital Selective Calling (DSC) facilities.

2.1 The general principles and basic features of DSC.
DSC messages; DSC attempt; call acknowledgement; call relay.

2.2 Types of call.
Distress call; all ships call; call to individual station; geographic area call; group call.

2.3 The Maritime Mobile Service Identity (MMSI) number system.
The nationality identification: Maritime Identification Digits (MID); ship

station numbers; coast station numbers.

2.4 Call categorisation and priority.
Distress; urgency; safety; ship business; routine.

2.5 Call telecommand and traffic information.
Distress alerts; other calls; working channel information.

2.6 VHF DSC facilities and usage.
The Channel 70 instant alert selector.
DSC data entry and display; updating vessel position; entering preset message; entering traffic information; reviewing received messages.
DSC watchkeeping functions and controls.

(C) Operational procedures of the GMDSS and detailed practical operation of GMDSS subsystems and equipment.

(C1) Search and rescue (SAR) procedures in the GMDSS.

1.1 Sea areas and access to GMDSS facilities.

1.2 The role of RCCs.

1.3 Organisation of search and rescue.

(C2) Distress, urgency and safety communication procedures in the GMDSS.

2.1 Distress communications via VHF DSC equipment.
DSC distress alert : The definition of distress alert, transmission of a distress alert, transmission of a shore-to-ship distress alert relay, transmission of a distress alert by a station not itself in distress.
Receipt and acknowledgement of DSC distress alert : Acknowledgement procedure, receipt and acknowledgement by a coast station, receipt and acknowledgement by a ship station.
Handling of distress alerts : Preparations for handling distress traffic, distress traffic terminology.
On-scene communications.
SAR operation.

2.2 Urgency and safety communications via DSC equipment.
The meaning of urgency and safety communications, procedures for DSC urgency and safety calls, urgency communications, safety communications.

(C3) Protection of distress frequencies.

3.1 Avoiding harmful interference: Avoidance of transmission of false alerts, status of Channel 70.

3.2 Transmissions during distress traffic.

3.3 Prevention of unauthorised transmissions.

3.4 Test protocols and procedures: Testing DSC equipment, radiotelephone test procedures.

3.5 Avoidance of transmission in VHF guard bands.

3.6 Procedures to follow when a false distress alert transmitted.

(C4) Maritime Safety Information (MSI).

4.1 The NAVTEX system:
Purpose and capabilities, including distress and safety functions.

(C5) Alerting and locating signals.

5.1 Purpose and definition.

5.2 Emergency Position Indicating Radiobeacon (EPIRBs):

Registration and coding.
Operation, including automatic and manual activation of:-
COSPAS-SARSAT 406 MHz EPIRB.
Inmarsat-E 1.6 GHz EPIRB.
VHF DSC EPIRB.
121.5 MHz homing function.
Mounting considerations.
Routine maintenance: testing, checking battery expiry date, cleaning of the float-free release mechanism.

5.3 Search and rescue transponder (SART).
Operation: Operating height, effect of radar reflector, range of SART transmitter.

(D) Operational procedures and regulations for VHF radiotelephone communications.

(Dl) Ability to exchange communications relevant to the safety of life at sea.

1.1 Distress communications:
Distress signal: the correct use and meaning of the signal MAYDAY.
Distress call, distress message.
Acknowledgement of a distress message: obligation to acknowledge a distress message, correct form of acknowledgement, action to be taken following acknowledgement.
Control of distress traffic:
The correct use and meanings of the signals:-
SEELONCE MAYDAY
SEELONCE DISTRESS
PRU-DONCE
SEELONCE FEENEE.
Transmission of a distress message by a station not itself in distress.
The correct use and meanings of the signal MAYDAY RELAY.

1.2 Urgency communications.
Urgency signal: the correct use and meaning of the signal PAN-PAN:
Urgency message:
Obtaining urgent medical advice through a coast station.

1.3 Safety communications.
Safety signal: the correct use and meaning of the signal SECURITE.
Safety message.
Special procedures for communication with appropriate national organisations on matters affecting safety.

1.4 Reception of MSI by VHF RT.

1.5 Awareness of the existence and use of the IMO Standard Marine Navigational Vocabulary.
Knowledge of the following basic signals:
ALL AFTER; ALL BEFORE; CORRECT; CORRECTION; IN FIGURES; IN LETTERS; I SAY AGAIN; I SPELL; OUT; OVER; RADIO CHECK; READ BACK; RECEIVED; SAY AGAIN; STATION CALLING; TEXT; TRAFFIC; THIS IS; WAIT; WORD AFTER; WORD BEFORE; WRONG.

1.6 Use of international phonetic alphabet.

(D2) Regulations, obligatory procedures and practices.

2.1 Awareness of the international documentation and availability of national publications.

2.2 Knowledge of the international regulations and agreements governing the maritime mobile service:

Requirement for ship station licence: regulations concerning control of the operation of radio equipment by the holder of an appropriate certificate of competence: national regulations concerning radio record keeping: preservation of the secrecy of correspondence: types of call and types of message which are prohibited.

(D3) Practical and theoretical knowledge of radiotelephone procedures.

3.1 Public correspondence and radiotelephone call procedures.

Method of calling a coast station by RT: Ordering for a manually switched link call, ending the call.

Calls to ships from coast stations: special facilities of calls, method of calling a coast station by DSC for general communications, selecting an automatic RT call.

3.2 Traffic charges.

International charging system.

Accounting Authority Identification Code (AAIC).

3.4 Practical traffic routines.

Correct use of callsigns.

Procedure for establishing communication: on intership, public correspondence, small craft safety and port operation and ship movement channels.

Procedure for unanswered calls and garbled calls.

Control of communications.

12.21 UK OPERATOR'S CERTIFICATES AND AUTHORITIES TO OPERATE

A person may not carry out the duties of an operator of a ships radio station licensed by the Secretary of State unless that person possess an appropriate certificate of competence and an Authority to Operate (ATO). All maritime Radio Operators Certificates of Competence issued by the Secretary of State for Trade and Industry or his predecessors are valid irrespective of the ITU Convention to which they may refer.

In previous years, ATOs attached to Maritime Radio Operators Certificates of Competence made reference to a Convention of the ITU and a new ATO was issued each time a new Convention came into force.

From December 1991 however, all ATOs (irrespective of the Convention to which they refer) issued by the Secretary of State for Trade and Industry or by his predecessors shall be regarded as being valid for use by their holders, in conformity with the type of certificate to which they are attached, on ships whose radio stations are licensed by the Secretary of State for Trade and Industry. Therefore updated ATOs will no longer be issued.

ATOs issued by the Secretary of State for Trade and Industry or by his predecessors marked "NOT VALID, NON-BRITISH SUBJECT" (or with a similar

nationality endorsement) are now valid for use by their holders, in conformity with the type of certificate to which they are attached, on ships whose radio stations are licensed by the Secretary of State for Trade and Industry. ATOs to replace those previously so endorsed will NOT be issued.

The Secretary of state for Trade and Industry recognises Certificates of Competence issued by certain other countries (most Commonwealth Countries, Hong Kong and the Republic of Ireland). Holders of such certificates may be issued with ATOs by the Secretary of State authorising them to operate the radio station, in conformity with the type of certificate held, on ships whose radio stations are licensed by the Secretary of State for Trade and Industry.

Certificates of Competence that are lost or stolen may be replaced on application; a charge will be levied for this service. All enquiries should be addressed to either:

for all certificates other than VHF

AMERC NAC (GMDSS)
c/o Wray Castle Limited
Bridge Mills
Stramongate
Kendal
LA9 2UB
NAC Telephone : +44 (0) 1539 742745
NAC FAX : +44 (0) 1539 742746

for VHF Only Certificates

Royal Yachting Association
RYA House
Romsey Road
Eastleigh
Hampshire S05 4YA
Telephone : +44 (0) 1703 627 400
FAX : +44 (0) 1703 629 924

12.22 LIST OF AMERC REGIONAL EXAMINATION CENTRES: UK

Aberdeen College
ASET
Gallowgate Centre
Aberdeen
AB25 1BN
Tel: +44 (0) 1224 612000
Fax: +44 (0) 1224 612001

Blackpool and The Fylde College
Fleetwood Nautical Campus
Broadwater,
Fleetwood
Lancashire
FY7 8JZ
Tel: +44 (0) 1253 352352
Fax: +44 (0) 1253 773014

City of Bristol College
Brunei Centre
Ashley Down
Bristol
BS7 9BU
Tel: +44 (0) 117 904 5000
Fax: +44 (0) 117 904 5050

Glasgow College of Nautical Studies
21 Thistle Street
Glasgow
G5 9XB
Tel: +44 (0) 141 565 2711
Fax: +44 (0) 141 565 2618

Hull College
School of Automotive Electronic Eng.
Queens Gardens
Hull
HU1 3DG
Tel: +44 (0) 1482 329 943
Fax: +44 (0) 1482 598 733

Jewel and Esk Valley College
24 Milton Road East
Edinburgh
EH15 2PP
Tel: +44 (0) 131 657 7240
Fax: +44 (0) 131 669 0386

Lairdside Maritime Centre
Liverpool John Moore's University
3 Vanguard Way
Campbelton Road
Birkenhead
CH41 9HX
Tel: +44 (0) 151 647 0494
Fax: +44 (0) 151 647 0498

Lowestoft College
St Peters Street
Lowestoft
Suffolk
NR32 2NB
Tel: +44 (0) 1502 583 521
Fax: +44 (0) 1502 500 031

North West Kent College
Gravesend Campus
Dering Way
Gravesend
Kent
DA12 2JJ
Tel: +44 (0) 1474 534 960
Fax: +44 (0) 1474 360 244

Plymouth College of Further Education
Faculty of Technology
Kings Road

Plymouth
Devon
PL1 5QG
Tel: +44 (0) 1752 305 388
Fax: +44 (0) 1752 305 399

Royal National Lifeboat Institution
West Quay
Poole
Dorset
BH15 1HZ
Tel: +44 (0) 1202 663 275
Fax: +44 (0) 1202 663 287

Southampton Institute of Higher Education
Faculty of Technology
East Park Terrace
Southampton
SO14 ORD
Tel: +44 (0) 23 8031 9000
Fax: +44 (0) 23 8033 4441

South Tyneside College
Faculty of Electrical and Electronic Engineering
St George's Avenue
South Shields
Tyne and Wear
NE34 6ET
Tel: +44 (0) 191 427 3550
Fax: +44 (0) 191 427 3907

12.22.1 List of AMERC Regional Examination Centres: Overseas

France

Freedom Yachting
7 Boulevard d'Aguillon
06600 Antibes
France
Tel: + 33 493 344 773
Fax: + 33 493 347 774

The Blue Water School
La Galerie du Port
8 Boulevard d'Aguillon
06600 Antibes
France
Tel: + 33 493 343 413
Fax: + 33 493 343 593

India. Madras (Chennai)
Academy of Maritime Education and Training
135 East Coast Road
Kanathur 603 112
India
Tel: + 91 4114 472 803
Fax: + 91 4114 472 804

India. Mumbai
Anglo Eastern Maritime Training Centre
Plot 50, Karmayog Building
Near Sona Udyog
Parsi Panchayat road
Andheri (E)
Mumbai 400 069
India
Tel: + 91 222 683 7007
Fax: + 91 222 683 7008

St Xavier's College
St Xavier's Technical Institute
Mahim Causeway
Bombay 400 016
India
Tel: + 91 222 445 5937
Fax: + 91 222 445 4482

T.S. Rahaman
S.M.Y. Seamen Welfare foundation
Jahaz Mahal Annexe
170k Samander Point Estate
Mumbai 400 018
India
Tel: + 91 222 493 8740
Fax:+ 91 222 495 0270

Italy
Tema Safety and Training s.r.l.
Via del Tratturello
Tarantino, 6
74100 Taranto
Italy
Tel: + 39 099 472 5898
Fax: + 39 099 473 5250

Sri Lanka
CINEC Maritime Campus
Colombo International Nautical and Engineering College
Millennium Drive
IT Park

Malambe
Sri Lanka
Tel: + 941 241 3500
Fax: + 941 241 3505

United States of America
International Yachtmaster Training and Deliveries
910 SE 17th Street
Ft. Lauderdale
Florida
FL 33316
United States of America
Tel: + 1 954 779 7764
Fax: + 1 954 779 7165

Maritime Professional Training
Marine Technical Institute
1915 S. Andrews Avenue
Ft. Lauderdale
Florida
FL 33316
United States of America
Tel: + 1 954 525 1014
Fax: + 1 954 764 0431

12.22.2 List of British Council and Lloyds Offices Overseas

India
Bombay (Mumbai)
British Council Division
British Deputy High Commission
Mittal Tower "C" Wing
Noriman Point
PO Box 11594
Bombay 400 021
India
Tel: + 91 222 282 3460
Fax: + 91 222 285 2024

Madras (Chennai)
Examination Services Manager
British Council Division
British Deputy High Commission
737 Anna Salai
Chennai 600 002
India
Tel: + 91 44 2852 5002
Fax: + 91 44 2852 3234

Italy
The British Council
Via F. Crispi 92
80121 Naples
Italy
Tel: + 39 081 66 7410
Fax: + 39 081 66 9563

Executive Language Service
The British Council
Via Quattro Fontane 20
00184 Roma
Italy
Tel: + 39 064 781 4212
Fax: + 39 064 781 4296

Sri Lanka
The British Council
49 Alfred House Gardens
Colombo 3
Sri Lanka
Tel: + 94 11 258 1171
Fax: + 94 11 258 7079

United States of America
Lloyds Miami
Lloyds Register North America
South East Area Office
100 N. Biscayne Boulevard
Suite 1200
Miami
Florida 33132
United States of America
Tel: + 1 305 577 6146
Fax: + 1 305 577 6146

12.23 ELECTRONIC NAVIGATIONAL EQUIPMENT MAINTENANCE CERTIFICATE (ENEM)

The mandatory requirement for certain UK ships to be provided with at least one officer or member of the crew adequately qualified to carry out radar maintenance was repealed by the Merchant shipping (Navigational equipment) Regulations 1993.

With effect from 1st August 1995, the statutory examinations leading towards the issue of the following Certificates were withdrawn:

(a) Radar Maintenance Certificate,
(b) Certificate in the Maintenance of Electronic Navigational Equipment (known as the ENEM Certificate).

No further certificates will be issued on behalf of the Department of Transport after this date.

However, a programme of training and certification is proposed by the Association of Marine Electronic and Radio Colleges (AMERC). The aim of this training course is to provide the knowledge and skill necessary to maintain the electronic navigational equipment commonly found on merchant ships and to fault-find on this equipment to component or printed circuit board level, depending on the constraints imposed by the design of the equipment and the resources normally available on board ship. The training includes the operational use of equipment, the objective being to enable the maintainer to appreciate the range of operational uses of electronic navigational data and the limitations and probable errors of positions derived from the use of such information as a primary basis for the evaluation of faults occurring in systems. Training of fault finding techniques includes the application of systems level knowledge, the use of test equipment including any built-in test equipment, an understanding of the advantages and disadvantages of modular replacement techniques, and the location of faults to component level. At least 50% of course time shall consist of practical exercises on approved equipment. The Marine Safety Agency recognises this programme as equivalent to training currently in place and hereby draws it to the attention of personnel on merchant ships.

Further details regarding these courses and certificates can be obtained directly from AMERC's National Administration Centre whose contact details appear in section 12.4 of this handbook.

12.24 MNTB ETO CERTIFICATE: PROPOSED SYLLABUS

To obtain a Merchant Navy Training Board ETO certificate candidates need to have successfully passed a GMDSS Radio Maintenance Certificate AND an Electronic Navigation Equipment Maintenance (ENEM) Certificate. The individual qualifications can be studied and examined separately. The Association of Marine Electronic and Radio Colleges (AMERC) will be responsible for conducting examinations for the GMDSS Radio Maintenance and the ENEM Certificate. The proposed syllabus was still under discussion at the time of going to press but essentially will be as that shown below.

Principles

1. **Electronic Navigation** Systems

The aim of this unit is to provide learners with an understanding of the operation of a range of marine electronic navigations systems. The underpinning principles of operation of each system are studied. An understanding of the system and sub systems that form each overall system is developed.

To achieve this unit a student must investigate:

Marine Radar and Automatic Radar Plotting Systems (ARPA)
Terrestrial and satellite position fixing systems
Ship speed and distance measuring systems and echo sounding systems
Automatic steering systems
Marine compass and control systems.

Detailed syllabus

1. Marine Radar and Automatic Radar Plotting Systems (ARPA)

Principles of radar: factors affecting range and bearing e.g. radar parameters (pulse length, PRF, HBW, VBW, duty cycle, antenna speed), radar range equation, antenna/target height, tropospheric refraction e.g. standard, sub and super refraction, ducting. Information display modes in use in marine radars e.g. relative motion (ship's head up, north up, course up), true motion.

Radar systems: block diagrams of systems and subsystems e.g. transmitter, receiver, video retiming, video correlation, azimuth systems, raster scan, time-base, range and bearing indicators, performance monitoring.

ARPA systems: block diagram of an ARPA, interfacing requirements e.g. speed log, gyrocompass, GPS, synthetic video, operation controls, ECDIS, principles of target acquisition and tracking.

2. Terrestrial and satellite position fixing and transponder systems

Hyperbolic navigation systems: principles of time and phase difference measurement. Loran C operating principles e.g. coding delay, secondary coding delay, chain identity, adjacent chain interference rejection. Block diagram of Loran C.

Satellite systems: principles of position fixing using GPS e.g. chart datum, satellite constellation, PRN codes, spread spectrum modulation, differential GPS navigation almanac. Block diagram of GPS receiver.

Automatic Identification Systems (AIS): principles of AIS e.g. TDMA, FSK, frequency allocation, regulations, potential errors. Block diagram of AIS.

3. Ship speed and distance measuring systems and echo sounding systems.

Propagation of sound in sea water: factor affecting the speed of sound in sea water e.g. temperature, pressure and salinity. Constant used for speed of sound in sea water for calibration purposes. Transmission losses e.g. spreading and absorption losses, "volume reverberation noise".

Ship speed and distance measurement devices: definitions of absolute and relative speeds and their applications. Principles of Doppler shift measurement, use of single and dual axis Janus arrays, transducers. Effects of change in sea water salinity and temperature on indicated reading and methods used to compensate. Measuring principles of electromagnetic log. Block diagram of electromagnetic log.

Echo sounding systems: principles of echosounding e.g. relating speed and time to the measurement of depth, CW and pulsed systems. Choice of transmission frequency, pulse width, PRF and peak pulse power. Block diagram of echosounder, transducers and their installation requirements.

4. Automatic steering systems

Regulations: rotational and translational movements and attitudes of vessel. regulations relating to main and auxiliary steering gear.

Electrohydraulic steering gear; action of electrohydraulic steering gear, rudder stock and rudder, non follow up (NFU), and follow up (FU) control actions. Block diagram of rudder control loop.

Autopilot: operating principles, feedback systems, multiloop control block diagram, application of three term control, effect of operator control settings on course keeping accuracy, integration with other navigation systems.

5. Marine compasses and control systems
Magnetic compass: principles of magnetic compass e.g. deviation, variation, compass adjustment. Construction of maritime magnetic compass, safe operating distances, lubber line.

Gyro compass: principle of free gyro, conversions to north seeking, damping in tilt, damping in azimuth, control and follow up systems, gyro errors. Block diagrams of follow up system and transmission system.

Repeaters; transmitting magnetic compass, synchros, stepper motors, optical systems.

To achieve assessment criteria a student must demonstrate an ability to:

1. Investigate Marine Radar and Automatic Radar Plotting Systems (ARPA).
Calculate system parameters to meet specified requirements.
Understands different modes of presentation.
Construct a block diagram of a marine radar/ARPA system or sub-system.
Explain target acquisition and tracking in ARPA.

2. Investigate terrestrial and satellite position fixing systems.
Explain the principles of hyperbolic navigation systems.
Construct the block diagram of a Loran C receiver.
Explain the principles of GPS.
Construct the block diagram of a GPS receiver and AIS transponder.

3. Investigate ship speed and distance measuring systems and echo sounding systems.
Explain the principles of Doppler speed measurement systems.
Explain the principles of an electromagnetic log.
Explain the principles of echosounding.
Construct the block diagram of a ship's log.
Construct the block diagram of a typical marine echosounder.

4. Investigate automatic steering systems.
Explain the principles of a marine automatic steering system.
Understand the regulations governing automatic steering system.
Construct the block diagram of a typical marine auto pilot.

5. Investigate marine compass and control systems.
Explain the construction and siting of marine magnetic compass.
Explain the principle of converting a free gyroscope into a marine gyro compass.
Construct the block diagram of gyro compass and repeater system.

Generating evidence

Evidence of outcome may be demonstrated in a variety, or mixture, of methods. Assignments, written tests/examination or workshop activities may be used to demonstrate competency. Evidence is likely to be at outcome level, however it may

be across units to provide flexibility and to demonstrate its relationship with allied topics.

Additional information

The unit should be linked with Navigation System Fault Diagnosis, see following sub section. The unit will also enforce and integrate many of the skills developed within other units in the programme. Students will require access to marine navigation systems equipment. The delivery of the unit will be a mixture of lectures and assignment work, however practical work must be adopted to reinforce the principles. It may be treated as a stand alone unit but may be integrated into other units in which case care must be taken with the tracking of outcomes.

2. Navigation Systems Fault Diagnosis

The aim of this unit is to enable students to develop confidence and expertise in analysing and rectifying fault conditions occurring in marine electronic navigations systems.

To achieve this unit a student must:

Compare operation and performance of equipment with manufacturer's specifications.

Compile operational data for a variety of operating conditions.

Analyse equipment operation and identify system malfunction.

1. Compare operation and performance of equipment with manufacturer's specifications

Operates: a range of electronic navigation equipment e.g. radar, ARPA, gyrocompass, echosounder, Loran C, GPS, autopilot, speed logs.

Performance: uses manufacturer's information and specifications, BITE and operational controls to determine whether equipment meets specification.

2. Compile operational data for a variety of operating conditions

Test equipment: safely uses a range of test equipment, e.g. DVM, oscilloscope, storage scope, logic probe, logic pulser.

Test data; records test data on a range of electronic navigation equipment e.g. radar, ARPA, gyrocompass, echosounder, Loran C, GPS, autopilot, speed logs, compares with manufacturer's data, evaluates response to operational controls.

3. *Analyse equipment operation and identify system malfunction*

Safety: carries out risk assessment, follows correct safety procedures.

Fault diagnosis: analyses fault symptoms, uses a range of fault finding strategies, e.g. input-output, output-input, half split, module and component isolation, where appropriate remove fault component and repair fault using component which meets manufacturer's specification.

Report: produces a technical report detailing fault symptoms, fault finding procedure and fault identity.

To achieve assessment criteria a student must demonstrate an ability to:

1. Compare operation and performance of equipment with manufacturer's specifications.

Correctly operate a range of navigation equipment.

Compare performance with equipment specification.
Use built in test functions to assess performance.
Assess system response to user inputs.

2. *Compile operational data for a variety of operating conditions.*
Operate correctly a range of test equipment.
Compile data from a number of equipment types.
Maintain a log of tasks performed.

3. *Analyse equipment operation and identify system malfunction.*
Be observant to follow safety procedures.
Diagnose faults on a range of electronic navigation equipment.
Diagnose and repair faulty equipment.
Produce technical report of fault symptoms.

Generating evidence

Evidence of outcome may be demonstrated in a variety, or mixture, of methods. Assignments, solutions to applied problems and results of unseen examinations. Evidence is likely to be at outcome level, however it may be across units to provide flexibility and to demonstrate its relationship with allied topics. Evidence may be accumulated by students building a portfolio of activities or by a tutor led combination of assignments or tests. Evidence must be both relevant and sufficient to justify the grade awarded.

Additional information

The unit should be linked with Navigation System Principles, see previous sub section. The unit will also enforce and integrate many of the skills developed within other units in the programme. Students will require access to marine navigation equipment and appropriate test equipment. Entry requirements for this unit are at the discretion of the teaching centre. The delivery of the unit will be as an integrated package with Navigation System Principles, in which case care must be taken with the tracking of outcomes.

12.25 GMDSS/ENEM MAINTENANCE CERTIFICATES: PROPOSED EXAMINATION STRUCTURE

The proposed structure of the examination for the qualification is as follows.

Principles

A written examination where candidates will have to answer 12 questions from the 14 set by the examination panel. The pass mark for this principles paper has been set at 50%.

GMDSS Radio Maintenance. Practical assessment.

The candidate carries out operation and performance testing (OPT) checks on a GMDSS station using test equipment and BITE. Candidates have to identify each item of equipment as being operational or defective. Candidates will also have to locate one fault to component, interconnection or board level as appropriate to the

equipment being serviced. They will also have to demonstrate testing on EPIRBs and SARTs and write a short report of equipment tests and fault location.

GMDSS Radio Maintenance. Underpinning knowledge.

For students on an HND programme, successful completion of the Radio Communications Unit should be accepted as an understanding of underpinning knowledge.

For non HND candidates a syllabus based on the four sections of this unit is prepared with a set of two questions per section. A candidate correctly answers one question selected randomly from each section either as a continuous assessment during a course or as an examination paper.

ENEM. Practical assessment.

For students on an HND programme, successful completion of the Navigation Systems Fault Diagnosis Unit should be accepted as an understanding of underpinning knowledge.

Assessment of non HND candidates consists of the candidate producing a logbook and a fault report sheet showing that they have successfully located three faults to component, interconnection or board level as appropriate to the equipment being serviced from each of the following equipment categories:

Radar and ARPA
Terrestrial and satellite position fixing systems
Ship speed and distance measuring systems and echo sounding systems
Automatic steering systems
Marine compasses

ENEM. Underpinning knowledge.

For students on an HND programme, successful completion of the Electronic Navigation Systems Unit should be accepted as an understanding of underpinning knowledge.

For non HND candidates a syllabus based on the five sections of this unit is prepared with a set of two questions per section. A candidate correctly answers one question selected randomly from each section either as a continuous assessment during a course or as an examination paper.

12.25.1 GMDSS/ENEM Maintenance Certificates: Entry requirements

Depending on the applicants experience and background the entry requirement to such course can vary as applicants may qualify for exemptions to parts of the course. Applicants are advised to contact individual training establishments for further information. Currently, (2004) only South Tyneside College is offering these courses although Lairdside Maritime Centre and Glasgow College of Nautical Studies intend running them in the future. See section 12.22 for contact addresses.

APPENDIX 1

IMO guidelines for the avoidance of false distress alerts

Resolution A.814(19) adopted on 23 November 1995 (Agenda item 10)*

THE ASSEMBLY,

RECALLING Article 15(j) of the Convention on the International Maritime Organisation concerning the functions of the Assembly in relation to regulations and guidelines concerning maritime safety and the prevention and control of marine pollution from ships,

CONSIDERING problems reported by Member Governments in regard to the proper operation of the GMDSS, in particular that false distress alerts are becoming a major obstacle to the efficient operation of search and rescue (SAR) services,

RECALLING that the GMDSS was developed on, the basis of resolution 6 of the International Conference on Maritime Search and Rescue, 1979, and that according to that resolution the GMDSS should provide, among other things, the essential radio elements of the international SAR plan,

NOTING that the excessive amount of false distress alerts imposes a considerable and unnecessary burden on Rescue Co-ordination Centres (RCCs), may have adverse effects on seafarers' confidence in the GMDSS, and could also have a potentially serious impact on real distress situations and on safety of life at sea,

BEING AWARE that, if a substantive reduction in the number of false distress alerts now occurring is not achieved in the near future, the quality and efficiency of SAR organisations may be jeopardised,

CONSIDERING that an urgent dissemination of some of the problems which have become evident to providers of rescue services would help to educate people and organisations involved and eventually contribute to a reduction in the number of false distress alerts,

CONSIDERING ALSO that Administrations, manufacturers, educators, users, providers of communications and rescue services, and all others concerned need guidance on ways and means of reducing false distress alerts,

HAVING CONSIDERED the recommendation made by the Maritime Safety Committee at its sixty-fifth session,

* Reproduced with kind permission of IMO, London.

(1) ADOPTS the Guidelines for Avoiding False Distress Alert out in the annex to the present resolution;
(2) URGES Governments to bring these Guidelines to the attention of all concerned.

Annex: Guidelines for avoiding false distress alerts

1 Administrations should:

.1 inform shipowners and seafarers about the implications of the rising number of false distress alerts;
.2 take steps to enable ships properly to register all GMDSS equipment, and ensure that this registration data is readily available to RCCs;
.3 consider establishing and using national enforcement measures to prosecute those who:

.3.1 inadvertently transmit a false distress alert without proper cancellation, or who fail to respond to a distress alert due to misuse or negligence;
.3.2 repeatedly transmit false distress alerts; and
.3.3 deliberately transmit false distress alerts;

.4 use the International Telecommunication Union violation reporting process for false distress alerts, or for failure to respond to a distress alert relayed from shore-to-ship;
.5 ensure that all relevant ship personnel know how GMDSS equipment operates, the importance of avoiding false distress alerts, the steps to be taken to prevent the transmission of such false distress alerts, and the procedures to be followed when a false distress alert has been transmitted;
.6 inform type-approval authorities of false distress alert problems, in order to draw their attention to the testing and alerting functions of radio equipment during the type approval process;
.7 urge companies installing radio equipment to ensure that relevant ship personnel are made familiar with the operation of the installed equipment;
.8 investigate the cause when a specific model of GMDSS equipment repeatedly transmits unwanted distress alerts, and inform the appropriate organizations accordingly;
.9 ensure that surveyors and inspectors are informed about GMDSS equipment, and particularly about how to operate and test it without transmitting a false distress alert; and
.10 require that GMDSS radio operators be appropriately certificated.

2 Manufacturers, suppliers and installers should:

.1 design equipment for distress alerting so that:

.1.1 it will not be possible to transmit a distress alert unintentionally;
.1.2 the panel for emergency operation is separated from the one for normal operation and is partially fitted with a cover, and the switches on the panel are clearly classified by colouring; and

.1.3 there are standardized arrangements of operation panels and operational procedures;

.2 design test features so that the testing of GMDSS equipment will not result in transmission of false distress alerts;

.3 ensure that any distress alert activation is indicated visually or acoustically, or both, and shows that the equipment is transmitting a distress alert until manually deactivated;

.4 ensure that the satellite EPIRB position on board, installations (including the release and activation mechanisms) and handling procedures preclude unwanted activation (designing the EPIRB so that when it is out of its bracket it must also be immersed in water to activate automatically, and so that, when operated manually, a two-step activation action is required);

.5 provide clear and precise operational instructions that are easy to understand (maintenance and operational instructions should be separated, and should be written both in English and in any other language deemed necessary);

.6 ensure that when any GMDSS equipment has been installed, the necessary instructions are given to ship personnel, drawing specific attention to operational procedures (a record should be kept that such instructions have been given); and

.7 ensure that supply and installation personnel understand how the GMDSS works, and the consequences of transmitting a false distress alert.

3 Trainers and educators should:

.1 ensure that maritime education centres are informed about false distress alert problems and their implications for SAR, the GMDSS, etc., and procedures to be followed if a false distress alert is transmitted, and include them in their teaching programmes;

.2 obtain and use actual case histories as examples;

.3 emphasize the need to avoid false distress alerts; and

.4 ensure that no inadvertent transmission of a false distress alert occurs when training on GMDSS equipment.

4 Companies, masters and seafarers should, as appropriate:

.1 ensure that all GMDSS certificated personnel responsible for sending a distress alert have been instructed about, and are competent to operate, the particular radio equipment on the ship;

.2 ensure that the person or persons responsible for communications during distress incidents give the necessary instructions and information to all crew members on how to use GMDSS equipment to send a distress alert;

.3 ensure that as part of each "abandon ship" drill, instruction is given on how emergency equipment should be used to provide GMDSS functions;

.4 ensure that GMDSS equipment testing is only undertaken under the supervision of the person responsible for communications during distress incidents;

.5 ensure that GMDSS equipment testing or drills are never allowed to cause false distress alerts;

.6 ensure that encoded identities of satellite EPIRBS, which are used by SAR personnel responding to emergencies, are properly registered in a database accessible 24 h a day or automatically provided to SAR authorities (masters should confirm that their EPIRBs have been registered with such a database, to help SAR services identify the ship in the event of distress and rapidly obtain other information which will enable them to respond appropriately);

.7 ensure that EPIRB, Inmarsat and DSC registration data is immediately updated if there is any change in information relating to the ship such as owner, name or flag, and that the necessary action is taken to reprogramme the ship's new data in the GMDSS equipment concerned;

.8 ensure that, for new ships, positions for installing EPIRBs are considered at the earliest stage of ship design and construction;

.9 ensure that satellite EPIRBs are carefully installed in accordance with manufacturers' instructions and using qualified personnel (sometimes satellite EPIRBs are damaged or broken due to improper handling or installation. They must be installed in a location that will enable them to float free and automatically activate if the ship sinks. Care must be taken to ensure that they are not tampered with or accidentally activated. If the coding has to be changed or the batteries serviced, manufacturers' requirements must be strictly followed. There have been cases where EPIRB lanyards were attached to the ship so that the EPIRB could not float free; lanyards are only to be used by survivors for securing the EPIRB to a survival craft or person in water);

.10 ensure that EPIRBs are not activated if assistance is already immediately available (EPIRBs are intended to call for assistance if the ship is unable to obtain help by other means, and to provide position information and homing signals for SAR units);

.11 ensure that, if a distress alert has been accidentally transmitted, the ship makes every reasonable attempt to communicate with the RCC by any means to cancel the false distress alert using the procedures given in the appendix;

.12 ensure that, if possible, after emergency use, the EPIRB is retrieved and deactivated; and

.13 ensure that when an EPIRB is damaged and needs to be disposed of, if a ship is sold for scrap, or if for any other reason a satellite EPIRB will no longer be used, the satellite EPIRB is made inoperable, either by removing its battery and, if possible, returning it to the manufacturer, or by demolishing it.

Note: If the EPIRB is returned to the manufacturer, it should be wrapped in tin foil to prevent transmission of signals during shipment.

Appendix: Instructions for mariners and others* on how to cancel a false distress alert

DSC

1 **VHF**

.1 switch off transmitter immediately;†

.2 switch equipment on and set to Channel 16; and

* Appropriate signals should precede these messages in accordance with the ITU Radio Regulations chapter NIX.

t This applies when the false alert is detected during transmission.

.3 make broadcast to "All Stations" giving the ship's name, call sign and DSC number, and cancel the false distress alert.

Example

All Stations, All Stations, All Stations
This is NAME, CALL SIGN,
DSC NUMBER, POSITION.

Cancel my distress alert of
DATE, TIME UTC,
= Master NAME, CALL SIGN,
DSC NUMBER, DATE, TIME UTC.

2 MF

.1 switch off equipment immediately;†
.2 switch equipment on and tune for radiotelephony transmission on 2,182 kHz; and
.3 make broadcast to "All Stations" giving the ship's name, callsign and DSC number, and cancel the false distress alert.

Example

All Stations, All Stations, All Stations
This is NAME, CALL SIGN,
DSC NUMBER, POSITION.

Cancel my distress alert of
DATE, TIME UTC,
= Master NAME, CALL SIGN,
DSC NUMBER, DATE, TIME UTC.

3 HF

As for MF, but the alert must be cancelled on all the frequency bands on which it was transmitted. Hence, in stage 2.2 the transmitter should tuned consecutively to the radiotelephony distress frequencies in the 4, 6, 8, 12 and 16 MHz bands, as necessary.

4 Inmarsat-C

Notify the appropriate RCC to cancel the alert by sending a distress priority message via the same CES through which the false distress alf was sent.

Example of message

NAME, CALL SIGN, IDENTITY NUMBER,
POSITION,

† This applies when the false alert is detected during transmission.

Cancel my Inmarsat-C distress
alert of DATE, TIME, UTC
= Master +

5 EPIRBs

If for any reason an EPIRB is activated accidentally, the ship should contact the nearest coast station or an appropriate coast earth station or RCC and cancel the distress alert.

6 General

6.1 Notwithstanding the above, ships may use any means available to them to inform the appropriate authorities that a false distress alert has been transmitted and should be cancelled.

6.2 No action will normally be taken against any ship or mariner for reporting and cancelling a false distress alert. However, in view of the serious consequences of false alerts, and the strict ban on their transmission, Governments may prosecute in cases of repeated violations.

APPENDIX 2

United Kingdom and Irish MRCC/MRSC contact details

Station	*Address*	*Telephone / fax / telex contact*
MRCC Falmouth	Pendennis Point Castle Drive Falmouth Cornwall TR11 4WZ	Tel: (0)1326 317575 Fax: (0)1326 318342 Tlx: +51 45560 Answerback: 45560 FALMCG G Inmarsat direct contact details: Inmarsat-A (AOR-E) 1441532 Inmarsat-C (AOR-E) 423200158 Inmarsat-C (AOR-W) 423200159
MRCC Aberdeen	Marine House Blaikies Quay Aberdeen AB11 5PB	Tel: (0)1224 592334 Fax: (0)1224 575920 Tlx: +51 73123 Inmarsat-A (AOR-E) 1441761
MRCC Clyde	Navy Buildings Eldon Street Greenock PA16 7QY	Tel: (0) 1475 729988 Fax: (0)1475 786955 Tlx: +51 777006
MRCC Dover	Langdon Battery Swingate Dover Kent CT15 5NA	Tel: (0)1304 210008 Fax: (0)1304 202137 Tlx: +51 96273
MRCC Swansea	Tutt Head Mumbles Swansea SA3 4EX	Tel: (0) 1792 366534 Fax: (0) 1792 369005 Tlx: +51 48202
MRCC Yarmouth	Havenbridge House Great Yarmouth Norfolk NR30 1HZ	Tel: (0) 1493 851338 Fax: (0) 1493 852307 Tlx: +51 97400
MRCC Dublin	Irish Marine Emergency Service Department of Marine Leeson Lane, DUBLIN	Tel: +353-1-6620922/3 Fax: +353-1-662795 Tlx: +500 93039

Aeronautical Rescue Coordination Centre

ARCC Kinloss Tel: (0)1343 836 022
Fax: (0)1309 678 308
Tlx: +51 75193

CONTACT NUMBERS OF MARITIME RESCUE SUB-CENTRES (MRSCs)

MRSC Station	*Phone number*	*Fax number*	*Telex*
Brixham	(0)1803 882704	(0)1803 882780	+51 42981
Portland	(0)1305 760439	(0)1305 - 760452	+51 417138
Solent	(0)23 9255 2100	(0)23 9255 1763	+51 869194
Thames	(0)1255 675518	(0)1255 675249	+51 98314
London	(0)20 8312 7380	(0)20 8312 7679	
Humber	(0)1262 672317	(0)1262 606915	+51 527351
Forth	(0)1333 450666	(0)1333 450725	+51 72440
Shetland	(0)1595 692976	(0)1595 694810	+51 75141
Stornoway	(0)1851 702013	(0)1851 704387	+51 75157
Belfast	(0)28 9146 3933	(0)28 9146 5886	+51 747833
Liverpool	(0)151 931 3341	(0)151 931 3347	+51 629448
Holyhead	(0)1407 762051	(0)1407 764373	+51 61124
Milford Haven	(0)1646 690909	(0)1646 692176	+51 48293
Jersey Radio	(0)1534 41121	(0)1534 499089	
St Peter Port Radio	(0)1481 720672	(0)1481 714177	
Alderney Radio	(0)1481 822620	(0)1481 822436	
Malin Head	+351 777 0103	+351 777 0221	+500 42072
Valentia	+351 66 9476109	+351 66 9476289	+500 73968

APPENDIX 3

Standard marine navigational vocabulary

MERCHANT SHIPPING NOTICE NO. M1252[1]

INTRODUCTION

This vocabulary has been compiled:

— to assist in the greater safety of navigation and of the conduct of ships.
— to standardize the language used in communication for navigation at sea, in port-approaches, in waterways and harbours.

These phrases are not intended to supplant or contradict the International Regulations for Preventing Collisions at Sea or special local rules or recommendations made by IMO concerning ships' routeing. Neither are they intended to supersede the International Code of Signals nor to supplant normal radiotelephone practice as set out in the ITU Regulations.

It is not intended that use of the vocabulary shall be mandatory, but rather that through constant repetition in ships and in training establishments ashore, the phrases and terms used will become those normally accepted and commonplace among seamen. Use of the contents of the vocabulary should be made as often as possible in preference to other wording of similar meaning.

In this way it is intended to become an acceptable "language", using the English tongue, for the interchange of intelligence between individuals of all maritime nations on the many and varied occasions when precise meanings and translations are in doubt, increasingly evident under modern conditions at sea.

The typographical conventions used throughout most of this vocabulary are as follows:

() brackets indicate that the part of the message enclosed within the brackets may be added where it is relevant.
/ oblique stroke indicates that the items on either side of the stroke are alternatives.
. . . dots indicate that the relevant information is to be filled in where the dots occur.

WHEN SPELLING IS NECESSARY, ONLY THE LETTER SPELLING TABLE CONTAINED IN THE INTERNATIONAL CODE OF SIGNALS, CHAPTER X, AND IN THE RADIO REGULATIONS SHOULD BE USED.

1 Reproduced with the kind permission of the Maritime and Coastguard Agency.

PART 1. GENERAL

1. Procedure/message markers

When it is necessary to indicate that phrases in this vocabulary are to be used, the following messages may be sent:

"Please use the Standard Marine Navigational Vocabulary."
"I will use the Standard Marine Navigational Vocabulary."

If necessary, external communication messages may be preceded by the following message markers:

QUESTION	indicates that the following message is of interrogative character.
ANSWER	indicates that the following message is the reply to a previous question.
REQUEST	indicates that the contents of the following message are asking for action from others with respect to the ship.
INFORMATION	indicates that the following message is restricted to observed facts.
INTENTION	indicates that the following message informs others about immediate navigational actions intended to be taken.
WARNING	indicates that the following message informs other traffic participants about dangers.
ADVICE	indicates that the following message implies the intention of the sender to influence the recipient(s) by a recommendation.
INSTRUCTION	indicates that the following message implies the intention of the sender to influence the recipient(s) by a regulation.

2. Standard verbs

Where possible, sentences should be introduced by one of the following verb forms:

IMPERATIVE

Always to be used when mandatory orders are being given

You must	Do not	Must I?
INDICATIVE	NEGATIVE	INTERROGATIVE
I require	I do not require	Do I require?
I am	I am not	Am I?
You are	You are not	Are you?
I have	I do not have	Do you have?
I can	I cannot	Can I? - *is it* Can you? *possible?*
I wish to	I do not wish to	Do you wish to?
I will — *future*	I will not - *future*	
You may	You need not	May I? - *permission*
Advise	Advise not	
There is	There is not	Is there? What is/are? Where is/are? When is/are?

Note: See section 1 — Message markers.

3. Responses

Where the answer to a question is in the affirmative, say:
"yes . . ." — followed by the appropriate phrase in full.
Where the answer to a question is in the negative, say:
"no . . . " — followed by the appropriate phrase in full.
Where the information is not immediately available but soon will be, say:
"Stand by".
Where the information cannot be obtained, say:
"No information".
Where a message is not properly heard, say:
"Say again".
Where a message is not understood, say:
"Message not understood".

4. Distress/urgency/safety messages

MAYDAY	(repeated three times)	is to be used to announce a distress message
PAN PAN	(repeated three times)	is to be used to announce an urgency message
SECURITE	(repeated three times)	is to be used to announce a safety message

5. Miscellaneous phrases

5.1 What is your name (and call sign)?
5.2 How do you read me?

I read you . . .	with signal strength . . .
(bad/1)	(1/barely perceptible)
(poor/2)	(2/weak)
(fair/3)	(3/fairly good)
(good/4)	(4/good)
(excellent/5)	(5/very good)

5.4 Stand by on channel . . .
5.5 Change to channel . . .
5.6 I cannot read you.
(Pass your message through vessel . . .).
(Advise try channel . . .).
5.7 I cannot understand you.
Please use the . . .
(Standard Marine Navigational Vocabulary).
(International Code of Signals).
5.8 I am passing a message for vessel . . .
5.9 Correction . . .
5.10 I am ready to receive y our message.
5.11 I am not ready to receive your message.
5.12 I do not have channel . . . Please use channel .

6. Repetition

If any parts of the message are considered sufficiently important to need safeguarding, use the word "repeat".

Examples: "You will load 163 repeat 163 tons bunkers."
"Do not repeat not overtake."

7. Position

When latitude and longitude are used, these shall be expressed in degrees and minutes (and decimals of a minute if necessary), north or south of the Equator and east or west of Greenwich.

When the position is related to a mark, the mark shall be a well-defined charted object. The bearing shall be in the 360 degree notation from true north and shall be that of the position FROM the mark.

Examples: "There are salvage operations in position 15 degrees 34 minutes north 61 degrees 29 minutes west."
"Your position is 137 degrees from Barr Head lighthouse distance two decimal four miles."

8. Courses

Always to be expressed in 360 degree notation from north (true north unless otherwise stated). Whether this is to TO or FROM a mark can be stated.

9. Bearings

The bearing of the mark or vessel concerned, is the bearing in the 360 degree notation from north (true north unless otherwise stated), except in the case of relative bearings. Bearings may be either FROM the mark or FROM the vessel.

Examples: "The pilot boat is bearing 215° from you."
"Your bearing is 127° from the signal station."

Note: Vessels reporting their position should always quote their bearing FROM the mark, as described in paragraph 7.

Relative bearings

Relative bearings can be expressed in degrees relative to the vessel's head or bow. More frequently this is in relation to the port or starboard bow.

Example: "The buoy is 030° on your port bow."

Relative D/F bearings are more commonly expressed in the 360 degree notation.

10. Distances

Preferably to be expressed in nautical miles or cables (tenths of a mile) otherwise in kilometres or metres, the unit always to be stated.

11. Speed

To be expressed in knots:

(a) without further notation meaning speed through the water; or
(b) "ground speed" meaning speed over the ground.

12. Numbers

Numbers are to be spoken:

"One-five-zero" for 150.
"Two point five" for 2.5.

13. Geographical names

Place names used should be those on the chart or Sailing Directions in use. Should these not be understood, latitude and longitude should be given.

14. Time

Times should be expressed in the 24 hour notation indicating whether UTC, zone time or local shore time is being used.

Note: In cases not covered by the above phraseology normal radiotelephone practice will prevail.

PART II. GLOSSARY

1. General

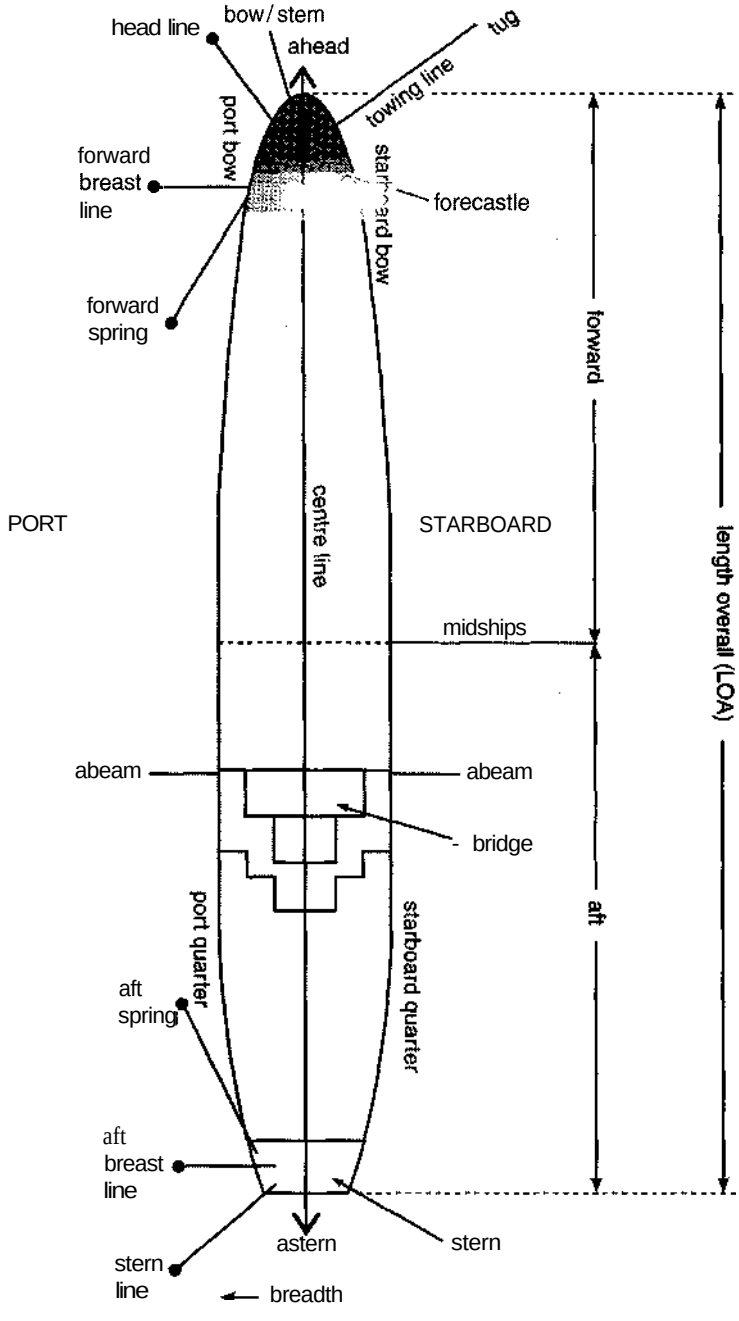
Air draught	Height of highest point of vessel's structure above waterline, e.g. radar, funnel, cranes, masthead.
Anchor position	Place where a specific vessel is anchored or is to anchor.
Calling-in-point (C.I.P.)	(see Way point).
"Correction"	An error has been made in this transmission, the corrected version is ...
Dragging (of anchor	An anchor moving over the sea bottom involuntarily because it is no longer preventing the movement of the vessel.
Dredging anchor	Vessel moving, under control, with anchor moving along the sea bottom.
Draught	Depth from waterline to vessel's bottom, maximum/ deepest unless otherwise specified.
Established	Brought into service, placed in position.
ETA	Estimated time of arrival.
ETD	Estimated time of departure.
Fairway	Navigable part of waterway.
Fairway speed	Mandatory speed in a fairway.
Foul (anchor)	Anchor has its own cable twisted around it or has fouled an obstruction.
Foul (propeller)	A line, wire, net, etc. is wound round the propeller.
Hampered vessel	A vessel restricted in her ability to manoeuvre by the nature of her work.
Icing	Formation of ice on vessels.
Inoperative	Not functioning.
Mark	General term for a navigational mark, e.g. buoy, structure or topographical feature which may be used to fix a vessel's position.

Offshore installation	Any offshore structure (e.g. a drilling rig, production platform, etc.) which may present a hazard to navigation.
Receiving point	A mark or place at which a vessel comes under obligatory entry, transit, or escort procedure (such as for port entry, canal transit or ice-breaker escort).
Reporting point	(see Way point).
Vessel crossing	A vessel proceeding across a fairway/traffic lane/route.
Vessel inward	A vessel which is proceeding from sea to harbour or dock.
Vessel leaving	A vessel which is in the process of leaving a berth or anchorage. (When she has entered the navigable fairway she will be referred to as an outward, inward, crossing or turning vessel.)
Vessel outward	A vessel which is proceeding from harbour or anchorage to seawards.
Vessel turning	A vessel making LARGE alteration in course, such as to stem the tide when anchoring, or to enter, or proceed, after leaving a berth, or dock.
Way point	A mark or place at which a vessel is required to report to establish its position. (Also known as reporting point or calling-in-point.)

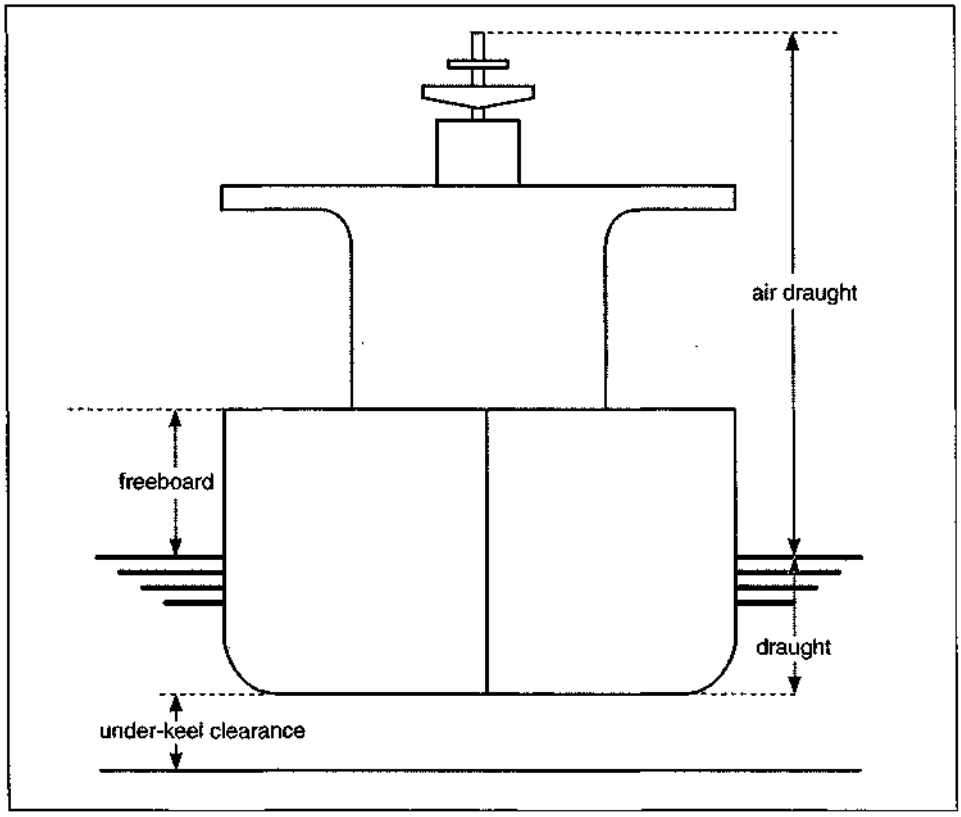
2. Terms used in ships' routeing

Attention is drawn to the following terms which are regularly used in communications regarding ships' routeing and are defined in the General Provisions on Ships' Routeing*:

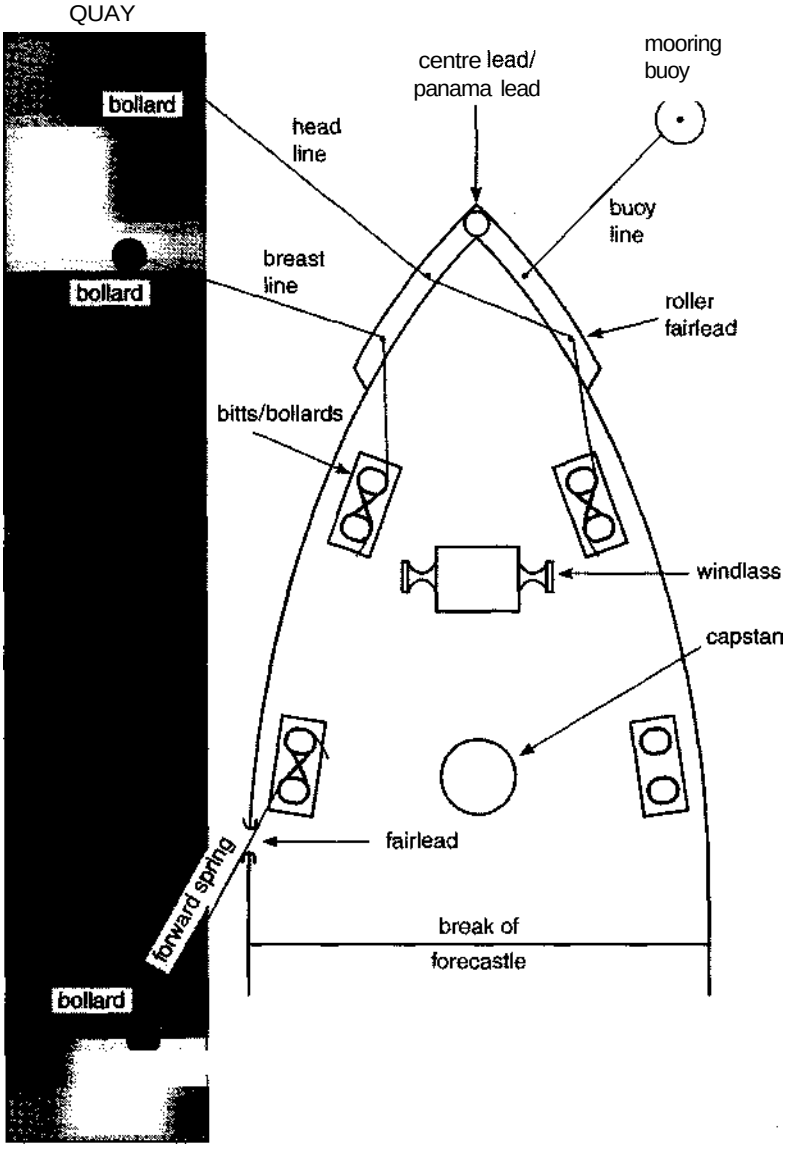
— Routeing system
— Traffic separation scheme
— Separation zone or line
— Traffic lane
— Roundabout
— Inshore traffic zone
— Two-way-route
— Recommended track
— Deep water route
— Precautionary area
— Area to be avoided
— Established direction of traffic flow
— Recommended direction of traffic flow


head line
bow/stem
ahead
tug
towing line
port bow
starboard bow
forward breast line
forecastle
forward spring
forward
length overall (LOA)
PORT
STARBOARD
centre line
midships
abeam
abeam
bridge
port quarter
starboard quarter
aft
aft spring
aft breast line
astern
stern
stern line
breadth

SKETCH 1



SKETCH 2


QUAY
bollard
bollard
bollard
centre lead/
panama lead
mooring
buoy
head
line
buoy
line
breast
line
roller
fairlead
bitts/bollards
windlass
capstan
fairlead
forward spring
break of
forecastle

SKETCH 3

PART III. PHRASE VOCABULARY FOR EXTERNAL COMMUNICATION

Chapter A - Dangers to navigation, warnings, assistance

1 Warnings

1.1 You are running into danger.
(Shallow water ahead of you)
(Submerged wreck ahead of you)
(Risk of collision imminent)
(Fog bank ahead of you)
(Bridge will not open)
(……).

1.2 Dangerous obstruction/wreck reported at ...

1.3 Unknown object(s) in position . . .

1.4 Floating ice in position . . .
(Considered hazardous to navigation).

1.5 Mine(s) reported in position . . .

1.6 Navigation is closed (prohibited) in area . . .

1.7 There has been a collision in position . . .
(keep clear)
(stand by to give assistance).

1.8 It is dangerous to . . .
(stop)
(remain in present position)
(alter course to starboard)
(alter course to port)
(approach close to my vessel)
(……).

1.9 Vessel . . . is aground in position . . .

1.10 Vessel . . . is on fire in position . . .

1.11 Large vessel leaving. Keep clear of approach channel.

1.12 Go to emergency anchorage.

1.13 Your navigation lights are not visible.

1.14 You are running aground.

1.15 Keep clear . . .
(I am jettisoning dangerous cargo)
(vessel is leaking inflammable cargo in position . . .)
(vessel is leaking noxious cargo in position . . .)
(vessel is leaking poisonous cargo in position . . .)
(you are crossing my nets)
(I have a long tow)
(you are heading towards my tow)
(you are heading towards a towing line)
(……).

2 Assistance

2.1 I need help . . .
(I am sinking)

(I am on fire)
(I have been in collision)
(I am aground)
(.....).

2.2 I am on fire and have dangerous cargo on board.

2.3 I am on fire . . .
(in the engine-room)
(in the hold)
(in the cargo tanks)
(in the accommodation)
(in the living spaces)
(.....).

2.4 I have lost a man overboard (at . . .). Help with search and rescue.

2.5 What is your position?
What is the position of the vessel in distress?

2.6 What assistance is required?

2.7 I require . . .
(a lifeboat)
(a helicopter)
(medical assistance*)
(fire-fighting assistance)
(a tug)
(tugs)
(.....).

* *Note:* Further messages should be made using the Medical Section of the International Code of Signals

2.8 I am coming to your assistance.

2.9 I expect to reach you at ... hours.

2.10 Send a ...
(boat)
(raft).

2.11 I am sending a boat to you.
I am sending a raft to you.

2.12 Make a lee for ...
(my vessel)
(the boat)
(the raft).

2.13 I will make a lee for . . .
(your vessel)
(the boat)
(the raft).

2.14 I cannot send a ...
(boat)
(raft).

2.15 I will attempt rescue by breeches-buoy.

2.16 Is it safe to fire a rocket?

2.17 It is safe to fire a rocket.
It is not safe to fire a rocket.

2.18 Take command of search and rescue.
2.19 I am in command of search and rescue.
Vessel . . . is in command of search and rescue.
2.20 Assistance is not required. You may proceed.
Assistance is no longer required. You may proceed.
2.21 You must keep radio silence in this area unless you have messages about the casualty.

Note: Further messages should be made using the International Code of Signals and/or the Merchant Ship Search and Rescue Manual (MERSAR).

Chapter B - General

3 Anchoring

3.1 I am at anchor (at . . .)•
3.2 I am heaving up anchor.
3.3 My anchor is clear of the bottom.
3.4 You may anchor . . .
(at . . . hours)
(in . . . position)
(until pilot arrives)
(until tug(s) arrive(s))
(until there is sufficient depth of water)
(.....).
You must anchor . . .
(at . . . hours)
(in . . . position)
(until pilot arrives)
(until tug(s) arrive(s))
(until there is sufficient water)
(..... ....).
3.5 Do not anchor (in position . . .).
3.6 Anchoring is prohibited.
3.7 I will anchor (at . . .).
3.8 Vessel . . . is at anchor (at . . .).
3.9 Are you dragging anchor?
Are you dredging anchor?
3.10 My anchor is dragging.
Your anchor is dragging.
3.11 Do not dredge anchor.
3.12 You must heave up anchor.
3.13 You must shorten your cable to . . . shackles.
3.14 My anchor is foul.
3.15 You are obstructing . . .
(the fairway)
(other traffic).
3.16 You must anchor in a different position . . .
3.17 You must anchor clear of the fairway.

3.18 What is the anchor position for me?
3.19 You are at anchor in the wrong position.
3.20 I have slipped my anchor (and cable) (and buoyed it) in position . . .
I have lost my anchor (and cable) (and buoyed it) in position . . .

4 Arrival, berthing and departure

4.1 Where do you come from?
What was your last port of call?
4.2 From what direction are you approaching?
4.3 What is your ETA (at . . .)?
What is your ETD (from . . .)?
4.4 My ETA (at . . .) is . . . hours.
My ETD (from . . .) is . . . hours.
4.5 Do not pass receiving point . . . until . . . hours.
4.6 What is your destination?
4.7 My destination is . . .
4.8 What are my berthing instructions?
What are my docking instructions?
4.9 Your berth is clear (at . . . hours).
Your berth will be clear (at . . . hours).
4.10 You will berth at . . .
You will dock at . . .
4.11 May I enter?
4.12 You may enter (at . . .hours).
4.13 May I proceed?
4.14 You may proceed (at . . . hours).
4.15 Is there any other traffic?
4.16 There is a vessel turning at . . .
There is a vessel manoeuvring at . . .
4.17 Vessel . . . will turn at . . .
4.18 Vessel . . . will leave . . . at . . . hours.
4.19 Vessel . . . is leaving . . .
4.20 Vessel . . . has left . . .
4.21 Vessel . . . has entered the fairway at . . .
4.22 Your orders are to . . .
Your orders are changed to . . .
4.23 Vessel . . . inward in position . . .
Vessel . . . outward in position . . .
4.24 Are you underway?
4.25 I am underway.
4.26 I am ready to get underway.
4.27 I am not ready to get underway.
4.28 You must get underway.
4.29 I am making way through the water.
4.30 I have steerage way.
I do not have steerage way.
4.31 Vessel in position (make fast).

4.32 Move ahead (. . . feet/metres).
Move astern (. . . feet/metres).

5 Course

5.1 What is your course?
5.2 My course is . . .
5.3 Your course is correct.
5.4 What course do you advise?
5.5 Advise you make course . . .
5.6 Advise you keep your present course.
5.7 You are steering a dangerous course — to *be followed by indication of danger or advice for further action.*
5.8 I am keeping my present course.
5.9 I cannot keep my present course.
5.10 I am altering course to . . .
5.11 I am altering course to . . .
(port/starboard)
(left/right).
5.12 Advise you alter course to . . . (at .. .).

6 Draught and air draught

6.1 What is your draught?
6.2 My draught is . . .
6.3 What is your draught forward?
What is your draught aft?
6.4 My draught forward is . . .
My draught aft is . . .
6.5 Vessel . . . is of deep draught.
6.6 Do you have any list?
6.7 I have a list to port of . . . degrees.
I have a list to starboard of . . . degrees.
6.8 Maximum permitted draught is . . .
6.9 What is your freeboard?
6.10 My freeboard is . . .
6.11 What is your air draught?
6.12 My air draught is . . .
6.13 Are you trimmed by the head/stern?
6.14 Are you on even keel?
6.15 What is your maximum draught now?

Note 1: When necessary it must be specified whether salt or fresh water draught is given.

Note 2: Air draught is the highest point of the vessel's structure above the waterline.

7(a) Fairway navigation

7.1 There is a vessel entering the fairway (at . . .).
7.2 There is a vessel leaving the fairway (at . . .).

7.25 Your station in convoy will be number . . .
7.26 Transit speed is … knots.
Convoy speed is … knots.
7.27 You must close up on vessel ahead of you.
7.28 Convoys will pass in area . . .
Vessels will pass in area . . .
7.29 You . . .must wait at …
You. . . must moor at . . .
Convoy . . . must wait at …
Convoy . . . must moor at …
7.30 What time may I enter the canal?
What time may I enter the lock?
7.31 You will enter canal at … hours.
You will enter lock at … hours.
7.32 You must wait for lock clearance at . . . (until . . . hours).

Note: See also section 4.

8 Manoeuvring

The use of these messages does not relieve vessels of their obligations to comply with local bye-laws and the International Regulations for Preventing Collisions at Sea.

8.1 I am altering my course to port.
I am altering my course to starboard.
8.2 I am keeping course and speed.
8.3 I am going astern.
8.4 I am not making way through the water.
8.5 What are your intentions?
8.6 Keep well clear of me.
8.7 I wish to overtake (. . .).
8.8 Do not overtake (. . .).
8.9 Ship astern . . . wishes to overtake (on your port side).
Ship astern . . . wishes to overtake (on your starboard side).
Vessel . . . wishes to overtake (on your port side).
Vessel . . . wishes to overtake (on your starboard side).
8.10 You may overtake (. . .).
8.11 Vessel . . . approaching an obscured area (. . .) approaching vessels acknowledge.
8.12 I am not under command.
8.13 I am a hampered vessel.
8.14 I am manoeuvring with difficulty. Keep clear of me.
8.15 Advise you alter course to port.
Advise you alter course to starboard.
I will alter course to port.
I will alter course to starboard
I cannot alter course to port.
I cannot alter course to starboard.
8.18 Advise you stop engines.
8.19 I will stop engines.

8.20 Do not pass ahead of me.
Do not pass astern of me.
8.21 Do not pass on my port side.
Do not pass on my starboard side.
8,22 I ... will overtake (. . .).
Vessel . . . will overtake (. . .).
8.23 Advise you pass ahead of me ...
Advise you pass astern of me . . .
Advise you pass ahead of vessel . . .
Advise you pass astern of vessel . . .
I will pass ahead of you . . .
I will pass astern of you . . .
I will pass ahead of vessel . . .
I will pass astern of vessel . . .
8.25 Wait for . . . to cross ahead of you.
8.26 I will wait for . . . to cross ahead of me.
8.27 Advise you pass north of ... vessel.
Advise you pass s outh of . . vessel.
Advise you pass east of . . vessel.
Advise you pass west of . vessel.
Advise you pass north of . . mark.
Advise you pass south of , . mark.
Advise you pass east of . . mark.
Advise you pass west of . mark.
8.28 I will pass north of . . . vessel.
I will pass south of . . . vessel.
I will pass east of ... vessel.
I will pass west of ... vessel.
I will pass north of . . . mark.
I will pass south of . . . mark.
I will pass east of . . . mark.
I will pass west of . . . mark.
8.29 Wait for . . . to clear (. . . mark) before entering fairway.
Wait for . . . to clear (. . . position) before entering fairway.
Wait for . . . to clear (. . . mark) before getting underway.
Wait for ... to clear (. . . position) before getting underway.
Wait for ... to clear (. . . mark) before leaving berth.
Wait for ... to clear (. . . position) before leaving berth.
8.30 I will wait for . . . to clear (. mark) before entering fairway.
I will wait for . . . to clear (. position) before entering fairway.
I will wait for . . . to clear (. mark) before getting underway.
I will wait for . to clear (. . . position) before getting underway.
I will wait for . to clear (. . . mark) before leaving berth.
I will wait for . to clear (. . . position) before leaving berth.

9 Pilotage

9.1 I require a pilot.
9.2 Do you require a pilot?

9.3 Is the pilot boat on station?
9.4 At what position can I take pilot?
9.5 You can take pilot at point . . . (at . . . hours).
You can take pilot near . . . (at . . . hours).
9.6 At what time will the pilot be available?
9.7 Must I take a pilot?
9.8 You may navigate by yourself or wait for pilot at . .
9.9 Pilot is coming to you.
9.10 Pilot boat is approaching your vessel.
9.11 You must rig pilot ladder on port side.
You must rig pilot ladder on starboard side.
9.12 Pilot ladder is rigged on port side.
Pilot ladder is rigged on starboard side.
9.13 Pilotage suspended for all vessels.
Pilotage suspended for small vessels.
Pilotage resumed for all vessels.
Pilotage resumed for small vessels.
9.14 You must rig gangway combined with pilot ladder.
9.15 Have a heaving line ready at pilot ladder.
9.16 Put lights on at pilot ladder position.
9.17 Give a slight heel to port/starboard.
9.18 Lower/lift up the pilot ladder/hoist.
9.19 Make a lee on the port/starboard side.
9.20 Stop the engines until the pilot boat is clear.
9.21 Keep the pilot boat on the port/starboard side.
9.22 Stop where you are and wait for the pilot.
9.23 The pilot will board you in . . . minutes/hours.

10 Position

10.1 What is your position?
10.2 What is my position?
10.3 My position is . . .
Your position is . . .
10.4 Your position is . . . degrees . . . miles from . . .
10.5 You are passing . . .
10.6 You are entering area . . .
10.7 What is your present position, course and speed?
10.8 My present position, course and speed is . . .
10.9 What is the course to . . . ?
10.10 The course to . . . is . . .
10.11 What is the course to reach you?
10.12 The course to reach me is . . .
10.13 Do not arrive at . . . before . . . hours.
10.14 Do not arrive at . . . after . . . hours.
10.15 Say again your position for identification.
10.16 Has your position been obtained by radar?
Has your position been obtained by decca?

Has your position been obtained by astronomical observation?
Has your position been obtained by . . . ?

10.17 My position has been obtained by radar.
My position has been obtained by decca.
My position has been obtained by astronomical observation.
My position has been obtained by …

11 Radar - ship-to-ship/shore-to-ship/ship-to-shore

11.1 Is your radar working?
11.2 My radar is working.
My radar is not working.
11.3 I do not have radar.
11.4 I have located you on my radar* (your position is … degrees . . . miles from . . .).

**Note*: This message may only be used when the vessel is positively identified.

11.5 I cannot locate you on my radar.
11.6 You must alter course for identification.
You must alter speed for identification.
11.7 I have altered course to … for identification.
I have altered speed to . . . for identification.
11.8 I have lost radar contact.
11.9 Have you altered course?
11.10 Report your position for identification.
11.11 Vessel ahead of you is on the same course.
11.12 You are getting closer to the vessel(s) ahead.
11.13 Your position is …
11.14 My position is …
11.15 What range scale are you using?
11.16 I am using . . . mile range scale.
11.17 Advise you change to larger range scale.
Advise you change to smaller range scale.
11.18 I require shore-based radar assistance.
11.19 Is shore-based radar assistance available?
11.20 Shore-based radar assistance is available.
Shore-based radar assistance is not available.
11.21 I am at way point. . . ., course . . ., speed . . .
I am at reporting point . . ., course . . ., speed . . .
I am at C.I.P. . . ., course . . ., speed . . .
I am approaching way point . . ., course . . ., speed . . .
I am approaching reporting point . . ., course . . ., speed . . .
I am approaching C.I.P. . . ., course . . ., speed . . .
11.22 I will stop at position . . . at … hours.
11.23 You are in the fairway.
11.24 Vessel on opposite course passing your port side.
Vessel on opposite course passing your starboard side.
11.25 Vessel is … miles/metres ahead on port bow.
Vessel is … miles/metres ahead on starboard bow.

11.26 Vessel ahead of you is on opposite course.
11.27 Vessel following will overtake you on port side.
Vessel following will overtake you on starboard side.
11.28 You are leaving my screen.
11.29 Do you want navigational assistance to reach the inshore pilot station?
11.30 I want navigational assistance to reach the inshore pilot station.
11.31 You are approaching - starboard/port limit of fairway.
— radar reference line of fairway.
11.32 Your position is buoy number . . . distance . . . metres port/starboard from reference line.
11.33 Your position is buoy number . . . distance . . . metres port/starboard from reference line . . . track . . . parallel with reference line.
11.34 Your position is buoy number . . . distance . . . metres port/starboard from reference line . . . track . . . closing reference line.
11.35 Your position is buoy number . . . distance . . . metres port/starboard from reference line . . . track . . . diverging from reference line.
11.36 Your position is distance . . . metres from intersection of reference line . . . and reference line . . .
11.37 Your position is distance . . . metres from intersection of reference line . . . and reference line . . . and distance . . . metres port/starboard from reference line . . .
11.38 My radar has become inoperative.

12 Navigational warnings

12.1 There is a dangerous wreck in position . . .
(marked by . . . showing . . .).
There is a dangerous rock in position . . .
(marked by . . . showing . . .).
There is a dangerous shoal in position . . .
(marked by . . . showing . . .).
12.2 There is a drifting mine reported in position . . .
12.3 There is a gas leakage (from fractured pipeline) in position . . .
12.4 There is a slick of oil in position . . . (extending . . .).
12.5 There are pipelaying operations in position . . .
There are cable-laying operations in position . . .
12.6 There are salvage operations in position . . .
There are oil clearance operations in position . . .
12.7 There are tankers transferring . . . in position . . .
12.8 There are current meters moored in position . . .
There are oceanographic instruments moored in position . . .
12.9 There is a derelict adrift in position . . . (at . . . hours).
12.10 There is a vessel with a difficult tow on passage from . . . to . . .
12.11 There is a drilling rig . . . (name) established in position . . .
There is an offshore installation . . . (name) established in position . . .
12.12 There is a . . . buoy in position . . . unlit.
There is another mark in position . . . unlit.
There is a . . . buoy in position . . . off station.
There is another mark in position . . . off station.

12.13 There is a … buoy (showing . . .) established in position.
There is another mark (showing . . .) established in position.
12.14 There is a … light in position . . . now showing . . .
There is a buoy in position . . . now showing . . .
There is another mark in position . . . now showing . . .
12.15 There is a vessel carrying out hydrographic survey in position . . .
There is a vessel carrying out seismic survey in position . . .
There is a vessel carrying out hydrographic survey in area . . .
There is a vessel carrying out seismic survey in area . . .
12.16 Abnormally low tides expected in … at … hours.
Abnormally low tides expected in … around . . . hours.
12.17 Decca Chain . . . red transmissions interrupted at . . ., check all lane numbers.
Decca Chain . . green transmissions interrupted at . . ., check all lane numbers.
Decca Chain . purple transmissions interrupted at . . ., check all lane numbers.
12.18 Vessels must keep clear of this area.
Vessels must keep clear of area indicated.
Vessels must avoid this area.
Vessels must avoid area indicated.
12.19 Vessels are advised to keep clear of this area.
Vessels are advised to avoid this area.
12.20 Vessels must navigate with caution.
12.21 There is a vessel not under command in position . . .
There is a vessel not under command in area . . .
12.22 There is a hampered vessel in position . . .
There is a hampered vessel in area . . .
12.23 Radio beacon service . . . has been discontinued.
12.24 Advise you keep clear of sea area . . . search and rescue in operation.
12.25 Route . . . has been suspended.
Traffic lane . . . has been suspended.
Route . . . has been discontinued.
Traffic lane . . . has been discontinued.
Route . . . has been diverted.
Traffic lane . . . has been diverted.

13 Routeing

13.1 May I enter traffic lane?
May I enter route?
13.2 You may enter traffic lane.
Do not enter traffic lane.
You may enter route.
Do not enter route.
13.3 You may enter traffic lane at position . . . at … hours.
You may enter route at position . . . at … hours.

13.4 I will enter traffic lane . . . at . . . hours.
I will enter route . . . at . . . hours.
13.5 You are not complying with traffic regulations.
13.6 You are not keeping to your correct traffic lane.
13.7 There is a vessel in position . . . on course . . . and speed . . . which is not complying with traffic regulations.
13.8 There is a vessel anchored ahead of you in position . . .
13.9 There is a vessel ahead obstructing your movements.
13.10 There is a hampered vessel in position . . . on course . . . and speed . . .
13.11 You will meet crossing traffic at . . .
13.12 There is a vessel crossing . . . traffic lane on course . . . and speed . . . in position . . .
13.13 There are many fishing vessels at . . .
13.14 Route . . . has been suspended.
Traffic lane . . . has been suspended.
Route . . . has been discontinued.
Traffic lane . . . has been discontinued.
Route . . . has been diverted.
Traffic lane . . . has been diverted.

14 Speed

14.1 What is your present speed?
What is your full speed?
14.2 My present speed is . . . knots.
My full speed is . . . knots.
14.3 What is your full manoeuvring speed?
14.4 My full manoeuvring speed is . . . knots.
14.5 You are proceeding at a dangerous speed.
14.6 Fairway speed is . . . knots.
14.7 You must reduce speed.
14.8 I am reducing speed.
14.9 You must increase speed.
14.10 I am increasing speed.
14.11 I cannot increase speed.
14.12 You must keep your present speed.
14.13 I am keeping present speed.
14.14 What speed do you advise?
14.15 Advise speed . . . knots

15 Tide and depth

15.1 What is the tide doing?
What is the tidal stream doing?
15.2 The tide is rising (it is . . . hours before high water).
The tide is rising (it is . hours after low water).
15.3 The tide is rising (it is . metres/feet below high water).
The tide is rising (it is . metres/feet above low water).

15.4 The tide is falling (it is . . . hours after high water).
The tide is falling (it is . . . hours before low water).
15.5 The tide is falling (it is . . . metres/feet below high water).
The tide is falling (it is . . . metres/feet above low water).
15.6 The tide is slack.
The tide is with you.
The tide is against you.
15.7 Present height of the tide above datum is . . . metres/feet at position . . .
15.8 Tide is (. . . metres/feet) above prediction.
Tide is (. . . metres/feet) below prediction.
15.9 The tide is . . . knots at . . .
The current is . . . knots at . . .
15.10 Tide is setting in direction . . .
15.11 In your present position you will be aground at low water.
15.12 Is there sufficient depth of water?
15.13 There is sufficient depth of water.
There is not sufficient depth of water.
15.14 My draught is . . . metres/feet. At what time may I enter . . . ?
My draught is . . . metres/feet. At what time may I pass . . . ?
15.15 Charted depths are decreased by . . . metres/feet due to state of the sea.
Charted depths are decreased by . . . metres/feet due to state of the winds.
15.16 Abnormally low tides expected in . . . at . . . hours.
Abnormally low tides expected in . . . around . . . hours.

16 Tropical storms

16.1 What is your latest tropical storm warning information?
16.2 Tropical storm centre (name) reported in . . .
16.3 What is the atmospheric pressure (and its change)?
What is the atmospheric pressure (at position . . .)?
What is the atmospheric pressure (at your position)?
16.4 The atmospheric pressure is . . . and its change is (. . .).
16.5 What is the position, direction and speed of the tropical storm centre (name)?
16.6 The tropical storm centre (name) was (at . . . hours) in position . . . moving . . . at . . . knots.
16.7 Tropical storm (name) at . . . hours was moving in direction . . . at . . . knots with maximum winds force . . .
Tropical storm (name) at . . . hours was moving in direction . . . at . . . knots with maximum winds speed . . .

17 Tugs

17.1 I require a tug.
I require . . . tugs.
17.2 Must I take tug(s)?
17.3 How many tugs must I take?
17.4 You must take . . . tug(s).
17.5 At what position will tug(s) meet me?

17.6 Tug(s) will meet you at (position . . .) (at . . . hours).
Tug(s) will meet you (near . . .) (at . . . hours).
17.7 Tug services suspended.
Tug services resumed.

18 Way points/reportingpoints/C.I.P.

18.1 (Vessel indicated) I am at way point . . .
(Vessel indicated) I am approaching way point . . .
18.2 (Vessel indicated) You are approaching way point . .
18.3 Report at next way point . . .
Report at next position . . .
18.4 Vessel . . . has reported at . . .
18.5 You must arrive at … at … hours.

19 Weather

19.1 What is the weather forecast (for area . . .)?
19.2 What is the wind direction and force (in area . . .)?
What is the wind direction and speed (in area . . .)?
19.3 Wind direction and force at … is …
Wind direction and speed at … is …
19.4 Is the wind expected to change?
19.5 The wind is backing and increasing.
The wind is veering and increasing.
The wind is backing and decreasing.
The wind is veering and decreasing.
19.6 Is the wind force expected to increase at . . .?
Is the wind speed expected to increase at . . .?
Is the wind force expected to decrease at . . .?
Is the w ind speed expected to decrease at . . .?
19.7 The wind at . . . will increase to force . . . within the next . . . hours.
The wind at … will decrease to force . . . within the next . . . hours.
The wind at … will increase to speed . . . within the next . . . hours.
The wind at . . . will decrease to speed . . . within the next . . . hours.
19.8 What is the visibility at . . .?
19.9 Visibility at … is … metres/miles.
19.10 Visibility is reduced by fog.
Visibility is reduced by rain.
Visibility is reduced by snow.
Visibility is reduced by dust.
Visibility is reduced by …
19.11 Is visibility expected to change?
19.12 Visibility is expected to increase to … metres/miles in … hours.
Visibility is expected to decrease to … metres/miles in … hours..
Visibility is expected to increase to … metres/miles by … hours.
Visibility is expected to decrease to … metres/miles by … hours.
19.13 What is the state of the sea at . . . ?
19.14 There is a sea of height . . . metres/feet from . . .
There is a swell of height . . . metres/feet from . . .

19.15 Are sea conditions expected to change within the next . . . .hours?
19.16 Sea is expected to increase within the next . . hours.
Swell is expected to increase within the next . hours.
Sea is expected to decrease within the next . hours.
Swell is expected to decrease within the next . . hours.
19.17 Can icing be expected at . . . ?
19.18 Icing may be expected to form slightly at . . .
Icing may be expected to form moderately at . . .
Icing may be expected to form severely at . . .
Icing may be expected to form very severely at . . .
19.19 Icing should not be expected at . . .
19.20 Are there any . . . warnings in operation for . . . ?
19.21 A warning of gales was issued at . . . hours starting . .
A warning of storms was issued at . . . hours starting .
19.22 What is the latest information about storm . . . ?

Chapter C - Special

20 Fishing

20.1 Navigate with caution small fishing boats are within . . . miles of me.
20.2 Is there fishing gear ahead of me?
20.3 You are heading towards fishing gear.
20.4 There are nets with buoys in this area.
20.5 Fishing gear has fouled my propeller.
20.6 You have caught my fishing gear.
20.7 Advise you recover your fishing gear.
20.8 Fishing in this area is prohibited.
20.9 You are approaching a prohibited fishing area.

21 Helicopters

21.1 Vessel . . . ready for helicopter.
21.2 (Vessel . . .) helicopter now proceeding to you.
21.3 My course and speed is . . . degrees . . . knots.
21.4 Identify yourself (by method indicated).
21.5 (Vessel . . .) I am now making identification signal.
21.6 You are identified.
21.7 Present relative wind direction and force is . . . degrees . . .
21.8 Keep the wind on starboard bow.
Keep the wind on port bow.
Keep the wind on starboard quarter.
Keep the wind on port quarter.
21.9 Indicate landing point.
Indicate contact point.
21.10 May I land on deck?
21.11 You may land on deck.
21.12 Do not land on deck.
21.13 Operation will be carried out using hoist.

21.14 Landing party ready to receive you.
21.15 I am landing.
I am commencing operation.
21.16 Do not make fast hoist.

22 Ice-breakers

If there is more than one vessel being assisted, ice-breaker commands concerning all the vessels in a convoy are to be immediately obeyed and repeated consecutively by each vessel in turn. Ice-breaker commands applying to a single vessel in a convoy are repeated only by that vessel.

When being assisted by an ice-breaker it is important to keep a continuous listening watch on the appropriate radiotelephone frequency and to keep a proper lookout for sound and visual signals.

	ICE-BREAKER	ASSISTED VESSEL(S)
1	Attention Attention all vessels Attention vessel . . .	Attention vessel . . . listening Attention . . . (convoy number) listening
2	Ice-breaker assistance is now coming (I will arrive in . . . hours).	Okay. Ice-breaker assistance is now coming (You will arrive in . . . hours)
3	Ice-breaker assistance is now commencing (Keep a continuous radiotelephone listening watch) (Your place in the convoy is . . .) (You will be followed by vessel . . .)	Okay. Ice-breaker assistance is now commencing (I am keeping a continuous radiotelephone listening watch) (My place in the convoy is . . .) (I will be followed by vessel . . .)
4	Go ahead; (follow me).	I am going ahead. (I am following you).
5	Proceed along the ice channel.	I am proceeding along the ice channel.
6	Do not follow me; Stop — *Reason may be given.*	I will not follow you. I am stopping.
7	Do not follow me. Proceed along the ice channel.	I will not follow you; I will proceed along the ice channel.
8	Slow down. - *Reason may be given.*	I am slowing down.
9	Increase your speed. Increase your engine power. — *Reason may be given.*	I am increasing my speed. I am increasing my engine power.
10	Reverse your engine . . . (slow astern) (half astern) (full astern).	I am reversing my engine . . . (slow astern) (half astern) (full astern).
11	Full ahead immediately.	I am going full ahead immediately.
12	Stop your vessel immediately. — *Reason may be given.*	I am stopping my vessel immediately.
13	Keep a distance of . . . cables/metres between vessels.	I will keep a distance of . . . cables/metres between vessels.

14	Shorten the distance between vessels to ... cables/metres.	I am shortening the distance between vessels to ... cables/metres.
15	Increase the distance between vessels to ... cables/metres.	I am increasing the distance between vessels to ... cables/metres.
16	Prepare to receive tow line.	I will prepare to receive tow line.
	Prepare to cast off tow line.	I will prepare to cast off tow line.
17	Be ready to receive tow line.	I am ready to receive tow line.
	Be ready to cast off tow line.	I am ready to cast off tow line.
18	Stop vessel to receive tow line.	I will stop and receive tow line.
	Stop vessel to cast off tow line.	I will stop and cast off tow line.
19	Ice-breaker assistance is Okay. suspended.*	Ice-breaker assistance is suspended.

**Note*: Reason/time may be given.

20	Stay where you are.	I will stay where I am.
21	Switch on bow searchlight.	I will switch on bow searchlight.
	Switch on stern searchlight.	I will switch on stern searchlight.
22	Ice-breaker . . . assistance is finished. Reason . . .	Thank you, Okay. Ice-breaker . . . assistance is finished. Reason . . .
	(There is open water ahead)	(There is open water ahead)
	(There is light ice condition ahead)	(There is light ice condition ahead)
	(Proceed by yourself)	(I will proceed by myself)
	(Ice-breaker . . . will assist you in further navigation)	(Ice-breaker . . . will assist me in further navigation)
	(Proceed by yourself to an area . . .)	(I am proceeding by myself to an area . . .)
	(Ice-breaker . . . will meet and escort you).	(Ice-breaker . . . will meet and escort me).

PART IV. PHRASE VOCABULARY FOR ON-BOARD COMMUNICATION

1 Standard wheel orders

All wheel orders given should be repeated by the helmsman and the officer of the watch should ensure that they are carried out correctly and immediately. All wheel orders should be held until countermanded. The helmsman should report immediately if the vessel does not answer the wheel.*

ORDER	MEANING	
1.1	Midships	Rudder to be held in the fore and aft position.
1.2	Port five	5° of port rudder to be held.
1.3	Port ten	10° of port rudder to be held.
1.4	Port fifteen	15° of port rudder to be held.
1.5	Port twenty	20° of port rudder to be held.

**Note*: When there is concern that the helmsman is inattentive he should be questioned and he should respond in accordance with the terminology In Part Ill, subparagraphs 5.1 and 5.2 of the Standard Marine Navigational Vocabulary: this is a reminder to the helmsman to mind his helm.

1.6	Port twenty-five	25° of port rudder to be held.
1.7	Hard-a-port	Rudder to be held fully over to port.
1.8	Starboard five	5° of starboard rudder to be held.
1.9	Starboard ten	10° of starboard rudder to be held.
1.10	Starboard fifteen	15° of starboard rudder to be held.
1.11	Starboard twenty	20° of starboard rudder to be held.
1.12	Starboard twenty-five	25° of starboard rudder to be held.
1.13	Hard-a-starboard	Rudder to be held fully over to starboard.
1.14	Ease to five	Reduce amount of rudder to 5° and hold.
1.15	Ease to ten	Reduce amount of rudder to 10° and hold.
1.16	Ease to fifteen	Reduce amount of rudder to 15° and hold.
1.17	Ease to twenty	Reduce amount of rudder to 20° and hold.
1.18	Steady	Reduce swing as rapidly as possible.
1.19	Steady as she goes	Steer a steady course on the compass heading indicated at the time of the order. The helmsman is to repeat the order and call out the compass heading on receiving the order. When the ship is steady on that heading, the helmsman is to call out: "Steady on ..."
1.20	Keep the buoy/mark/beacon on port/starboard side.	
1.21	Report if she does not answer the wheel.	
1.22	Finished with the wheel.	

When the officer of the watch requires a course to be steered by compass, the direction in which he wants the wheel turned should be stated followed by each numeral being said separately, including zero, for example:

ORDER	COURSE TO BE STEERED
"Port, steer one eight two"	182°
"Starboard, steer zero eight two"	082°
"Port, steer three zero five"	305°

On receipt of an order to steer, for example, 182°, the helmsman should repeat it and bring the ship round steadily to the course ordered. When the ship is steady on the course ordered, the helmsman is to call out:
"Steady on one eight two".

The person giving the order should acknowledge the helmsman's call out.

If it is desired to steer on a selected mark the helmsman should be ordered to: "Steer on ... buoy . . . (mark) . . . (beacon)".

The helmsman should repeat the order and when steady on the mark call out: "Steady on ... buoy . . . (mark) . . . (beacon)".

The person giving the order should acknowledge the helmsman's call out.

2 Standard engine orders

Any engine order given should be repeated by the person operating the bridge telegraph(s) and the officer of the watch should ensure the order is carried out correctly and immediately.

ORDER	MEANING **(see note *)**	
2.1	**Full ahead**	Maximum manoeuvring engine revolutions for ahead propulsion
2.2	Half ahead	Revolutions as Indicated in ship's orders.

2.3	Slow ahead	Revolutions as Indicated In ship's orders.
2.4	Dead slow ahead	Revolutions as indicated in ship's orders.
2.5	Stop engine(s)	No engine revolutions.
2.6	Dead slow astern	Revolutions as indicated in ship's orders.
2.7	Slow astern	Revolutions as Indicated in ship's orders.
2.8	Half astern	Revolutions as Indicated in ship's orders.
2.9	Full astern	Revolutions as indicated in ship's orders.
2.10	Emergency full ahead/ astern	Revolutions as Indicated In ship's orders.
2.11	Stand by engine	Engine-room personnel fully ready to manoeuvre and bridge manned with personnel to relay engine orders.
2.12	Finished with engines	Movement of engine(s) no longer required.

In ships fitted with twin propellers, the word "both" should be added to all orders affecting both shafts, e.g. "Full ahead both", and "Slow astern both", except that the words "Stop all engines" should be used, when appropriate. When required to manoeuvre twin propellers independently, this should be Indicated, i.e. "Full ahead starboard", "Half astern port", etc.
Where bow thrusters are used, the following orders are used:

ORDER	MEANING	
2.13	Bow thrust full (half) to port	Ship's head to move to port with power as specified.
2.14	Bow thrust full (half) to starboard	Ship's head to move to starboard withpower as specified.
2.15	Stern thrust full (half) to port	Ship's stern to move to port with power as specified.
2.16	Stern thrust full (half) to starboard.	Ship's stern to move to starboard with power as specified.
2.17	Bow (stern) thrust stop	No bow (stern) thrust revolutions.

*Note: For vessels with variable pitch propellers, the meaning of the order would include the combination of pitch and revolutions as indicated in ship's orders.

3 Pilotage

3.1 Have heaving line ready at pilot ladder.
3.2 Put on lights at pilot ladder position.
3.3 Stand by the pilot ladder.
3.4 Lower/lift up the pilot ladder/hoist.
3.5 Make a lee on the port/starboard side.
3.6 Stop the engines until the pilot boat is clear.
3.7 Keep the pilot boat on the port/starboard side.
3.8 Rig pilot ladder on port/starboard side.
3.9 Pilot ladder is rigged on port/starboard side.
3.10 Rig gangway combined with pilot ladder on port/starboard side.
3.11 Gangway is rigged combined with pilot ladder on port/starboard side.

4 Manoeuvring

4.1 May I have your manoeuvring data?
4.2 What is the diameter of the turning circle?
4.3 What are the advance and transfer distance in a crash stop?

4.4 How long does it take from hard-a-port to hard-a-starboard?
4.5 Is the turning effect of the propeller very strong?
4.6 Where is the whistle control?
4.7 What notice is required to reduce from full sea speed to manoeuvring revolutions?
4.8 Is there an automatic pilot?
4.9 Is the automatic pilot reliable?
4.10 Is the helmsman experienced?
4.11 Does the helmsman understand English?
4.12 Give . . . short/long blast(s) on the whistle.
4.13 Put a man on lookout.
4.14 Keep a speed of ... knots.
4.15 What is her (manoeuvring) speed at full/half/slow/dead slow ahead?
4.16 Her (manoeuvring) speed at full/half/slow/dead slow ahead is ... knots.
4.17 Fairway speed is ... knots.
4.18 What is full sea speed?
4.19 Full sea speed is ... knots.

5 Propulsion system

5.1 Is the engine a diesel/turbine?
5.2 Is the engine-room manned or is the engine on bridge control?
5.3 How long does it take to change the engine from ahead to astern?
5.4 How long does it take to start the engine from stopped?
5.5 Is there extra power available in emergency while manoeuvring?
5.6 Do you have a controllable pitch propeller?
5.7 Do you have a right- or left-hand propeller?
5.8 Does she have a single screw/twin screws?
5.9 Are there bow/stern thrusters?
5.10 What is her maximum manoeuvring power ahead/astern?
5.11 What are her maximum revolutions ahead/astern?
5.12 What is the maximum power ahead/astern in an emergency?
5.13 Do the (twin) propellers turn inboard/outboard when going ahead?

6 Anchoring

Going to anchor

6.1 Is/are the anchor(s) ready for dropping?
6.2 Anchor(s) is/are ready for dropping.
6.3 Walk out the anchor(s).
6.4 We are going to an anchorage.
6.5 Have the starboard/port/both anchor(s) ready.
6.6 We will use/drop the port/starboard/both anchor(s).
6.7 Put one/two/three . . . shackles in the water/In the pipe/on deck.
6.8 Walk back the port/starboard/both anchor(s) one/one and a half shackle(s).
6.9 We will let go the port/starboard/both anchor(s) . . . shackle(s) and dredge it.
6.10 Drop port/starboard/both anchor(s).
6.11 Pay out the cable(s).
6.12 Check the cable(s).

6.13 Hold on the port/starboard/both cable(s).
6.14 Where is the cable leading?
6.15 Is the anchor/are the anchors holding?
6.16 Has she been brought up?
6.17 She has been brought up in position . . .
6.18 Switch on the anchor lights.
6.19 Hoist/lower the anchor ball.
6.20 Check the anchor position by bearings.

Leaving the anchorage

6.21 How much cable is out?
6.22 Stand by to heave up.
6.23 Put the windlass in gear.
6.24 Where is the cable leading?
6.25 The Cable(s) is/are leading ahead/astern/to port/to starboard/round the bow/up and down,
6.26 Heave in the port/starboard/both cable(s).
6.27 Is there much weight on the cable?
6.28 There is a lot of weight/too much weight/no weight on the cable.
6.29 Stop heaving.
6.30 How many shackles are left (to come in)?
6.31 There is a turn/there are turns in the cable(s).
6.32 The anchor is/cables are clear.
6.33 Anchor(s) is/are — clear of the water
— home
– foul
6.34 Anchor(s) has/have been secured.

7 Berthing/unberthing

General

7.1 Is the propeller clear?
7.2 Keep the propeller clear.
7.3 The propeller is/is not clear.
7.4 Stop the (controllable pitch) propeller.
7.5 Have fenders ready forward/aft.
7.6 There are fenders on the berth.

Berthing

1.1 We will be berthing port/starboard side.
7.8 We will be mooring to a buoy/buoys ahead and astern.
7.9 We will be mooring alongside/to dolphins.
7.10 Put out . . . head/stern/breast lines.
7.11 Put out . . . spring(s) fore/aft.
7.12 We will use . . . shore head/stern/breast lines/wires.
7.13 Do you have tension winches forward/aft on all/some lines?

7.14 Have heaving lines ready forward and aft.
7.1 5 Send a heaving line/the head line/the stern line/the breast line ashore to the boatmen/linesmen.
7.16 The boatmen/linesmen will use shackles/lashings to secure the mooring.
7.17 Give the boatmen/linesmen some shackles/lashings for the lines.
7.18 Have a messenger line ready.
7.19 Use the centre/panama lead forward/aft.
7.20 Use the bow/port/starboard quarter lead.
7.21 Heave on the . . . line(s)/spring.
7.22 Pick up the slack on the . . . Iine(s)/spring.
7.23 Heave away.
7.24 Stop heaving.
7.25 Slack away the . . . line(s)/spring.
7.27 Hold on the … line(s)/spring.
7.28 Heave in easy/heave alongside.
7.29 Keep the lines tight.
7.30 Do not break the lines.
7.31 Report the forward/stern distance to …
7.32 We have to move . . . metres ahead/astern.
7.33 We are in position.
7.34 Make fast forward and aft.

Unberthing

7.35 Are the engines ready?
7.36 How long does it take to have the engines ready?
7.37 We are/we are not ready to get underway.
7.38 Prepare to let go.
7.39 Single up to … line(s) and . . . spring(s) forward and aft.
7.40 Slack away and head line/stern line/breast line.
7.41 Hold on the head line/stern line/breast line.
7.42 Slack away the forward/aft spring.
7.43 Hold on to the forward/aft spring.
7.44 Heave on the head line/stern line/forward/aft spring.
7.45 Heave in the head line/stern line/forward/aft spring.
7.46 Let go everything forward/aft.
7.47 Keep someone forward to stand by the anchor.
7.48 Let go head line/stern line/towing line/spring.

8 Radar

8.1 Is your radar working?
8.2 I do not have radar.
8.3 Does your radar have any blind sectors?
8.4 My radar has blind sectors from . . . to … degrees and from . . . to … degrees.
8.5 Switch the radar to … mile range.
8.6 Switch your radar to:
— relative head-up
— relative north-up
— relative course-up

true north-up
• true course-up

9 Tugs

9.1 We will use . . . tug(s).
9.2 Stand by to make fast tug(s).
9.3 Make . . . tug(s) fast forward/on the port/starboard bow.
9.4 Make . . . tug(s) fast aft/on the port/starboard quarter.
9.5 The forward tug will make fast alongside on the port/starboard side.
9.6 The after tug(s) will make fast alongside on the port/starboard side.
9.7 Make . . . tug(s) fast on each bow/quarter.
9.8 The tugs will pull/push.
9.8 Use the ship's lines.
9.10 Use the tug's lines.
9.11 Use the centre/panama lead.
9.12 Use the fairlead/panama lead on the:
— port side/starboard side/amidships
— port bow/starboard bow
— port quarter/starboard quarter.
9.13 Send two lines to the tug.
9.14 Send a heaving line to the tug.
9.15 Lower the towing line to the tug/water.
9.16 Slack away the towing line.
9.17 Make fast the towing line.
9.18 Put/do not put the eye of the towing line on the bitts.
9.19 Stand well clear of the towing line.
9.20 Stand by to let go the tug(s).
9.21 Let go the tug(s).
9.22 The towing line has parted.

10 Draught and air draught

10.1 Is she trimmed by the head/stern?
10.2 Is she on even keel?
10.3 What is her maximum draught now?
10.4 What is her draught forward/aft?
10.5 What is her air draught?

Note 1: When necessary it must be specified whether salt or fresh water draught is given.

Note 2: "Air draught" is the height of the highest point of the vessel above the water line (see sketch 2).

APPENDIX 4

International Morse Code signals and Phonetic Alphabet

LETTERS

a . _	n _ .
a . ___ . _	n _____ . ___
a . _ . _	o _____
b _____	6 _____ .
c _ . _ .	p _____
ch _____	q _____ . _
d _ . .	r . _ .
e .	s . . .
e . . _ . .	t _
f _____	u . . _
g _____	ü . . ___
h ____	v . . . _
i	w . ___
j . _____	x _ . . _
k _ . _	y _____
1 _____	z _____ . .
m ____	

Spacing and length of the signals:
(a) a dash is equal to three dots;
(b) the space between the signals forming the same letter is equal to one dot;
(c) the space between two letters is equal to three dots;
(d) the space between two words is equal to seven dots.

FIGURES

1	6
2	7
3	8
4	9
5	0

In routine repetitions, if there can be no misunderstandings in consequence of the presence together of figures and letters or groups of letters, figures may be rendered by means of the following abbreviated signals:

1 . _	6 _____
2	7 _____
3 _____	8
4 _____	9
5	0

PUNCTUATION AND OTHER SIGNS

Full stop (period)	(.)	· – · – · –
Comma	(,)	– – · · – –
Colon or division mark	(:)	– – – · · ·
Question mark (note of interrogation or request for repetition of a transmission not understood)	(?)	· · – – · ·
Apostrophe	0	· – – – – ·
Hyphen or dash or subtraction sign	(-)	– · · · · –
Fraction bar or division sign	(/)	– · · – ·
Lefthand bracket (parenthesis)	[(]	
Righthand bracket ((parenthesis)	[)]	– · – – · –
Inverted commas (quotation marks) before and after the words	(“ ”)	· – · · – ·
Double hyphen	(=)	– · · · –
Understood		· · · – ·
Error		· · · · · · · ·
Cross or addition sign		· – · – ·
Invitation to transmit		– · –
Wait		· – · · ·
End of work		· · · – · –
Commencing signal		– · – · –
Multiplication sign		

In order to avoid all possible confusion in transmitting fractional numbers, the fraction must be preceded or followed, as the case may be, by a dash.

Examples: for 2% transmit 2-0/0 and not 20/0
for $4\frac{1}{2}$% transmit 4-1/2-0/0 and not 41/20/0
for $\frac{3}{4}$8 transmit f-8 and not |8

The minute sign (') and the second sign (") shall be transmitted by means of the apostrophe sign, transmitted once for the minute sign and twice for the second sign.

PHONETIC ALPHABET AND FIGURE CODE

1. When it is necessary to spell out call signs, service abbreviations and words, the following letter spelling table shall be used:

Letter to be transmitted	Word to be used	Spoken as*	Letter to be transmitted	Word to be used	Spoken as*
A	Alfa	AL FAH	N	November	NO VEM BER
B	Bravo	BRAH VOH	O	Oscar	OSS CAH
C	Charlie	CHAR LEE or SHAR LEE	P	Papa	PAH PAH
			Q	Quebec	KEH BECK
D	Delta	DELL TAH	R	Romeo	ROW ME OH
E	Echo	ECK OH	S	Sierra	SEE AIR RAH
F	Foxtrot	FOKS TROT	T	Tango	TANG GO
G	Golf	GOLF	U	Uniform	YOU NEE FORM or 00 NEE FORM
H	Hotel	HOH TELL			
I	India	IN DEE AH	V	Victor	VIK TAH
J	Juliett	JEW LEE ETT	W	Whiskey	WISS KEY
K	Kilo	KEY LOH	X	X-ray	ECKS RAY
L	Lima	LEE MAH	Y	Yankee	YANG KEY
M	Mike	MIKE	Z	Zulu	ZOO LOO

*Each syllable should be equally emphasised.

2. When it is necessary to spell out figures or marks, the following table shall be used.

Figure or mark to be transmitted	Code word to be used	Spoken as*
0	NADAZERO	NAH-DAH-ZAY-ROH
1	UNAONE	OO-NAH-WUN
2	BISSOTWO	BEES-SOH-TOO
3	TERRATHREE	TAY-RAH-TREE
4	KARTEFOUR	KAR-TAY-FOWER
5	PANTAFIVE	PAN-TAH-FIVE
6	SOXISIX	SOK-SEE-SIX
7	SETTESEVEN	SAYTAY-SEVEN
8	OKTOEIGHT	OK-TOH-AIT
9	NOVENINE	NO-VAY-NINER
Decimal point	DECIMAL	DAY-SEE-MAL
Full stop	STOP	STOP

*Each syllable should be equally emphasised.

APPENDIX 5

International telephone dialling codes

Afghanistan	93	China	86
Albania	355	Colombia	57
Algeria	213	Comoros	269
Andorra	376	Congo PR	242
Angola	244	Cook Island	682
Anguilla	1264	Costa Rica	506
Antigua and Barbuda	1268	Croatia	385
Argentina	54	Cuba	53
Armenia	374	Cyprus	357
Aruba	297	Cyprus (Northern)	90
Ascension Islands	247	Czechoslovakian Republic	420
Australia	61	Denmark	45
Austria	43	Diego Garcia	246
Azerbaijan	994	Djibouti	253
Azores	351	Dominica	1767
Bahamas	1242	Dominican Republic	1809
Bahrain	973	Ecuador	593
Bangladesh	880	Egypt AR	20
Barbados	1246	El Salvador	503
Belarus	375	Equatorial Guinea	240
Belgium	32	Eritrea	291
Belize	501	Estonia	372
Benin	229	Ethiopia	251
Bermuda	1441	Falkland Islands	500
Bhutan	975	Faeroe Islands	298
Bolivia	591	Fiji	679
Bosnia Herzegovina	387	Finland	358
Botswana	267	France	33
Brazil	55	French Guiana	594
Brunei Darussalam	673	French Polynesia	689
Bulgaria	359	Gabon	241
Burkina	226	Gambia	220
Burma	95	Georgia	995
Burundi	257	Germany	49
Cambodia	855	Ghana	233
Cameroon	237	Gibraltar	350
Canada	1	Greece	30
Canary Islands	34	Greenland	299
Cape Verde Islands	238	Grenada	1473
Cayman Islands	1345	Guadaloupe	590
Central African Rep.	236	Guam	671
Chad	235	Guatemala	502
Chile	56	Guinea	224

Guinea Bissau	245
Guyana	592
Haiti	509
Honduras	504
Hong Kong	852
Hungary	36
Iceland	354
India	91
Indonesia	62
Iran	98
Iraq	964
Ireland	353
Israel	972
Italy	39
Ivory Coast	225
Jamaica	1876
Japan	81
Jordan	962
Kazakhstan	7
Kenya	254
Kiribati	686
Korea PDR (North)	850
Korea Republic (South)	82
Kuwait	965
KyrgyzsTan	996
LaoS People's Republic	856
Latvia	371
Lebanon	961
Lesotho	266
Liberia	231
Libya	218
Liechtenstein	423
Lithuania	370
Luxembourg	352
Macao	853
Macedonia	389
Madagascar	261
Malawi	265
Malaysia	60
Maldive Islands	960
Mali	223
Malta	356
Marshall Islands	692
Martinique	596
Mauritania	222
Mauritius	230
Mayotte	269
Mexico	52
Micronesia	691
Moldova	373
Monaco	377
Mongolia	976
Montserrat	1664
Morocco	212
Mozambique	258
Myanmar (Burma)	95
Namibia	264
Nauru Island	674
Nepal	977
Netherlands	31
Netherlands Antilles	599
New Caledonia	687
New Zealand	64
Nicaragua	505
Niger	227
Nigeria	234
Niue Island	683
Norfolk Island	672
N. Mariana Islands	670
North Korea	850
Northern Marianas	1670
Norway	47
Oman	968
Pakistan	92
Palau	680
Panama	507
Papua New Guinea	675
Paraguay	595
Peru	51
Philippines	63
Poland	48
Portugal	351
Puerto Rico	1787
Qatar	974
Reunion	262
Romania	40
Russian Federation	7
Rwanda	250
St Helena	290
St Kitts-Nevis	1869
St Lucia	1758
St Pierre/Miquelon	508
St Vincent/Grenadines	1784
Samoa (US)	684
Samoa (Western)	685
San Marino	378
Sao Tome/Principe	239
Saudi Arabia	966
Senegal	221
Seychelles	248
Sierra Leone	232
Singapore	65
Slovak Republic	421
Slovenia	386

Solomon Islands	677
Somali DR	252
S. Africa	27
South Korea	82
Spain	34
Sri Lanka	94
Sudan	249
Surinam	597
Swaziland	268
Sweden	46
Switzerland	41
Syria	963
Taiwan	886
Tajikistan	7
Tanzania	255
Thailand	66
Togo	228
Tokelau	690
Tonga	676
Trinidad & Tobago	1868
Tunisia	216
Turkey	90
Turkmenistan	993
Turks & Caicos Is.	1649
Tuvalu	688
Uganda	256
Ukraine	380
United Arab Emirates	971
United Kingdom	44
Uruguay	598
USA	1
Uzbekistan	7
Vanuatu	678
Vatican City	39
Venezuela	58
Vietnam	84
Virgin Is. (British)	1284
Western Samoa	685
Yemen AR	967
Yugoslavia	381
Zaire	243
Zambia	260
Zimbabwe	263

Telephone codes for contacting ships via Inmarsat

Atlantic Ocean Region (East)	871
Pacific Ocean Region	872
Indian Ocean Region	873
Atlantic Ocean Region (West)	874
Any Ocean Region	870

APPENDIX 6

International telex dialling codes

Afghanistan	79
Albania	604
Algeria	408
American Samoa	770
Andorra	590
Angola	991
Anguilla	391
Antigua and Barbuda	393
Argentina	33
Armenia	684
Aruba	303
Ascension Islands	939
Australia	71
Australian external territories	766
Austria	47
Azerbaijan	784
Bahamas	297
Bahrain	490
Bangladesh	780
Barbados	392
Belarus	681
Belgium	46
Benin	972
Bermuda	290
Bhutan	890
Bolivia	309
Bosnia	600
Botswana	962
Brazil	38
British Virgin Islands	292
Brunei	809
Bulgaria	67
Burkina Faso	978
Burundi	903
Cambodia	807
Cameroon	970
Canada	21
Canada (TWX)	26
Cape Verde Islands	993
Cayman Islands	293
Central America	37
Central African Republic	971
Chad	976
Chile	342/343/344/346
China	85
Christmas Island	766
Cocos Keeling Island	766
Colombia	35
Congo PR	981
Cook Island	772
Costa Rica	376
Croatia	599
Cuba	28
Cyprus	605
Czechoslovakian Republic	66
Denmark	55
Diego Garcia	938
Djibouti	979
Dominica	394
Dominican Republic	201/202
Ecuador	308
Egypt AR	91
El Salvador	373
Equatorial Guinea	999
Estonia	537
Ethiopia	980
Falkland Islands	306
Faeroe Islands	502
Fiji	701
Finland	957
France	42
French Polynesia	702
Gabon	973
Gambia	996
Georgia	683
Germany	41
Ghana	94
Gibraltar	405
Greece	601
Greenland	503
Grenada	395
Guadeloupe	299
Guam	700
Guatemala	372
Guiana	300
Guinea	995

Guinea Bissau	969
Guyana	295
Guyane Francaise	300
Haiti	203
Hawaii (RCA)	704
Hawaii (WI1)	708
Hawaii (WUH)	709
Hawaii (ITT)	705
Honduras	374
Hong Kong	802
Hungary	61
India	81
Indonesia	73
Iran	88
Iraq	491
Ireland	500
Israel	606
Italy	43
Ivory Coast	983
Jamaica	291
Japan	72
Kazakhstan	785
Kenya	987
Kiribati	761
Korea PDR	899
Korea Republic	801
Kuwait	496
Kyrgystan	788
Laos People's Dem. Rep.	804
Latvia	538
Lebanon	494
Lesotho	963
Liberia	997
Libya	901
Liechtenstein	45
Lithuania	539
Luxembourg	402
Macao	808
Macedonia	597
Madagascar	986
Madeira	404
Malawi	904
Malaysia	84
Maldive Islands	896
Mali	985
Malta	403/406
Marshall Islands	765
Martinique	298
Mauritania	974
Mauritius	966
Micronesia	764
Moldova	682
Monaco	42
Mongolia	800
Montserrat	396
Mozambique	992
Myanmar	83
Namibia	908
Nauru Island	775
Nepal	891
Netherlands	44
Netherlands Antilles	390
New Caledonia	706
New Zealand	74
Nicaragua	375
Niger	975
Nigeria	905
Niue Island	776
Northern Mariana Islands	760
Norway	56
Oman	498
Pakistan	82
Palau	763
Panama	379
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Paraguay	305
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Poland	63
Portugal	404
Puerto Rico (AACR)	206
Puerto Rico (MCI/WUI)	205
Qatar	497
Reunion	961
Romania	65
Russian Federation	64
Rwanda	909
St Helena	960
St Kitts and Nevis	397
St Lucia	398
St Pierre/Miquelon	204
St Vincent/Grenadines	399
Samoa (US)	770
Samoa (Western)	779
San Marino	505
Saudi Arabia	495
Senegal	906
Serbia and Montenegro	62

Seychelles	965
Sierra Leone	998
Singapore	87
Slovak Republic	66
Solomon Islands	778
Somalia DR	900
S. Africa	95/962/968
Spain	52
Sri Lanka	803
Sudan	984
Surinam	304
Swaziland	964
Sweden	54
Switzerland	45
Syria	492
Taiwan	769
Tajikistan	787
Tanzania	989
Togo	977
Tokelau	762
Tonga	777
Trinidad & Tobago	294
Tunisia	409
Turkey	607
Turkmenistan	789
Turks & Caicos Is.	296
Tuvalu	774
Uganda	988
Ukraine	680
United Arab Emirates	893
United Kingdom & N. Ireland	51
Uruguay	32
USA	23/230/255/231
Uzbekistan	786
Vanuatu	771
Vatican City	5042
Venezuela	31
Vietnam	805
Virgin Is. (British)	292
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Wallis & Futuna Islands	707
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List of country MIDs

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201	Albania
202	Andorra
203	Austria
204	Azores
205	Belgium
206	Belarus Republic
207	Bulgaria
208	Vatican City
209	Cyprus
210	Cyprus
211	Germany
212	Cyprus
213	Georgia
218	Germany
219	Denmark
224	Spain
227	France
228	France
230	Finland
231	Faeroe Islands
232	UK and N. Ireland
233	UK and N. Ireland
234	UK and N. Ireland
236	Gibraltar
237	Greece
238	Croatia
239	Greece
242	Morocco
243	Hungary
244	Netherlands
245	Netherlands
246	Netherlands
247	Italy
249	Malta
250	Ireland
251	Iceland
252	Liechtenstein
253	Luxembourg
254	Monaco
255	Madeira
256	Malta
257	Norway
258	Norway
259	Norway
261	Poland
263	Portugal
264	Romania
265	Sweden
268	San Marino
269	Switzerland
270	Czechoslovak Republic
271	Turkey
272	Ukraine
273	Russia
274	Former Macedonian Rep.
275	Latvia
276	Estonia
277	Lithuania
278	Slovenia
279	Yugoslavia
301	Anguilla
303	Alaska
304	Antigua and Barbuda
306	Netherlands Antilles
307	Aruba
308	Bahamas
309	Bahamas
310	Bermuda
312	Belize
314	Barbados
316	Canada
319	Cayman Islands
321	Costa Rica
323	Cuba
325	Dominican Commonwealth
327	Dominican Republic
329	Guadaloupe
330	Grenada
331	Greenland
332	Guatemala

MID	*Country*
334	Honduras
336	Haiti
338	Hawaii
339	Jamaica
341	St. Kitts & Nevis
343	Saint Lucia
345	Mexico
347	Martinique
348	Montserrat
350	Nicaragua
352	Panama
353	Panama
354	Panama
355	Panama
356	Panama
358	Puerto Rico
359	El Salvador
361	St. Pierre and Miquelon
362	Trinidad and Tobago
364	Turks and Caicos Islands
366	U.S.A.
367	U.S.A.
368	U.S.A.
376	St. Vincent and Grenadines
378	British Virgin Islands
379	U.S. Virgin Islands
401	Afghanistan
403	Saudi Arabia
405	Bangladesh
408	Bahrain
410	Bhutan
412	China
416	Taiwan
417	Sri Lanka
419	India
422	Iran
423	Azerbaijan
425	Iraq
428	Israel
431	Japan
438	Jordan
440	Korean Republic
445	People's Republic of Korea
447	Kuwait
450	Lebanon
453	Macao
455	Maldives
457	Mongolia
459	Nepal
461	Oman
463	Pakistan
466	Qatar
468	Syria
470	United Arab Emirates
473	Yemen Arab Republic
475	Yemen People's Republic
477	HongKong
501	Adelie Land
503	Australia
506	Myanmar (Burma)
508	Brunei Darussalem
510	Micronesia
511	Palau
512	New Zealand
514	Cambodia
516	Christmas Island
518	Cook Islands
201	Albania
202	Andorra
203	Austria
204	Azores
205	Belgium
206	Belarus Republic
207	Bulgaria
208	Vatican City
209	Cyprus
210	Cyprus
211	Germany
212	Cyprus
213	Georgia
214	Moldova
215	Malta
218	Germany
219	Denmark
220	Denmark
224	Spain
226	France
227	France
228	France
230	Finland
231	Faroe Islands
232	UK and N. Ireland
233	UK and N. Ireland
234	UK and N. Ireland
235	UK and N. Ireland
236	Gibraltar
237	Greece
238	Croatia
239	Greece
242	Morocco
243	Hungary

MID	*Country*
244	Netherlands
245	Netherlands
246	Netherlands
247	Italy
248	Malta
249	Malta
250	Ireland
251	Iceland
252	Liechtenstein
253	Luxemborg
254	Monaco
255	Madeira
256	Malta
257	Norway
258	Norway
259	Norway
261	Poland
263	Portugal
264	Romania
265	Sweden
267	Slovak Republic
268	San Marino
269	Switzerland
270	Czechoslovak Republic
271	Turkey
272	Ukraine
273	Russia
274	Former Macedonian Rep.
275	Latvia
276	Estonia
277	Lithuania
278	Slovenia
279	Yugoslavia
301	Anguilla
303	Alaska
304	Antigua and Barbuda
306	Netherlands Antilles
307	Aruba
308	Bahamas
309	Bahamas
310	Bermuda
311	Bahamas
312	Belize
314	Barbados
316	Canada
319	Cayman Islands
321	Costa Rica
323	Cuba
325	Dominican Commonwealth
327	Dominican Republic
329	Guadeloupe
330	Grenada
331	Greenland
332	Guatemala
334	Honduras
336	Haiti
338	Hawaii
339	Jamaica
341	St. Kitts & Nevis
343	Saint Lucia
345	Mexico
347	Martinique
348	Montserrat
350	Nicaragua
351	Panama
352	Panama
353	Panama
354	Panama
355	Panama
356	Panama
357	Panama
358	Puerto Rico
359	El Salvador
361	St. Pierre and Miquelon
362	Trinidad and Tobago
364	Turks and Caicos Islands
366	U.S.A.
367	U.S.A.
368	U.S.A.
369	U.S.A.
375	St. Vincent and Grenadines
376	St. Vincent and Grenadines
377	St. Vincent and Grenadines
378	British Virgin Islands
379	U.S. Virgin Islands
401	Afghanistan
403	Saudi Arabia
405	Bangladesh
408	Bahrain
410	Bhutan
412	China
413	China
416	Taiwan
417	Sri Lanka
419	India
422	Iran
423	Azerbaijan
425	Iraq
428	Israel
431	Japan
432	Japan
434	Turkmenistan

MID	*Country*
436	Kazakstan
438	Jordan
440	Korean Republic
441	Korean Republic
445	People's Republic of Korea
447	Kuwait
450	Lebanon
455	Maldives
457	Mongolia
459	Nepal
461	Oman
463	Pakistan
466	Qatar
468	Syria
470	United Arab Emirates
473	Yemen Arab Republic
475	Yemen People's Republic
477	Hong Kong
501	Adelie Land
503	Australia
506	Myanmar
508	Brunei Darussalem
510	Micronesia
511	Palau
512	New Zealand
514	Cambodia
515	Cambodia
516	Christmas Island
518	Cook Islands
520	Fiji
523	Cocos Keeling Islands
525	Indonesia
529	Kiribati
531	Lao People's Republic
533	Malaysia
536	Northern Mariana Islands
538	Marshall Islands
540	New Caledonia
542	Niue Island
544	Nauru
546	French Polynesia
548	Philppines
553	Papua New Guinea
555	Pitcairn Island
557	Solomon Islands
559	American Samoa
561	Samoa
563	Singapore
564	Singapore
567	Thailand
570	Tonga
572	Tuvalu
574	Viet Nam
576	Vanuata
578	Wallis and Futuna Islands
601	South Africa
603	Angola
605	Algeria
607	St. Paul and Amsterdam Is.
608	Ascension Island
609	Burundi
610	Benin
611	Botswana
612	Central African Republic
613	Cameroon
615	Congo
616	Comoros
617	Cape Verde
618	Crozet Archipelago
619	Cote d'Ivoire
621	Djibouti
622	Egypt
624	Ethiopia
625	Eritrea
626	Gabonese Republic
627	Ghana
629	Gambia
630	Guinea-Bissau
631	Equatorial Guinea
632	Guinea Republic
633	Burkina Faso
634	Kenya
635	Kerguelen Islands
636	Liberia
642	Libya
644	Lesotho
645	Mauritius
647	Madagascar
649	Mali
650	Mozambique
654	Mauritania
655	Malawi
656	Niger
657	Nigeria
659	Namibia
660	Reunion
661	Rwandese Republic
662	Sudan
663	Senegal
664	Seychelles
665	St. Helena
666	Somali Democratic Republic

MID	*Country*
667	Sierra Leone
668	Sao Tome and Principe
669	Swaziland
670	Chad
671	Togolese Republic
672	Tunisia
674	Tanzania
675	Uganda
676	Republic of Congo
677	Tanzania
678	Zambia
679	Zimbabwe
701	Argentine Republic
710	Brazil
720	Bolivia
725	Chile
730	Colombia
735	Ecuador
740	Falkland Islands (Malvinas)
745	Guiana
750	Guyana
755	Paraguay
760	Peru
765	Suriname
770	Uruguay
775	Venezuela

IN ORDER BY COUNTRY

Country	*MID*	*Country*	*MID*
Adelie Land	501	Bulgaria	207
Afghanistan	401	Burkina Faso	633
Alaska	303	Burundi	609
Albania	201	Cambodia	514
Algeria	605	Cambodia	515
American Samoa	559	Cameroon	613
Andorra	202	Canada	316
Angola	603	Cape Verde	617
Anguilla	301	Cayman Islands	319
Antigua and Barbuda	304	Central African Republic	612
Argentine Republic	701	Chad	670
Aruba	307	Chile	725
Ascension Island	608	China	412
Australia	503	China	413
Austria	203	Christmas Island	516
Azerbaijan	423	Cocos Keeling Islands	523
Azores	204	Colombia	730
Bahamas	308	Comoros	616
Bahamas	309	Congo	615
Bahamas	311	Congo Democratic Republic	676
Bahrain	408	Cook Islands	518
Bangladesh	405	Costa Rica	321
Barbados	314	Cote d'Ivoire	619
Belarus Republic	206	Croatia	238
Belgium	205	Crozet Archipelago	618
Belize	312	Cuba	323
Benin	610	Cyprus	209
Bermuda	310	Cyprus	210
Bhutan	410	Cyprus	212
Bolivia	720	Czechoslovak Republic	270
Botswana	611	Denmark	219
Brazil	710	Denmark	220
British Virgin Islands	378	Djibouti	621
Brunei Darussalem	508	Dominican Commonwealth	325

Country	*MID*	*Country*	*MID*
Dominican Republic	327	Kazakstan	436
Ecuador	735	Kenya	634
Egypt	622	Kerguelen Islands	635
El Salvador	359	Kiribati	529
Equatorial Guinea	631	Korean People's Republic	445
Eritrea	625	Korean Republic	440
Estonia	276	Korean Republic	441
Ethiopia	624	Kuwait	447
Falkland Islands (Malvinas)	740	Lao People's Republic	531
Faroe Islands	231	Latvia	275
Fiji	520	Lebanon	450
Finland	230	Lesotho	644
Former Macedonian Rep.	274	Liberia	636
France	226	Libya	642
France	227	Liechtenstein	252
France	228	Lithuania	277
French Polynesia	546	Luxemborg	253
Gabonese Republic	626	Madagascar	647
Gambia	629	Madeira	255
Georgia	213	Malawi	655
Germany	211	Malaysia	533
Germany	218	Maldives	455
Ghana	627	Mali	649
Gibraltar	236	Malta	215
Greece	237	Malta	248
Greece	239	Malta	249
Greenland	331	Malta	256
Grenada	330	Marshall Islands	538
Guadeloupe	329	Martinique	347
Guatemala	332	Mauritania	654
Guiana	745	Mauritius	645
Guinea Republic	632	Mexico	345
Guinea-Bissau	630	Micronesia	510
Guyana	750	Moldova	214
Haiti	336	Monaco	254
Hawaii	338	Mongolia	457
Honduras	334	Montserrat	348
Hong Kong	477	Morocco	242
Hungary	243	Mozambique	650
Iceland	251	Myanmar	506
India	419	Namibia	659
Indonesia	525	Nauru	544
Iran	422	Nepal	459
Iraq	425	Netherlands	244
Ireland	250	Netherlands	245
Israel	428	Netherlands	246
Italy	247	Netherlands Antilles	306
Jamaica	339	New Caledonia	540
Japan	431	New Zealand	512
Japan	432	Nicaragua	350
Jordan	438	Niger	656

Country	*MID*	*Country*	*MID*
Nigeria	657	St. Kitts & Nevis	341
Niue Island	542	St. Paul and Amsterdam Is.	607
Northern Mariana Islands	536	St. Pierre and Miquelon	361
Norway	257	St. Vincent and Grenadines	375
Norway	258	St. Vincent and Grenadines	376
Norway	259	St. Vincent and Grenadines	377
Oman	461	Sudan	662
Pakistan	463	Suriname	765
Palau	511	Swaziland	669
Panama	351	Sweden	265
Panama	352	Switzerland	269
Panama	353	Syria	468
Panama	354	Taiwan	416
Panama	355	Tanzania	674
Panama	356	Tanzania	677
Panama	357	Thailand	567
Papua New Guinea	553	Togolese Republic	671
Paraguay	755	Tonga	570
Peru	760	Trinidad and Tobago	362
Philppines	548	Tunisia	672
Pitcairn Island	555	Turkey	271
Poland	261	Turkmenistan	434
Portugal	263	Turks and Caicos Islands	364
Puerto Rico	358	Tuvalu	572
Qatar	466	U.S. Virgin Islands	379
Reunion	660	U.S.A.	366
Romania	264	U.S.A.	367
Russia	273	U.S.A.	368
Rwandese Republic	661	U.S.A.	369
Saint Lucia	343	Uganda	675
Samoa	561	UK and N. Ireland	232
San Marino	268	UK and N. Ireland	233
Sao Tome and Principe	668	UK and N. Ireland	234
Saudi Arabia	403	UK and N. Ireland	235
Senegal	663	Ukraine	272
Seychelles	664	United Arab Emirates	470
Sierra Leone	667	Uruguay	770
Singapore	563	Vanuata	576
Singapore	564	Vatican City	208
Slovak Republic	267	Venezuela	775
Slovenia	278	Viet Nam	574
Solomon Islands	557	Wallis and Futuna Islands	578
Somali Democratic Republic	666	Yemen Arab Republic	473
South Africa	601	Yemen People's Republic	475
Spain	224	Yugoslavia	279
Sri Lanka	417	Zambia	678
St. Helena	665	Zimbabwe	679

APPENDIX8

GMDSS distress and safety frequencies

Band	*DSC*	*RT*	*Telex*
MF	2 187.5	2 182.0	2 174.5
HF 4 MHz	4 207.5	4 125.0	4 177.5
HF 6 MHz	6 312.0	6 215.0	6 268.0
HF 8 MHz	8 414.5	8 291.0	8 376.5
HF 12 MHz	12 577.0	12 290.0	12 520.0
HF 16 MHz	16 804.5	16 420.0	16 695.0
VHF	Channel 70	Channel 16	No facility

APPENDIX 9

List of web sites

Organisation	Web address
Admiralty Lists of Radio Signals	*www.ukho.gov.uk*
Association of Marine Electronic and Radio Colleges (AMERC)	*www.amerc.ac.uk*
Australia, GMDSS site	*www.gmdss.com.aus*
BT	*www.globalmobile.bt.com*
Compuship, (magazine)	*www.compuship.net*
COSPAS-SARSAT, London	*www.cospas-sarsat.org*
GLONASS satellite information	*www. rssi. ru/SFCSIC/SFCIC_ main.html*
Hydrographic Office	*www.ukho.gov.uk*
International Chamber of Commerce. (Gives weekly piracy reports)	*www.icc-ccs.org*
International Maritime Organisation	*www.imo.org*
Inmarsat	*www.inmarsat.org*
Ionospheric predictions	*www.radio-portal.org*
International Telecommunication Union, Geneva	*www.itu.int*
Maritime Coastguard Agency ("M" notices etc.)	*www.mcga.gov.uk*
Meteorological Office	*www. metoffice.gov. uk*
Nautical Institute	*www.nautinst.org*
NELS Coordinating Agency, Norway	*www.odin.dep.no/fid/publ/nels*
Office of Communication	*www.ofcom.org.uk*
Radio Surveys (bookings)	*www.marconi-marine.com*
Safety at Sea (magazine)	*www.dmgworldmedia.com*
	www.sas-intl.com
Ship Radio Licensing Centre	*www.radiolicencecentre.co.uk*
US Coastguard	*www.navcen.uscg.mil*

Equipment manufacturers

ACR Electronics Inc	*www.acrelectronics.com*
Avmar	*www.avmar.co.uk*
Comast Mobile Communications	*www.comsat.com*
Digital Antena	*www.digitalantenna.com*
EMS Technologies	*www.emssatcom.com*
Furuno USA	*www.furunousa.com*
Icom America	*www.icomamerica.com*

ICC	*www.iccwbo.org*
ICS	*www.icselectronics.co.uk*
Innovation Technology Ltd (406 MHz EPIRB guard receiver supplier)	*www.stanguard.com*
IridiumJapanese Radio Corporation (JRC)	*ww.jrcamerica.com*
Jotron	*www.jotron.com*
K.Hughes	*www.kelvinhughes.co.uk*
Maritel	*www.mariteline.com*
McMurdo	*www.mcmurdo.co.uk*
Nera, Norway	*www.nera.no*
Northern Airborne Technologies	*www.nat-inc.com*
National Marine Electronics Association	*www.nmea.org*
Pains-Wessex Safety Systems	*www.pwss.org*
Prosar Technologies	*www.prosartech.com*
Raymarine (Light Marine)	*www.raymarine.com*
Raytheon High Seas	*www.raytheonmarine.de*
Ross Engineering	*www.rossdsc.com*
Sailor (SP Radio ASF)	*www.sailor.dk*
Satellite Phones	*www. iridium.com*
Sea and Datamarine	*www.sea-dmi.com*
Skanti, Denmark	*www.skanti.dk*
Sperry Marine	*www.sperry-marine.com*
Standard Horizon	*www.yaesuusa.com*
Stratos	*www.stratos.ca*
Thrane and Thrane	*www.tt.dk*
Tideland signal	*www.tidelandsignal.com.sg*
Transas Navigation	*www.transas.com*
Trimble Navigation	*www.trimble.com*

Land Earth Station Operators

Eik, Norway	*www.eik.com*
Frence Telecom	*www.inmarsat.francetelecom.fr*
Greece	*www.otesat.gr*
Australia	*www.telsta.com.au*
Sentosa, Singapore Telecoms Company	*www.singtel.com*

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